Transformer-Based Modeling for Speech Quality Prediction in Telecommunications

vorgelegt von
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Measure what can be measured, and make measurable what cannot be measured.

– Galileo Galilei

Abstract

This thesis investigates the use of pretrained transformer-based foundation models for non-intrusive speech quality prediction, with a primary focus on telecommunications applications. Speech quality is modeled as a multidimensional perceptual construct comprising overall quality and the established perceptual dimensions of noisiness, discontinuity, coloration, and loudness. Traditionally, speech quality assessment has relied on subjective listening tests; over time, this led to the development of instrumental reference-based and signal-processing-based methods, and more recently to machine-learning-based models that aim to approximate human perception.

Motivated by the success of transformer-based foundation models across a wide range of speech and audio tasks, this work examines their applicability to non-intrusive speech quality assessment. Five widely used pretrained models—wav2vec 2.0, HuBERT, WavLM, Whisper, and the Audio Spectrogram Transformer (AST)—are first evaluated in a zero-shot setting to analyze the extent to which their learned representations encode perceptually relevant information. Based on this analysis, AST is selected as the backbone for further model development.

Two modeling strategies are subsequently explored: a multi-task framework that jointly predicts overall speech quality and all four perceptual dimensions, and a single-target (dimension-specific) approach in which each perceptual attribute is modeled independently. To support robust training and evaluation, several new speech quality databases are introduced to expand the coverage of perceptual degradations. The proposed models are evaluated across a large and diverse set of speech quality databases and are further subjected to independent external evaluation within the context of the International Telecommunication Union (ITU-T) standardization process for the P.SAMD work item.

Finally, the dimension-specific models are applied to an out-of-domain task involving the assessment of quality and naturalness of synthetic speech, demonstrating their broader applicability. The results highlight both the potential and the limitations of foundation models for perceptually aligned speech quality prediction and identify directions for future research.

Zusammenfassung

Diese Dissertation untersucht den Einsatz vortrainierter, transformerbasierter Foundation-Modelle für die nicht-intrusive Sprachqualitätsvorhersage mit einem Schwerpunkt auf Anwendungen in der Telekommunikation. Sprachqualität wird dabei als ein multidimensionales perzeptives Konstrukt modelliert, das sich aus der Gesamtqualität sowie den etablierten perzeptiven Dimensionen Rauschhaftigkeit (Noisiness), Diskontinuität, Klangfärbung (Coloration) und Lautheit zusammensetzt. Traditionell basierte die Sprachqualitätsbewertung auf subjektiven Hörtests; im Laufe der Zeit wurde sie durch instrumentelle referenzbasierte und signalverarbeitungsbasierte Verfahren ergänzt und in jüngerer Zeit durch maschinelle Lernverfahren erweitert, die darauf abzielen, die menschliche Wahrnehmung zu approximieren.

Motiviert durch den Erfolg transformerbasierter Foundation-Modelle in einer Vielzahl von Sprach- und Audiowiedergabeaufgaben untersucht diese Arbeit deren Anwendbarkeit für die nicht-intrusive Sprachqualitätsbewertung. Zunächst werden fünf weit verbreitete vortrainierte Modelle — wav2vec 2.0, HuBERT, WavLM, Whisper und der Audio Spectrogram Transformer (AST) — in einer Zero-Shot-Evaluation analysiert, um zu untersuchen, in welchem Umfang ihre gelernten Repräsentationen perzeptiv relevante Informationen kodieren. Auf Basis dieser Analyse wird AST als Grundlage für die weitere Modellentwicklung ausgewählt.

Anschließend werden zwei Modellierungsstrategien untersucht: ein Multi-Task-Ansatz, der die Gesamtsprachqualität sowie alle vier perzeptiven Dimensionen gemeinsam vorhersagt, sowie ein Single-Target-Ansatz (dimensionsspezifisch), bei dem jede perzeptive Eigenschaft unabhängig modelliert wird. Zur Unterstützung einer robusten Trainings- und Evaluationspipeline werden mehrere neue Sprachqualitätsdatenbanken eingeführt, um die Abdeckung perzeptiver Beeinträchtigungen zu erweitern. Die vorgeschlagenen Modelle werden auf einer großen und heterogenen Sammlung von Sprachqualitätsdatenbanken evaluiert und zusätzlich einer unabhängigen externen Bewertung im Rahmen des Standardisierungsprozesses der International Telecommunication Union (ITU-T) für das Arbeitselement P.SAMD unterzogen.

Abschließend werden die dimensionsspezifischen Modelle auf eine domänenfremde Aufgabe zur Bewertung der Qualität und Natürlichkeit synthetischer Sprache angewendet, wodurch ihre weitergehende Anwendbarkeit demonstriert wird. Die Ergebnisse verdeutlichen sowohl das Potenzial als auch die Grenzen von Foundation-Modellen für eine perzeptiv ausgerichtete Sprachqualitätsvorhersage und zeigen vielversprechende Richtungen für zukünftige Forschung auf.

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Finally, I would like to express my deepest gratitude to my family and friends for their unwavering support, patience, and encouragement throughout this journey. Their belief in me made this work possible.

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Acronyms

ACR Absolute Category Rating
AMR Adaptive Multi-Rate Codec
AMR-NB Adaptive Multi-Rate Narrowband Codec
AMR-WB Adaptive Multi-Rate Wideband Codec
ANIQUE Auditory Non-intrusive Quality Estimation
ASR Automatic Speech Recognition
AST Audio Spectrogram Transformer
BERT Bidirectional Encoder Representations from Transformers
BGN Background Noise
CI Confidence Interval
COL Coloration
DIAL Diagnostic Instrumental Assessment of Listening Quality
DIS Discontinuity
DTLN Deep Two-Stage Loss Network
EVS Enhanced Voice Services
FB Fullband
FM Foundation Model
HuBERT Hidden-Unit Bidirectional Encoder Representations from Transformers
ITU International Telecommunication Union
ITU-T International Telecommunication Union – Telecommunication Standardization Sector
LLM Large Language Model
LOT Listening Only Test
ML Machine Learning
MOS Mean Opinion Score
NB Narrowband
NISQA Non-Intrusive Speech Quality Assessment
NOI Noisiness
NR Noise Reduction
P.SAMD Single-Ended Assessment of Multiple Dimensions
PCC Pearson Correlation Coefficient
PESQ Perceptual Evaluation of Speech Quality
PL Packet Loss
POLQA Perceptual Objective Listening Quality Assessment

PSQM	Perceptual Speech Quality Measurement
QoE	Quality of Experience
QoS	Quality of Service
RMSE	Root Mean Square Error
RMSE*	Epsilon-adjusted Root Mean Square Error
SSL	Self-Supervised Learning
TOSQA	Telecommunications Objective Speech Quality Assessment
TTS	Text-to-Speech
VC	Voice Conversion
ViSQOL	Virtual Speech Quality Objective Listener
WB	Wideband

Chapter 1
Introduction

1.1 Motivation

Speech communication systems are deeply embedded in modern life, supporting everyday interactions through mobile telephony, internet-based conferencing, and an increasing range of synthetic and assistive speech technologies. In the aftermath of the COVID-19 pandemic, reliance on telecommunication systems has increased substantially, with remote work, education, and social interaction becoming persistent components of daily life. As a result, speech technologies now mediate a significant portion of human communication at scale.

In this context, the perceived quality of transmitted speech has become a critical factor influencing intelligibility, naturalness, listener comfort, and overall user experience. Even moderate degradations in speech quality can negatively affect communication efficiency and increase cognitive listening effort, particularly in prolonged or professional interactions. Consequently, the reliable assessment and prediction of speech quality remains a central challenge in the design, evaluation, and deployment of contemporary speech communication systems.

The established gold standard for assessing speech quality remains subjective listening tests, in which human listeners rate degraded signals using the Mean Opinion Score (MOS) or perceptual dimensions such as noisiness, coloration, discontinuity, and sub-optimal loudness (Wältermann, 2012). While reliable, these tests are costly, time-consuming, and unsuitable for continuous or large-scale monitoring.

To overcome these limitations, objective instrumental measures have been developed and standardized. Reference-based methods such as ITU-T Recommendation P.863 (2018) (POLQA) provide accurate quality predictions but require access to a clean reference signal, which is unavailable in most real-world applications. Non-intrusive approaches such as ITU-T Recommendation P.563 (2004) eliminate this requirement, yet are limited in bandwidth, predictive accuracy, and diagnostic resolution. More recently, deep learning-based models such as NISQA (Mittag et al., 2021) have demonstrated improved non-intrusive performance, but they remain sensitive to domain shifts and lack standardized robustness across heterogeneous datasets and conditions.

In parallel, the field of machine learning (ML) has undergone a paradigm shift from narrowly task-specific models toward large transformer-based foundation models that are pretrained on broad, non-domain-specific data across modalities such as text, audio, and images. These models are typically trained using multiple self-supervised or

weakly supervised objectives and released as general-purpose representations that can be adapted to a wide range of downstream tasks with limited task-specific modification. Prominent examples include large language models (LLMs) trained on diverse text corpora like BERT (Devlin et al., 2019), foundation models for speech and audio processing like wav2vec (Baevski et al., 2020b), HuBERT (Hsu et al., 2021), WavLM (Chen et al., 2021), AST (Gong et al., 2021), and Whisper (Radford et al., 2023), as well as multimodal models trained jointly on vision and language data (Radford et al., 2021). In speech and language processing, such models have shown strong transferability across tasks including language translation, speech recognition, speaker characterization, and paralinguistic analysis. This development raises a central question for speech quality assessment: whether general-purpose speech representations can be leveraged to achieve robust, non-intrusive, and perceptually meaningful quality prediction across diverse speech communication systems and conditions.

1.2 Objectives and Research Questions

The objective of this work is to develop a robust, non-intrusive speech quality assessment model based on transformer-based pretrained foundation models. The proposed approach aims to predict both overall speech quality and established perceptual dimensions, namely noisiness, discontinuity, coloration, and sub-optimal loudness.

Within the current landscape of ML and speech technologies, this objective gives rise to the following research questions:

- **RQ1:** How can transformer-based pretrained foundation models be adapted for accurate and reliable non-intrusive prediction of overall speech quality?

- **RQ2:** To what extent can foundation model representations support the prediction of perceptual speech quality dimensions, and how consistently do these predictions reflect subjective human judgments?

- **RQ3:** Does joint multi-task learning of overall quality and perceptual dimensions improve prediction accuracy and robustness compared to single-task modeling?

- **RQ4:** How well do foundation model-based speech quality predictors generalize across heterogeneous datasets, domains, and recording conditions, including independent hidden datasets used in ITU-T standardisation?

- **RQ5:** Can a foundation model-based speech quality predictor be effectively transferred to out-of-domain assessment tasks without extensive retraining or task-specific engineering?

1.3 Thesis Structure

The remainder of this thesis is structured as follows:

Chapter 2 provides the background on speech quality assessment, covering both human perceptual evaluation and objective instrumental methods. It reviews the state of the art in reference-based and non-intrusive models and outlines the evolution toward data-driven approaches. In addition, it introduces relevant deep learning architectures, with a focus on transformer-based encoders and pretrained foundation models for speech processing. This chapter establishes the conceptual and methodological foundations for all subsequent research questions.

Chapter 3 describes the methodological framework of this work. It details the speech quality databases used, the criteria and process for selecting suitable model architectures, and the evaluation methodology employed throughout the thesis. This chapter supports the experimental investigation of all the research questions.

Chapter 4 investigates the applicability of pretrained foundation models to speech quality assessment in a zero-shot transfer learning setting. It compares multiple foundation models on speech quality prediction tasks and analyzes their strengths and limitations. The findings of this chapter primarily address RQ1 and RQ4, and motivate the selection of a candidate model for further adaptation.

Chapter 5 focuses on adapting the selected foundation model for multi-task speech quality prediction. The proposed model jointly predicts overall quality (MOS) and the perceptual dimensions of noisiness, discontinuity, coloration, and sub-optimum loudness. This chapter addresses RQ2 and RQ3 by evaluating perceptual alignment and the benefits of joint modeling.

Chapter 6 examines single-task modeling as an alternative to the multi-task approach. It analyzes whether task-specific optimization improves predictive performance and robustness compared to joint modeling, thereby further informing RQ3.

Chapter 7 evaluates the transferability of the proposed speech quality prediction framework to out-of-domain applications. Specifically, it investigates model performance on text-to-speech naturalness assessment and on the quality prediction of synthetic speech in voice conversion scenarios. This chapter primarily addresses RQ5.

Finally, Chapter 8 summarizes the main findings and contributions of this work, discusses its limitations, and outlines directions for future research, with reference to all the research questions.

1.4 Prior Contributions

Over the course of this doctoral research, I contributed to both the academic and standardization communities in the area of speech quality assessment:

- ITU-T Recommendation (in preparation): *Single-ended Assessment of Multiple Dimensions (P.SAMD)*. Since 2022, I have contributed multiple technical submissions to the corresponding ITU-T Study Group 12 work item and discussed them during the ITU-T Study Group 12 meetings in Geneva, Switzerland. The speech quality

model developed in this work was evaluated within the ITU-T standardisation process and selected as the basis for the forthcoming Recommendation. At the time of writing, the Recommendation text is under preparation within Study Group 12, where I contribute as one of the main authors.

– **ITU-T Contribution SG12-C022 (2022).** *On Non-intrusive Speech Quality Assessment Models.* Technical report, International Telecommunication Union (ITU-T), Geneva, Switzerland.

  This contribution shared the outcome of the *Non-Intrusive Objective Speech Quality Assessment* track of the ConferencingSpeech 2022 Challenge, held in conjunction with Interspeech 2022, where I was responsible for preparing the baseline models. I discussed the submitted models' relevance to the P.SAMD work item, and identified transfer learning as a promising direction for non-intrusive speech quality assessment.

– **ITU-T Contribution SG12-C023 (2022).** *Some Considerations on How to Finalize P.SAMD.* Technical report, International Telecommunication Union (ITU-T), Geneva, Switzerland.

  In this contribution, I helped extract and consolidate key methodological insights derived from the ConferencingSpeech 2022 Challenge results. I translated these findings into concrete technical considerations and proposals to support the finalization of the P.SAMD study item.

– **ITU-T Contribution SG12-C082 (2023).** *Update on the Ongoing Work on P.SAMD.* Technical report, International Telecommunication Union (ITU-T), Geneva, Switzerland.

  In this work, I investigated and demonstrated the feasibility of using a pretrained wav2vec 2.0 model for non-intrusive prediction of overall speech quality. I designed and conducted evaluations on held-out test databases to assess generalization performance, thereby providing empirical evidence for the suitability of pretrained foundation models as candidate components for P.SAMD.

– **ITU-T Contribution SG12-C151 (2023).** *Update and Proposal on the P.SAMD Study Item.* Technical report, International Telecommunication Union (ITU-T), Geneva, Switzerland.

  In this contribution, I collaborated with multiple parties to align on the design of three new speech quality databases intended for training and testing candidate P.SAMD models. My responsibilities included contributing to the definition of dataset requirements and use cases to support robust model development and evaluation.

– **ITU-T Contribution SG12-C239 (2024).** *Progress of P.SAMD Model.* Technical report, International Telecommunication Union (ITU-T), Geneva, Switzerland.

  In this work, I led the fine-tuning of a pretrained wav2vec 2.0 model for non-intrusive prediction of overall speech quality and the perceptual dimensions of noisiness, discontinuity, coloration, and loudness. While the achieved performance did not yield substantial improvements over the baseline model, the analysis informed subsequent architectural choices and motivated the investigation of Audio Spectrogram Transformer (AST)–based models in future work.

– **ITU-T Contribution SG12-TD605 (2024).** *Detailed Progress of P.SAMD model.* Technical report, International Telecommunication Union (ITU-T), Geneva, Switzerland.

  In this work, I investigated the performance of the baseline NISQA, wav2vec 2.0, AST, and an ensemble of all three models for non-intrusive prediction of overall speech quality and the perceptual dimensions. It was seen that while the ensemble often outperformed the individual models, the AST model was sometimes better than the ensemble.

– **ITU-T Contribution SG12-C251 (2024).** *Finalising the P.SAMD Model.* Technical report, International Telecommunication Union (ITU-T), Geneva, Switzerland.

  In this contribution I presented the initial results of fine-tuning the pretrained AST model to predict the overall speech quality, investigating the order of using high-quality subjective databases and instrumentally-labelled datasets.

– **ITU-T Contribution SG12-C277 (2024).** *P.SAMD Databases.* Technical report, International Telecommunication Union (ITU-T), Geneva, Switzerland.

  In this contribution, I compiled and documented the databases used for training and validation of the P.SAMD model.

– **ITU-T Contribution SG12-C035 (2025).** *P.SAMD Model Description and Performance.* Technical report, International Telecommunication Union (ITU-T), Geneva, Switzerland.

  In this contribution, I presented the AST-based speech quality model that predicts the overall speech quality and the four perceptual dimensions. Each target dimension was treated as an independent task and five AST-based modules were trained for each task.

– **ITU-T Contribution SG12-C044 (2025).** *Proposal for New Work Items Related to the Quality of Speech Generated with Machine Learning Techniques.* Technical report, International Telecommunication Union (ITU-T), Geneva, Switzerland.

In this contribution, I helped in proposing for new work to be performed in SG12 on the quality of speech generated with machine-learning techniques, especially techniques based on large generative models, such as text-to-speech synthesis, speech coding, speech enhancement, speaker transformation, and anonymization.

– **ITU-T Contribution SG12-C076 (2025).** *Draft Recommendation for P.SAMD Work Item.* Technical report, International Telecommunication Union (ITU-T), Geneva, Switzerland.

In this contribution, I led the preparation of the ITU-T Recommendation draft document towards the standardization of the P.SAMD model.

– **ITU-T Contribution SG12-C078 (2025).** *Progress of the P.SAMD Candidate Model Towards Standardization.* Technical report, International Telecommunication Union (ITU-T), Geneva, Switzerland.

In this contribution, I reported and analyzed the results of multiple fine-tuning strategies investigated to improve the performance of an AST-based speech quality prediction model. Based on independent internal evaluations, the selected model demonstrated promising performance, and the findings informed proposals by partnering parties for the development of new test databases.

- **Wardah, W.**, Lobato, T., Schäfer, M., Spang, R., and Möller, S. (2026). *On the Perceptual Relevance of the Lombard Effect for Speech Quality Assessment*. Proceedings of the 52nd Annual Conference on Acoustics (DAGA).

  This work investigates how Lombard speech affects perceived speech quality, with implications for both subjective assessment and objective quality modeling. The study will be presented at the DAGA 2026 conference in Dresden, Germany. I conceptualised this study and designed the experiment fully. I designed the speech material and led my student group to create recordings of 6 speakers under various conditions, including the presence and absence of noise and lombard effect in the speech. I then conducted a listening test with 25 participants, obtaining overall quality, loudness, noisiness, and listening effort ratings. I wrote the paper and present the findings at the conference.

- **Wardah, W.**, Spang, R., Barriac, V., Reimes, J., Llagostera, A., Berger, J., and Möller, S. (Wardah et al., 2025b). *SQ-AST: A Transformer-Based Model for Speech Quality Prediction*. Proceedings of *Interspeech 2025*, pp. 2335–2339.

  The work was presented and discussed at the Interspeech conference in Rotterdam, Netherlands. In this work, I carried out the model development task, wrote the paper, and presented at the conference. My colleagues provided the test datasets and conducted the evaluation of the model.

- **Wardah, W.**, Vergari, M., Spang, R., and Möller, S. (Wardah et al., 2025c). *Perceptual Drift in Speech Quality: Re-assessing Listener Judgments a Decade Later*. Proceedings of the 51st Annual Conference on Acoustics (DAGA), pp. 1287–1290.

  This work investigates long-term changes in subjective speech quality judgments by comparing contemporary listener ratings with historical data. In this study, I conceptualized the experiment and conducted the subjective listening test. I was also responsible for writing the paper, and presenting this work at the DAGA 2025 conference in Copenhagen, Denmark.

- **Wardah, W.**, Büyüktaş, T., Shchegelskiy, K., Möller, S., and Spang, R. (Wardah et al., 2025a). *Language Barriers: Evaluating Cross-Lingual Performance of CNN and Transformer Architectures for Speech Quality Estimation*. Proceedings of the ISCA/ITG Workshop on Diversity in Large Speech and Language Models.

  This work examines the cross-lingual generalization of convolutional and transformer-based models for non-intrusive speech quality prediction. I co-conceptualized the study and prepared the models for training and evaluation. The study was presented as a poster at the ISCA/ITG Workshop, in Berlin, Germany.

- **Wardah, W.**, Spang, R., Schmitt, V., and Möller, S. (Wardah et al., 2024). *Dissecting Speech Quality: Investigating the Interplay Between Overall Quality and Dimensional Perceptions*. Proceedings of the 50th Annual Conference on Acoustics (DAGA), pp. 1444–1446.

  This work analyzes the relationship between overall speech quality judgments and individual perceptual dimensions, providing empirical insight into their interaction. I conceptualized this study, then wrote the paper and presented it at the DAGA 2024 conference in Hannover, Germany.

- **Wardah, W.** and Möller, S. (Wardah and Möller, 2023). *Predicting Perceived Noisiness Ratings in Speech Signals Using Transformer-Based Pretrained Language Models*. Proceedings of the 49th Annual Conference on Acoustics (DAGA), pp. 546–548.

  This work presents an early investigation into the use of transformer-based pretraining. I conducted this study, wrote the paper, and presented the findings at the DAGA 2023 conference in Hamburg, Germany.

- Yi, G., Xiao, W., Xiao, Y., Naderi, B., Möller, S., **Wardah, W.**, Mittag, G., Cutler, R., Zhang, Z., Williamson, D. S., Chen, F., Yang, F., and Shang, S. (Yi et al., 2022). *Conferencingspeech 2022 Challenge: Non-intrusive Objective Speech Quality Assessment (NISQA) Challenge for Online Conferencing Applications*. Proceedings of *Interspeech 2022*, pp. 3308–3312.

  I contributed as a co-author and member of the organizing committee, and was

responsible for the development of the baseline models used in the Conferencingspeech 2022 Challenge. The work was presented in Incheon, Korea.

- Schuh, B, **Wardah, W.**, Naderi, B., Michael, T., and Moeller, S. (Schuh et al., 2023). *Hearing Impairment in Crowdsourced Speech Quality Assessments: Its Effect and Screening with Digit Triplet Hearing Test*. Speech Communication. Proceedings of the 15th ITG Conference, in Aachen, Germany.

  In this study I helped plan the experiment and then structured and contributed to writing the paper with my coauthors.

- Pirklbauer, J., Sach, M., Fluyt, K., Tirry, W., **Wardah, W.**, Möller, S., and Fingscheidt, T. (Pirklbauer et al., 2023).*Evaluation Metrics for Generative Speech Enhancement Methods: Issues and Perspectives*. Speech Communication; Proceedings of the 15th ITG Conference, in Aachen, Germany.

  In this work, I contributed to the conceptualization of the problem and to the development of ideas for defining appropriate metrics to assess the quality of synthetic speech.

Chapter 2
State of the Art

This chapter introduces the scientific fundamentals of speech quality and reviews the technological state-of-the-art in speech quality assessment and modeling. It begins with the principles of human perception of speech quality and the methods used to capture perceptual judgments in subjective listening tests. The chapter then presents the evolution of objective instrumental approaches, from traditional reference-based methods to non-intrusive models. Next, it surveys the transition toward data-driven and deep learning-based methods, including recent advances in transformer-based foundation models for learning transferable speech representations. The chapter concludes by identifying open challenges and limitations in current approaches, thereby motivating the methodological framework and model development presented in Chapter 3.

2.1 Speech Quality Preliminaries

This section establishes the conceptual foundations required for the subsequent discussion of speech quality assessment methods. It covers human perception of speech quality, transmission-related impairments, and the principles of subjective listening-only tests (LOTs). The section also introduces perceptual speech quality dimensions beyond overall quality ratings.

2.1.1 Human Perception of Speech Quality

Human assessment of speech quality emerges from a sequence of auditory and cognitive processes that together shape a listener's overall judgment. As described in psychoacoustic theory, listeners do not perceive speech as a set of isolated physical signals, but as auditory events formed through internal perceptual processing (Blauert, 1997; Zwicker and Fastl, 1999). The resulting percept is evaluated against listener-specific reference points, which are shaped by prior experience and current expectations. Consequently, speech quality cannot be observed directly from the acoustic signal itself, but is accessed through subjective descriptions and judgments provided by listeners. This conceptual distinction between physical sound events, internal auditory representations, and descriptive responses is illustrated in Figure 2.1.

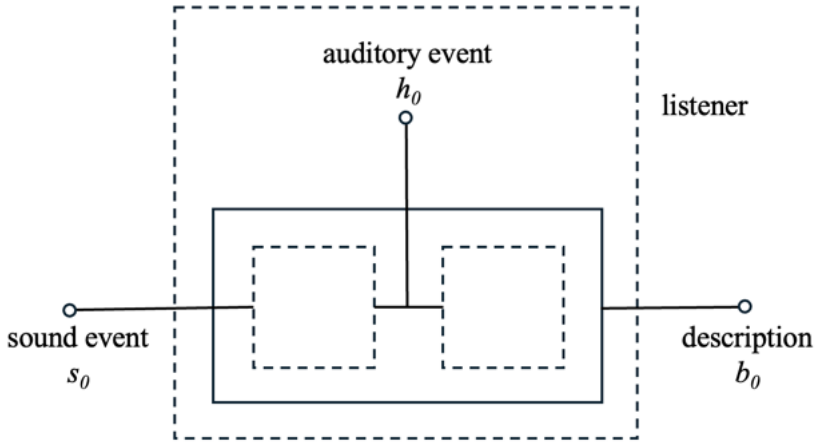


Fig. 2.1: Conceptual model of auditory perception adapted from Blauert (1997). A physical sound event s_0 gives rise to an internal auditory event h_0 within the listener, shaped by auditory processing and cognitive interpretation. The perceived event is subsequently described or judged by the listener as b_0. The dashed boundaries indicate that the auditory event and its interpretation are internal to the listener and not directly observable, highlighting the distinction between physical stimulus, perceptual representation, and descriptive response.

In telecommunications, subjective judgments of speech quality are commonly framed within the concept of Quality of Experience (QoE), which characterizes a user's delight or annoyance with a service (Möller, 2000; Raake and Egger, 2014). QoE represents the outcome of a communication system as perceived by the user and extends beyond purely technical performance measures. In contrast, network-level Quality of Service (QoS) parameters such as bandwidth, delay, jitter, and packet loss describe system behavior but are not sufficient to predict perceived quality. A communication system may exhibit excellent QoS while still resulting in poor QoE if it increases listening effort, violates user expectations, or interferes with task execution.

QoE is inherently subjective and context-dependent. It is shaped by an interplay of system-related factors, contextual conditions, and user-related characteristics. These include both relatively stable traits, reflecting individual differences between listeners, and transient states, such as attention, fatigue, or situational engagement. Individuality constitutes a systematic source of variability in subjective quality ratings, such that inter-listener differences are not merely random noise but reflect meaningful and structured perceptual differences (Spang, 2023). These aspects underline the fundamental challenge of speech quality assessment: perceived quality cannot be inferred from signal properties alone, but emerges from the interaction between technical conditions, context, and the individual listener.

2.1.2 Quality Impairments in Speech Transmission

Speech communication systems encompass a wide range of transmission technologies, including landline telephony, mobile cellular networks, and over-the-top (OTT) voice-over-IP (VoIP) services. While these systems differ in their technical realization, they share a common goal: the reliable transmission of speech signals under varying network and device constraints. In practice, speech is almost always subject to degradations introduced during acquisition, encoding, transmission, and playback.

Modern speech communication systems are predominantly digital and rely on packet-switched transmission. Speech signals are encoded into frames or packets, transmitted over the network, and reconstructed at the receiver. Transmission-related impairments arise when packets are delayed, lost, or corrupted, or when bandwidth constraints and low-bitrate coding limit signal fidelity. To mitigate these effects, communication systems employ mechanisms such as jitter buffering, packet-loss concealment, noise reduction, and gain control. While these techniques aim to stabilize transmission, they may themselves introduce perceptible artifacts.

Common transmission-induced impairments include restricted audio bandwidth, codec distortions, packet or frame loss, jitter-related time-scale modifications, and background or circuit noise. These impairments alter the spectral and temporal structure of the speech signal and manifest perceptually as increased noisiness, discontinuities, unnatural coloration, or inappropriate loudness. Importantly, different impairments affect different perceptual attributes and do not necessarily degrade speech quality in the same way.

As a result, perceived speech quality reflects the combined influence of multiple transmission effects rather than a single dominant factor. Some degradations primarily affect intelligibility, while others increase listening effort or reduce naturalness without rendering speech unintelligible. This many-to-one relationship between physical impairments and perceived quality complicates both subjective assessment and objective prediction.

2.1.3 Perceptual Quality Dimensions

Perceived speech quality arises from the integration of multiple auditory impressions rather than from isolated signal properties. Consequently, overall quality judgments reflect the combined influence of several perceptual cues rather than a single underlying attribute (Jekosch, 2005). To enable systematic analysis of such judgments, a range of approaches has been proposed to decompose perceived speech quality into perceptual dimensions that capture distinct aspects of experienced degradation.

In the context of speech communication assessment, early standardized methodologies largely focused on a single global quality judgment. While effective for benchmarking overall performance of a communication channel, such approaches offer limited insight into the perceptual mechanisms underlying quality impairments. This limitation motivated the development of multidimensional perspectives on speech quality, aiming to relate overall judgments to specific auditory characteristics of the speech signal and its transmission conditions.

Within the International Telecommunication Union's standardization sector (ITU-T), this perspective was reflected in successive efforts to structure perceptual assessment beyond a single overall scale. Building on earlier diagnostic concepts, later frameworks demonstrated that multidimensional quality judgments can be obtained reliably when perceptual attributes are carefully defined and limited in number. In particular, the ITU-T Recommendation P.835 (2003), ITU-T Recommendation P.806 (2014), and the proposed P.MULTI (ITU-T Contribution SG12-C0228, 2011) frameworks introduced perceptual quality dimensions targeting complementary aspects of perceived impairment, including temporal degradations of the speech signal (e.g., slow and fast discontinuities), spectral distortions affecting timbral balance (e.g., coloration), characteristics of background noise (e.g., noise level and variability), as well as overall loudness and overall quality. Collectively, these works established a practical conceptual foundation for structured perceptual decomposition of speech quality.

A more formal and compact representation of perceptual speech quality was proposed by Wältermann, who described perceived quality as a point in a multidimensional perceptual space spanned by a small number of auditory dimensions (Wältermann et al., 2010). Each dimension corresponds to a distinct perceptual attribute contributing to overall quality and is treated, for analytical convenience, as approximately orthogonal to the others. The perceptual dimensions were empirically derived through large-scale subjective listening experiments involving narrowband (NB; 300–3400 Hz) and mixed narrowband/wideband (WB; 100–7000 Hz) speech conditions (Wältermann et al., 2006). Using pairwise similarity judgments, multidimensional scaling, semantic differential methods, and subsequent factor-analytic techniques, the authors demonstrated that perceived speech quality in modern telecommunication systems can be described by three dominant dimensions: noisiness, coloration, and discontinuity. These dimensions account for a substantial proportion of the variance observed in overall quality judgments and provide a compact yet perceptually meaningful basis for speech quality modeling. In the original studies, listening levels were normalized to a preferred reference level, thereby excluding level-related effects from the perceptual space. Subsequent work by Côté et al. (2007) showed that perceived loudness constitutes an additional perceptual factor influencing speech quality judgments and should therefore be included as a further dimension.

The perceptual quality dimensions considered in this work are described below.

Noisiness The *noisiness* (NOI) dimension characterizes the extent to which background noise or other additive distortions are perceptually present in a speech signal. It reflects the audibility of non-speech components that mask or interfere with the target speech and thereby reduce perceived quality.

Discontinuity The *discontinuity* (DIS) dimension characterizes the presence of temporal interruptions or irregularities in the speech signal. It reflects perceptually salient breaks in temporal continuity, such as dropouts, abrupt signal changes, or intermittently emerging musical tones. These artifacts are commonly caused by packet or frame loss, jitter-related effects, or aggressive time-domain processing, and are perceived as disruptions of the natural temporal flow of speech.

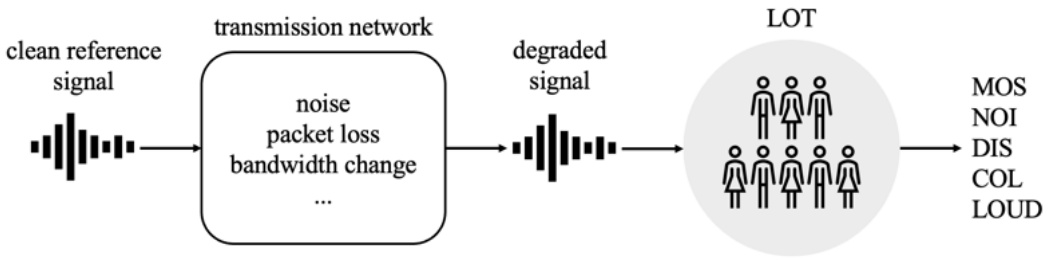


Fig. 2.2: Overview of a listening-only test (LOT) procedure for subjective speech quality assessment. A clean reference speech signal is processed by a transmission network introducing degradations such as noise, packet loss, or bandwidth limitations, resulting in a degraded signal. This signal is presented to a group of listeners under controlled conditions, who provide subjective ratings. The outcome is a pooled score derived by averaging multiple raters' scores. This produces an overall quality score (MOS) often extended to ratings of perceptual dimensions, namely noisiness (NOI), discontinuity (DIS), coloration (COL), and loudness (LOUD).

Coloration The *coloration* (COL) dimension characterizes perceptual changes in the spectral balance or timbre of speech that alter its natural sound without introducing additive noise. Such effects commonly arise from bandwidth limitations or frequency-selective transmission and coding.

Loudness The *loudness* (LOUD) dimension characterizes deviations in the perceived intensity of speech relative to an expected or preferred listening level. Both excessively low and excessively high levels can reduce listening comfort and degrade perceived quality.

Figures 2.4 and 2.5 illustrate how degradations along these perceptual dimensions manifest in the speech signal.

2.1.4 Subjective Listening Tests

Listening-only tests (LOTs) are the most widely used methodology for assessing perceived speech quality in a controlled and reproducible manner. In LOTs, listeners evaluate pre-recorded speech signals under predefined conditions, without interacting with a communication system or performing any task beyond listening. This experimental design isolates auditory perception from higher-level interaction effects and allows speech quality to be studied as a purely perceptual phenomenon. At the same time, the absence of bidirectional interaction limits ecological validity, as real-world communication typically involves conversational dynamics, turn-taking, and task-dependent listener engagement.

In a LOT, listeners provide subjective quality judgments, corresponding to the descriptive response b_0 in the auditory event model shown in Figure 2.1. By presenting identical stimuli to multiple listeners under controlled playback conditions, LOTs en-

able systematic and comparable evaluations across systems, degradation conditions, and datasets. As a result, they form the methodological basis of most standardized speech quality assessment procedures.

In many LOTs, listeners summarize their overall quality impression using a single scalar rating on an ordinal scale. Figure 2.2 illustrates this procedure. Overall speech quality is most commonly assessed using the Absolute Category Rating (ACR) scale, as specified in ITU-T Recommendation P.800 (1996) for controlled laboratory experiments and ITU-T Recommendation P.808 (2021) for crowdsourced listening tests. Other scales include the Degradation Category Rating (DCR) and Comparison Category Rating (CCR), however, they are not used in this work. In the ACR method, listeners assign a quality score using the five-point scale shown in Table 2.1. The arithmetic mean of these ratings across listeners yields the Mean Opinion Score (MOS), which provides a robust and widely accepted measure of perceived overall speech quality.

Table 2.1: Absolute Category Rating (ACR) scale used for subjective speech quality assessment.

Score	Label
5	Excellent
4	Good
3	Fair
2	Poor
1	Bad

Ratings for perceptual quality dimensions are typically obtained using procedures analogous to the ACR method defined in ITU-T Recommendation P.800 (1996), but implemented on continuous rather than discrete five-point scales. Multidimensional subjective assessment procedures are specified in ITU-T Recommendation P.806 (2014), and the perceptual dimensions considered in this work are reflected in ITU-T Recommendation P.804 (2017) through bipolar antonym-pair rating scales for conversational speech analysis. In practice, listeners evaluate each perceptual dimension independently, commonly using graphical sliders or visual analogue scales, as illustrated in Figure 2.3.

Continuous rating scales are often preferred over discrete category scales because they provide higher sensitivity and resolution in subjective judgments. To facilitate comparability with conventional ACR-based MOS values, ratings obtained on an extended continuous scale ranging from 0 to 7 can be mapped to the five-point ACR scale in the range of 1 to 5 as presented in Table 2.1. A polynomial mapping function for this purpose was proposed by Köster et al. (2015) and is given by

$$\widehat{\mathrm{MOS}}_{\mathrm{ACR}} = 1 + 0.1907 \cdot \mathrm{MOS}_{\mathrm{EC}} + 0.2368 \cdot \mathrm{MOS}_{\mathrm{EC}}^2 - 0.0262 \cdot \mathrm{MOS}_{\mathrm{EC}}^3, \qquad (2.1)$$

where $\mathrm{MOS}_{\mathrm{EC}}$ denotes the rating on the extended continuous scale and $\widehat{\mathrm{MOS}}_{\mathrm{ACR}}$ the corresponding estimate on the ACR MOS scale.

An illustrative example of the perceptual degradations described above is shown in Figures 2.4 and 2.5. The speech samples and ratings are drawn from the `NISQA_TEST_P501` corpus (Mittag et al., 2021), annotated via crowdsourcing according to

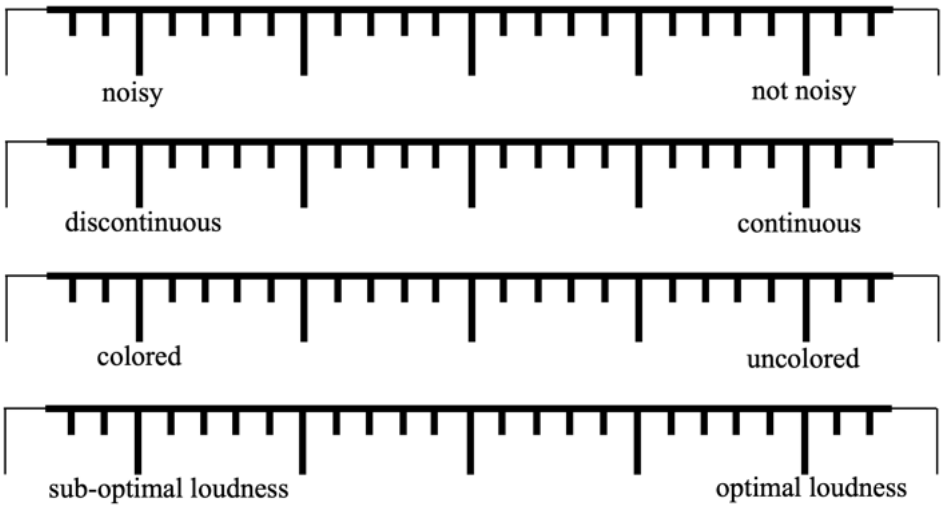


Fig. 2.3: Illustration of the extended continuous scales for obtaining ratings for perceptual dimensions of speech quality. Higher scale values correspond to reduced impairment or more optimal perceptual quality for each dimension.

ITU-T Recommendation P.808. The topmost waveform and spectrogram correspond to a clean fullband (FB; 20-20,000 Hz) speech signal with high subjective overall quality and perceptual ratings close to the upper end of the scale across all dimensions. As degradations are introduced, perceived quality decreases in a systematic and dimension-specific manner. For instance, the addition of white noise at 12 dB SNR reduces the noisiness rating from 4.60 to 1.39. Temporal clipping of 20% leads to a decrease in the discontinuity rating from 4.55 to 1.54. Band-pass filtering between 500-2500 Hz causes the coloration rating to drop from 4.47 to 1.80, while attenuating the signal level by 20 dB reduces the loudness rating from 4.55 to 2.11.

Nuances and Limitations of Subjective Listening Tests

Subjective listening tests provide direct access to human perception of speech quality, but their outcomes are inherently influenced by factors beyond the acoustic properties of the test stimuli. ITU-T Recommendation P.1401 (2020) emphasizes that listener selection, experimental design, scale interpretation, and test context all contribute to variability in subjective ratings, even under standardized procedures. While methodological choices such as the use of anchor conditions and consistent protocols reduce unwanted variance, residual bias and uncertainty cannot be fully eliminated.

Beyond experimental design, subjective quality judgments arise from an internal quality-formation process involving perception, evaluation, and the assignment of a rating on the response scale. Acoustic signals are first transformed into auditory events, which are then evaluated against expectations of clarity, naturalness, and communicative adequacy before being mapped onto a rating scale. Variability is introduced at each of these stages, and repeated ratings of the same stimulus by the same listener may differ due to perceptual noise, attentional fluctuations, or momentary cognitive state.

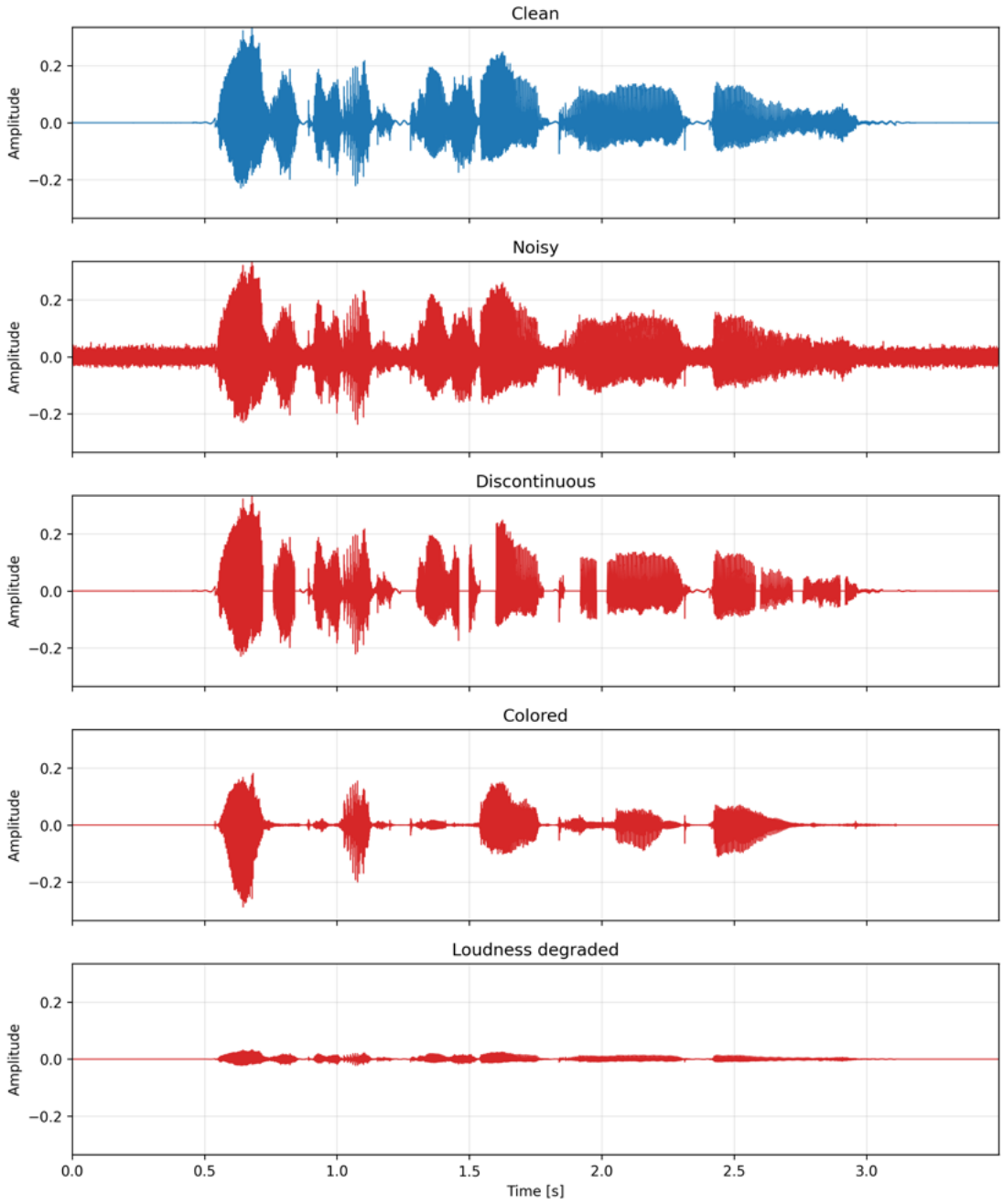


Fig. 2.4: Waveform examples illustrating typical degradations associated with perceptual quality dimensions. The top panel shows the clean fullband reference, followed by versions degraded in *noisiness* (white noise added at 12 dB SNR), *discontinuity* (time clipping of 20%), *coloration* (band-pass filtering between 500–2500 Hz), and *loudness* (level attenuation by 20 dB). The same speech sample (*female 2* from the `NISQA_TEST_P501` corpus (Mittag et al., 2021)) was used across all conditions, speaking the sentence "*Rice is often served in round bowls.*"

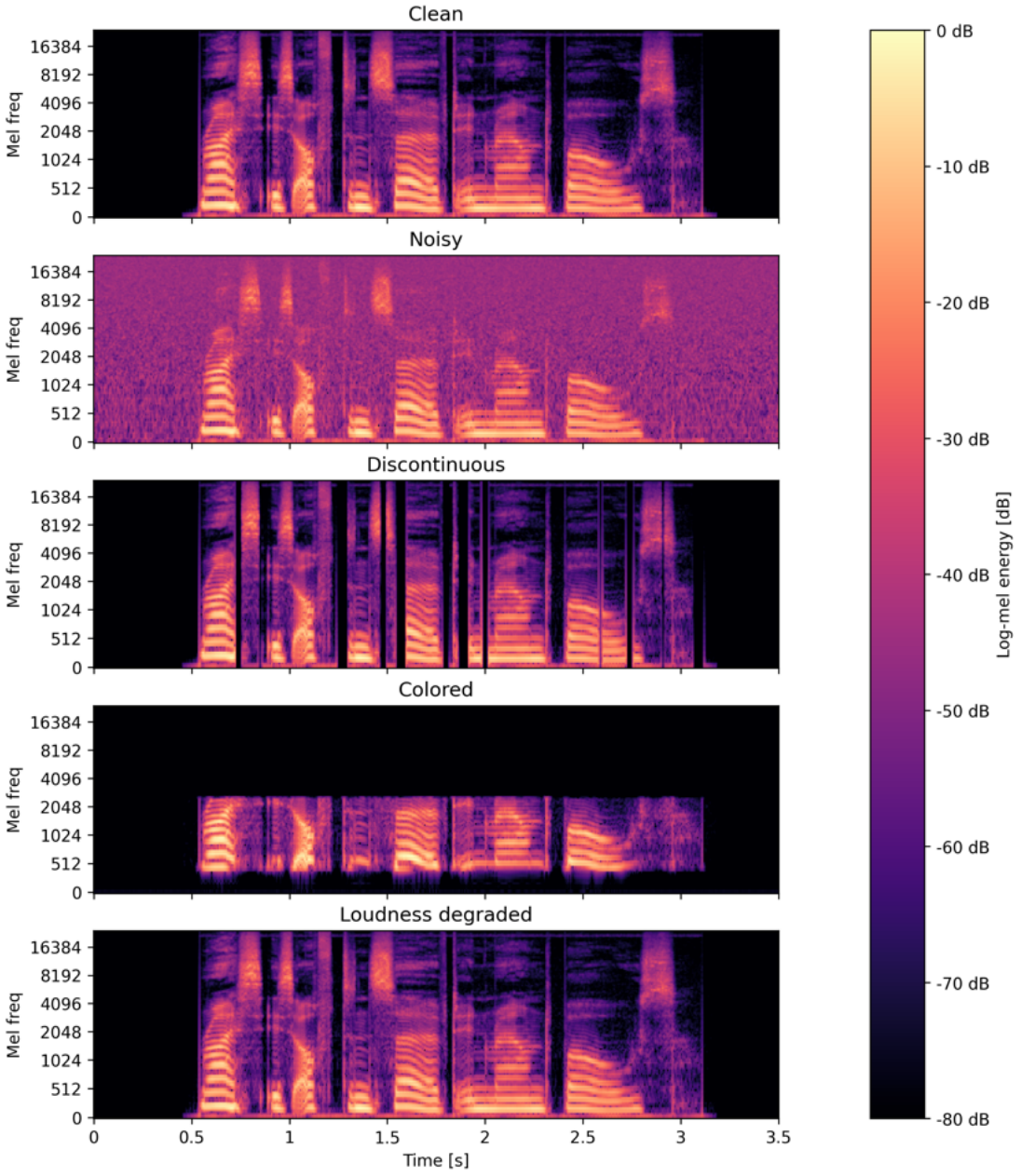


Fig. 2.5: Mel-spectrogram representations of the same speech samples shown in Figure 2.4, illustrating spectral and temporal effects of typical perceptual degradations. The top panel shows the clean fullband reference, followed by degradations in *noisiness* (white noise added at 12 dB SNR), *discontinuity* (time clipping of 20%), *coloration* (band-pass filtering between 500–2500 Hz), and *loudness* (level attenuation by 20 dB). All samples correspond to *female 2* from the `NISQA_TEST_P501` corpus (Mittag et al., 2021), speaking the sentence "*Rice is often served in round bowls.*"

Listener judgments are further influenced by contextual dependencies. Short-term sequential effects can bias the perception of a stimulus based on previously presented samples, while session-level effects shape how rating categories are interpreted relative to the overall quality distribution. Consequently, subjective quality scores should be regarded as estimates with inherent uncertainty rather than exact ground-truth values.

When subjective listening tests are conducted in crowd-based settings, additional factors can influence the fidelity of the collected quality ratings, even when the procedures strictly adhere to ITU-T Recommendation P.808 (2021). These factors include conceptual challenges in test design, variability in listener reliability, the choice of incentive and payment schemes used to motivate participation, as well as hidden influence factors arising from the largely uncontrolled listening environment. Furthermore, the statistical treatment and aggregation of the collected ratings play a critical role in mitigating these effects and ensuring robust and interpretable results.

These characteristics underline both the strengths and limitations of subjective listening tests. While indispensable for capturing human perception, their cost, variability, and limited scalability motivate the development of objective speech quality models that aim to approximate perceptual judgments with greater consistency and efficiency.

2.2 Instrumental Speech Quality Assessment

Instrumental speech quality assessment methods aim to predict perceived speech quality using computational models, thereby reducing the need for time-consuming and costly subjective listening tests. Such models are essential for large-scale evaluation, system development, and real-time monitoring of speech communication services. Over the past decades, a variety of instrumental approaches have been proposed, differing in their assumptions, required inputs, and degree of access to the communication system.

Following the established taxonomy summarized by Möller et al. (2011), instrumental speech quality models can be broadly categorized into parametric and signal-based approaches. The following subsections review these classes of models and discuss their respective strengths and limitations.

2.2.1 Parametric Models

Parametric speech quality models estimate perceived quality from transmission and system parameters obtained through network planning or monitoring, without analysing the speech signal itself. These models describe the impact of individual impairments on conversational quality using analytically defined relationships and are therefore computationally efficient and well suited for network design and optimization tasks.

A prominent example is the E-model, standardized in the ITU-T Recommendation G.107 (2015), which predicts conversational speech quality by combining impairment factors related to delay, packet loss, codec characteristics, and echo effects into a scalar quality estimate. Due to its interpretability and low computational complexity, the E-model has been widely adopted in telecommunication engineering. Originally developed

for narrowband telephony, it has subsequently been extended to wideband and fullband scenarios (ITU-T Recommendation G.107.1, 2015; ITU-T Recommendation G.107.2, 2019).

However, because parametric models do not operate on the acoustic signal, they cannot capture signal-specific perceptual degradations or account for complex interactions between impairments. As a result, their applicability is limited when fine-grained perceptual quality assessment or signal-level diagnostics are required, motivating the development of signal-based approaches.

2.2.2 Signal-Based Models

Signal-based speech quality models estimate perceived quality by analysing audio signals measured at electrical or acoustic interfaces of the transmission chain. By operating directly on the speech signal, these models are able to capture signal-specific degradations and their perceptual consequences more explicitly than parametric approaches. As a result, signal-based models provide a closer link between acoustic distortions and subjective quality judgments.

Depending on the availability of a clean reference signal, signal-based approaches can be further categorized into *intrusive* (*reference-based* or *double-ended*) and *non-intrusive* (*no-reference* or *single-ended*) methods. The following subsections review these two classes and discuss their respective capabilities and limitations.

2.2.2.1 Intrusive (Reference-Based) Models

Intrusive, or reference-based, instrumental models estimate perceived speech quality by directly comparing a degraded signal with a corresponding clean reference, as illustrated in Figure 2.6. By quantifying perceptual differences between the two signals through analytically defined transformations, these models provide a deterministic estimate of listening quality. Reference-based methods have played a central role in standardization and remain essential tools for codec development, system benchmarking, and controlled laboratory evaluation.

An early milestone in this class is ITU-T Recommendation P.861 (1996), which introduced the Perceptual Speech Quality Measure (PSQM) as the first standardized intrusive model for narrowband codec evaluation. This approach was subsequently extended and formalized in ITU-T Recommendation P.862 (2001), known as the Perceptual Evaluation of Speech Quality (PESQ). PESQ incorporates preprocessing, temporal alignment, auditory transformation, and perceptual disturbance measures to produce a MOS-like quality estimate and became the de facto standard for narrowband telephony assessment.

As speech communication systems evolved beyond narrowband telephony toward wideband, super-wideband (SWB; 50–14,000 Hz), and fullband (FB; 20–20,000 Hz) operation, the limitations of PESQ motivated the development of more advanced reference-based instrumental models. An important intermediate research contribution in this evolution is the DIAL model (Cóté et al., 2010), which targeted super-wideband speech and anticipated several design concepts later incorporated into fullband-capable instrumental standards.

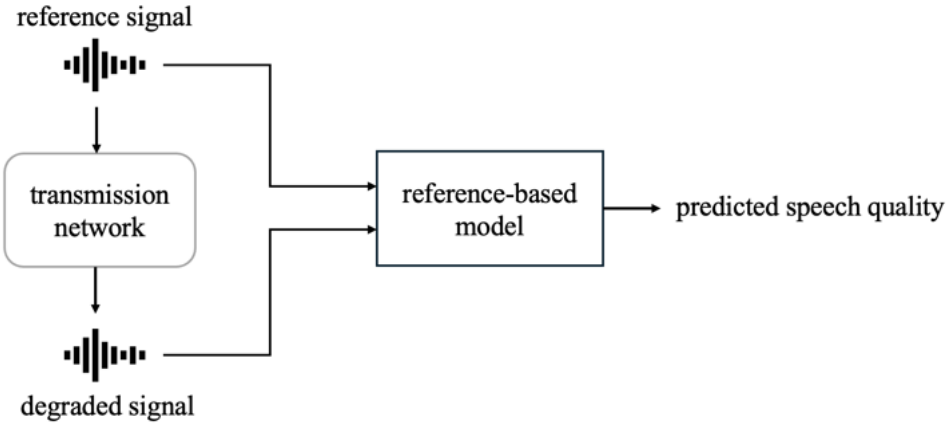


Fig. 2.6: Conceptual overview of intrusive (reference-based) speech quality assessment. A clean reference signal and its corresponding degraded signal, obtained after transmission through the communication network, are jointly processed by a reference-based model. The model compares both signals to estimate perceived speech quality, typically expressed as a MOS-like score.

The Perceptual Objective Listening Quality Prediction (POLQA) model, standardized in ITU-T Recommendation P.863 (2018), introduced improved temporal alignment procedures and perceptual processing stages to robustly handle bandwidth transitions, nonlinear distortions, and modern coding schemes. POLQA demonstrates strong correspondence with subjective listening results across a wide range of transmission scenarios, including packet-based and mobile networks. It should be noted that POLQA is a proprietary model, with reference implementations and calibration parameters available only through licensed distributions.

More recent extensions of the POLQA framework reflect the increasing complexity of modern speech services. While earlier instrumental methods were limited to predicting overall quality in terms of a single MOS value, ITU-T Recommendation P.863.2 (2022) augments the original POLQA listening-quality prediction by estimating multiple perceptual quality dimensions—noisiness, discontinuity, coloration, and loudness—alongside the overall MOS, thereby acknowledging the multidimensional nature of perceived speech quality.

An early reference-based perceptual model that extends intrusive quality prediction to explicit perceptual dimensions is the Diagnostic Instrumental Assessment of Listening quality (DIAL) model (Côté, 2011). DIAL is an intrusive model that uses both a clean reference and its corresponding degraded signal to estimate perceived overall speech quality together with the four perceptual dimensions of noisiness, coloration, discontinuity, and loudness. The model builds on a Core instrumental predictor related to earlier frameworks such as TOSQA (Berger, Jens, 1998) and supplements it with dedicated dimension estimators that quantify degradations associated with specific perceptual attributes. In DIAL, the Core model output and the dimension-specific quality features are combined using a non-parametric k-Nearest Neighbours (kNN) approach, which serves as a lightweight machine learning mechanism for aggregating perceptual indicators into a single overall quality estimate.

A complementary intrusive formulation places stronger emphasis on perceptual similarity as the basis for quality prediction. The Virtual Speech Quality Objective Listener (ViSQOL) (Hines et al., 2012) estimates speech quality by comparing reference and degraded signals in a perceptually motivated representation space, yielding a measure of perceptual distance between the two signals. While this formulation is framed differently, it shares the same underlying principle as earlier standardized models such as PESQ and POLQA, which likewise perform reference-based comparisons after auditory-inspired transformations, temporal alignment, and masking-related processing. The primary distinction lies not in whether perceptual modeling is employed, but in how perceptual differences are summarized and mapped to quality judgments. In ViSQOL, perceptual similarity measures are used directly as features, and subsequent extensions—including ViSQOLAudio and ViSQOL v3—employ supervised machine learning to map these measures to subjective MOS, rather than relying solely on fixed disturbance aggregation rules (Hines et al., 2015; Chinen et al., 2020).

Progressing toward data-driven approaches, Mittag (2022) proposed a double-ended machine learning framework that combines reference-based comparison with learned feature representations. In this approach, Mel-spectrogram segments of the reference and degraded signals are processed by a Siamese convolutional neural network with self-attention, producing comparable latent representations for both signals. The reference features are subsequently aligned to the degraded features using a cosine similarity measure combined with a hard-attention mechanism. To explicitly capture perceptual differences, the aligned reference features, degraded features, and an additional distance feature vector are concatenated and jointly processed. The fused representation is then passed through a second self-attention layer followed by temporal pooling to produce the final quality prediction.

Taken together, intrusive speech quality models can be understood as reference-based approaches that estimate perceived quality by computing a perceptually motivated distance between a reference and a degraded signal. Classical standardized models such as PSQM, PESQ, and POLQA predominantly rely on deterministic, rule-based processing pipelines, in which auditory-inspired transformations, temporal alignment, and masking-related mechanisms are combined with fixed procedures for aggregating perceptual differences into an overall quality estimate. More recent approaches, including ViSQOL and data-driven double-ended models, retain this reference-based perceptual distance paradigm but increasingly replace fixed aggregation rules with learned representations and mappings derived from subjective listening data.

Despite their strong validity and transparency under controlled conditions, all intrusive models fundamentally depend on the availability of a clean reference signal. This requirement restricts their applicability in many practical scenarios, such as live telephony, conferencing, streaming services, and generative speech applications, where reference signals are unavailable. These limitations have motivated growing interest in non-intrusive, single-ended approaches, which aim to predict perceived speech quality directly from the degraded signal alone.

2.2.2.2 Non-Intrusive (Single-Ended) Models

Non-intrusive, or single-ended, instrumental models estimate perceived speech quality directly from the degraded signal without access to a clean reference, as illustrated in Figure 2.7. Such approaches are essential in practical deployment scenarios where only the received audio is observable, including network monitoring, streaming services, conferencing applications, voice assistants, and generative speech systems. Compared to intrusive methods, single-ended assessment is inherently more challenging, as the model must infer both the presence and perceptual relevance of degradations solely from the observed signal. At the same time, this setting closely mirrors the human listening condition, in which quality judgments are likewise formed based only on the signal available at the receiver.

The first standardized single-ended speech quality model was ITU-T Recommendation P.563 (2004), developed for narrowband telephony. P.563 employs a rule-based signal-processing framework that extracts hand-crafted indicators of perceptual degradation, including noise, coding artifacts, temporal irregularities, and loudness effects, which are combined into a predicted listening-quality score. The model demonstrated the feasibility of reference-free quality estimation and enabled large-scale network monitoring without access to clean reference signals. However, its applicability is limited to narrowband conditions and specific transmission scenarios, and its performance degrades for wideband speech and modern processing pipelines.

In parallel, several non-intrusive models were proposed outside the ITU-T standardization framework, most notably the Auditory Non-Intrusive Quality Estimation (ANIQUE) model (Kim, 2005) and its successor ANIQUE+ (Kim and Tarraf, 2007). ANIQUE adopted a rule-based auditory processing approach using temporal-envelope and modulation-domain features to characterize perceptual degradations. ANIQUE+ retained this auditory foundation while introducing simple machine learning (ML) components, including multilayer perceptrons for distortion and mute detection, which

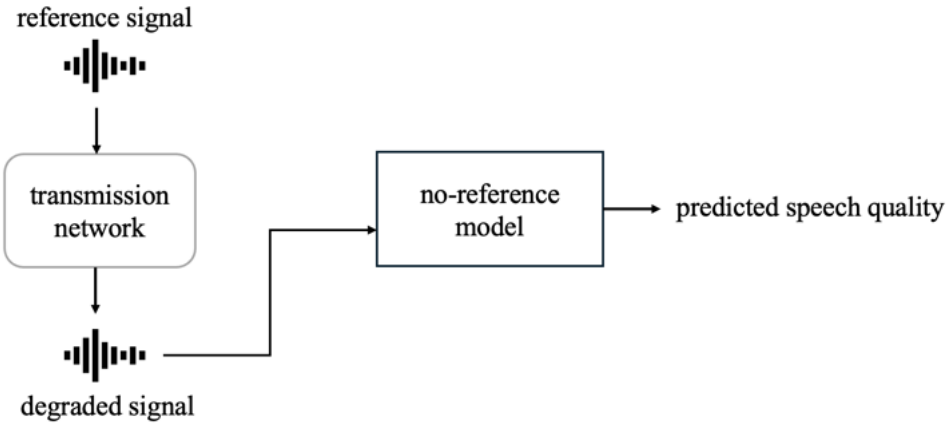


Fig. 2.7: Conceptual overview of non-intrusive (no-reference) speech quality assessment. Only the degraded speech signal observed at the receiver is available to the model, which estimates perceived speech quality directly from the signal without access to a clean reference.

substantially improved prediction accuracy under narrowband conditions. Published as an American National Standard (ANSI/ATIS, 2006), ANIQUE+ served as an influential benchmark in the early development of single-ended speech quality assessment and was frequently referenced within ITU-T Study Group 12 as a complementary non-intrusive approach.

2.2.3 Data-Driven Speech Quality Models

Data-driven approaches to speech quality assessment emerged as an alternative to fully rule-based non-intrusive models, enabling quality estimation through statistical learning from annotated data. Early work focused on classical machine-learning techniques that operated on hand-crafted acoustic and perceptual features. Among the earliest and most influential examples, Gaussian Mixture Models (GMMs) were employed to model the statistical structure of clean speech features and to quantify deviations introduced by transmission impairments. As generative models, GMMs enabled likelihood-based consistency scoring across speech segments and formed the basis of early machine-learning-based speech quality estimators, most notably in the work of Falk and Chan (2006). These approaches laid the groundwork for subsequent regression-based and neural-network models.

Building on this foundation, later systems combined perceptual front-ends with supervised learning to better capture complex interactions between signal features and perceived quality. A prominent milestone was ANIQUE+, which retained an auditory modeling framework while replacing deterministic quality mappings with multilayer perceptrons trained on large collections of MOS-labelled speech. This marked a decisive transition from analytically defined rules toward data-driven estimation, enabling improved prediction accuracy under narrowband conditions. Related efforts explored alternative regression techniques, including decision-tree-based methods and support vector regression, further expanding the role of machine learning in non-intrusive quality assessment.

The advent of deep learning substantially accelerated this shift. Non-intrusive speech quality models increasingly moved away from expert-designed features toward end-to-end learning from time-frequency representations. Models such as Quality-Net (Fu et al., 2018) and MOSNet (Lo et al., 2019) demonstrated that convolutional and recurrent neural networks can directly regress MOS from spectrogram-based inputs, achieving significantly improved correlation with subjective listening quality ratings. Variants of these architectures were subsequently applied to text-to-speech and voice-conversion systems, where perceived naturalness and expressiveness are closely tied to overall quality.

As non-intrusive quality prediction matured, research increasingly shifted toward domain-specific deep-learning models trained on large-scale MOS datasets and tailored to particular speech-processing applications. Systems such as UTMOS (Saeki et al., 2022) and MOSA-Net (Zezario et al., 2023) extend the MOSNet paradigm by employing more expressive convolutional, recurrent, and attention-based components and by targeting specific application domains, including text-to-speech, voice conversion, and neural speech enhancement. Trained on extensive subjective datasets capturing

synthesis- and processing-related artifacts, these models achieve strong correlation with human judgments within their targeted domains. However, their reliance on fully supervised learning and domain-specific training data limits robustness and generalization to unseen degradations or novel signal conditions, reinforcing the need for more transferable speech representations.

A related but complementary effort addressed quality assessment for a specific class of degradations introduced by neural speech enhancement. The DNSMOS model (Reddy et al., 2021), developed within the Deep Noise Suppression (DNS) Challenge, predicts perceived speech quality and signal integrity using deep neural networks trained on large-scale crowdsourced MOS datasets. Designed for real-time, non-intrusive monitoring, DNSMOS is particularly effective at capturing perceptual artifacts introduced by modern neural enhancement pipelines. The model consists of four convolution neural network layers followed by three dense layers, and predicts the overall quality MOS from log Mel-spectrograms inputs.

The NISQA framework (Non-Intrusive Speech Quality Assessment) (Mittag et al., 2021) represents a significant advancement in deep-learning-based speech quality prediction by introducing a unified non-intrusive model capable of estimating both overall speech quality and multiple perceptual dimensions. Specifically, NISQA predicts the overall quality MOS alongside the perceptual quality dimensions of noisiness, coloration, discontinuity, and loudness. The model combines convolutional processing for local feature extraction with attention-based mechanisms for temporal integration and is trained on large, heterogeneous datasets covering wideband and fullband speech with 48kHz sampling rate. These datasets encompass a broad range of degradations, including codec artifacts, packet-based transmission impairments, and speech enhancement effects. By jointly modeling overall quality and its underlying perceptual dimensions, NISQA demonstrated that non-intrusive approaches can achieve state-of-the-art performance across diverse conditions while also providing diagnostically meaningful insight into the nature of perceived degradations. Owing to its wide adoption and close alignment with the objectives of this work, the multidimensional NISQA model is used as a baseline in the present study. An overview of the NISQA architecture is shown in Figure 2.8.

Recently, auditory large language models (LLMs) have also been explored for predicting the quality of speech (Wang et al., 2025). In contrast to task-specific models trained for a single perceptual target, auditory LLMs provide a flexible framework for jointly addressing multiple evaluation aspects, such as overall quality, speaker similarity, and preference judgments, through task-specific prompting and fine-tuning. Beyond scalar quality prediction, these models can generate natural language descriptions of perceptual attributes, offering improved interpretability. Experimental results across multiple speech quality datasets indicate that auditory LLMs achieve competitive performance compared to specialized small models, while enabling unified multi-task prediction and descriptive quality assessment within a single model architecture.

In summary, data-driven speech quality assessment has evolved from classical statistical learning on hand-crafted features to deep neural architectures that learn perceptually relevant representations directly from data. Despite substantial progress, most existing models are trained in a fully supervised manner on task-specific or domain-limited datasets, which restricts their ability to generalize across recording conditions,

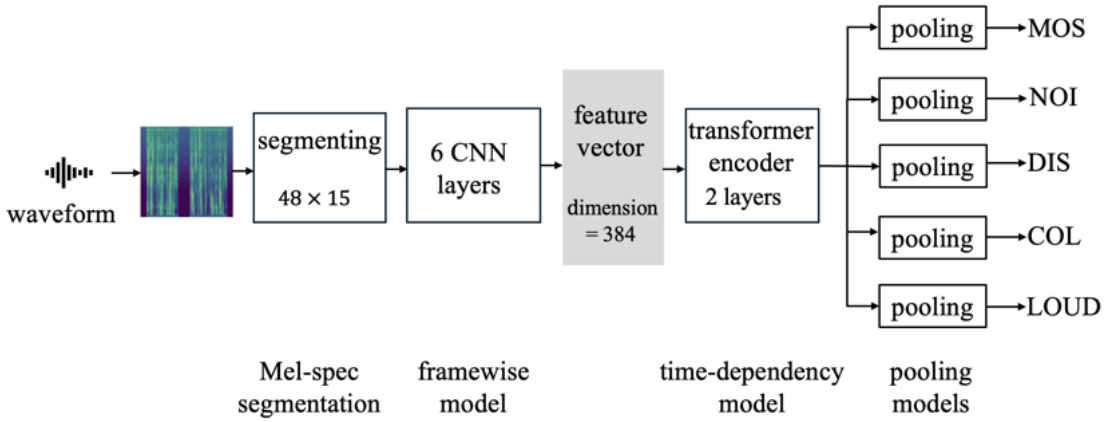


Fig. 2.8: Overview of the NISQA architecture. The input waveform is first transformed into a Mel spectrogram and segmented into overlapping time–frequency patches of size 48×15 frames (150 ms). Each segment is processed independently by a deep framewise convolutional network, yielding a sequence of 384-dimensional local perceptual feature vectors. Temporal dependencies across the full utterance are subsequently modeled by a lightweight Transformer encoder with two layers, operating on a reduced latent dimensionality. Finally, task-specific temporal attention pooling modules aggregate the sequence into utterance-level representations for predicting overall speech quality (MOS) as well as the perceptual dimensions of noisiness (NOI), discontinuity (DIS), coloration (COL), and loudness (LOUD).

languages, speakers, and distortion types and conditions not represented during training. As a result, performance often degrades under unseen transmission conditions, novel processing artefacts, or cross-database evaluation. These limitations motivate the adoption of large-scale pretraining and transfer learning using foundation models and self-supervised speech representations, which are discussed in Section 2.3.

2.3 Transformer-Based Foundation Models for Speech

This section provides an overview of transformer-based foundation models for speech and introduces the key methodological concepts required to understand their use in downstream applications. The discussion focuses on self-supervised representation learning and transfer learning as enabling paradigms for reusing large-scale pre-trained speech representations, in conjunction with transformer-based encoder architectures. Building on this conceptual background, representative models, including wav2vec 2.0, HuBERT, WavLM, Whisper, and the Audio Spectrogram Transformer (AST), are reviewed, with emphasis on their relevance to speech quality assessment.

2.3.1 Foundation Models and Transfer Learning

This subsection introduces the key concepts underlying modern speech foundation models, with a focus on self-supervised representation learning, large-scale pre-training, and transfer learning through fine-tuning or feature reuse. Together, these concepts provide the methodological basis for adapting general-purpose speech representations to downstream tasks with limited labeled data and form the conceptual foundation for the models discussed in the subsequent subsections.

Terminology and Adaptation Strategies

In the context of transfer learning, several adaptation strategies and architectural components are commonly distinguished. *Pre-training* refers to the initial training of a model on large-scale datasets, typically using self-supervised or weakly supervised objectives (discussed later in this section), with the goal of learning general-purpose *representations*. In this context, representations denote latent feature encodings that capture salient characteristics of the input signal, such as acoustic structure, temporal dynamics, or higher-level abstractions. When represented as fixed-dimensional vectors, these representations are commonly referred to as *embeddings*.

Transfer learning denotes the reuse of such pre-trained representations or embeddings for a downstream task that differs from the original training objective, either by directly employing them as fixed features or by further adapting them through task-specific training.

When applying a pre-trained model to a downstream task, the encoder may be *frozen*, meaning that its parameters are kept fixed and only additional task-specific layers are trained. In this case, the pre-trained encoder acts as a fixed feature extractor. In contrast, *fine-tuning* refers to updating some or all parameters of the pre-trained encoder during downstream training, allowing the representations to adapt to the target task. The term *adaptation* is used more generally to describe any modification of a pre-trained model for a downstream application, including partial fine-tuning, task-adaptive training, or architectural extensions.

A *mapping head* (or task head) denotes a lightweight, task-specific network component—typically composed of linear or shallow neural layers—that maps the high-dimensional representations produced by the pre-trained encoder to the desired output space, such as a scalar MOS value or multiple perceptual quality dimensions. In speech quality assessment, such mapping heads enable the reuse of shared representations while tailoring the final prediction to perceptual targets.

In this context, *single-task learning* refers to training a model to predict a single target variable, such as overall speech quality, using one task-specific mapping head. By contrast, *multi-task learning* involves jointly predicting multiple related target variables—such as overall quality and several perceptual dimensions—using a shared encoder and multiple task-specific heads. Multi-task learning introduces an inductive bias that encourages the shared representations to capture information relevant across tasks, while single-task learning allows representations to specialize for a specific prediction objective. Both formulations are considered in this work as alternative strategies for adapting pre-trained speech representations to speech quality assessment.

Self-Supervised Learning

Self-supervised learning (SSL) provides the methodological foundation for modern large-scale representation learning. Rather than relying on human-annotated labels, SSL leverages the intrinsic structure of the data by defining surrogate prediction tasks, where the supervision signal is derived directly from the input itself. In speech processing, two classes of SSL objectives have proven particularly influential: masked prediction and contrastive prediction. In masked prediction objectives, portions of the input or intermediate representations are masked, and the model is trained to infer the missing information from surrounding context. This paradigm was popularized in natural language processing (NLP) through masked language modeling (MLM) in BERT (Devlin et al., 2019), and was later adapted to continuous speech representations in self-supervised speech models. In contrastive prediction, the model is trained to discriminate true future or masked target representations from a set of negative samples, given a contextual representation of the input. This objective was first introduced to speech representation learning through contrastive predictive coding (CPC) (van den Oord et al., 2018).

These objectives encourage models to learn representations that capture both local acoustic structure and longer-range temporal dependencies directly from raw audio, without relying on explicit linguistic annotations. The resulting representations encode a broad range of information, including phonetic content, prosody, speaker characteristics, and temporal coherence, as demonstrated in several self-supervised speech models such as wav2vec 2.0, HuBERT, and WavLM (Baevski et al., 2020b; Hsu et al., 2021; Chen et al., 2021).

Transfer Learning in Speech Processing

Recent advances in speech processing have been driven by the emergence of large transformer-based foundation models pre-trained on diverse and large-scale speech corpora. Training such models from scratch typically requires massive amounts of unlabeled data, specialised expertise, and substantial computational resources, which are beyond the reach of most research groups. As a result, transfer learning has become a central paradigm in modern speech research, enabling knowledge acquired during large-scale pre-training to be reused and adapted for downstream tasks. The availability of pre-trained foundation models—developed and released by large industrial and academic consortia—has therefore become a key enabler for downstream research, allowing learned speech representations to be reused, fine-tuned, or analysed without incurring the cost of large-scale pre-training (see Figure 2.9).

In contrast to task-specific models trained from scratch, foundation models learn general-purpose speech representations through self-supervised or weakly supervised training objectives, facilitating transfer across a wide range of downstream tasks. In the speech domain, such models have achieved state-of-the-art performance in applications including automatic speech recognition, speaker characterization, and audio classification. Although not originally developed for speech quality assessment, the representations learned by these models encode rich acoustic and temporal information that is closely related to perceptual speech attributes. This observation motivates the central

research question of this thesis: whether pre-trained speech representations can be effectively leveraged through transfer learning to enable reliable, non-intrusive prediction of perceived speech quality. In this work, transfer learning is explored using both frozen and fine-tuned representations, as well as single-task and multi-task formulations, to assess how different adaptation strategies affect speech quality prediction performance.

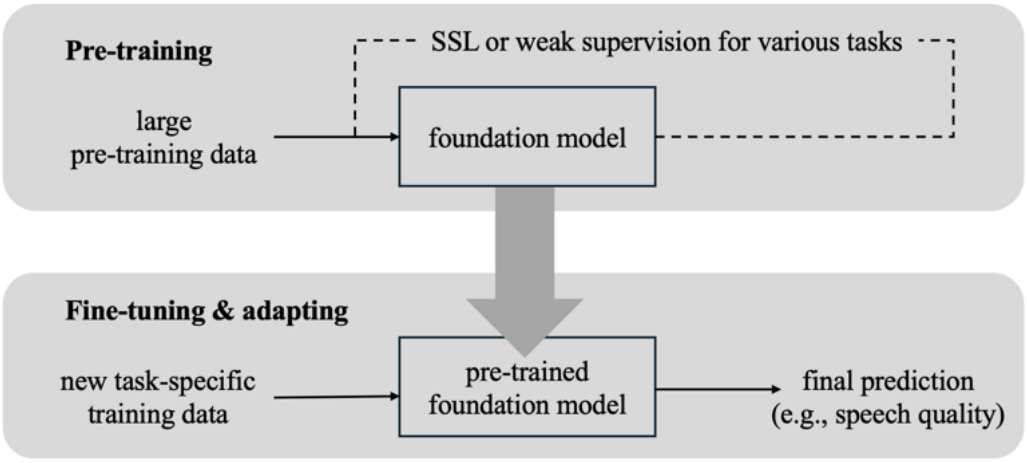


Fig. 2.9: Conceptual illustration of the foundation model paradigm in speech processing. During pre-training, large-scale unlabeled speech data are used to train a foundation model using self-supervised or weakly supervised learning objectives. The resulting pre-trained model can then be reused and adapted in a second stage by fine-tuning on task-specific data to perform new prediction objectives, such as speech quality assessment, without requiring training from scratch.

Transformer Encoders

At the core of these foundation models is the transformer architecture. It was introduced by Vaswani et al. (Vaswani et al., 2017) and marked a fundamental shift in sequence modeling by replacing recurrence and convolution with a fully attention-based mechanism. Prior sequence models, such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs), were constrained by limited parallelism and difficulties in capturing long-range dependencies, particularly for long and variable-length sequences. Transformers address these limitations through self-attention, which enables each element of a sequence to attend directly to all others, allowing global contextual dependencies to be modeled efficiently and in parallel.

A transformer is composed of stacked encoder and decoder blocks, each built from multi-head self-attention layers and position-wise feed-forward networks, augmented with residual connections and layer normalization. In representation learning settings, the encoder plays a central role by producing contextualized latent representations in which each time step integrates information from the full input sequence. This encoder-centric design has proven especially effective for large-scale pre-training, as exempli-

fied by MLM in BERT (Devlin et al., 2019) and later adopted in speech representation learning. The self-attention mechanism enables each element of an input sequence to adaptively weight information from all other elements, allowing global contextual dependencies to be modeled without relying on sequential processing. This capability is particularly advantageous for speech signals, which exhibit long-range temporal structure and variability across multiple time scales.

In this work, the term *hidden (representation) space* refers to the fixed-dimensional internal vector space of a neural network in which embeddings and contextualized representations reside; embeddings denote vector representations prior to contextual modeling, while contextualized representations are embeddings that have been updated through mechanisms such as self-attention to incorporate information from the surrounding context.

The following subsections review prominent transformer-based speech foundation models and discuss their relevance for perceptual speech quality assessment.

2.3.2 wav2vec 2.0

wav2vec 2.0 (Baevski et al., 2020b) represents a seminal milestone in self-supervised speech representation learning. While SSL had previously achieved major success in text-based NLP (Devlin et al., 2019) and had been explored for speech in earlier work (van den Oord et al., 2018; Baevski et al., 2020a), wav2vec 2.0 was the first approach to demonstrate that large-scale self-supervised pre-training directly on raw audio waveforms can rival and even surpass fully supervised ASR systems. This result established SSL as a viable alternative to annotation-intensive training paradigms in speech processing.

Architecturally, wav2vec 2.0 consists of three main components, as illustrated in Figure 2.10. First, a convolutional feature encoder transforms the raw audio waveform into a sequence of latent acoustic representations while substantially reducing temporal res-

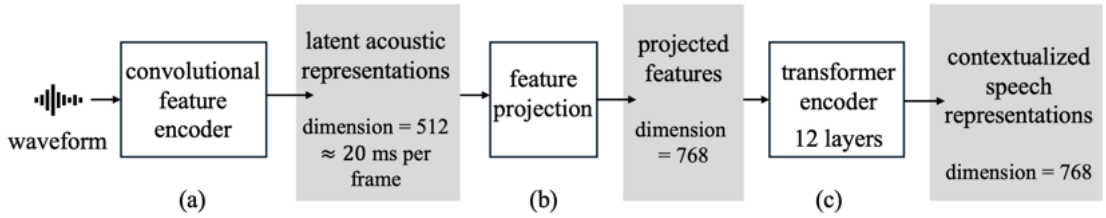


Fig. 2.10: Architecture of the *wav2vec 2.0* representation learning backbone. (a) A multi-layer convolutional feature encoder maps the raw audio waveform (sampled at 16 kHz) into a sequence of latent acoustic representations by learning local spectro-temporal patterns while substantially reducing temporal resolution, from 16,000 samples to 49 frames, corresponding to approximately 20 ms per frame. (b) A feature projection module normalizes and maps these representations into the Transformer hidden space. (c) A multi-layer Transformer encoder contextualizes the projected features over time using self-attention, producing high-level contextualized speech representations.

olution (Figure 2.10(a)). This feature encoder operates directly on waveform amplitude values using a stack of one-dimensional convolutional layers. In the first layer, convolutional kernels of size 10 samples are applied with a stride of 5 samples, corresponding to a 50% overlap between consecutive windows and capturing very local waveform structure at sub-millisecond time scales. Importantly, no hand-crafted time–frequency representation such as a mel-spectrogram is used; instead, all acoustic features are learned directly from the raw waveform. Subsequent convolutional layers do not operate on raw samples, but on feature vectors whose receptive fields already span multiple waveform samples. As a result, successive strided convolutions hierarchically combine increasingly larger portions of the original signal, yielding a sequence of latent acoustic representations with an effective temporal resolution of approximately 20 ms per frame. Second, a feature projection module (Figure 2.10(b)) normalizes the latent acoustic representations and maps them into the Transformer hidden space. This projection operates independently on each time step, aligning the dimensionality and statistical properties of the convolutional features with the requirements of the Transformer encoder, without introducing additional temporal context. Third, a Transformer encoder contextualizes the projected features over time using self-attention, enabling the modeling of long-range temporal dependencies and producing contextualized speech representations (Figure 2.10(c)). In its base configuration, wav2vec 2.0 employs a 12-layer Transformer encoder with a hidden dimension of 768 and comprises approximately 95 million parameters, illustrating the scale at which self-supervised pre-training becomes effective for learning high-level speech representations.

wav2vec 2.0 is pre-trained in a fully self-supervised manner on large-scale unlabeled speech corpora, such as LibriSpeech (Panayotov et al., 2015), using raw audio waveforms sampled at 16 kHz. During pre-training, the convolutional feature encoder produces a sequence of latent acoustic representations, of which contiguous segments are randomly masked. The Transformer encoder is trained using a contrastive objective with masked latent representations, requiring it to identify the true latent representation corresponding to each masked segment from a set of negative samples drawn from the same utterance. This objective encourages the model to learn discriminative speech representations that integrate both local acoustic structure and broader temporal context. To stabilize training and prevent representational collapse, a vector quantization module discretizes the latent targets using a finite codebook that promotes the emergence of phoneme-like acoustic units.

The representations learned by wav2vec 2.0 have been widely adopted across downstream speech processing tasks. Most prominently, wav2vec 2.0 achieves strong performance in ASR when fine-tuned with limited labeled data, substantially reducing annotation requirements (Baevski et al., 2020b). Beyond ASR, wav2vec 2.0 representations have been successfully applied to speaker recognition and verification (Fan et al., 2020) and speech emotion recognition (Pepino et al., 2021). These applications illustrate the broad transferability of self-supervised speech representations learned directly from raw audio.

2.3.3 HuBERT

HuBERT (*Hidden Unit BERT*) (Hsu et al., 2021) extends the masked prediction paradigm originally introduced in NLP (Devlin et al., 2019) to speech representation learning by incorporating discrete pseudo-labels derived from unsupervised clustering. In contrast to wav2vec 2.0, which relies on a contrastive objective, HuBERT formulates pre-training as a classification task over latent acoustic units, allowing the model to learn phoneme-like abstractions without access to human annotations.

Architecturally, HuBERT follows a structure similar to wav2vec 2.0 (Figure 2.10), also comprising approximately 95 million parameters, consisting of a convolutional feature encoder that maps the raw waveform to a sequence of latent acoustic representations, followed by a Transformer encoder that contextualizes these representations over time. The key difference lies in the training objective. Prior to pre-training, an offline clustering algorithm (e.g., k-means) is applied to frame-level acoustic features to generate discrete pseudo-labels, which serve as prediction targets.

During pre-training, contiguous segments of the latent representation sequence are masked, and the Transformer encoder is trained to predict the corresponding cluster assignments from the surrounding context. Because these targets are obtained through unsupervised clustering, the model is encouraged to discover structure inherent to the acoustic signal rather than relying on externally defined linguistic labels. The clustering procedure can be iteratively refined by recomputing pseudo-labels from progressively improved representations, leading to more consistent and discriminative latent units.

By replacing contrastive prediction with cluster-based masked classification, HuBERT provides a stable and structured training signal that promotes the emergence of linguistically meaningful representations. Empirically, HuBERT has been shown to achieve strong performance on phoneme recognition and ASR tasks, particularly in low-resource settings (Hsu et al., 2021). These properties make HuBERT a valuable point of comparison when analysing how different self-supervised objectives shape the acoustic and linguistic content encoded in speech representations.

2.3.4 WavLM

WavLM (Chen et al., 2021) extends the architectural and training principles of wav2vec 2.0 (Baevski et al., 2020b) and HuBERT (Hsu et al., 2021) by explicitly targeting robustness to noise and channel variability and by substantially enlarging the pre-training corpus. The model is designed to learn general-purpose speech representations that remain stable across diverse acoustic conditions and transfer effectively to a wide range of downstream tasks, including ASR, diarisation, speaker verification, and paralinguistic analysis.

Architecturally, WavLM follows the same cluster-based masked prediction paradigm as HuBERT, combining a convolutional feature encoder with a Transformer encoder trained to predict discrete pseudo-labels for masked segments. The WavLM base model employs the same convolutional feature encoder and 12-layer Transformer backbone as wav2vec 2.0 and HuBERT, resulting in a comparable model size of approximately 95 million parameters. It added a gated relative position bias for the Transformer structure

to better capture the sequence ordering of the input speech. Its primary contributions lie in the pre-training strategy rather than the network structure. First, WavLM introduces a denoising-based pre-training objective in which the input waveform is deliberately corrupted using additive noise or reverberation, while prediction targets are derived from clean acoustic features. This encourages the model to learn representations that are invariant to environmental and channel-related distortions. Second, WavLM is pre-trained on a substantially larger and more heterogeneous corpus, comprising tens of thousands of hours of speech across multiple languages, speakers, and recording conditions.

As a result of these design choices, WavLM learns representations that encode phonetic structure, long-range temporal context, and speaker-related characteristics, while promoting invariance to background noise and channel variability for content-oriented tasks. From a speech quality perspective, this emphasis on invariance does not necessarily align with the requirements of perceptual quality prediction, where sensitivity to degradations is essential. WavLM therefore serves as an informative reference model for analysing how self-supervised Transformer representations balance invariance and degradation sensitivity, and how different adaptation strategies are required to recover quality-relevant cues under heterogeneous degradation conditions.

2.3.5 Whisper

Whisper (Radford et al., 2023) represents a distinct paradigm within the landscape of speech foundation models, combining large-scale weakly supervised training with an encoder-decoder Transformer architecture. In contrast to waveform-based self-supervised models such as wav2vec 2.0, HuBERT, and WavLM, Whisper is trained directly on paired speech-text data using a sequence-to-sequence objective. The model is pre-trained on approximately 680,000 hours of multilingual and multi-condition speech, encompassing a wide range of speakers, recording environments, noise conditions, and channel characteristics.

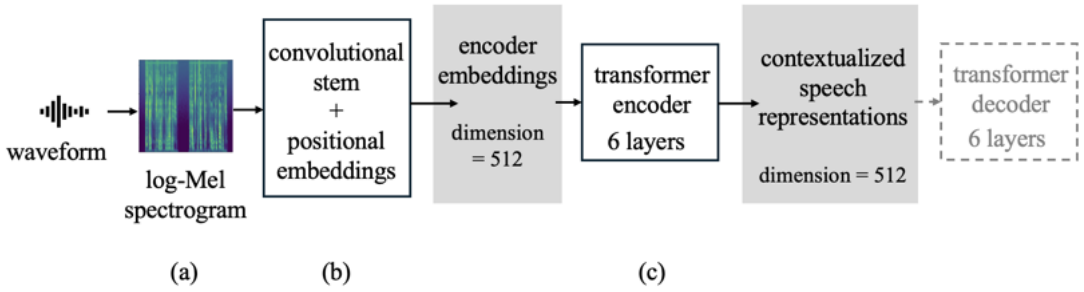


Fig. 2.11: Overview of the Whisper Base architecture. (a) The input waveform is transformed into a fixed 80-bin log-Mel spectrogram using a hand-crafted time-frequency front-end. (b) A shallow convolutional stem maps the spectrogram frames into a 512-dimensional embedding space and adds learned positional embeddings. (c) A 6-layer Transformer encoder produces contextualized speech representations. A Transformer decoder (dashed) is used during sequence-to-sequence training but omitted when Whisper is employed as an upstream feature encoder.

Architecturally, Whisper operates on fixed log-Mel spectrogram inputs rather than raw waveforms, as illustrated in Figure 2.11. The input audio, sampled at 16 kHz, is first transformed into an 80-bin log-Mel spectrogram using a 25 ms analysis window and a 10 ms frame shift. A shallow convolutional stem then maps these log-Mel frames into a 512-dimensional embedding space while downsampling the temporal resolution by a factor of two. The resulting encoder embeddings are processed by a 6-layer Transformer encoder, which produces contextualized acoustic representations. During training, a Transformer decoder generates text autoregressively via cross-attention to the encoder states, enabling end-to-end speech recognition and translation.

When used as an upstream representation model, only the Whisper encoder is retained. In this setting, the encoder produces frame-level embeddings that jointly encode linguistic content and acoustic context. Although Whisper is optimized primarily for ASR and translation, the scale and diversity of its weakly supervised training data expose the encoder to substantial acoustic variability, encouraging the emergence of representations that generalize beyond narrowly defined ASR conditions. The Whisper base upstream encoder model contains approximately 21 million parameters.

From a speech quality and perceptual modeling perspective, Whisper constitutes an informative reference model rather than a task-aligned quality predictor. Owing to its large-scale weakly supervised training on highly diverse and noisy real-world data, Whisper's encoder representations capture robust information related to intelligibility, temporal structure, and coarse background conditions. However, this robustness is primarily optimized to support transcription and generalization across domains, rather than sensitivity to perceptual degradations themselves. Consequently, Whisper exemplifies a class of large-scale acoustic encoders in which data diversity and weak supervision promote invariance to nuisance factors, making it a valuable model for analysing how robustness-oriented representations encode—or suppress—quality-relevant cues under realistic speech degradation scenarios.

2.3.6 AST

The Audio Spectrogram Transformer (AST) (Gong et al., 2021) adapts the Transformer architecture to audio processing by operating directly on time–frequency representations rather than raw waveforms. Unlike waveform-based self-supervised encoders and sequence-based spectrogram models such as Whisper, AST treats a log-Mel spectrogram as a two-dimensional input and applies self-attention to spectro-temporal patches. This design introduces a distinct inductive bias in which temporal and spectral structure are modeled jointly rather than sequentially.

Architecturally, the AST consists of three main components, as illustrated in Figure 2.12. First, a fixed front-end converts the raw audio waveform into a 128-bin log-Mel spectrogram (Figure 2.12(a)) using a 25 ms analysis window and a 10 ms frame shift. Unlike waveform-based encoders, AST does not learn acoustic features from raw amplitude values; instead, it operates entirely on this spectrogram representation.

Second, the log-Mel spectrogram is partitioned into fixed-size time-frequency patches (Figure 2.12(b)) of size 16×16. Each patch spans a local region in both time and frequency, with overlapping patches extracted across the spectrogram. The

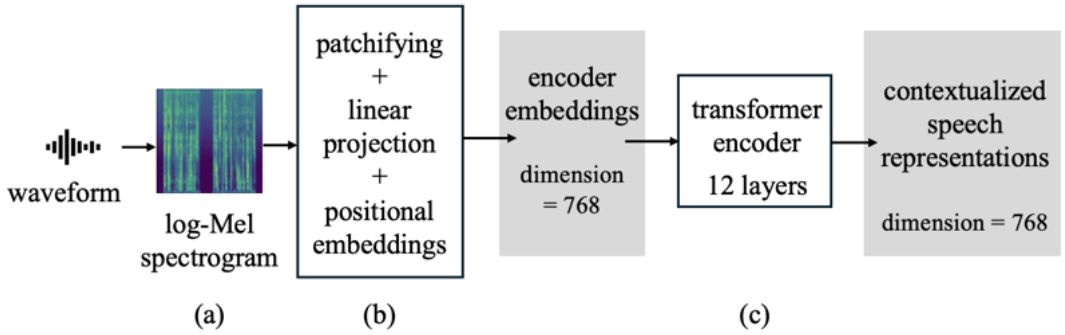


Fig. 2.12: Overview of the Audio Spectrogram Transformer (AST) architecture. (a) The input waveform is converted into a fixed log-Mel spectrogram using a hand-crafted time-frequency front-end. (b) The spectrogram is partitioned into fixed-size time-frequency patches, which are flattened, linearly projected into a latent embedding space, and augmented with learned positional embeddings; a classification token (`[CLS]`) is prepended to aggregate global information. (c) A multi-layer Transformer encoder processes the patch embeddings using self-attention to produce contextualized spectro-temporal representations.

patches are flattened and mapped to a latent embedding via a linear projection. Learned positional embeddings are added to encode the spatial location of each patch along both axes, thereby preserving the two-dimensional structure of the spectrogram. While a convolutional operation is used to implement the patch embedding efficiently, it serves solely as a tokenization mechanism rather than as a hierarchical feature extractor.

Third, a Transformer encoder processes the resulting sequence of patch embeddings using multi-head self-attention (Figure 2.12(c)). Self-attention operates jointly across time and frequency, enabling the model to capture long-range spectro-temporal dependencies without imposing an explicit temporal hierarchy. In its base configuration, AST employs a 12-layer Transformer encoder with a hidden dimension of 768 and comprises approximately 86 million parameters.

AST is obtained via transfer learning from vision Transformers (ViT) (Dosovitskiy et al., 2021). Specifically, the Transformer encoder is initialized with weights from the ViT model pre-trained on ImageNet data, and subsequently fine-tuned on large-scale audio classification datasets, most notably AudioSet sampled at 16 kHz. This fine-tuning is performed using supervised or weakly supervised objectives on log-Mel spectrogram inputs, adapting the attention mechanisms learned from visual data to acoustic time-frequency structure.

The representations learned by AST emphasize joint spectro-temporal structure, including both spectral balance and temporal modulation patterns. Unlike waveform-based self-supervised encoders that primarily target phonetic abstraction and linguistic structure, AST embeddings remain closely tied to the physical and perceptual characteristics of the acoustic signal. From a speech-quality modeling perspective, these properties are particularly relevant. The patch-based representation combined with global self-attention allows localized distortions to be integrated into a holistic representation, reflecting how listeners form overall quality judgments from distributed perceptual cues. Moreover, operating on log-Mel spectrograms aligns the model input with perceptually

motivated representations that are widely used in psychoacoustics and speech-quality research. As such, AST provides a representative example of transformer architectures that operate directly on time–frequency representations and highlights design principles that are relevant for perceptual quality modeling.

2.3.7 Implications for Speech Quality Modeling

The review of speech quality assessment methods and transformer-based foundation models highlights both substantial progress and limitations in current approaches. These observations directly motivate the research questions addressed in this thesis.

First, while recent transformer-based pre-trained models have demonstrated strong representational power across a wide range of speech tasks, their adaptation to non-intrusive speech quality prediction remains underexplored. This motivates **RQ1**, which investigates how such models can be effectively leveraged and adapted to achieve accurate and reliable quality predictions without access to reference signals.

Second, existing data-driven speech quality models predominantly focus on predicting a single overall quality score, despite the multidimensional nature of human quality perception. The availability of rich foundation model representations raises the question of whether perceptually meaningful quality dimensions can be inferred directly from these embeddings and how closely such predictions align with subjective human judgments, as addressed in **RQ2**.

Third, the literature reveals limited consensus on whether modeling overall quality and perceptual dimensions jointly offers systematic benefits over isolated prediction tasks. This gap motivates **RQ3**, which examines the potential advantages of multi-task learning for improving robustness and consistency in speech quality assessment.

Fourth, the heterogeneity of datasets, recording conditions, and application domains remains a central challenge in speech quality modeling. Although foundation models are often credited with improved generalization, their effectiveness in cross-dataset and cross-domain quality assessment scenarios, including independent evaluation settings relevant to standardization, has not been thoroughly established. This observation underpins **RQ4**.

Finally, the growing diversity of speech-related assessment tasks raises the question of whether foundation model-based speech quality predictors can be transferred to novel or out-of-domain tasks without extensive retraining or task-specific engineering. Addressing this challenge is the focus of **RQ5**.

2.4 Summary

This chapter reviewed the scientific foundations and technological state of the art underlying modern speech quality assessment. It began by outlining the principles of human perception of speech quality, emphasizing that perceived quality emerges from the interaction of auditory processing, cognitive evaluation, and contextual factors rather than from acoustic properties alone. Building on this perspective, the chapter introduced the concept of perceptual quality dimensions and discussed how overall quality judgments can be decomposed into distinct dimensions, namely: noisiness, discontinuity, coloration, and loudness, that together enable more diagnostic and interpretable quality assessment.

The chapter introduced how human subjective ratings of speech quality are obtained using subjective listening tests. It then surveyed the evolution of instrumental approaches to speech quality prediction. Classical reference-based models, including PSQM, PESQ, and POLQA, were presented as historically dominant solutions that estimate perceptual degradation through explicit comparison between a degraded signal and a clean reference. Their perceptual motivation, signal-alignment strategies, and central role in standardization were discussed alongside their inherent limitations, most notably their reliance on reference signals and restricted applicability in real-world monitoring scenarios. This was followed by a review of non-intrusive instrumental models, which aim to predict perceived quality directly from the received signal. Early rule-based and statistical approaches, such as ITU-T Recommendation P.563, were discussed in the context of the telecommunication environments for which they were originally designed and have been reported in the literature to exhibit limited robustness when applied to modern speech-processing pipelines.

Subsequently, the chapter examined the transition toward data-driven and deep-learning-based speech quality models. Classical ML approaches marked a shift toward supervised learning from subjective listening data, while modern deep neural architectures further reduced reliance on hand-crafted features through end-to-end representation learning. Recent models demonstrated that non-intrusive prediction of both overall speech quality and perceptual dimensions is feasible across a wide range of distortions, including those introduced by speech enhancement, coding, and synthesis systems.

The latter part of the chapter discussed SSL and transformer-based architectures for speech representation learning. It introduced the general principles underlying SSL and transformer models and reviewed widely used pre-trained speech foundation models, including wav2vec 2.0, HuBERT, WavLM, Whisper, and the AST. These models were presented with respect to their core architectural characteristics, training paradigms, and input representations. Despite their demonstrated success in speech recognition, representation learning, and related downstream tasks, these foundation models have not been systematically investigated for non-intrusive prediction of perceptual quality dimensions of speech alongside overall speech quality. This gap motivates the investigation of foundation-model-based representations for speech quality modeling pursued in this thesis. The following chapter introduces the methodological framework of this thesis, detailing the datasets, evaluation protocols, and experimental design used to investigate how selected foundation-model encoders can be adapted for the prediction of overall speech quality and perceptual dimensions.

Chapter 3
Methodology

This chapter establishes the methodological foundation of the thesis. It formalizes the task of non-intrusive speech quality prediction, defines the experimental assumptions under which pretrained speech foundation models are investigated, and specifies the learning and evaluation protocols employed throughout the work. In addition, it describes the datasets used and the evaluation metrics adopted for performance assessment. Building on the conceptual background and prior work reviewed in the preceding chapters, the following section formally defines the prediction task considered in this thesis.

3.1 Problem Definition

In this thesis, non-intrusive speech quality prediction is treated as a supervised regression task operating on a single observed speech signal. Conceptual aspects of speech quality perception, listening test methodology, and quality dimensions have been reviewed in Chapter 2; the present section focuses solely on the formal task definition used for modeling and evaluation.

Let $x(t)$ denote a single-channel speech waveform of finite duration, with utterance lengths typically ranging from approximately 4 to 12 seconds, representing the output of a communication or processing system. The objective is to learn a mapping

$$f_\theta : x(t) \rightarrow \hat{y},$$

where $\hat{y} \in \mathbb{R}^K$ denotes the predicted quality target vector and θ the model parameters. Depending on the experimental setting, $\hat{y}$ may represent a scalar quality estimate (e.g., $K = 1$ for overall MOS) or a multidimensional vector of perceptual quality dimensions (e.g., $K > 1$ for noisiness, discontinuity, coloration, and loudness).

The ground truth, or target, speech quality labels used in this thesis are primarily derived from subjective listening experiments and correspond to aggregated human judgments. In addition, proxy labels obtained from established instrumental speech quality models are used when human ratings are unavailable or when conducting additional listening experiments is impractical due to time or resource constraints. As a result, the learning problem is characterized by continuous-valued targets, inter-listener

variability, and varying levels of label fidelity across datasets. The same task formulation is applied consistently across all in-domain experiments, independent of specific degradation types, transmission conditions, or application domains.

3.2 Data

This work draws on a large and heterogeneous collection of speech quality databases obtained from prior subjective listening experiments and related evaluation efforts conducted at the Quality and Usability Lab, Technische Universität Berlin, and its partners. In general, a speech quality database consists of a set of speech samples generated under controlled or naturally occurring conditions and annotated with perceptual quality judgments collected from human listeners, typically following ITU guidelines (ITU-T Recommendation P.800, 1996; ITU-T Recommendation P.808, 2021). Many of the databases used in this work were originally collected in the context of developing and evaluating instrumental speech quality assessment models standardized by the ITU, such as ITU-T Recommendation P.863 (2018). As several of these databases date back multiple decades, potential temporal effects in subjective speech quality judgments were explored in prior work (Wardah et al., 2025c); however, a detailed analysis of this phenomenon is beyond the scope of the present thesis. It should be noted that the database collection has grown to its current size over the course of this thesis, hence, not all these databases were available during every phase of this work.

An overview of the individual databases, including their size, language, and the number of votes per file for each perceptual dimension, is provided in Appendix A. The dataset collection comprises over one hundred speech quality databases and spans nearly two hundred thousand speech files in total, where each speech file has a duration ranging from approximately 4 to 12 seconds. The speech material covers a diverse set of languages, including English, German, French, Swedish, Italian, Czech, Finnish, Mandarin, and Japanese. Across datasets, speech samples are associated with a wide range of acoustic conditions, including different transmission impairments, noise environments, bandwidth constraints, and processing artifacts. Depending on the data collection paradigm, conditions may be applied systematically to reference speech material or arise implicitly in live or recorded communication scenarios. As a result, individual datasets differ substantially in size, structure, number of speakers, and condition coverage.

Subjective human ratings obtained from controlled listening tests are regarded as the gold-standard label source in this thesis. However, not all databases were subjectively labeled across all perceptual dimensions, and unfortunately, conducting additional large-scale listening experiments to obtain missing perceptual dimension labels was not feasible within the scope of this work. To address this limitation and to maximize dataset size, proxy labels generated by established instrumental speech quality models are used selectively. These include both reference-based approaches, where a clean reference signal is available, and single-ended approaches, where only the degraded signal is accessible. Such proxy labels are used as supervision signals during training but are not treated as equivalent to human perceptual judgments for evaluation purposes.

Given the diversity of datasets and the varying fidelity of available labels, uniform experimental conditions and homogeneous perceptual label distributions are not assumed across databases. Variability in listening test protocols and data characteristics is instead treated as an inherent property of the data. A shared task formulation and evaluation framework is applied across datasets, with model evaluation relying exclusively on fully subjectively labeled data.

3.3 Experiment Design

This section describes the experimental setups used in this thesis. All experiments are based on a common task formulation and evaluation protocol, while differing in how pre-trained transformer models are used and adapted. Specifically, the experiments cover zero-shot evaluation, supervised in-domain training, and out-of-domain fine-tuning, which are presented in the following subsections. Across all experimental settings considered in this thesis, the same regression head architecture is employed to map learned speech representations to quality targets. Differences between zero-shot, single-task, and multi-task learning formulations arise solely from which components of the model are updated during training—namely, whether the encoder parameters are kept frozen or optimized jointly with the regression heads.

Zero-shot Evaluation

First, a zero-shot evaluation setting is used to assess the suitability of pre-trained speech representations for the speech quality domain. Representations extracted from several transformer-based foundation models, including wav2vec 2.0, HuBERT, WavLM, Whisper, and AST, are evaluated using a regression head trained on a limited development subset, while all foundation model parameters remain frozen.

The regression head implements a nonlinear mapping that provides a task-specific readout required to convert pre-trained representations into continuous speech quality estimates, without updating the underlying encoder representations. The evaluation setup is illustrated in Figure 3.1. Performance is assessed on a held-out test set that remains unseen during model selection. This evaluation serves to identify a suitable pre-trained representation for subsequent experiments and is described in Chapter 4.

Multi-task Learning

Next, the selected pre-trained representation is adapted for speech quality prediction using a multi-task learning formulation. In this setting, a shared transformer encoder is jointly optimized with task-specific prediction heads to estimate overall speech quality as well as the four perceptual quality dimensions. In addition, the model is adapted to operate on speech quality datasets sampled at 48 kHz, in contrast to the 16 kHz sampling rate used during pre-training of the foundation models. The multi-task formulation allows the shared representation to be optimized jointly across related speech quality

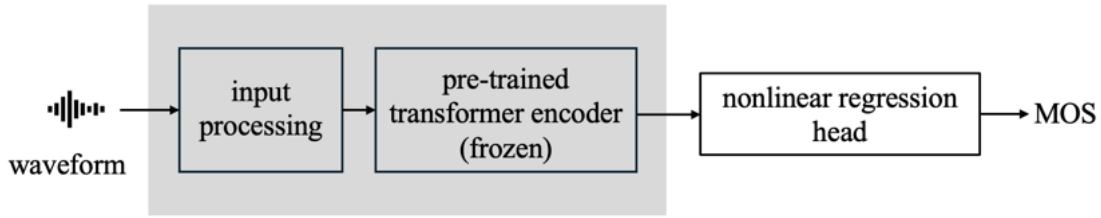


Fig. 3.1: An overview of the zero-shot evaluation setup used in this work. Pre-trained transformer encoders are kept fixed and combined with a nonlinear regression head that maps the extracted representations to speech quality estimates.

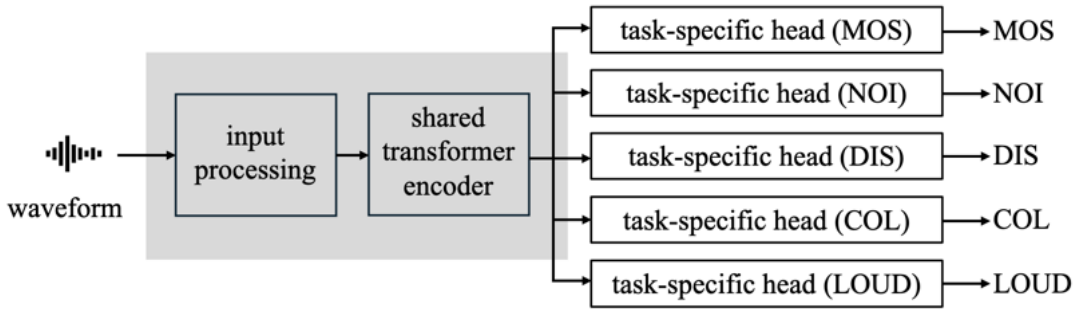


Fig. 3.2: Overview of the proposed multi-task speech quality modeling framework. The foundation model serves as a shared transformer backbone whose parameters are updated using supervision from all target dimensions, while independent task-specific heads are used for each speech quality target and are updated only with respect to their corresponding losses.

targets while maintaining separate prediction heads for each dimension. Each task-specific head is trained using its own regression objective, and the overall training objective is formed by combining the individual task losses. The multi-task setup is illustrated in Figure 3.2 and is described in Chapter 5.

Single-task Learning

Finally, the selected pre-trained representation that was considered in the multi-task setup is adapted using a single-task learning formulation, where each model is optimized to predict one speech quality target at a time. In total, five separate models are trained, corresponding to overall speech quality and the four perceptual quality dimensions of noisiness, discontinuity, coloration, and sub-optimal loudness. The single-task setup is illustrated in Figure 3.3 and is described in Chapter 6.

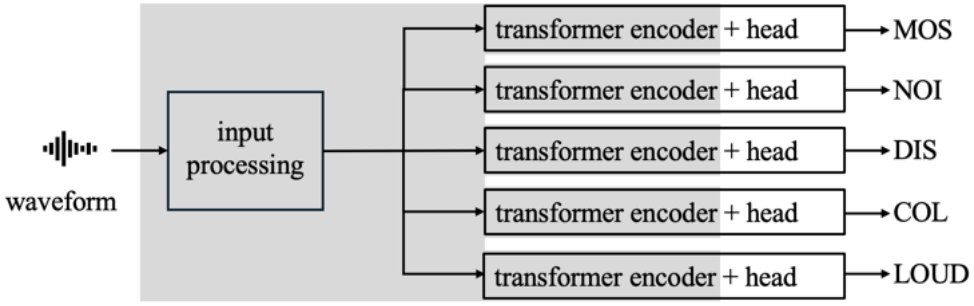


Fig. 3.3: Overview of the single-task inference setup used in this work. Each speech quality dimension is predicted by an independently trained transformer-based model consisting of an encoder and a task-specific head. While the models are trained separately, they are deployed in parallel during inference to jointly produce a multidimensional quality estimate.

Out-of-Domain Adaptation

Finally, the speech quality prediction model from the previous task of this work is evaluated and adapted on out-of-domain (OoD) data to assess its generalization beyond the primary telecommunication domain. In this setting, a speech quality pre-trained model is fine-tuned end-to-end on synthetic speech data representing previously unseen conditions. This experimental setup is described in Chapter 7.

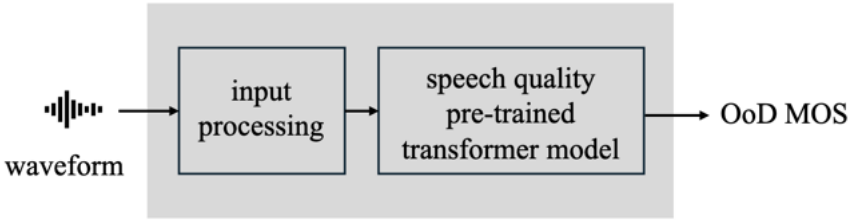


Fig. 3.4: An overview of the OoD adaptation setup used in this work. A transformer-based speech quality model pretrained on in-domain data is fine-tuned end-to-end on out-of-domain speech signals to assess generalization and adaptation to synthetic speech conditions.

3.4 Evaluation Metrics

The evaluation methodology used in this thesis follows established best practices for instrumental speech quality assessment as summarized in ITU-T Recommendation P.1401 (2020). Model performance is assessed, where possible, exclusively on datasets with available subjective listening test labels, such that reported metrics reflect agreement with human perceptual judgments.

Evaluation is performed on a per-database basis. This avoids confounding effects arising from differences in listening test design, rating scales, and listener populations across datasets, and allows performance to be interpreted relative to the conditions under which the subjective data were collected.

Metrics are computed at two levels of aggregation. File-level metrics quantify prediction accuracy across individual speech samples. Condition-level metrics are obtained by averaging predictions and subjective ratings across files sharing the same transmission or degradation condition, thereby reducing variance due to speaker and content effects. Both levels of analysis are reported.

Due to intra- and inter-listener variability, subjective quality ratings are not treated as absolute reference values. Instead, evaluation focuses on the relative ordering of quality between samples within a database, which has been shown to be stable between repeated listening experiments. Consequently, before the metric computation, a monotonic mapping between predicted scores and subjective ratings is applied, as recommended in ITU-T P.1401, to account for scale differences inherent to subjective listening tests and speech quality databases in general. Within each database, prediction accuracy is quantified using the Pearson correlation coefficient (PCC), the root mean square error (RMSE), and the epsilon-adjusted root mean square error (RMSE*). These metrics are described in the following subsections.

3.4.1 Post-Mapping of Predicted Scores

Following the recommendations of ITU-T P.1401, a post-mapping function is applied to instrumental model predictions prior to the computation of error-based performance metrics. The purpose of this mapping is to compensate for the systematic scale and offset differences between the predicted scores and subjective ratings. Such post-mapping improves numerical agreement with subjective scores without altering the underlying prediction model.

First Order Mapping

Let y denote the subjective quality score and $\hat{y}$ the corresponding instrumental model prediction. The simplest and most commonly used mapping is a first-order affine function,

$$y_1 = a_1 + b_1 \hat{y}, \tag{3.1}$$

where the parameters a_1 and b_1 are estimated by least-squares minimization of the squared error between y and y_1. This mapping corrects for global bias and scale mismatch and is commonly used to calibrate instrumental predictions to the subjective rating scale. Algorithm 1 presents the procedure used in this work to obtain the parameters a_1 and b_1. In the implementation, the least-squares solution is computed using `numpy.linalg.lstsq`. Once a_1 and b_1 are determined, Equation 3.1 is applied to all instrumental predictions to obtain the final mapped scores.

Algorithm 1 First-order (affine) post-mapping by least squares

Require: Target scores $\mathbf{y} \in \mathbb{R}^N$, Instrumental predictions $\hat{\mathbf{y}} \in \mathbb{R}^N$
Ensure: Mapping parameters a_1 (intercept) and b_1 (slope)
1: Remove pairs $(y_i, \hat{y}_i)$ containing NaN values to obtain cleaned vectors of length M
2: **if** $M < 2$ **then**
3: **return** `None, None` ▷ Insufficient data for regression
4: Construct design matrix $\mathbf{X} \in \mathbb{R}^{M \times 2}$ where $\mathbf{X}_{i,:} = [1, \hat{y}_i]$
5: Solve $[a_1, b_1]^\top \leftarrow \arg\min_{\mathbf{w} \in \mathbb{R}^2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$
6: **return** a_1, b_1

Third-Order Mapping

More flexible monotonic mappings may be employed when additional curvature is required to align predicted scores with subjective ratings. A commonly used higher-order alternative is the monotonic part of a third-order polynomial,

$$y_3 = a_3 + b_3\hat{y} + c_3\hat{y}^2 + d_3\hat{y}^3, \tag{3.2}$$

where the coefficients are optimized under a monotonicity constraint. Such mappings allow for nonlinear calibration while enforcing a non-decreasing relationship between predicted and subjective scores, thereby largely preserving the ordering of predictions. In the implementation, the constrained optimization is solved using `scipy.optimize.minimize` with the `SLSQP` method. For the monotone third-order mapping (Algorithm 2), a grid size of $G = 50$ is used to enforce the monotonicity constraint. Once the coefficients a_3, b_3, c_3, and d_3 are determined, the mapping defined in Equation 3.2 is applied to all instrumental predictions to obtain the final mapped scores.

It is emphasized that these mapping functions do not enhance the intrinsic perceptual modeling capability of the instrumental model. Instead, they provide an optimal calibration of predicted scores to the subjective rating scale within a predefined functional form. Consequently, correlation-based metrics such as the Pearson correlation coefficient remain invariant under first-order affine post-mapping, whereas absolute error measures such as RMSE and RMSE* are reduced.

3.4.2 Absolute Prediction Error (RMSE)

The accuracy of instrumental quality predictions is quantified using the root mean square error (RMSE), which measures the absolute deviation between subjective ratings and mapped model predictions. RMSE reflects the numerical agreement between predicted and subjective scores after post-mapping and is therefore sensitive to residual prediction errors that remain after global bias and scale differences have been compensated.

Let $y(i)$ denote the subjective quality score of speech sample i, and let $y_m(i)$ denote the mapped prediction for the same sample, where y_m may correspond to a first-order or higher-order monotonic mapping (e.g., y_1 or y_3). The prediction error for sample i is defined as

Algorithm 2 Monotone third-order (cubic) post-mapping by constrained least squares

Require: Target scores $\mathbf{y} \in \mathbb{R}^N$, predictions $\hat{\mathbf{y}} \in \mathbb{R}^N$, grid size G
Ensure: Coefficients a_3, b_3, c_3, d_3 for $y_3 = a_3 + b_3\hat{y} + c_3\hat{y}^2 + d_3\hat{y}^3$
Remove pairs $(y_i, \hat{y}_i)$ containing NaN values to obtain vectors of length M
if $M < 4$ **then**
    return `None`
Set $x_{\min} \leftarrow \min_i \hat{y}_i$, $x_{\max} \leftarrow \max_i \hat{y}_i$
if $x_{\min} = x_{\max}$ **then**
    return `None`
Construct grid $\mathcal{G} = \{g_1, \ldots, g_G\}$ linearly spaced in $[x_{\min}, x_{\max}]$
Construct design matrix $\mathbf{X} \in \mathbb{R}^{M\times 4}$ with $\mathbf{X}_{i,:} = [1, \hat{y}_i, \hat{y}_i^2, \hat{y}_i^3]$
Define objective $J(\mathbf{w}) \leftarrow \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$ with $\mathbf{w} = [a_3, b_3, c_3, d_3]^\top$
Enforce monotonicity constraints on $\mathcal{G}$:

$$b_3 + 2c_3 g_k + 3d_3 g_k^2 \geq 0 \quad \forall g_k \in \mathcal{G}$$

Initialize $\mathbf{w}_0 \leftarrow \arg\min_{\mathbf{w}} J(\mathbf{w})$ ▷ unconstrained least squares
Solve $\mathbf{w}^* \leftarrow \arg\min_{\mathbf{w}} J(\mathbf{w})$ subject to the constraints (SLSQP)
if solver fails **then**
    return `None`
return a_3, b_3, c_3, d_3 from $\mathbf{w}^*$

$$\mathrm{Perror}(i) = y(i) - y_m(i), \tag{3.3}$$

where the index i enumerates the speech samples within a database.

The RMSE is then computed as

$$\mathrm{RMSE} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N} \mathrm{Perror}(i)^2}, \tag{3.4}$$

where N denotes the total number of samples. The normalization by $N - 1$ yields an unbiased estimator of the error variance under the assumption of independent samples.

Following ITU-T P.1401, the RMSE is approximately characterized by a chi-squared distribution $\chi^2(n)$, where the number of degrees of freedom n is given by

$$n = N - d, \tag{3.5}$$

with d denoting the number of free parameters of the applied mapping function. For example, $d = 2$ for a first-order (linear) mapping and $d = 4$ for a third-order polynomial mapping.

3.4.3 Epsilon-Insensitive RMSE (RMSE*)

As discussed above, the absolute prediction error is quantified using the prediction error (Perror) and summarized by the RMSE. However, subjective quality ratings are inherently affected by statistical uncertainty due to finite listener populations and intra- and inter-listener variability. To explicitly account for this uncertainty, ITU-T P.1401 introduces the epsilon-insensitive RMSE, denoted as RMSE*.

The RMSE* metric extends the conventional RMSE by incorporating the confidence interval of the subjective quality scores. Small deviations between predicted and subjective scores that fall within the uncertainty range of the subjective measurement are not penalized. As a result, RMSE* reflects only those prediction errors that exceed the expected variability of the subjective data.

Let $y(i)$ denote the subjective mean opinion score (MOS) of sample i, and let $y_m(i)$ denote the mapped instrumental prediction. The modified prediction error is defined as

$$\mathrm{Perror}(i) = \max\left(0, |y(i) - y_m(i)| - \epsilon(i)\right), \tag{3.6}$$

where $\epsilon(i)$ represents the uncertainty margin of the subjective score for the sample i. According to ITU-T P.1401, $\epsilon(i)$ is defined as the half-width of the 95 % confidence interval of the MOS.

The epsilon-insensitive RMSE is then computed analogously to the conventional RMSE, using the modified prediction error defined in Equation 3.6.

By construction, RMSE* discounts prediction errors that are statistically insignificant relative to the uncertainty in subjective scores. This enables a more robust and significance-aware comparison of instrumental models, particularly when subjective ratings exhibit substantial confidence intervals.

The PCC quantifies the strength of the linear association between subjective ratings and instrumental model predictions and is invariant to affine transformations of the predicted scores. In contrast, the RMSE measures the absolute numerical agreement between the calibrated predictions and subjective scores. RMSE* further accounts for the statistical uncertainty of subjective ratings by discounting prediction errors that fall within the confidence interval of the listening test. Together, these metrics provide complementary perspectives on rank consistency, numerical accuracy, and statistical significance.

Reproducibility and Code Availability

All scripts used for model training and evaluation across the experiments presented in this work are publicly available.[1] The repository contains the code corresponding to all chapters of this work and enables reproduction of the reported results, subject to dataset licensing constraints.

3.5 Summary

This chapter established the methodological foundation of the thesis. It formally defined non-intrusive speech quality prediction as a supervised regression task operating on a single observed speech signal and clarified the assumptions under which pretrained transformer-based foundation models are investigated. The task formulation was defined independently of specific degradation types or application domains, enabling consistent experimentation across heterogeneous datasets.

[1] `https://github.com/WafaaWardah/speech-quality-modeling`

A large and diverse collection of speech quality databases was introduced, comprising over 100 data sets collected under varying listening test protocols, languages, and acoustic conditions. Subjective human ratings were treated as the primary ground-truth reference, while proxy labels generated by established instrumental models were used selectively for supervision when subjective data were unavailable. Variability across datasets was explicitly acknowledged as an inherent property of the data rather than an artifact to be eliminated.

The chapter then described the experimental designs employed throughout the thesis, including zero-shot evaluation of pretrained representations, supervised multi-task learning across multiple perceptual quality dimensions, single-task learning for dimension-specific modeling, and out-of-domain adaptation to assess generalization beyond the primary telecommunication domain. These experimental paradigms provide a structured framework for analyzing representation quality, learning strategies, and transfer behavior in subsequent chapters.

Finally, a standards-compliant evaluation methodology was defined according to ITU-T P.1401. Monotonic mapping functions were introduced to align predicted scores with subjective ratings while preserving rank order, and three complementary performance metrics (PCC, RMSE, and RMSE*) were specified to capture rank consistency, calibrated numerical accuracy, and statistical significance relative to subjective uncertainty.

Together, the definitions, datasets, experimental protocols, and evaluation methodology presented in this chapter form a coherent and reproducible framework for the empirical analyses that follow.

Chapter 4
Zero-Shot Comparison

This chapter investigates the suitability of pretrained speech foundation models for non-intrusive speech quality prediction under a zero-shot evaluation setting. The objective is to assess the intrinsic transferability of learned speech representations without task-specific fine-tuning of the encoder parameters. To this end, representations extracted from several transformer-based models are evaluated using a downstream regression mapping trained on a limited development subset, while all pretrained model parameters remain frozen. Performance is assessed on held-out test data with subjective listening test labels, enabling a controlled comparison across architectures and pre-training strategies. The results of this chapter are used to identify representations that are well-aligned with perceptual speech quality and to motivate the modeling choices explored in subsequent chapters.

4.1 Introduction

The zero-shot evaluation protocol adopted in this chapter assesses the transferability of pretrained speech representations to the speech quality domain while minimizing task-specific adaptation of the pretrained encoders. Each pretrained foundation model is treated as a fixed feature extractor: all encoder parameters are frozen and no representation-level adaptation to the speech quality task is performed. An overview of this evaluation setup is illustrated in Figure 3.1 in the preceding chapter.

The foundation models considered in this work consist of transformer-based encoders pretrained on large-scale speech or audio datasets using task-specific or SSL objectives. These were introduced in Section 2.3. Wav2vec 2.0, HuBERT, and WavLM operate directly on raw audio waveforms and are pretrained using SSL objectives on large-scale unlabeled speech corpora. Whisper and the AST, in contrast, operate on time-frequency representations. Whisper processes log-Mel spectrogram inputs and is pretrained in a supervised or weakly supervised manner on large multilingual speech datasets for ASR. AST likewise operates on log-Mel spectrograms, which are framed and patchified prior to transformer encoding, and is obtained by fine-tuning a Vision Transformer initialized from ImageNet pretraining on the AudioSet audio classification task.

In this chapter, the evaluation is restricted to overall perceived speech quality. While the broader objective of this thesis includes the prediction of multiple perceptual quality dimensions, these are intentionally not considered in the zero-shot setting, as the present analysis aims to establish a principled baseline for representation suitability before introducing more complex prediction tasks.

Because pretrained representations are not inherently aligned with perceptual speech quality conventions, such as the ACR scale ranging from 1 to 5, a task-specific adaptor is required. This adaptor is implemented as a downstream regression mapping that translates the model-specific representation space into an output space aligned with subjective speech quality labels. Importantly, this mapping serves solely as a lightweight readout and does not modify the underlying pretrained representations, providing a controlled and practically feasible approximation of zero-shot transfer to the speech quality domain.

4.2 Method

To achieve the objective of this chapter, the evaluation protocol is structured into two stages:

(i) a downstream regression adaptor is trained for each pretrained encoder to enable compatibility with speech quality labels;
(ii) the fixed representations produced by the pretrained encoders are evaluated on held-out test databases, using the trained adaptor, to assess their suitability for speech quality modeling in the zero-shot setting.

The following subsections elaborate on how these stages are realized in practice. All pretrained foundation model encoders evaluated in this work are instantiated using the publicly available reference implementations provided by the Hugging Face `transformers` library.

4.2.1 Downstream Mapping

All zero-shot experiments use pretrained speech foundation models as frozen encoders in combination with a uniform downstream regression mapping. The mapping operates on fixed-length encoder representations and is applied identically across all foundation model architectures. Its structure and training configuration are specified below.

Structure

For the waveform-based encoders (wav2vec 2.0, HuBERT, and WavLM), the frozen encoder produces a sequence of hidden states of shape $[B, S, 768]$, where B denotes the batch size and S the encoder-specific sequence length after temporal downsampling performed by the convolutional feature encoder. To obtain a fixed-length representation

(per input signal), mean pooling is applied over the temporal dimension. When an attention mask is available at the waveform level, it is converted to a feature-level mask aligned with the encoder output sequence length, and pooling is performed as a length-normalized masked mean over valid frames. This ensures that padding introduced by variable-length inputs does not influence the resulting representation.

Whisper produces a sequence of hidden states of shape $[B, S, 512]$. In this case, an utterance-level representation is obtained by unmasked mean pooling over the temporal dimension, yielding a fixed-length vector in $\mathbb{R}^{512}$. For AST, the encoder directly provides fixed-length representation via its internal pooling mechanism. Specifically, the `pooler_output` returned by the pretrained AST encoder is used as the fixed-dimensional representation passed to the downstream mapping, and no additional temporal aggregation is applied.

Training

All models evaluated in this chapter operate on audio sampled at 16 kHz. Consequently, all input signals are resampled to 16 kHz where necessary before feature extraction. The fixed-length representations produced by the frozen transformer encoders are mapped to scalar speech quality predictions using a uniform downstream regression head.

The downstream mapping consists of a single fully connected linear layer with scalar output followed by a sigmoid activation function, constraining predictions to the unit interval. For wav2vec 2.0, HuBERT, WavLM, and the AST, the linear layer maps from $\mathbb{R}^{768}$ to $\mathbb{R}$, resulting in 769 trainable parameters. For Whisper-based representations, the corresponding mapping operates on $\mathbb{R}^{512}$ and contains 513 trainable parameters. Apart from this dimensionality adjustment, the downstream mapping architecture is identical across all models.

Let $h \in \mathbb{R}^D$ denote the fixed-length representation extracted from the frozen pretrained encoder given an input speech signal $x(t)$. The downstream regression adaptor implements a mapping

$$g_\phi : h \rightarrow \hat{y}_{\text{scaled}},$$

where $\hat{y}_{\text{scaled}} \in [0, 1]$ denotes the predicted, normalized speech quality score and ϕ represents the trainable parameters of the adaptor.

Training of the downstream mapping is performed in a supervised manner using subjective speech quality labels defined on a five-point ACR scale ranging from 1 to 5. For optimization, target values are linearly scaled to the unit interval according to

$$y_{\text{scaled}} = \frac{y - 1}{4}. \tag{4.1}$$

The regression mapping is trained to predict scaled quality scores in $[0, 1]$. During evaluation, predictions are mapped back to the original quality scale using the inverse transformation

$$\hat{y} = 4\,\hat{y}_{\text{scaled}} + 1. \tag{4.2}$$

Training of the downstream mapping uses the mean squared error loss between predicted and target scores per speech file. Optimization is performed with the AdamW optimizer using a learning rate of 10^{-3} and a weight decay of 10^{-5}. A reduce-on-plateau learning rate scheduler is applied, reducing the learning rate by a factor of 0.5 when validation loss no longer improves, with a minimum learning rate of 10^{-6}. Training is performed for up to 100 epochs, with early stopping based on validation loss using a patience of 10 epochs, and the best-performing checkpoint is retained. All training procedures are implemented in PyTorch.

4.2.2 Datasets

The zero-shot comparison of pretrained foundation model representations is conducted using a subset of the datasets employed in this work, with a complete overview provided in Appendix A. To support training and model selection of the downstream mapping, two larger datasets are used for training and two smaller datasets are used for validation. In addition, six datasets are held out entirely from this process and used exclusively for evaluating the pretrained representations in the zero-shot setting. The individual datasets and their roles within the evaluation protocol are described below.

Training and Validation Datasets

Training and validation of the downstream mapping adaptor are performed using four databases drawn from the NISQA corpus (Mittag et al., 2021). The training split comprises 11,020 speech files, while the validation split contains 2,700 speech files. These datasets are used exclusively to learn the mapping from fixed pretrained encoder representations to subjective speech quality scores.

The selected databases reflect relatively recent data collection efforts and cover a broad range of contemporary telecommunication artifacts and degradation conditions. This ensures that the downstream mapping is trained under conditions representative of modern speech communication scenarios, while remaining agnostic to the specific test databases used for evaluation. All four databases contain English speech.

NISQA_TRAIN_SIM and NISQA_VAL_SIM

The `NISQA_TRAIN_SIM` database is used as part of the training set, while `NISQA_VAL_SIM` is used for validation. These databases comprise a broad range of artificially generated distortions applied to clean speech signals, including additive white Gaussian noise, signal-correlated MNRU noise, randomly sampled background noise, spectral filtering (low-pass, high-pass, band-pass, and arbitrary filters with randomly selected cutoff frequencies), amplitude clipping, and speech level modifications. In addition, the datasets include a variety of speech coding conditions, covering multiple bitrate configurations of AMR-NB, G.711, G.722, EVS, and Opus, as well as codec tandem and triple-tandem setups. Packet-loss impairments with both random and bursty loss patterns are also represented, along with combinations of multiple distortion types. Clean source

speech for the simulated datasets is drawn from four corpora: Librivox audiobook recordings included in the DNS-Challenge dataset, the TSP speech database, and the AusTalk corpus. Noise signals are sourced from the DNS-Challenge dataset, which aggregates recordings from AudioSet, Freesound (CC0 subset), and the DEMAND database.

NISQA_TRAIN_LIVE and NISQA_VAL_LIVE

The `NISQA_TRAIN_LIVE` database is used for training, while `NISQA_VAL_LIVE` is used for validation. These live databases comprise recordings obtained under real transmission scenarios. Clean speech signals are played back over a loudspeaker and transmitted either via a fixed-line VoIP-to-mobile phone call or through a laptop-to-laptop Skype connection, with the received signal recorded on the destination device. During recording, additional real-world disturbances are deliberately introduced, including variations in playback volume and loudspeaker orientation, keyboard typing, and environmental noise such as open windows. This design captures a range of realistic acoustic and transmission effects that are difficult to simulate reliably and complements the simulated distortion conditions used elsewhere in the training data.

Held-Out Evaluation Datasets

After training the downstream mapping adaptor, six databases are used exclusively for evaluating the pretrained foundation models. These datasets are held out entirely from training and validation and are selected to cover a broad range of degradation conditions, recording setups, and data sources. This diversity is intended to provide a rigorous assessment of representation transferability under heterogeneous and previously unseen evaluation conditions.

NISQA_TEST_NSC

The `NISQA_TEST_NSC` (NSC) database comprises 240 German speech files covering 60 distinct degradation conditions. It includes both simulated distortions, such as various speech codecs, additive background noise, packet-loss patterns, and amplitude clipping, and live communication scenarios, including calls conducted via Skype, Zoom, and Google Meet, as well as mobile-to-landline connections. The preparation of the simulated conditions follows a similar procedure as used for the `NISQA_TRAIN_SIM` and `NISQA_VAL_SIM` databases, but with different speech material. Subjective speech quality annotations were collected via crowdsourcing in accordance with ITU-T Recommendation P.808. The database is part of the NISQA corpus (Mittag et al., 2021).

NISQA_TEST_P501

The NISQA_TEST_P501 (P501) dataset consists of 240 English speech files covering 60 distinct degradation conditions. It is a controlled speech quality evaluation corpus comprising a diverse set of both simulated and live communication impairments. Simulated degradations include various speech codecs, additive background noise, packet-loss patterns, and amplitude clipping. In addition, the dataset contains live call recordings conducted via Skype, Zoom, and WhatsApp, as well as recordings captured over mobile networks. Subjective quality annotations were obtained through crowdsourcing in accordance with ITU-T Recommendation P.808. Clean source speech signals are taken from Annex C of the ITU-T Recommendation P.501 dataset. Background noise signals used for the simulated degradations are sourced from the same collections as those used for the NISQA_TRAIN_SIM and NISQA_VAL_SIM databases, but the speech material is different. This database is also part of the NISQA corpus (Mittag et al., 2021).

FB_TUB_DEV

The FB_TUB_DEV (DEV) dataset is a German speech quality corpus comprising 96 speech files and 24 distinct conditions, including both simulated degradations and device- and application-specific recording scenarios. Simulated distortions include signal-correlated noise (MNRU), additive noise, level attenuation, band-pass filtering, and temporal clipping at different severities. In addition, the dataset contains recordings captured using multiple consumer mobile devices (iPhone 6, iPhone 6S, Moto G, and Samsung Note 4) under various usage scenarios, including direct speech recording, telephony, and voice-over-IP applications such as Skype and Google Hangouts. Subjective quality ratings were collected following standardized listening test procedures in the laboratory. This is an internal database created at the Quality and Usability Lab at TU-Berlin in Germany by Steinmetz (2020).

SWB_501_SWISSQUAL

The SWB_501_SWISSQUAL (501_SQ) dataset is a controlled laboratory speech quality corpus developed in accordance with the ITU-T Recommendation P.800 (1996) and the ITU-T Recommendation P.863 (2018) requirements specification. It comprises 200 German speech samples and covers 50 distinct degradation conditions. The dataset includes a broad range of controlled signal processing and transmission impairments, such as additive background noise, signal-correlated noise (MNRU), level attenuation, band-pass filtering, temporal clipping, and numerous codec-based conditions spanning narrowband and wideband speech (e.g., AMR-NB, AMR-WB, EVRC, EFR, and AAC-LC). This database is obtained from the lab of SwissQual AG in Switzerland.

Orange_C122_DB01

The `Orange_C122_DB01` (DB01) dataset is a controlled laboratory speech quality corpus designed to assess the perceptual effects of bandwidth limitation, noise, and modern speech coding technologies. The dataset includes signals at multiple bandwidths, conditions with signal-correlated noise (MNRU), and a broad range of codec-based degradations covering EVS, Opus, AMR-WB, AMR-NB, and AMR-HR across different bit rates and bandwidth modes. In total, the corpus comprises 864 speech files distributed over 36 distinct conditions. Subjective quality ratings were obtained using standardized laboratory listening tests. This database is obtained from Orange Innovations in France and is described in the ITU-T Contribution SG12-C122 (2023).

SUB_TUB_1

The `SUB_TUB_1` (TUB1) dataset is a controlled German speech quality evaluation corpus comprising 214 speech files distributed across 54 distinct conditions. It covers all bandwidth transmission conditions and combines clean reference signals with a structured set of signal-processing and transmission degradations including additive background noise at multiple signal-to-noise ratios (babble, car, and street noise), signal-correlated noise (MNRU), bandwidth limitation with varying cutoff frequencies, level attenuation and enhancement, and temporal signal clipping. In addition, the dataset includes codec-based conditions representative of practical communication systems, such as G.711, G.729, G.722, AMR, AMR-WB, and EVS, as well as combinations of coding, noise, and level distortions. Subjective quality ratings were obtained using standardized listening test procedures. This is an internal database created at the Quality and Usability Lab at TU-Berlin in Germany.

4.3 Results and Discussion

This section presents the results of the zero-shot evaluation described in the preceding sections. Model performance is assessed on subjective speech quality datasets that remain entirely unseen during downstream training and model selection. All results are obtained using frozen pretrained encoders in combination with an identical downstream regression mapping architecture, with model-specific parameters learned during training. Consequently, observed performance differences can be attributed to the transferability of the pre-trained representations rather than to effects of task-specific adaptation.

4.3.1 Downstream Training Behaviour

The training dynamics of the downstream regression mapping are examined prior to evaluating performance on held-out test sets. Figure 4.1 shows the training and validation loss curves for all pretrained encoders under the zero-shot evaluation setting.

Across all models, a rapid decrease in training loss is observed during the first few epochs, followed by stable convergence. This indicates that the regression mapping can be fitted efficiently on top of all frozen representations and that optimization does not pose a limiting factor. No signs of unstable training is observed across models.

Despite similar convergence behaviour, systematic differences in the final loss values emerge. In particular, the training loss converges to lower values for Whisper, AST, and HuBERT than for wav2vec 2.0, with WavLM exhibiting intermediate behaviour. This suggests differences in how readily speech quality information can be linearly extracted from the respective representations. The validation loss curves further highlight these differences. While all models reach a plateau after a small number of epochs, wav2vec 2.0 exhibits consistently higher validation loss and stronger fluctuations compared to the other encoders. In contrast, Whisper, AST, and HuBERT show smoother validation curves and lower final validation loss, indicating more stable generalization of the downstream mapping.

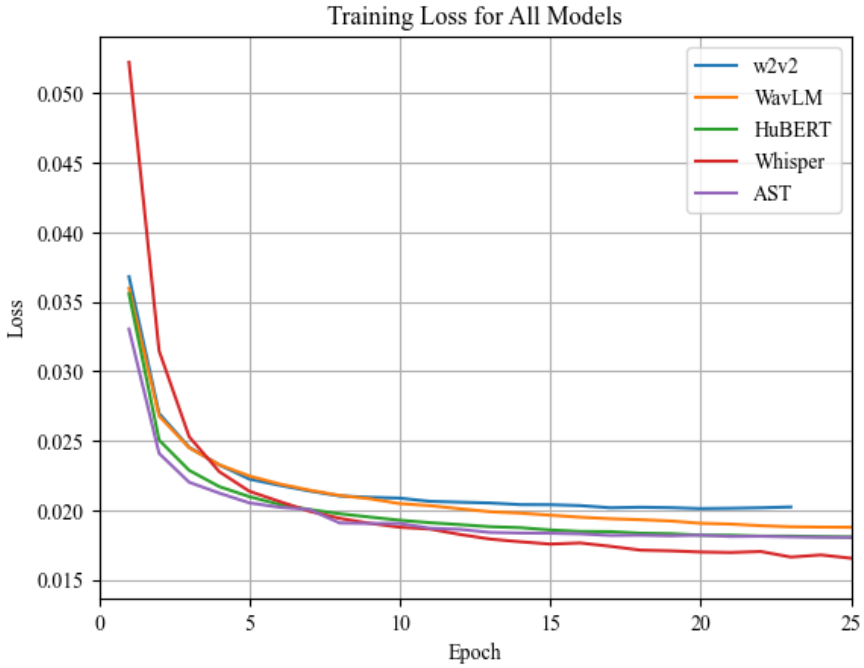
Given that the encoder parameters are frozen and the downstream regression architecture is identical across models, the observed differences in convergence level and validation behaviour can be attributed to differences in the pretrained representations rather than to optimization effects. These findings motivate the subsequent analysis of zero-shot performance on held-out test sets, where the impact of representation quality on speech quality prediction is evaluated directly.

4.3.2 Zero-Shot Evaluation on Held-Out Datasets

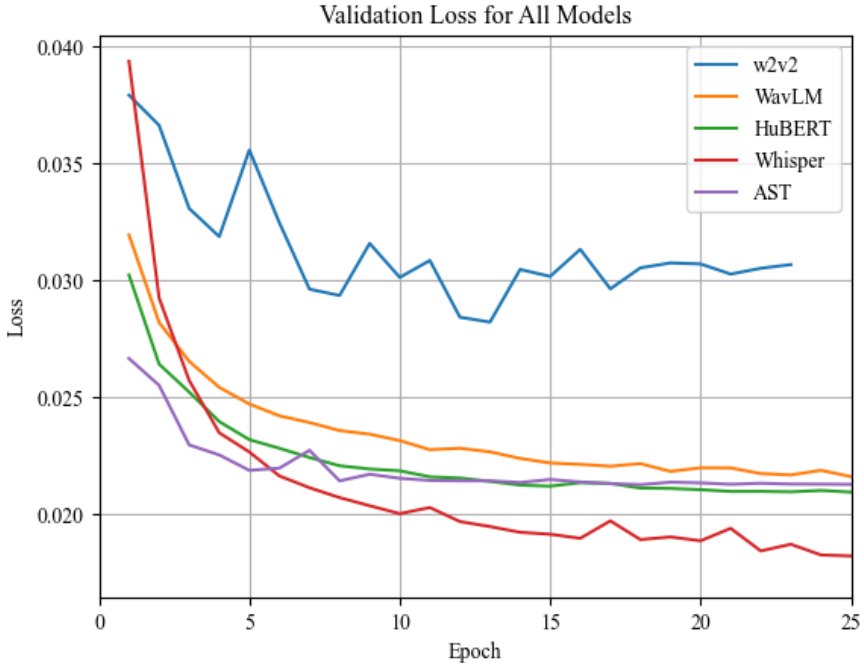
This section reports the zero-shot evaluation results obtained on held-out speech quality datasets. All models are evaluated using frozen pretrained encoders combined with their respective downstream regression mapping adaptors, enabling a controlled comparison of pretrained representations with respect to their suitability for speech quality prediction.

Results are reported on a per-database basis to avoid confounding effects due to differences in listening test design, rating scales, and listener populations. Table 4.1 summarizes file-level performance in terms of Pearson correlation coefficient (r), RMSE, and RMSE* (see Section 3.4 for definitions). The task-specific NISQA speech quality model is included for reference.Table 4.2 reports the corresponding condition-level results, where predictions and subjective scores are averaged per degradation condition. The task-specific NISQA (MOS-only) speech quality model is included for reference. It processes speech signals at 48 kHz, in contrast to the evaluated foundation models considered in this chapter.

Across all held-out evaluation datasets, all pretrained encoders achieve consistently positive correlations with subjective quality MOS at both file and condition level, as illustrated in Fig. 4.3. This indicates that pretrained speech representations encode information that is relevant to perceived speech quality, even though the models were not optimized for this task during pretraining.



(a) Training loss.



(b) Validation loss.

Fig. 4.1: Training and validation loss curves of the downstream regression mapping for all pretrained encoders under the zero-shot evaluation setting. Encoder parameters are frozen, and an identical nonlinear regression mapping architecture is trained for each model to assess representation suitability independently of encoder optimization.

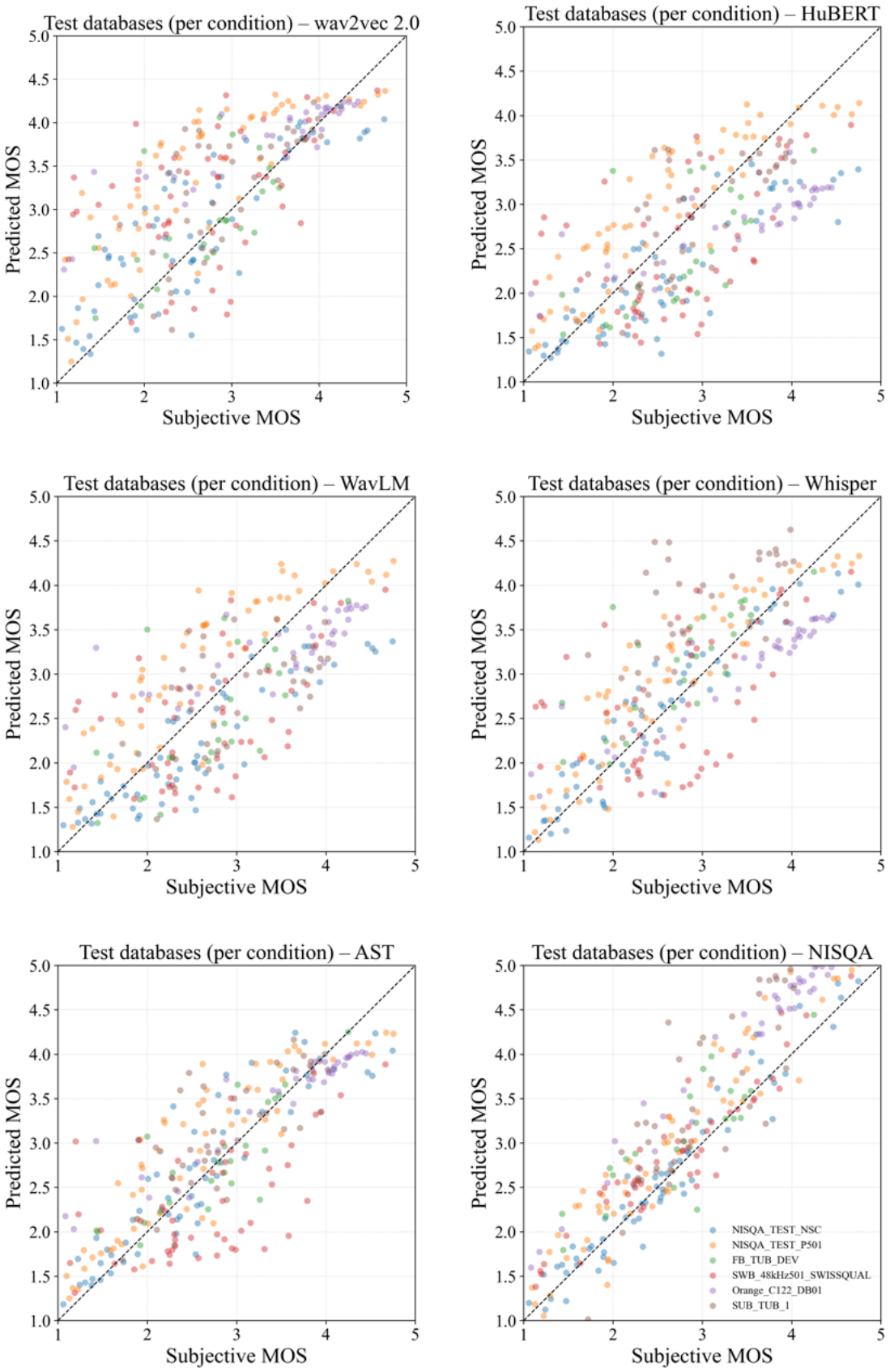


Fig. 4.3: Predicted MOS versus subjective MOS pooled across all held-out test databases for different frozen pretrained encoders and the NISQA speech quality model.

Table 4.1: Zero-shot performance on held-out test databases evaluated at the file level. Pearson correlation (r) is computed on raw predictions prior to mapping. RMSE is reported after first-order mapping, and RMSE* after third-order monotonic mapping. The task-specific NISQA speech quality model is included for reference. For the zero-shot models, the lowest RMSE* values per dataset are highlighted in bold.

Model	Metric	NSC	P501	DEV	501_SQ	DB01	TUB1	Avg.
wav2vec 2.0	r	0.785	0.785	0.606	0.402	0.764	0.638	0.664
	RMSE	0.590	0.632	0.612	0.788	0.688	0.515	0.637
	RMSE*	0.360	0.397	0.363	**0.535**	0.387	0.334	0.396
HuBERT	r	0.823	0.852	0.648	0.486	0.699	0.685	0.699
	RMSE	0.541	0.535	0.585	0.752	0.763	0.487	0.610
	RMSE*	0.364	0.346	0.354	0.542	0.463	0.321	0.398
WavLM	r	0.852	0.853	0.654	0.454	0.569	0.518	0.650
	RMSE	0.499	0.532	0.582	0.766	0.877	0.572	0.638
	RMSE*	0.315	0.325	0.347	0.547	0.547	0.395	0.413
Whisper	r	0.900	0.885	0.724	0.470	0.741	0.557	0.713
	RMSE	0.415	0.476	0.530	0.759	0.716	0.556	0.575
	RMSE*	**0.225**	**0.284**	0.313	0.536	0.423	0.384	**0.361**
AST	r	0.833	0.826	0.756	0.504	0.770	0.688	0.730
	RMSE	0.527	0.575	0.503	0.743	0.680	0.486	0.586
	RMSE*	0.348	0.404	**0.277**	**0.535**	**0.386**	**0.320**	0.378
NISQA	r	0.929	0.904	0.747	0.907	0.870	0.794	0.858
	RMSE	0.353	0.437	0.511	0.362	0.527	0.407	0.433
	RMSE*	0.181	0.274	0.268	0.166	0.250	0.250	0.232

Within this overall capability, however, clear performance differences emerge between the evaluated models. Among the frozen encoders, AST and Whisper consistently achieve the strongest zero-shot performance across datasets, particularly at the condition level. AST attains the highest average condition-level correlation (see Table 4.2), while Whisper yields the lowest average RMSE* at the file level (see Table 4.1).

Waveform-based self-supervised models, including wav2vec 2.0, HuBERT, and WavLM, demonstrate competitive but consistently lower performance. Their limitations become especially apparent on heterogeneous datasets, suggesting reduced robustness to variability unrelated to linguistic content.

Role of Pretraining Objectives and Input Representations

The strong zero-shot performance of AST and Whisper suggests that representations derived from time-frequency inputs combined with supervised or weakly supervised objectives are particularly well aligned with perceptual speech quality. AST benefits from fine-tuning on AudioSet, which exposes the model to a broad range of acoustic events and degradations that overlap with quality-relevant cues. Whisper, although trained

Table 4.2: Zero-shot performance on held-out test databases evaluated at the condition level. Predictions and subjective scores are averaged per condition prior to metric computation. Pearson correlation (r), RMSE, and RMSE* are reported following the same mapping procedure as in Table 4.1. The task-specific NISQA speech quality model is included for reference. For the zero-shot models, the lowest RMSE* values per dataset are highlighted in bold.

Model	**Metric**	**NSC**	**P501**	**DEV**	**501_SQ**	**DB01**	**TUB1**	**Avg.**
wav2vec 2.0	r	0.817	0.815	0.679	0.408	0.901	0.674	0.715
	RMSE	0.539	0.567	0.514	0.742	0.419	0.480	0.544
	RMSE*	0.422	0.427	0.448	0.653	0.342	0.428	0.454
HuBERT	r	0.864	0.885	0.652	0.508	0.850	0.717	0.746
	RMSE	0.470	0.455	0.531	0.700	0.508	0.453	0.519
	RMSE*	0.435	0.353	0.429	0.629	0.409	0.423	0.446
WavLM	r	0.888	0.881	0.679	0.498	0.813	0.567	0.721
	RMSE	0.429	0.463	0.514	0.705	0.561	0.535	0.535
	RMSE*	0.385	0.354	0.449	0.640	0.494	0.498	0.470
Whisper	r	0.930	0.908	0.756	0.476	0.917	0.600	0.765
	RMSE	0.343	0.410	0.458	0.715	0.384	0.519	0.472
	RMSE*	**0.273**	**0.308**	0.406	0.632	0.330	0.489	0.406
AST	r	0.896	0.892	0.818	0.592	0.934	0.763	0.816
	RMSE	0.416	0.443	0.402	0.655	0.343	0.420	0.447
	RMSE*	0.373	0.342	**0.328**	**0.584**	**0.274**	**0.388**	**0.382**
NISQA	r	0.970	0.949	0.839	0.936	0.984	0.827	0.918
	RMSE	0.228	0.309	0.380	0.286	0.170	0.365	0.290
	RMSE*	0.194	0.265	0.324	0.240	0.101	0.343	0.245

for ASR, is optimized for robustness across diverse acoustic conditions, languages, and recording environments. This appears to translate favorably to speech quality prediction, despite the absence of explicit perceptual supervision.

File-Level Versus Condition-Level Performance

All models exhibit improved performance when evaluated at the condition level rather than the file level, as typically expected. This behaviour reflects the reduction of variance introduced by speaker identity, linguistic content, and recording-specific factors when predictions and subjective ratings are averaged per condition.

The more pronounced condition-level gains observed for AST and Whisper suggest that these representations capture degradation-related structure more consistently across samples. This finding aligns with established practices in instrumental speech quality assessment, where condition-level analysis is commonly used to assess perceptual trends.

Comparison with a Task-Specific Speech Quality Model

The task-specific NISQA model substantially outperforms all frozen pretrained encoders across evaluation metrics and datasets. This performance gap is expected, as NISQA is explicitly trained on large-scale subjective speech quality data and incorporates task-specific representation learning. The comparison serves to contextualize the zero-shot results rather than to benchmark absolute performance. It illustrates the performance ceiling imposed by strict zero-shot constraints and confirms that dedicated speech quality models remain necessary for high-accuracy prediction.

Nevertheless, the comparatively strong zero-shot performance of AST and Whisper indicates that modern foundation models can provide highly informative representations that approach task-specific performance when combined with minimal adaptation.

4.3.3 Limitations

Several limitations of the present analysis should be acknowledged. First, the downstream regression mapping is intentionally restricted to a single linear layer to approximate zero-shot transfer. While this design ensures a controlled comparison, it may underestimate the potential of pretrained representations under limited fine-tuning or more expressive adaptation mechanisms.

Second, the evaluation is restricted to overall perceived speech quality. Other perceptual dimensions, such as noisiness, coloration, discontinuity, and loudness, are not considered in this chapter and may exhibit different sensitivity to pretrained representations.

Third, the downstream mapping is trained exclusively on English speech from the NISQA corpus. Although evaluation includes both English and German datasets, language effects cannot be fully disentangled from dataset-specific characteristics (Wardah et al., 2025a).

Fourth, other variants of the foundation models exist, but were not used here. The *base* versions of all the models are used in this work.

Finally, zero-shot performance is inherently constrained by the mismatch between pretraining objectives and perceptual speech quality. The observed performance gap to task-specific models should therefore not be interpreted as a limitation of the representations themselves, but as a consequence of the intentionally restrictive evaluation setting.

Implications for Subsequent Chapters

The results of this chapter motivate the selection of AST-based representations for further investigation in subsequent chapters. AST exhibits strong and consistent zero-shot performance, stable aggregation behaviour, and favorable generalization across heterogeneous datasets. These properties make it a suitable backbone for exploring multi-task learning, task-specific adaptation, and out-of-domain generalization in speech quality prediction.

4.4 Summary

This chapter evaluated five transformer-based pretrained foundation models, namely wave2vec 2.0, HuBERT, WavLM, Whisper, and AST, for non-intrusive speech quality prediction under a zero-shot transfer learning setting. All encoders were kept frozen and combined with an identical downstream mapping, enabling a controlled comparison of representational suitability for speech quality assessment.

All evaluated models exhibited a measurable ability to predict perceived speech quality, as reflected by consistently positive correlations with subjective MOS across all held-out datasets. This demonstrates that pretrained speech representations encode information relevant to perceptual quality without task-specific encoder adaptation, addressing RQ1 at the representation level.

At the same time, substantial performance differences were observed across architectures and pretraining strategies. Models operating on time–frequency representations and trained with supervised or weakly supervised objectives, in particular AST and Whisper, achieved stronger and more consistent zero-shot performance than waveform-based self-supervised models. These results motivate the selection of AST as a candidate backbone for subsequent model adaptation.

Evaluation on multiple heterogeneous and fully unseen datasets revealed pronounced variability in absolute performance across domains. However, the relative ranking of models remained largely stable, indicating that observed differences primarily reflect representational robustness rather than dataset-specific effects. This finding directly addresses RQ4 with respect to cross-domain generalization.

Finally, comparison with a task-specific speech quality model confirmed the expected performance gap between zero-shot predictors and models trained explicitly on subjective quality data. This gap reflects the inherent limitations of strict zero-shot transfer and establishes a principled baseline for the task-specific modeling approaches investigated in subsequent chapters.

Chapter 5
Multi-Task Speech Quality Modeling

Building on the zero-shot comparison presented in Chapter 4, this chapter adapts the Audio Spectrogram Transformer (AST) for non-intrusive speech quality prediction under a multi-task learning setting. Following its selection as a perceptually aligned representation, AST is first adapted to operate on 48 kHz speech signals used in the target datasets and is subsequently extended to jointly predict overall quality and multiple perceptual dimensions. In addition to mean opinion score (MOS), the model estimates noisiness, discontinuity, coloration, and sub-optimal loudness using task-specific regression heads operating on a shared latent representation. Performance is evaluated on held-out test data, enabling an analysis of perceptual alignment and task interactions under joint modeling. The results of this chapter address Research Questions RQ2 and RQ3.

5.1 Introduction

In this work, overall perceived quality is modeled as a continuous regression target derived from subjective listening test scores. In addition to overall quality, listening test methodologies provide ratings for multiple perceptual dimensions, such as noisiness, discontinuity, coloration, and loudness. These dimensions capture complementary aspects of quality perception and may be influenced by overlapping acoustic cues and degradation mechanisms.

In the preceding chapter, a zero-shot evaluation was used to identify pretrained speech representations that exhibit strong perceptual alignment without task-specific adaptation. Based on this analysis, the AST was selected as a suitable representation backbone for further modeling. The present chapter builds on this outcome by adapting AST to the target signal domain and extending it to a multi-task learning setting, in which overall quality and multiple perceptual dimensions are predicted jointly.

The focus is on controlled adaptation and empirical analysis, enabling the effects of shared representations and task coupling to be examined under realistic evaluation conditions.

This chapter first details the adaptation of AST to the 48 kHz signal domain, followed by the specification of the proposed multi-task model architecture and training setup. The resulting model is then evaluated and discussed with respect to perceptual alignment and task interactions.

5.2 Method

This section outlines the methodology used to implement and evaluate the proposed multi-task speech quality model. It specifies the adaptation of AST to fullband speech signals, the model architecture, and the training setup employed in the experiment.

5.2.1 Adapting AST to 48 kHz Speech Signals

The AST was originally pretrained using audio sampled at 16 kHz, reflecting the characteristics of many large-scale audio corpora commonly used for representation learning. Such corpora are often optimized for linguistic and semantic tasks, for which a 16 kHz sampling rate is generally sufficient to capture phonetic content while enabling scalable data collection and processing. In contrast, the speech quality datasets considered in this work predominantly contain super-wideband and fullband signals, with sampling rates up to 48 kHz, as is customary in contemporary telecommunication evaluation scenarios. Rather than downsampling these signals to match the pretraining configuration, AST is adapted to operate directly on 48 kHz input in order to preserve higher-frequency signal components that may contribute to perceived quality, particularly for degradations affecting timbre, coloration, and overall naturalness.

The adaptation is implemented exclusively at the feature-extraction stage. Log-Mel spectrograms are computed from 48 kHz waveforms using the same Mel filterbank dimensionality as in the original AST configuration. This preserves the frequency-axis dimensionality expected by the pretrained model while extending the analyzed bandwidth and increasing the temporal density of the time–frequency representation.

Because higher sampling rates yield a larger number of time frames for a given utterance duration, a fixed upper bound on temporal extent is imposed by truncating the extracted feature sequence to a maximum length of 1024 frames. This choice is consistent with the original AST design, which was pretrained under the same temporal constraint. As a result, the adaptation to 48 kHz input does not introduce a more restrictive sequence length than that already assumed by the pretrained model, nor does it modify the transformer's positional embedding capacity or attention structure.

In the implemented data pipeline, each utterance is represented as a tensor of shape $[B, T, F]$, where B denotes the batch size, $F = 128$ the number of Mel frequency bins, and $T \leq 1024$ the number of time frames. With batch size $B = 1$, this results in feature tensors of shape $[1, 1024, 128]$, accompanied by an attention mask that distinguishes valid frames from padding for shorter signals. These log-Mel features are then processed by AST's patch embedding module, which applies a two-dimensional convolutional projection with kernel size 16×16 and stride 10×10 over the time–frequency plane. This operation partitions the spectrogram into overlapping time–frequency patches and maps each patch to a 768-dimensional embedding, producing a sequence of patch tokens that, combined with positional encoding, serves as input to the transformer encoder.

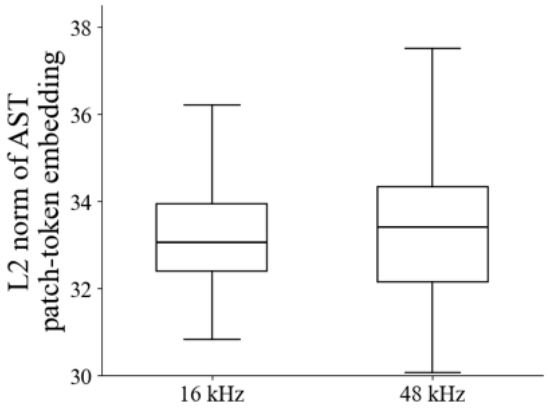


Fig. 5.1: Distribution of L2 norms of AST patch-token embeddings for the same speech utterance processed at 16 kHz and 48 kHz. The embeddings are extracted from the audio file `c00001_3_640_2_7_001-ch6-speaker_seg49.wav` from the NISQA_TRAIN_SIM database (Mittag et al., 2021). Overlapping medians and interquartile ranges indicate that adapting the input feature extraction to 48 kHz preserves the numerical scale of the pretrained AST representation.

Representation-Scale Consistency

To verify that the adaptation to higher sampling rates does not introduce unintended distributional shifts in the pretrained representation, the numerical scale of the resulting patch-token embeddings was examined. As illustrated in Figure 5.1, the distributions of L2 norms for embeddings extracted from the same utterance processed at 16 kHz and 48 kHz exhibit strongly overlapping medians and interquartile ranges. This observation indicates that the internal representation scale of AST remains stable under the proposed input adaptation and provides reassurance that subsequent learning stages operate on a numerically consistent feature space.

Importantly, the patchification strategy, projection dimensionality, and transformer architecture are retained exactly as in the pretrained AST model. Consequently, the adaptation affects only the signal-domain front-end used to generate the time–frequency representation, while all subsequent stages, from patch embedding to self-attention and pooling, operate under their original design assumptions. This controlled adaptation enables AST to process fullband speech signals in their native sampling rate while maintaining strict compatibility with the pretrained model configuration.

Building on this adapted representation backbone, the next section introduces the proposed multi-task model architecture used to jointly predict overall speech quality and multiple perceptual dimensions.

5.2.2 Model Architecture

The proposed multi-task predictor uses AST as a shared representation backbone and attaches separate task-specific regression heads for each perceptual speech quality dimension target. Given an input log-Mel representation $X \in \mathbb{R}^{T\times 128}$, the pretrained AST encoder produces a pooled utterance-level embedding $h \in \mathbb{R}^{768}$ via its pooling head. This pooled embedding serves as a shared latent representation for all tasks.

Five independent linear output layers are applied to the shared embedding h, one per target dimension: overall quality (MOS), noisiness, discontinuity, coloration, and loudness. Each task-specific head implements a linear mapping of the form

$$\hat{y}_k = \sigma(W_k h + b_k)\,, \tag{5.1}$$

where $W_k \in \mathbb{R}^{1\times 768}$ and $b_k \in \mathbb{R}$ denote the weights and bias of the k-th output head, $\sigma(\cdot)$ is a sigmoid nonlinearity, and $k \in \{\text{MOS}, \text{NOI}, \text{DIS}, \text{COL}, \text{LOUD}\}$. The individual predictions are concatenated to form a five-dimensional output vector

$$\hat{y} = \left[\hat{y}_{\text{MOS}}, \hat{y}_{\text{NOI}}, \hat{y}_{\text{DIS}}, \hat{y}_{\text{COL}}, \hat{y}_{\text{LOUD}}\right]^{\top} \in \mathbb{R}^5. \tag{5.2}$$

The overall structure of the proposed multi-task model follows the shared-encoder, multi-head architecture introduced in Chapter 3 (see Figure 3.2), with AST serving as the shared transformer backbone and independent regression heads for each target dimension. During training, all task-specific heads are active simultaneously, and the shared encoder is optimized jointly using supervision from all perceptual dimensions for which annotations are available.

5.2.3 Training Objective and Optimization

All target variables are scaled to the unit interval prior to training using the same linear transformation defined in Chapter 4. Specifically, subjective ratings originally defined on a five-point ACR scale are mapped to $[0, 1]$ for optimization and subsequently transformed back to the original scale for evaluation. This scaling strategy is applied consistently across all target dimensions, ensuring that the outputs of the sigmoid-activated task heads are compatible with the normalized training targets.

Joint Multi-Task Loss Formulation

Model training is formulated as a supervised multi-task regression problem with one loss term per target dimension. For each task $k \in \{\text{MOS}, \text{NOI}, \text{DIS}, \text{COL}, \text{LOUD}\}$, a mean squared error (MSE) loss is defined between the predicted value $\hat{y}_{i,k}$ and the corresponding scaled ground-truth label $y_{i,k}$,

$$\mathcal{L}_k = \frac{1}{|\Omega_k|} \sum_{i\in\Omega_k} \left(\hat{y}_{i,k} - y_{i,k}\right)^2, \tag{5.3}$$

where Ω_k denotes the subset of samples in the mini-batch for which valid labels are available for task k. Missing annotations are handled by masking and do not contribute to the corresponding task loss.

Gradient Aggregation and Parameter Updates

Gradients from all available task losses are accumulated within each mini-batch and a single optimizer update is applied per iteration. For each target dimension, the error signal produced by the corresponding task-specific head is backpropagated through that head and the shared AST encoder. As a result, each task contributes gradients to the shared representation, while updating only its own output head. The transformer encoder is shared across all tasks and therefore receives gradient contributions from each loss term for which valid labels are present. Each task-specific output head is updated only when its corresponding loss is defined. This training strategy enables joint learning across perceptual dimensions while allowing for incomplete annotations in individual tasks.

The overall training objective is given by the sum of all task-specific losses available in a mini-batch,

$$\mathcal{L} = \sum_{k \in \mathcal{K}} \mathcal{L}_k, \tag{5.4}$$

where $\mathcal{K}$ denotes the set of target dimensions with valid annotations for the current batch.

Model optimization is performed using the RAdam optimizer with a fixed learning rate of 10^{-6} and a weight decay of 10^{-5}. Training is conducted with early stopping based on validation loss using a patience of five epochs, and the best-performing model checkpoint is retained. In the reported experiments, the optimal model was obtained after 11 training epochs. All training procedures are implemented in PyTorch.

5.2.4 Data

The training set comprises approximately 88 databases containing about 144k speech files, while the validation set includes approximately 22 databases with about 22k speech files. All audio signals are processed at a sampling rate of 48 kHz and converted to log-Mel representations, as described in Section 5.2.1. The databases used for training and validation are listed in Appendix C.

Most datasets provide subjective ratings for overall speech quality (MOS), whereas ratings for individual perceptual dimensions are available for a smaller subset of databases. Detailed database information is provided in Appendix A.

The training and validation datasets cover a broad range of degradation conditions representative of realistic telecommunication and multimedia transmission scenarios. These include coding distortions introduced by NB, WB, and SWB speech codecs; transmission impairments such as packet loss, frame erasures, and jitter; additive background noise and competing speech at varying signal-to-noise ratios; level variations and loudness inconsistencies; as well as signal processing artifacts arising from noise suppression, echo control, bandwidth limitation, and neural speech enhancement systems. In addition, the datasets comprise both simulated and live recordings captured across diverse devices, acoustic environments, and network conditions. This diversity ensures that the multi-task model is exposed to perceptually heterogeneous degradations affecting different quality dimensions in distinct ways.

Held-Out Evaluation Set

Evaluation is performed on a set of six held-out test databases that are disjoint from the training and validation datasets. The evaluation set comprises all the Test datasets from the NISQA corpus (Mittag et al., 2021), including `NISQA_TEST_NSC` (NSC), `NISQA_TEST_P501` (P501), `NISQA_TEST_FOR` (FOR), and `NISQA_TEST_LIVETALK` (LIVE), as well as two additional datasets, `SUB_TUB_2` (TUB_2) and `Vin726` (Vin726), which were not used in earlier chapters.

All evaluation databases are annotated with subjective quality labels, with the exception of `Vin726`. The NISQA test databases and `SUB_TUB_2` were obtained from listening tests and provide subjective ratings for overall quality and perceptual dimensions. In contrast, `Vin726` is a French speech quality database for which overall quality MOS labels were derived using the reference-based POLQA algorithm (ITU-T Recommendation P.863), as provided by the database creators. This dataset is included exclusively for evaluation purposes to broaden the diversity of test conditions beyond the NISQA benchmark and to assess model behavior under alternative label generation assumptions.

Together, these databases cover a wide range of degradation types, recording conditions, and perceptual characteristics, providing a challenging and diverse evaluation scenario for multi-dimensional speech quality prediction.

5.3 Results and Discussion

This section presents and discusses the experimental results obtained with the multi-task AST model on the held-out evaluation databases. The analysis focuses on overall speech quality (MOS) as well as the perceptual dimensions of noisiness, discontinuity, coloration, and loudness, evaluated at the condition level. Condition-level scatter plots for all evaluation databases and perceptual dimensions are provided in Appendix D to illustrate database-specific prediction behavior.

5.3.1 Overall Speech Quality

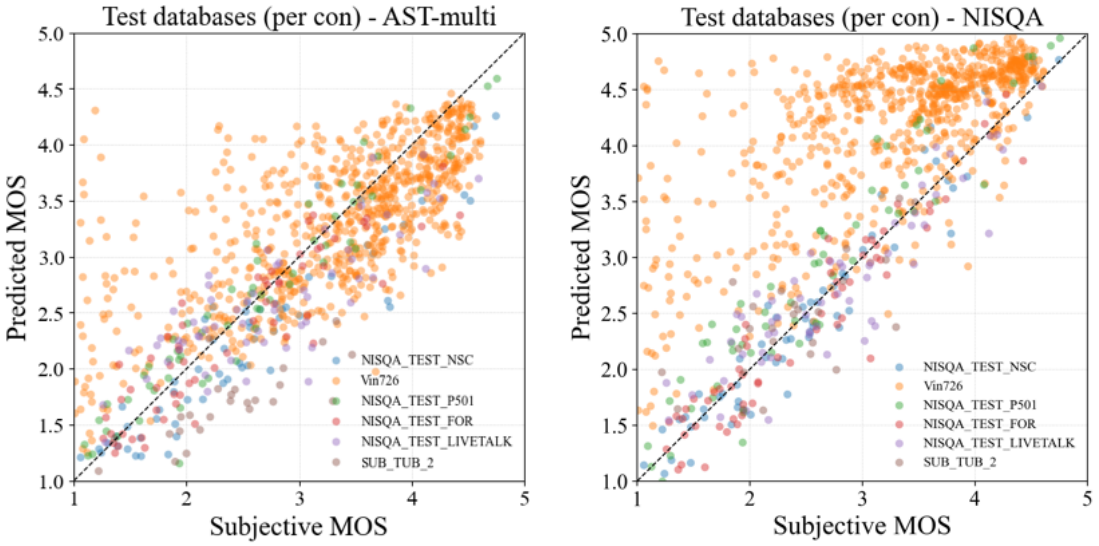
Table 5.1 reports the condition-level MOS prediction performance of the multi-task AST model on the held-out evaluation databases. Across the NISQA test sets, the model achieves high Pearson correlation coefficients, indicating a strong monotonic relationship between predicted and subjective MOS. This demonstrates that the pretrained AST representation preserves perceptually relevant information for overall speech quality assessment when adapted via supervised multi-task learning.

Absolute prediction errors vary across databases, reflecting differences in degradation characteristics and recording conditions. Nevertheless, the relative ordering of degradation conditions is largely preserved, and stable performance is observed across diverse evaluation scenarios, as illustrated by the pooled scatter plots in Figure 5.2. These results establish that AST-based representations can be successfully adapted for MOS prediction in a multi-task setting, providing perceptually aligned quality estimates

Table 5.1: Condition-level MOS prediction performance of the multi-task AST model on held-out test databases. Pearson correlation (r) is computed on raw predictions, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping.

Model	Metric	NSC	FOR	LIVE	P501	TUB_2	Vin726
multi-task AST	r	0.938	0.907	0.811	**0.960**	**0.782**	**0.695**
	RMSE	0.324	0.348	0.478	0.274	0.360	0.640
	RMSE*	0.294	0.302	0.405	0.236	0.303	—
NISQA	r	**0.970**	**0.949**	**0.902**	0.949	0.694	0.661
	RMSE	0.228	0.260	0.353	0.309	0.416	0.668
	RMSE*	0.194	0.216	0.283	0.265	0.368	—

across heterogeneous test data. For reference, Table 5.1 also reports the corresponding results obtained with NISQA, which provides a point of comparison under an alternative multi-task modeling approach.



(a) (b)

Fig. 5.2: Condition-level MOS scatter plots pooled across all held-out test databases. Subfigure (a) shows predictions from the multi-task AST model, and subfigure (b) shows predictions from the NISQA model. For the `Vin726` database, MOS labels are derived from the instrumental POLQA model (ITU-T Recommendation P.863) rather than from subjective listening tests.

5.3.2 Prediction of Perceptual Dimensions

Noisiness

Table 5.2 summarizes the condition-level prediction performance for the noisiness dimension. The multi-task AST model achieves consistently high Pearson correlation across the majority of evaluation databases, with $r \geq 0.92$ on NISQA_TEST_NSC, NISQA_TEST_FOR, NISQA_TEST_P501, and SUB_TUB_2, and moderately lower correlation on NISQA_TEST_LIVETALK. This pattern indicates that noisiness is strongly represented within the shared AST embedding space and can be reliably recovered under joint multi-task supervision across diverse degradation conditions.

Table 5.2: Condition-level noisiness prediction performance of the multi-task AST model on held-out test databases. Pearson correlation (r) is computed on raw predictions, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping.

Model	Metric	NSC	FOR	LIVE	P501	TUB_2
multi-task AST	r	0.936	0.919	**0.852**	**0.951**	**0.924**
	RMSE	0.279	0.277	0.377	0.289	0.218
	RMSE*	0.251	0.247	0.279	0.223	0.175
NISQA	r	**0.960**	**0.946**	0.759	0.949	0.660
	RMSE	0.222	0.228	0.470	0.296	0.427
	RMSE*	0.191	0.200	0.301	0.260	0.381

Discontinuity

Table 5.3 reports the condition-level prediction performance for the discontinuity dimension. Compared to noisiness, correlation values are consistently lower and exhibit greater variability across databases, indicating that discontinuity constitutes a more challenging perceptual attribute to predict. The multi-task AST model nevertheless achieves strong condition-level alignment on several evaluation sets, with correlation coefficients above $r = 0.90$ on NISQA_TEST_NSC and NISQA_TEST_FOR.

Performance degrades more noticeably on databases characterized by higher heterogeneity and live transmission degradations, such as NISQA_TEST_LIVETALK and SUB_TUB_2. However, even in these more challenging scenarios, the multi-task AST model maintains substantially higher correlation than the NISQA baseline, which exhibits markedly reduced alignment for discontinuity on the same database. This indicates that the shared AST representation retains sensitivity to temporal artifacts under adverse conditions, whereas performance differences are primarily attributable to database characteristics rather than a complete loss of perceptual structure. Overall, while discontinuity is less consistently encoded than noisiness, the results demonstrate that joint multi-task training enables the AST-based model to recover meaningful information related to temporal impairments across diverse evaluation conditions.

Table 5.3: Condition-level discontinuity prediction performance of the multi-task AST model on held-out test databases. Pearson correlation (r) is computed on raw predictions, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping.

Model	Metric	NSC	FOR	LIVE	P501	TUB_2
multi-task AST	r	0.928	0.908	0.779	0.893	**0.718**
	RMSE	0.380	0.393	0.448	0.401	0.332
	RMSE*	0.333	0.345	0.367	0.334	0.215
NISQA	r	**0.956**	**0.965**	**0.831**	**0.908**	0.348
	RMSE	0.299	0.245	0.397	0.373	0.447
	RMSE*	0.259	0.198	0.312	0.325	0.394

Coloration

Table 5.4 reports the condition-level prediction performance for the coloration dimension. As observed for discontinuity, correlation values vary across databases, indicating sensitivity of coloration prediction to database characteristics and degradation profiles. The multi-task AST model achieves moderate to strong alignment on the NISQA test sets, with correlation coefficients in the range $r \approx 0.78$–0.90, demonstrating that spectral coloration effects are captured by the shared representation under joint training.

On the SUB_TUB_2 database, the multi-task AST model maintains substantially higher correlation than the NISQA baseline, which exhibits a pronounced drop in alignment for coloration. This mirrors the behavior observed for discontinuity and indicates greater robustness of the AST-based representation to database-specific variability for spectral distortions. Overall, while coloration is less consistently predicted than noisiness, the results confirm that joint multi-task learning enables recovery of meaningful spectral quality information under heterogeneous evaluation conditions.

Table 5.4: Condition-level coloration prediction performance of the multi-task AST model on held-out test databases. Pearson correlation (r) is computed on raw predictions, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping.

Model	Metric	NSC	FOR	LIVE	P501	TUB_2
multi-task AST	r	0.833	0.900	0.789	0.812	**0.618**
	RMSE	0.427	0.310	0.392	0.442	0.303
	RMSE*	0.386	0.275	0.337	0.382	0.254
NISQA	r	**0.932**	**0.941**	**0.875**	**0.913**	0.369
	RMSE	0.280	0.240	0.309	0.310	0.359
	RMSE*	0.249	0.206	0.258	0.246	0.313

Loudness

Table 5.5 reports the condition-level prediction performance for the loudness dimension. On NISQA_TEST_NSC, NISQA_TEST_FOR, and NISQA_TEST_P501, the multi-task AST model achieves high correlation, indicating that large-scale loudness differences are captured by the learned representation.

On SUB_TUB_2, subjective loudness scores are confined to a narrow range between approximately 3.0 and 4.2, while the corresponding AST predictions lie between roughly 3.0 and 3.8. In this low-variance setting, the limited dynamic range constrains relative ordering of conditions, resulting in low and unstable correlation for the AST-based model. In contrast, the NISQA baseline maintains a moderate correlation on this database, highlighting that loudness remains the most challenging perceptual dimension for the AST-based multi-task model.

Table 5.5: Condition-level loudness prediction performance of the multi-task AST model on held-out test databases. Pearson correlation (r) is computed on raw predictions, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping.

Model	Metric	NSC	FOR	LIVE	P501	TUB_2
multi-task AST	r	0.933	0.943	0.608	0.924	-0.455
	RMSE	0.339	0.246	0.403	0.330	0.259
	RMSE*	0.308	0.207	0.311	0.252	0.261
NISQA	r	**0.964**	**0.961**	**0.709**	**0.955**	**0.685**
	RMSE	0.249	0.204	0.358	0.256	0.212
	RMSE*	0.189	0.170	0.294	0.222	0.142

Across perceptual dimensions, systematic performance patterns are observed for the multi-task AST model. Noisiness exhibits the highest and most stable correlation across databases, whereas discontinuity, coloration, and loudness show increased variability and stronger sensitivity to dataset characteristics. Databases associated with reduced correlation for one dimension frequently exhibit similar degradation for others, indicating that performance differences are primarily driven by database-specific factors rather than isolated task failures. This consistent cross-dimensional behavior suggests that the shared AST representation encodes common perceptual structure, while the task-specific heads govern sensitivity to individual quality attributes under joint training.

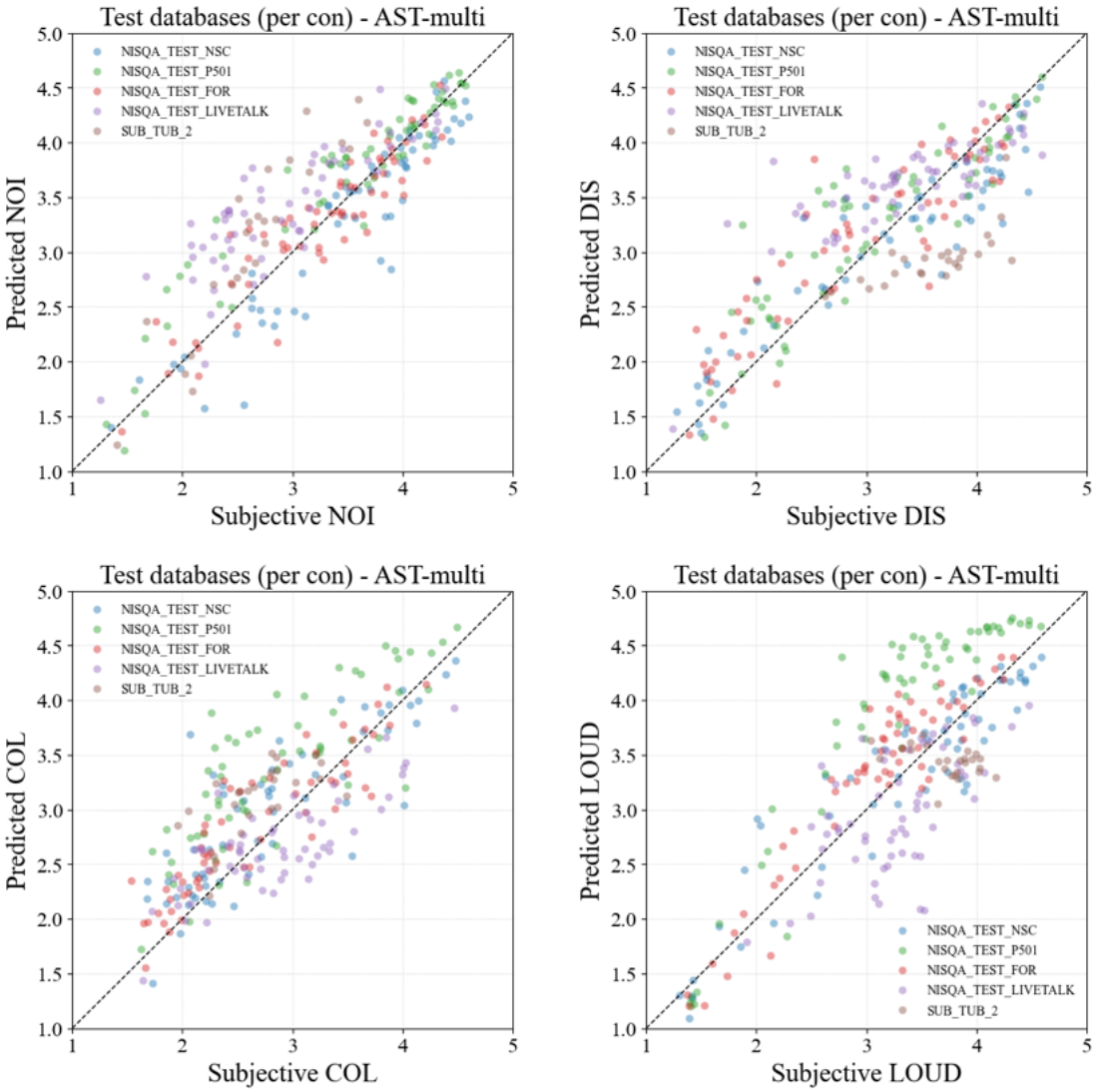


Fig. 5.3: Condition-level scatter plots of the multi-task AST model predictions for the perceptual quality dimensions of noisiness, discontinuity, coloration, and loudness across all evaluation databases.

5.3.3 Implications

The results of this chapter indicate that pretrained foundation model representations can support the joint prediction of overall speech quality and multiple perceptual dimensions, albeit with varying degrees of robustness across attributes. Noisiness emerges as the most reliably encoded dimension, while discontinuity and coloration exhibit increased sensitivity to database characteristics, and loudness remains challenging regarding particular databases. Importantly, performance variations tend to manifest coherently across dimensions, suggesting that database-specific factors play a larger role than isolated task failures. These findings imply that joint multi-task learning provides a structured lens for analyzing perceptual quality, and motivate further investigation into task-specific adaptation and domain robustness in subsequent chapters.

5.4 Summary

This chapter investigated the adaptation of the Audio Spectrogram Transformer to multi-task non-intrusive speech quality prediction. AST was first adapted to operate directly on 48 kHz speech signals without modifying the pretrained encoder architecture, ensuring compatibility with fullband telecommunication data. A shared-encoder, multi-head model was then introduced to jointly predict overall quality (MOS) and the perceptual dimensions of noisiness, discontinuity, coloration, and loudness.

Experimental results on held-out evaluation databases demonstrate that AST-based representations support perceptually aligned prediction of overall speech quality and multiple perceptual dimensions under joint multi-task learning, thereby addressing RQ2. Alignment varies across dimensions: noisiness exhibits the highest and most stable correspondence with subjective judgments, discontinuity and coloration show moderate but coherent behavior with increased database sensitivity, and loudness proves more challenging on datasets with limited loudness variation. Across dimensions, performance variations are primarily driven by database characteristics rather than isolated task failures, indicating consistent and structured behavior under joint modeling.

With respect to RQ3, the joint multi-task formulation reveals structured cross-dimensional behavior and consistent database-dependent performance patterns, indicating that shared representation learning enables simultaneous modeling of overall quality and perceptual attributes without destabilizing individual predictions. While joint learning does not uniformly improve accuracy across all dimensions and datasets, it provides a robust framework for analyzing perceptual interactions and supports unified speech quality modeling within a single architecture. The dimension-specific performance variations observed under joint training motivate a subsequent investigation of single-task AST models for each perceptual dimension.

Chapter 6
Single-Task Modeling

This chapter examines single-task learning as an alternative modeling strategy for non-intrusive speech quality assessment based on transformer-based pretrained representations. In contrast to joint multi-task learning, each perceptual target is optimized independently, allowing task-specific supervision to guide model adaptation without interaction between competing objectives.

The analysis builds on the methodological framework established in the preceding chapters and extends it to additional datasets that became available during the course of this work, including data introduced in the context of the ITU-T P.SAMD standardization activity. These datasets broaden the range of acoustic and transmission conditions considered and enable an examination of model generalization under increased heterogeneity.

Using consistent training and evaluation protocols, this chapter compares single-task model variants and assesses their robustness across multiple evaluation databases. Selected model configurations are further subjected to independent evaluation by ITU-T partners on previously unseen data as part of the standardization process.

6.1 Introduction

This section introduces the single-target modeling approach considered in this chapter and outlines its development and evaluation. Building on the transformer-based architecture and experimental framework established in earlier chapters, the focus here is on training and assessing independent models for each perceptual speech quality dimension, resulting in a coherent set of dimension-specific models that can be jointly deployed at inference time.

The selected single-target models are then evaluated and compared with the zero-shot and jointly trained multi-task approaches presented in previous chapters, as well as with external benchmark databases used in the context of ITU-T standardization activities. Comparative results across perceptual dimensions and evaluation databases provide insight into the behavior of AST-based foundation models under diverse acoustic and transmission conditions.

6.2 Method

This section defines the methodological framework used for single-task speech quality modeling and evaluation. It specifies the model architecture, training procedure, inference configuration, and evaluation protocol adopted in this chapter. All methodological components are described independently of dataset composition and model variants, ensuring reproducibility and controlled comparison with the multi-task formulation examined in Chapter 5.

6.2.1 Model Architecture

The proposed single-task model builds upon the AST foundation model described in Chapters 4 and 5, reusing the same encoder architecture and input representation. Accordingly, this section focuses exclusively on the task-specific components and architectural aspects relevant to single-task optimization.

The AST backbone is coupled with a single task-specific regression head, yielding one model per target perceptual dimension. As in the multi-task setting, the model operates on a log-Mel spectrogram representation $X \in \mathbb{R}^{T\times 128}$ and produces a pooled utterance-level embedding

$$h \in \mathbb{R}^{768}$$

via the pretrained AST pooling head. This embedding serves as the latent representation for downstream quality prediction.

In contrast to the multi-task formulation, each prediction target is modeled independently. For a given task $k \in \{\text{MOS}, \text{NOI}, \text{DIS}, \text{COL}, \text{LOUD}\}$, a single linear output layer is applied to the embedding h:

$$\hat{y}_k = \sigma(W_k h + b_k)\,, \tag{6.1}$$

where $W_k \in \mathbb{R}^{1\times 768}$ and $b_k \in \mathbb{R}$ denote the weights and bias of the task-specific regression head, and $\sigma(\cdot)$ is a sigmoid nonlinearity. Each model produces a scalar prediction $\hat{y}_k \in \mathbb{R}$ corresponding to one perceptual target. Following the same procedure as in Chapters 4 and 5, subjective ratings defined on a five-point ACR scale are linearly mapped to $[0, 1]$ for training and transformed back to the original scale for evaluation, ensuring compatibility with the sigmoid-activated output.

Across all single-task models, the AST encoder architecture, input representation, and output mapping are identical to those used in the multi-task setting. However, in contrast to the shared-encoder multi-task formulation, the encoder parameters are optimized independently for each task. As a result, differences between single-task models arise from both task-specific supervision and independently optimized encoder and regression head parameters.

6.2.2 Training Procedure and Inference Configuration

The single-target modeling approach results in five independently trained models, corresponding to the prediction of overall speech quality (MOS), noisiness, discontinuity, coloration, and sub-optimal loudness. While each model is trained separately, the resulting task-specific encoders and regression heads can be combined at inference time into a unified prediction pipeline that accepts a single speech signal as input and produces a set of five perceptual quality scores, as conceptualized in Figure 3.3.

Training Procedure

Five independent AST-based models are initialized using the same pretrained encoder weights and are fine-tuned separately for the overall speech quality and each perceptual speech quality dimension. This training strategy enables task-specific adaptation while retaining the general acoustic representations learned during large-scale pretraining.

Model optimization is carried out using the AdamW optimizer with a learning rate of 10^{-6} and a weight decay of 10^{-5}. Training employs early stopping based on validation loss, and the best-performing model checkpoint is retained for each task. All training procedures are implemented in PyTorch.

Inference Configuration

During inference, the input speech waveform is first processed by the AST feature extractor to compute a log-Mel spectrogram, followed by the same patchification, linear projection, and positional encoding steps used during training. This preprocessing yields a shared input representation that is compatible with all task-specific AST models.

For each perceptual dimension (and the overall speech quality), the input representation is passed through its corresponding independently trained AST encoder and regression head to produce a scalar quality estimate. While the front-end feature extraction is identical across tasks, encoder and regression head parameters are not shared. As a result, inference is performed independently for each perceptual dimension, and the final output consists of a set of five dimension-specific quality predictions.

6.2.3 Train-Validation Data Split Regimes

The development of the proposed single-target models was influenced by the evolving ITU-T standardization objectives for the P.SAMD model. In particular, three additional speech quality databases were introduced after the initial development phase to expand the coverage of discontinuity and loudness impairments in the training data. Their inclusion enabled the investigation of alternative training strategies aimed at improving generalization to unseen and heterogeneous evaluation data. To this end, three distinct train-validation split regimes are considered for selecting robust dimension-specific model checkpoints. A complete overview of all databases is provided in Appendix A.

Table 6.1: Overview of the train–validation data split regimes evaluated for the AST-based single-target framework.

Configuration	Description	Train Set	Validation Set	Test Set
$C1_tC2_v$	The pre-trained AST model is fine-tuned using the same training and validation database partition as employed for the multi-task AST experiments presented in Chapter 5. This configuration results in the published SQ-AST model (Wardah et al., 2025b).	C.1	C.2	C.5
$C3_tC4_v$	The pre-trained AST model is fine-tuned using the full set of available training databases, that is the training set C.1 plus 3 additional databases, and a reduced validation set curated for strictly subjective overall MOS labels and some objective ITU-T Recommendation P.863.2 (2022) perceptual dimension labels.	C.3	C.4	C.5
$C1_tC2_v \rightarrow C3_tC2_v$	The model from the $C1_tC2_v$ regime is further fine-tuned using the full set of available training databases, that is the training set C.1 plus 3 new databases, and the original validation set that was employed for the multi-task AST experiments.	C.1,C.3	C.2	C.5

All evaluated regimes employ the same AST-based model architecture and optimization settings described in Section 6.2. The regimes differ only in the composition of the training and validation data and in whether models are trained from the pretrained AST weights or fine-tuned from previous internally trained checkpoints. An overview of the evaluated training regimes is provided in Table 6.1, where each regime is defined by a distinct combination of training data coverage and validation setup.

Together, these regimes enable a controlled comparison between full retraining and targeted fine-tuning under identical architectural and evaluation conditions. The resulting model variants are analyzed in terms of prediction performance and robustness in the following sections.

The best-performing dimension-specific checkpoints, in terms of predicted and subjective label correlation per file, obtained from the training regimes described in Table 6.1, are selected to form the set of transformer encoders and regression heads used in the single-target speech quality prediction model henceforth. The performance of these models, together with their comparison to the zero-shot and multi-task models introduced in previous chapters, is presented and discussed in the following section.

6.3 Results and Discussion

This section presents the results obtained with the single-target (dimension-specific) modeling framework under the training regimes introduced in Section 6.2.3. Based on these results, the best-performing dimension-specific checkpoints are selected and assembled into the final single-target speech quality prediction model. The resulting model set is subsequently evaluated on both internal and external test databases and compared to the zero-shot, multi-task, and baseline models.

6.3.1 Checkpoint Selection Across Training Regimes

For internal model selection, file-level Pearson correlation (r) between predicted and subjective scores is used to identify the best-performing checkpoint and training regime for each perceptual dimension. This criterion is applied consistently across all candidate single-target models, as it directly reflects the strength of the linear association between predicted and subjective scores at the utterance level and provides a stable and interpretable basis for comparing competing training regimes. Results are reported separately for overall speech quality and each perceptual dimension in Tables 6.2 to 6.6.

The $C1_tC2_v \rightarrow C3_tC2_v$ training regime achieves the highest performance for overall speech quality (MOS) as well as for the perceptual dimensions of noisiness and discontinuity (see Tables 6.2–6.4). This regime builds on the initially developed $C1_tC2_v$ model, corresponding to the SQ-AST model (Wardah et al., 2025b), and further fine-tunes it using an expanded training set that includes three newly introduced databases targeting discontinuity and sub-optimal loudness impairments. The original validation set used in the multi-task AST experiments is retained.

Notably, despite the increased representation of discontinuity-related conditions in the expanded training data, the relative performance gain for discontinuity prediction is smaller than for the other affected dimensions (see Table 6.7). This suggests that additional data alone may be insufficient to yield proportional improvements for this perceptual attribute.

For the coloration dimension, the $C1_tC2_v$ training configuration, corresponding to the SQ-AST model (Wardah et al., 2025b), achieves the highest average file-level correlation between predicted and subjective coloration scores across the internal test databases, as shown in Table 6.5. In contrast to the other perceptual dimensions, the extended training regimes do not yield consistent improvements for coloration. This observation suggests that the additional training data introduced in the alternative regimes does not provide complementary information relevant to this perceptual attribute.

Table 6.2: File-level r on test sets for overall quality MOS across training regimes.

Training Regime	NSC	P501	FOR	LIVE	TUB_2	Avg.
$C1_tC2_v$	0.828	0.918	0.825	0.853	0.816	0.848
$C3_tC4_v$	0.891	0.913	0.867	0.840	0.810	0.864
$C1_tC2_v \rightarrow C3_tC2_v$	0.877	0.930	0.867	0.847	0.836	**0.872**

Table 6.3: File-level r on test sets for *Noisiness* across training regimes.

Training Regime	NSC	P501	FOR	LIVE	TUB_2	Avg.
$C1_tC2_v$	0.889	0.913	0.845	0.760	0.832	0.848
$C3_tC4_v$	0.897	0.896	0.856	0.718	0.821	0.838
$C1_tC2_v \rightarrow C3_tC2_v$	0.902	0.910	0.837	0.789	0.825	**0.853**

Table 6.4: File-level r on test sets for *Discontinuity* across training regimes.

Training Regime	NSC	P501	FOR	LIVE	TUB_2	Avg.
$C1_tC2_v$	0.851	0.862	0.816	0.724	0.652	0.781
$C3_tC4_v$	0.862	0.845	0.804	0.733	0.632	0.775
$C1_tC2_v \rightarrow C3_tC2_v$	0.858	0.849	0.823	0.759	0.626	**0.783**

Table 6.5: File-level r on test sets for *Coloration* across training regimes.

Training Regime	NSC	P501	FOR	LIVE	TUB_2	Avg.
$C1_tC2_v$	0.802	0.763	0.764	0.739	0.567	**0.727**
$C3_tC4_v$	0.769	0.751	0.820	0.744	0.519	0.720
$C1_tC2_v \rightarrow C3_tC2_v$	0.770	0.715	0.797	0.774	0.555	0.722

Table 6.6: File-level r on test sets for *Loudness* across training regimes.

Training Regime	NSC	P501	FOR	LIVE	TUB_2	Avg.
$C1_tC2_v$	0.797	0.861	0.731	0.612	-0.068	0.587
$C3_tC4_v$	0.872	0.878	0.841	0.549	-0.054	**0.617**
$C1_tC2_v \rightarrow C3_tC2_v$	0.856	0.872	0.811	0.558	-0.047	0.610

In contrast, the $C3_tC4_v$ training regime achieves the highest file-level Pearson correlation for the loudness dimension, as shown in Table 6.6. This suggests that the inclusion of additional training data in this regime is beneficial for loudness prediction. However, performance varies substantially across test databases, and the SUB_TUB_2 database remains challenging for all evaluated training regimes.

Overall, loudness exhibits greater sensitivity to both dataset characteristics and training strategy than the other perceptual dimensions. Average correlation values are lower, and near-zero or negative correlations are observed on some test sets. In particular, the SUB_TUB_2 database predominantly contains conditions with near-optimal loudness, resulting in a narrow range of subjective scores. This limited variability constrains the observable correlation and makes reliable prediction more difficult.

Selected Checkpoints

Based on the results presented in Tables 6.2 to 6.6, a single representative model is selected for each perceptual target dimension. Each model is trained independently to predict one perceptual dimension, while sharing a common architectural design and training framework. Collectively, these models form the single-target (dimension-specific) model set considered in the subsequent comparisons.

For each perceptual dimension, the training regime yielding the strongest and most consistent performance is selected based on the average file-level Pearson correlation (r) across all evaluated internal test databases. In addition to identifying the selected regime, the relative change in average correlation compared to the baseline $C1C2$ training regime is reported in order to quantify the impact of incorporating additional training data and subsequent fine-tuning. An overview of the resulting dimension-specific models, their corresponding training regimes, and the associated performance change is provided in Table 6.7.

Table 6.7: Selected single-target (dimension-specific) models and their corresponding training regimes for each perceptual dimension. The reported Δr indicates the change in average file-level Pearson correlation relative to the baseline $C1C2$ training regime.

Perceptual dimension	**Training regime**	Δr **(vs.** $C1C2$**)**
Overall quality	*C1C2→C3C2*	0.024
Noisiness	*C1C2→C3C2*	0.005
Discontinuity	*C1C2→C3C2*	0.002
Coloration	*C1C2*	-
Loudness	*C3C4*	0.030

6.3.2 Overall Quality MOS Prediction

Model performance is evaluated using established speech quality assessment metrics, following the evaluation principles adopted in the preceding chapters. Predictions are generated on held-out test datasets that are not used during training or model optimization, ensuring an unbiased assessment of generalization performance.

Performance of this single-target model set is compared to the performance of the NISQA model (Mittag et al., 2021), the Multi-Task model from Chapter 5, and the Zero-Shot model from Chapter 4. The databases used for this comparison consist of all the heldout test databases from the previous chapter. This includes the four databases from the NISQA Corpus (Mittag et al., 2021) and the SUB_TUB_2 database.

To diversify the evaluation set and test for generalization, the held out test databases from the *ConferencingSpeech 2022 Challenge* (Yi et al., 2022) Interspeech Conference is also used. This consists of three databases with subjective overall quality MOS scores obtained via crowdsourcing according to ITU-T Recommendation P.808 (2021), namely, the TUB_IS22_DB1, Tencent, and PSTN (Mittag et al., 2020). The TUB_IS22_DB1 (IS22) database was created using clean English conversation speech material processed

with multiple degradations including different codecs, bandwidths, single or multiple background noises, and packet loss, ensuring no speaker overlap. The Tencent database includes speech conditions with reverberation and without reverberation. In the without-reverberation condition, the speech clips exhibit the simulated impairments which are very often encountered in online conferences. In the with-reverberation condition, simulated impairments and live recording speech clips are both present. The PSTN database contains speech collected from real Skype calls recorded through the SkypeIn voicemail service.

The following subsections present the performance results of the three foundation-based models investigated in this work, together with the baseline NISQA model. Results are organized by perceptual target dimension, beginning with overall speech quality (MOS), followed by noisiness, discontinuity, coloration, and sub-optimum loudness.

For each dimension, performance is further analyzed across the individual test databases in order to examine dataset-dependent behavior and to identify conditions that influence model performance.

Table 6.8 summarizes the performance of the zero-shot, multi-task, single-target (dimension-specific), and the baseline NISQA models for predicting overall speech quality (MOS) across eight held-out test databases. The NISQA model achieves the highest correlation values and lowest error metrics on two databases, while the single-target model attains the best performance on the remaining six databases, indicating strong generalization across heterogeneous evaluation conditions. A bar chart illustrating model performance in terms of Pearson correlation (r) is shown in Figure 6.1.

Table 6.8: Condition-level performance of the four models for overall speech quality (MOS) on held-out test databases. Pearson correlation (r) is computed on raw predictions, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping.

Model	Metric	NSC	FOR	LIVE	P501	TUB_2	IS22	Ten	PSTN
Zero-Shot	r	0.896	0.831	0.760	0.892	0.728	0.794	0.742	0.477
	RMSE	0.416	0.460	0.530	0.443	0.396	0.369	0.782	0.282
	RMSE*	0.373	0.424	0.463	0.342	0.324	0.213	0.621	0.078
Multi	r	0.938	0.907	0.811	0.960	0.782	0.906	0.809	0.619
	RMSE	0.324	0.348	0.478	0.274	0.360	0.257	0.686	0.252
	RMSE*	0.294	0.302	0.405	0.236	0.303	0.164	0.508	0.058
Single	r	0.949	0.929	**0.921**	**0.963**	**0.877**	**0.915**	**0.825**	**0.630**
	RMSE	0.294	0.307	0.317	0.263	0.278	0.244	0.660	0.249
	RMSE*	0.257	0.265	0.244	0.223	0.238	0.137	0.474	0.057
NISQA	r	**0.970**	**0.949**	0.902	0.949	0.694	0.843	0.764	0.509
	RMSE	0.228	0.260	0.353	0.309	0.416	0.326	0.754	0.276
	RMSE*	0.194	0.216	0.283	0.265	0.368	0.195	0.605	0.080

The following figures present scatter plots of predicted versus subjective overall MOS for the Zero-Shot AST model, the Multi-Task model, the Single-Target model set, and the NISQA baseline model evaluated on the held-out test databases. Pearson correlation

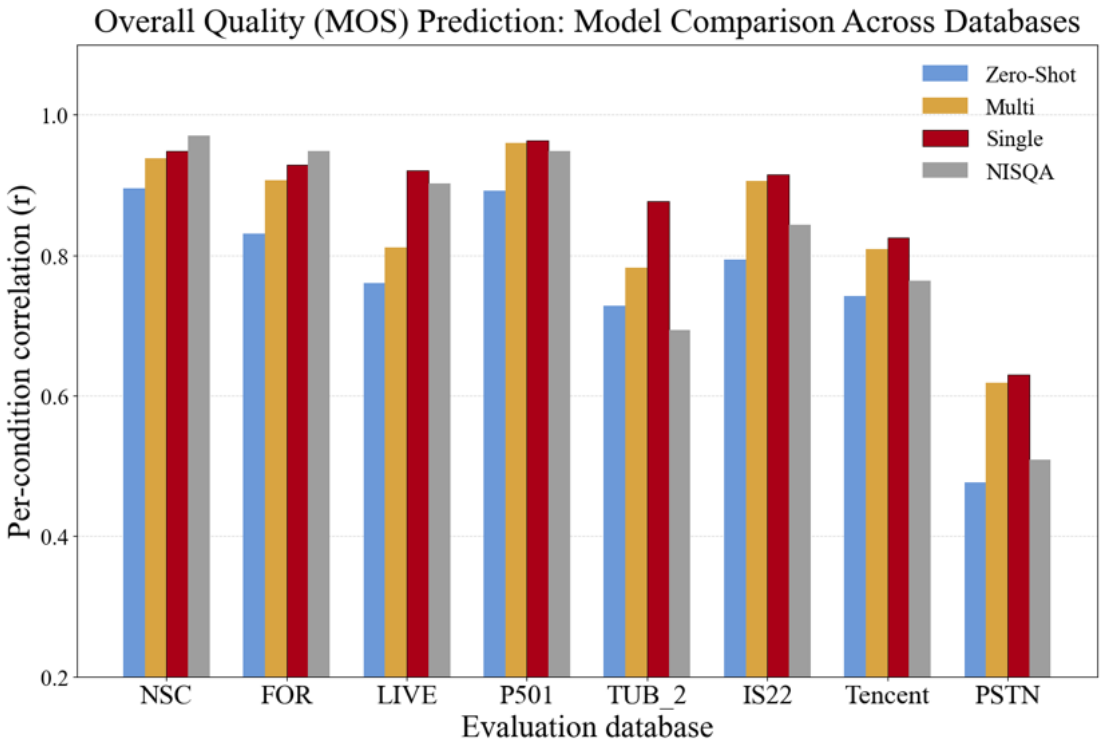


Fig. 6.1: Per-condition Pearson correlation coefficients (r) between predicted and subjective overall MOS for four model variants evaluated across multiple speech quality databases. Correlations are computed at the condition level, reflecting agreement across degradation conditions rather than individual utterances.

is computed on the raw model predictions, and the scatter plots are shown prior to the application of any mapping functions used for RMSE and RMSE* computation. A tolerance band of ±1 MOS unit around the ideal prediction line is included to highlight conditions that are particularly challenging for the models.

NISQA_TEST_NSC

The NISQA_TEST_NSC dataset comprises simulated speech degradations covering a range of codecs, background noise conditions, packet loss, and clipping artifacts. In addition, it includes live communication scenarios involving platforms such as Skype, Zoom, and Google Meet, as well as mobile-to-landline transmission conditions.

Figure 6.2 presents scatter plots of predicted versus subjective overall MOS for all four evaluated models on the NISQA_TEST_NSC database. Overall, all models demonstrate strong performance on this dataset, including the zero-shot model, which attains a Pearson correlation of $r = 0.896$. Of the 60 conditions contained in the dataset, only two fall outside the defined ±1 MOS tolerance band. These correspond to the EVS codec at 24.4 kbps with 15% bursty packet loss and the Opus codec at 25.0 kbps with 25% bursty packet loss, for which the zero-shot model overpredicts overall MOS by at least 1.12 points.

The multi-task and single-target models exhibit comparable performance, with both underpredicting overall MOS for conditions involving P50MNRU at 25 dB and the Google Meet *Good* condition. In addition, the multi-task model shows increased sensitivity to time clipping at 2%, where larger underpredictions are observed. Among all evaluated models, the NISQA baseline model achieves the highest correlation for overall quality prediction, with $r = 0.970$.

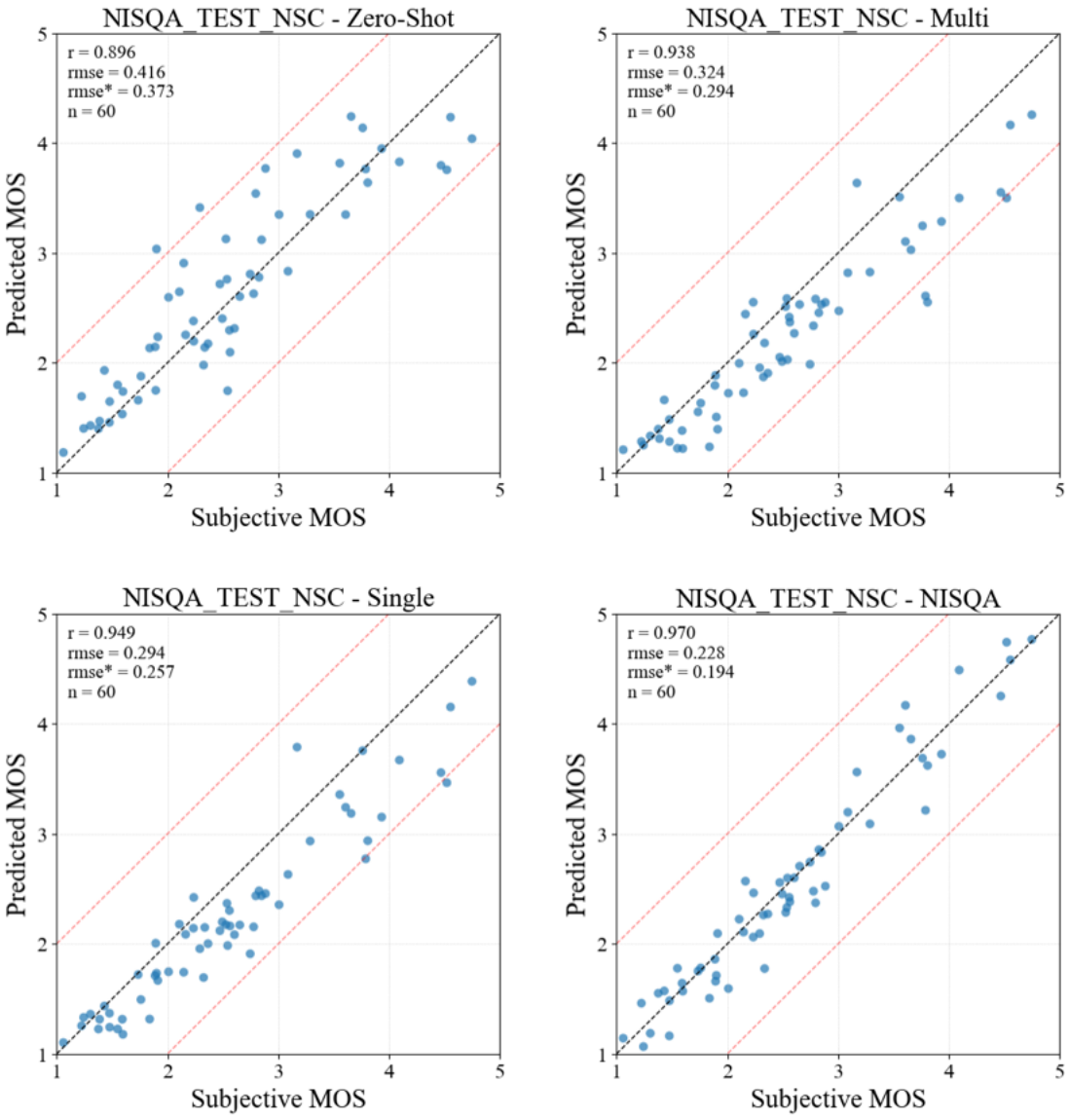


Fig. 6.2: Predicted versus subjective overall MOS for the Zero-Shot AST model, the Multi-Task AST model, the Single-Target AST model set, and the NISQA reference model evaluated on the NISQA_TEST_NSC database.

NISQA_TEST_FOR

The NISQA_TEST_FOR dataset similarly comprises simulated speech degradations involving different codecs, background noise conditions, packet loss, and clipping artifacts, as well as live communication scenarios including WhatsApp, Zoom, and Discord.

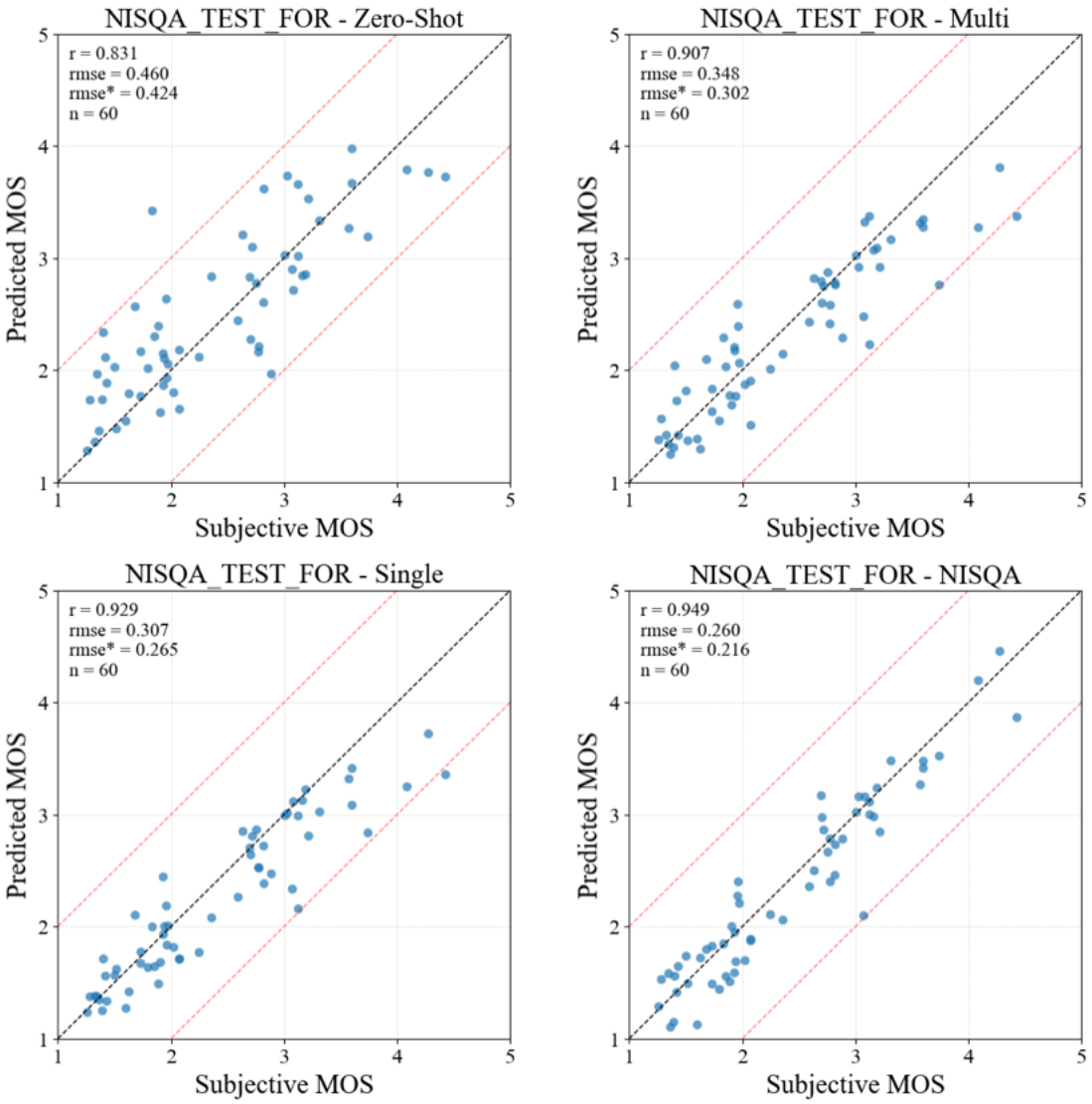


Fig. 6.3: Predicted MOS versus subjective MOS of the Zero-Shot AST model, the Multi-Task AST model, the Single-Target AST model set, and the NISQA speech quality model on the NISQA_TEST_FOR database.

Overall, all evaluated models demonstrate strong performance across the 60 conditions in this dataset, including the zero-shot model, which attains the lowest correlation among the models with $r = 0.831$, as shown in the top left plot in Figure 6.3. As observed previously, the Opus codec at 15.0 kbps with 18% bursty packet loss remains particularly challenging, with the zero-shot model overpredicting overall MOS by up to 1.58 points.

The multi-task and single-target AST models exhibit a tendency to underpredict the clean speech condition in this dataset by approximately one MOS unit. In contrast, the NISQA baseline model achieves the highest correlation of $r = 0.97$ and predicts all conditions within the ±1 MOS tolerance band.

NISQA_TEST_LIVETALK

The NISQA_TEST_LIVETALK dataset consists of recordings of real telephone and VoIP calls captured across a range of acoustic environments and thus reflects realistic, in-the-wild communication conditions.

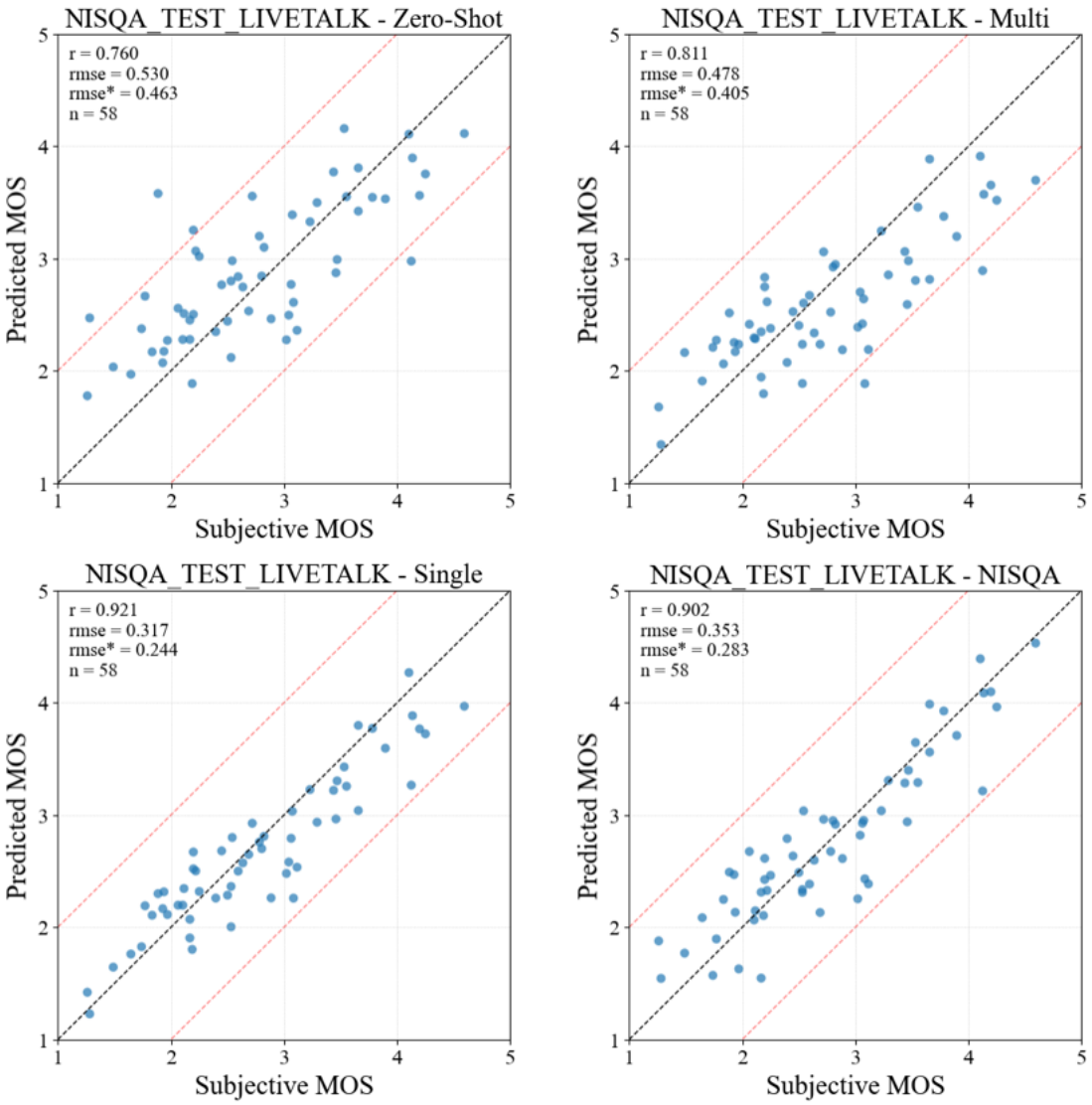


Fig. 6.4: Predicted MOS versus subjective MOS of the Zero-Shot AST model, the Multi-Task AST model, the Single-Target AST model set, and the NISQA speech quality model on the NISQA_TEST_LIVETALK database.

The scatter plots in Figure 6.4 show that the zero-shot model finds this dataset particularly challenging. Specifically, it tends to overpredict overall MOS for conditions involving P50MNRU at 25 dB, time clipping at 20%, and street noise recorded on a mobile phone with elevated speaking effort, while underpredicting the condition involving a 20 dB signal level attenuation.

In contrast, both the single-target AST model and the NISQA baseline model produce predictions that remain within the ±1 MOS tolerance band across all evaluated conditions. Among these, the single-target model achieves the most robust overall performance, with a Pearson correlation of $r = 0.921$.

NISQA_TEST_P501

The NISQA_TEST_P501 dataset comprises simulated speech degradations involving different codecs, background noise conditions, packet loss, and clipping artifacts. In addition, it includes live communication scenarios such as Skype, Zoom, and WhatsApp, as well as recordings captured over mobile networks. Scatter plots of predicted versus subjective overall MOS for all four evaluated models are shown in Figures 6.5 and 6.6.

Overall, all models perform well on this database, with the lowest correlation observed for the zero-shot model at $r = 0.892$. The zero-shot model tends to overpredict overall MOS for conditions involving the EVS codec at 24.4 kbps with 15% bursty packet loss and 25% random packet loss, the Opus codec at 25.0 kbps with 15% bursty packet loss, as well as for degraded Zoom call conditions. In contrast, the remaining three models predict all 60 conditions within the ±1 MOS tolerance band.

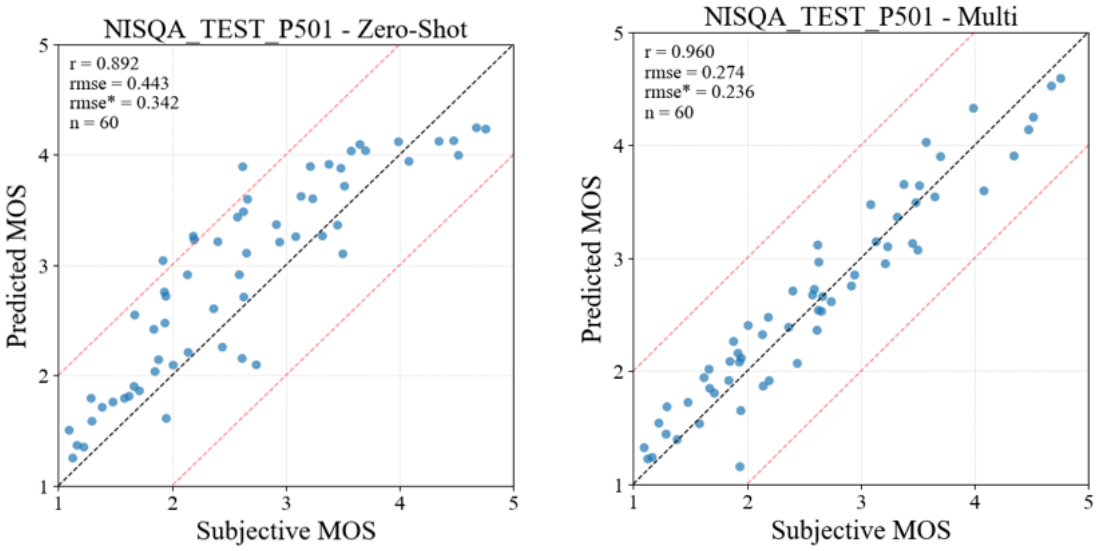


Fig. 6.5: Predicted MOS versus subjective MOS of the Zero-Shot AST model and the Multi-Task AST model on the NISQA_TEST_P501 database.

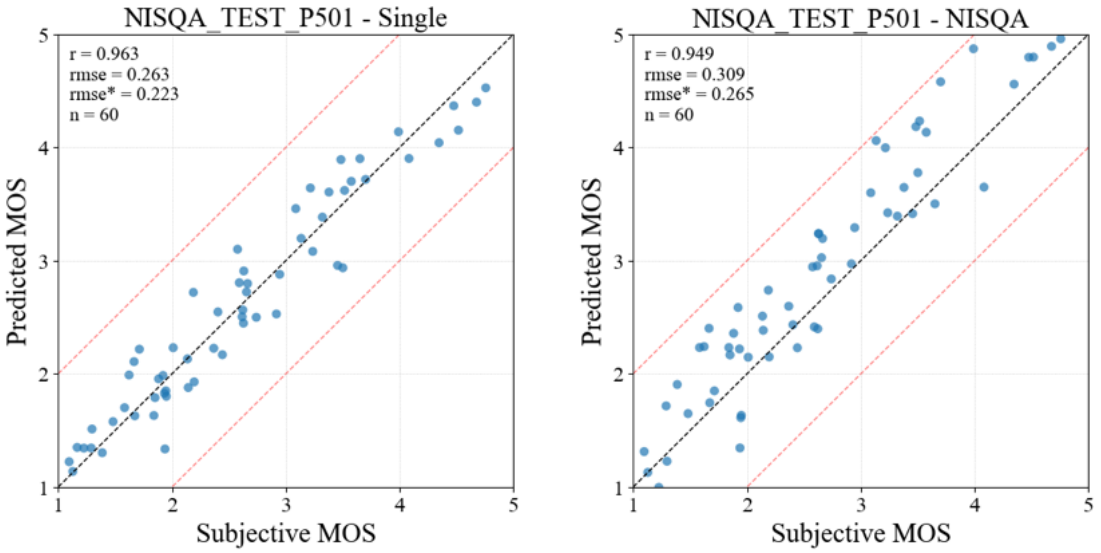


Fig. 6.6: Predicted MOS versus subjective MOS of the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_P501 database.

Notably, the NISQA baseline model exhibits a tendency to slightly overpredict overall MOS across most conditions in this dataset and is outperformed by both the multi-task and single-target models. Among all evaluated models, the single-target model achieves the most robust performance, with a Pearson correlation of $r = 0.963$.

SUB_TUB_2

The SUB_TUB_2 database comprises German handset speech recorded under a variety of real-world noise environments, including office, traffic, vehicle, and public space scenarios. Signals are processed using EVS super-wideband and AMR-WB codecs in combination with a range of noise suppression algorithms.

Across this database, all evaluated models outperform the NISQA baseline, which achieves a Pearson correlation of $r = 0.694$. Notably, even the zero-shot model attains a higher correlation than the baseline. For all models, none of the 25 evaluated conditions fall outside the ±1 MOS tolerance band, indicating generally well-aligned predictions. Figure 6.7 presents scatter plots for all models.

The multi-task model shows underprediction for conditions involving background noise without noise suppression, and for in-car recordings processed with neural noise reduction. In addition, both the multi-task and single-target models underpredict a challenging outdoor traffic intersection condition processed with a learned noise suppression method. Among all evaluated models, the single-target model achieves the strongest overall performance on this database, with a Pearson correlation of $r = 0.877$.

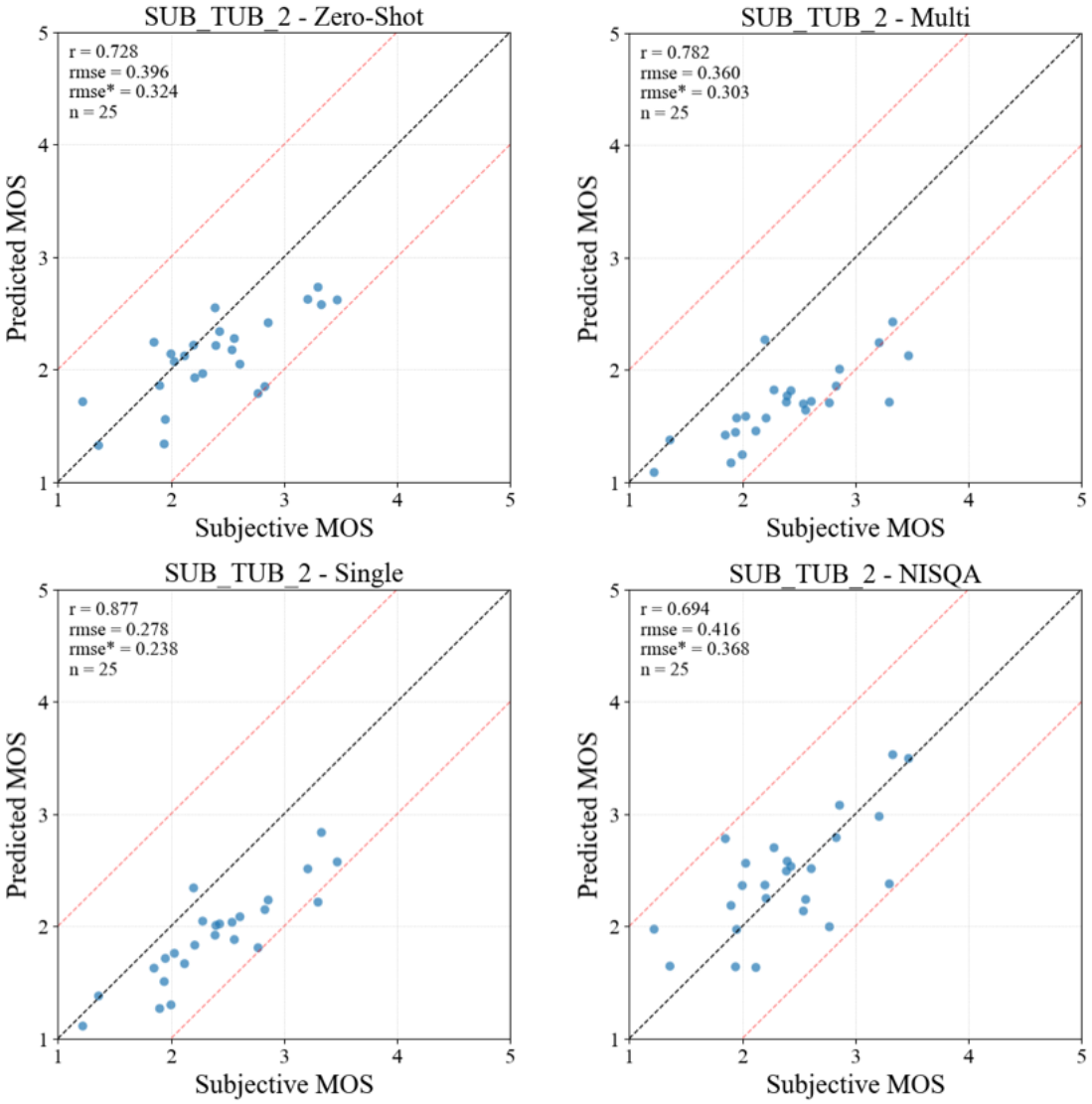


Fig. 6.7: Predicted MOS versus subjective MOS of the Zero-Shot AST model, the Multi-Task AST model, the Single-Target AST model set, and the NISQA speech quality model on the SUB_TUB_2 database.

TUB_IS22_DB1

This database forms part of the *ConferencingSpeech 2022 Challenge* (Yi et al., 2022) at Interspeech and consists of single-speaker conversational speech excerpts subjected to synthetic degradation conditions. These include variations in codec type, bandwidth, background noise (single and multiple sources), and packet loss scenarios. As individual condition descriptions are not available for this dataset, the scatter plots in Figure 6.8 additionally include the first-order mapping function to facilitate visual identification of overall performance trends.

Across the 62 conversation-like conditions, all evaluated models exhibit positive correlation between predicted and subjective overall MOS. The zero-shot model attains the lowest correlation with $r = 0.794$. Both the multi-task and single-target AST-based

models outperform the NISQA baseline, with the single-target model achieving the highest correlation of $r = 0.915$. Notably, a consistent downward shift is observed across all models, indicating a general tendency to underpredict overall speech quality for this dataset.

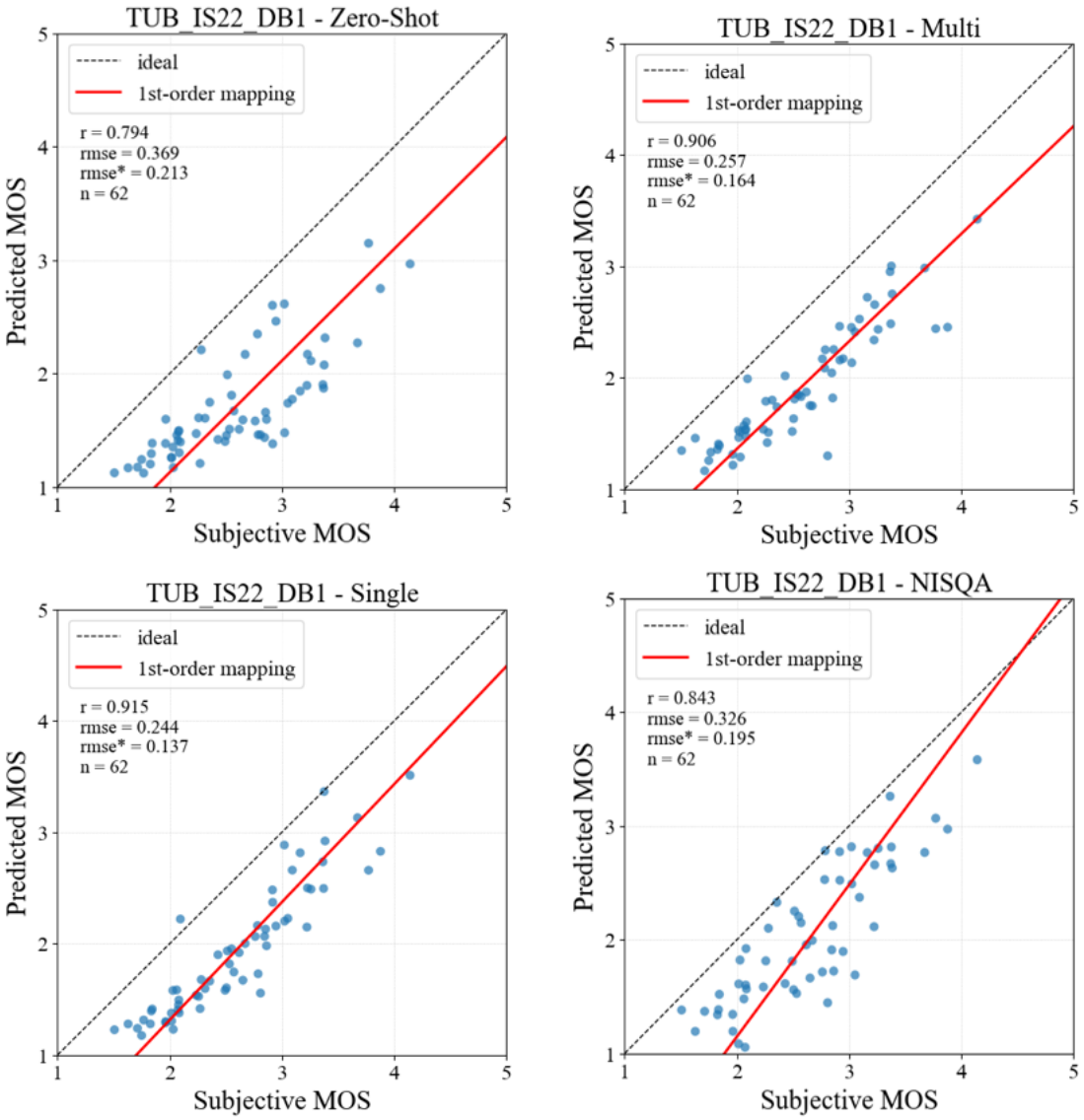


Fig. 6.8: Predicted MOS versus subjective MOS of the Zero-Shot AST model, the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the TUB_IS22_DB1 database.

Tencent Corpus

This database is also part of the *ConferencingSpeech 2022 Challenge* (Yi et al., 2022) at Interspeech. It comprises speech samples recorded under both reverberant and non-reverberant conditions. In the non-reverberant subset, speech signals are affected

by simulated impairments commonly encountered in online conferencing scenarios. In the reverberant subset, both simulated impairments and recordings from live reverberant environments are included.

Figure 6.9 presents scatter plots of predicted versus subjective overall MOS for all evaluated models. Conditions with and without reverberation are distinguished by color coding, and a third-order monotonic mapping function is overlaid to facilitate visual assessment of performance trends.

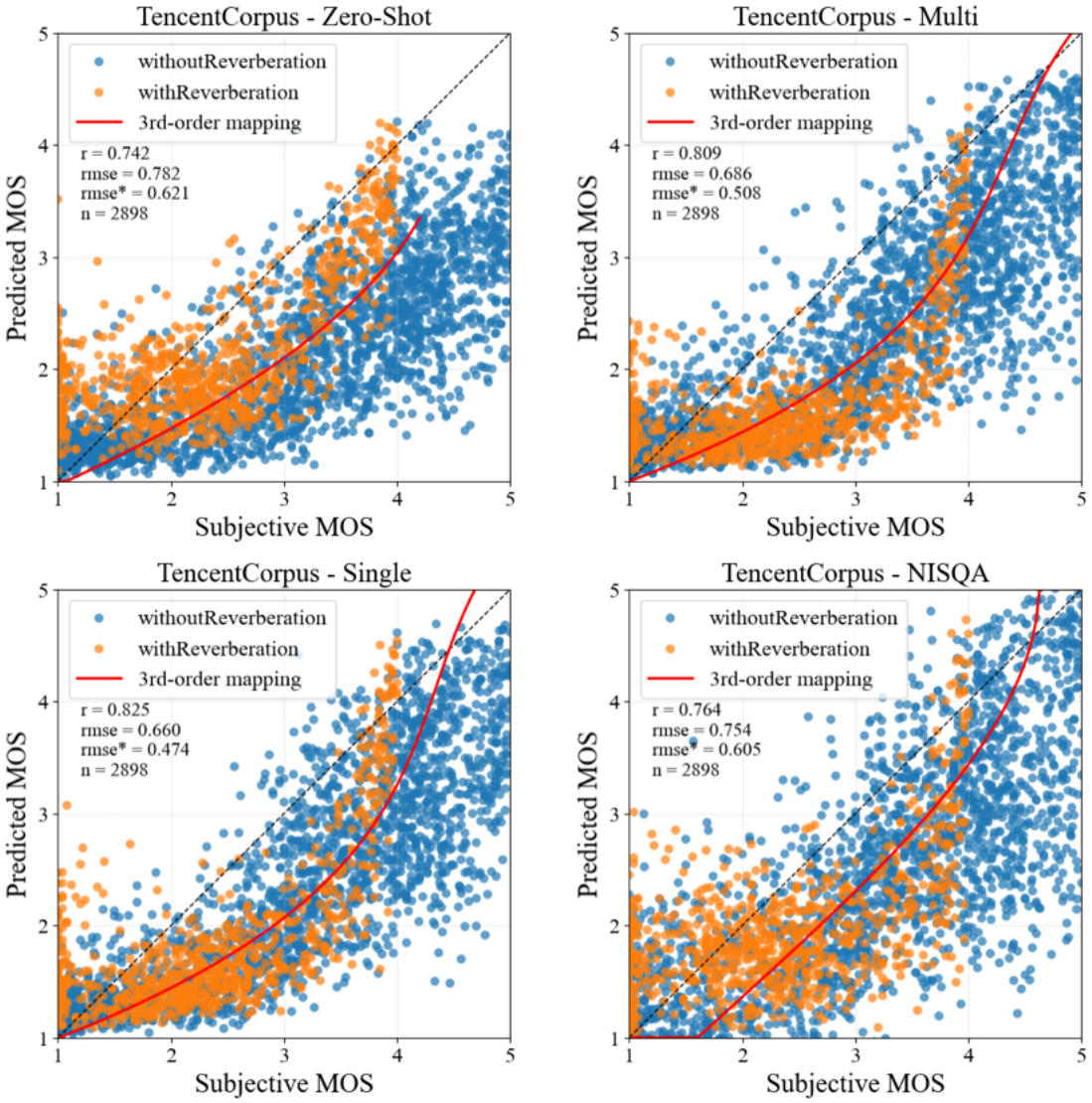


Fig. 6.9: Predicted MOS versus subjective MOS of the Zero-Shot AST model, the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the Tencent database.

All models exhibit positive correlation between predicted and subjective MOS, with the zero-shot model achieving the lowest correlation of $r = 0.742$. Notably, the zero-shot model tends to saturate its predictions and does not predict MOS values above approximately 4.2, despite subjective scores reaching the upper bound of the five-

point ACR scale. Both the multi-task and single-target AST-based models outperform the NISQA baseline, with the single-target model attaining an overall correlation of $r = 0.825$.

When analyzed separately by reverberation condition, the single-target model achieves a correlation of $r = 0.839$ for non-reverberant speech and $r = 0.691$ for reverberant speech. In comparison, the NISQA baseline achieves correlations of $r = 0.786$ and $r = 0.586$ for the non-reverberant and reverberant conditions, respectively.

PSTN Corpus

The PSTN corpus is also part of the *ConferencingSpeech 2022 Challenge* (Yi et al., 2022) at Interspeech and contains speech samples collected from real Skype calls recorded via the SkypeIn voicemail service (Mittag et al., 2020). Scatter plots of predicted versus subjective overall MOS for this database are shown in Figures 6.10 and 6.11. In addition, a first-order mapping function is overlaid to facilitate visual identification of global performance trends.

This dataset proves challenging for all evaluated models, with the zero-shot model achieving a correlation of $r = 0.477$. Both the multi-task and single-target AST-based models outperform the NISQA baseline. Among all models, the single-target model attains the strongest performance, achieving the highest correlation and lowest error, with $r = 0.630$. A consistent downward shift is observed across all models, indicating a general tendency to underpredict overall speech quality for this corpus.

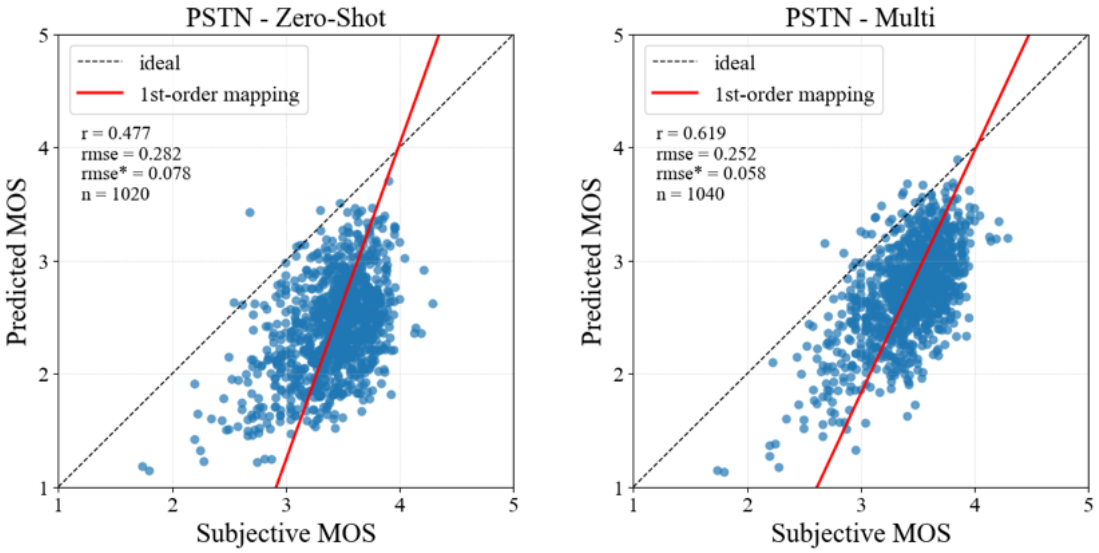


Fig. 6.10: Predicted MOS versus subjective MOS of the Zero-Shot AST model and the Multi-Task AST model on the PSTN database.

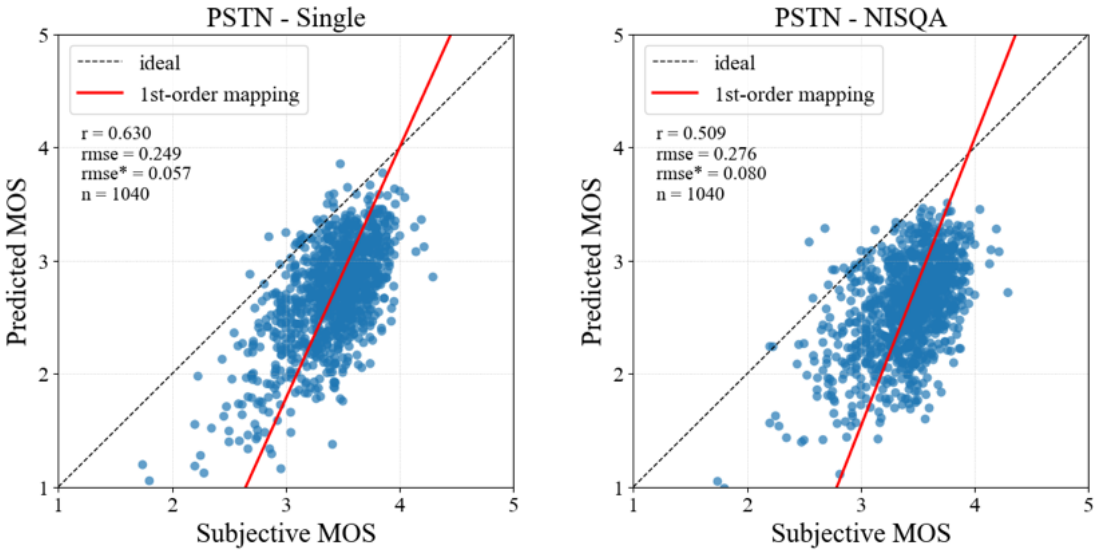


Fig. 6.11: Predicted MOS versus subjective MOS of the Single-Target AST model set and the NISQA speech quality model on the PSTN database.

Discussion: Overall Speech Quality (MOS) Prediction

Across the diverse set of held-out evaluation databases, consistent trends emerge regarding the behavior and relative strengths of the four evaluated models for overall speech quality (MOS) prediction. Taken together, the results highlight the impact of task-specific fine-tuning, data diversity, and architectural choices on generalization performance under heterogeneous acoustic and transmission conditions.

The zero-shot AST model demonstrates surprisingly strong performance across several databases, achieving moderate to high correlations even without task-specific training. This is particularly evident on the NISQA_TEST_NSC, NISQA_TEST_FOR, and NISQA_TEST_P501 datasets, where correlations exceed $r = 0.8$. These results confirm that pretrained transformer representations encode perceptually relevant information for speech quality assessment. However, the zero-shot model exhibits systematic limitations, including overprediction under severe packet loss, saturation effects at high subjective quality levels (notably in the Tencent corpus), and increased sensitivity to realistic distortions such as reverberation and PSTN transmission effects. These behaviors indicate that zero-shot representations alone are insufficient to reliably capture the full range of degradations encountered in real-world communication scenarios.

Both the multi-task and single-target AST-based models consistently improve upon the zero-shot baseline across all datasets. The gains are most pronounced on databases characterized by realistic or compound distortions, such as NISQA_TEST_LIVETALK, SUB_TUB_2, the Tencent corpus, and the PSTN corpus. In these settings, task-specific supervision enables the models to better align predictions with subjective judgments, reducing systematic bias and improving robustness. Nevertheless, the multi-task model occasionally exhibits increased sensitivity to specific impairment types, such as time clipping or clean-speech conditions, suggesting that joint optimization across perceptual dimensions may introduce competing gradients that slightly compromise MOS-specific calibration.

Among all evaluated approaches, the single-target (dimension-specific) model achieves the most consistent and robust performance for overall MOS prediction. It attains the highest correlation on the majority of databases, including NISQA_TEST_LIVETALK, NISQA_TEST_P501, SUB_TUB_2, TUB_IS22_DB1, the Tencent corpus, and the PSTN corpus. This behavior indicates that optimizing the encoder and regression head exclusively for overall quality allows the model to better exploit perceptually relevant cues without interference from auxiliary objectives. The improved performance is particularly evident on challenging datasets with realistic recording conditions, reverberation, or telephony artifacts, where the single-target model consistently reduces underprediction and avoids saturation effects.

The NISQA baseline model remains highly competitive and achieves the strongest performance on selected databases, notably NISQA_TEST_NSC and NISQA_TEST_FOR. However, across several other databases—especially those involving reverberation, diverse noise suppression pipelines, or real PSTN transmission—the learned transformer-based models, and in particular the single-target variant, surpass the baseline in terms of correlation and robustness. This suggests that foundation-model-based approaches offer superior adaptability to unseen and heterogeneous conditions when appropriately fine-tuned.

A common trend observed across multiple datasets, including TUB_IS22_DB1 and the PSTN corpus, is a general downward shift in predictions, indicating systematic underestimation of subjective MOS. This effect is shared by all models to varying degrees and likely reflects dataset-specific rating distributions, conversational context effects, or scale usage differences inherent to the underlying subjective tests. While mapping functions partially address this issue in error-based metrics, the persistence of this trend in raw predictions highlights the importance of calibration-aware training strategies and diverse evaluation data.

In summary, the results demonstrate that while pretrained transformer representations already encode substantial perceptual information for speech quality assessment, task-specific fine-tuning is essential for robust MOS prediction across realistic communication scenarios. Among the evaluated approaches, the single-target AST-based model provides the best overall trade-off between performance, robustness, and generalization, consistently outperforming the zero-shot and multi-task variants and, in several cases, surpassing the established NISQA baseline. These findings support the use of dimension-specific training as an effective strategy for non-intrusive modeling of overall speech quality in heterogeneous real-world conditions.

6.3.3 Noisiness Prediction

For the perceptual quality dimensions, five held-out test databases are available for evaluation. As the zero-shot model is trained exclusively for overall speech quality (MOS) prediction, it is excluded from the following performance analysis. Table 6.9 summarizes the performance of the multi-task, single-target (dimension-specific), and NISQA baseline models for noisiness (NOI) prediction across the test databases that provide perceptual dimension annotations.

Table 6.9: Condition-level performance of the three models for noisiness (NOI) on held-out test databases. Pearson correlation (r) is computed on raw predictions, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping.

Model	Metric	NSC	FOR	LIVE	P501	TUB_2
Multi	r	0.936	0.919	0.852	**0.951**	**0.924**
	RMSE	0.279	0.277	0.377	0.289	0.218
	RMSE*	0.251	0.247	0.279	0.223	0.175
Single	r	0.949	0.897	**0.855**	0.944	0.898
	RMSE	0.251	0.310	0.374	0.310	0.250
	RMSE*	0.215	0.276	0.260	0.249	0.186
NISQA	r	**0.960**	**0.946**	0.759	0.949	0.660
	RMSE	0.223	0.228	0.470	0.296	0.427
	RMSE*	0.191	0.200	0.301	0.260	0.381

Overall, the NISQA model achieves the highest correlation values and lowest error metrics on two of the evaluated databases. The multi-task model attains the best performance on two databases, while the single-target model performs best on one database.

The following figures present scatter plots of predicted versus subjective noisiness for the the Multi-Task model, the Single-Target model set, and the NISQA baseline model evaluated on the held-out test databases. Pearson correlation is computed on the raw model predictions, and the scatter plots are shown prior to the application of any mapping functions used for RMSE and RMSE* computation. A tolerance band of ± 1 MOS unit around the ideal prediction line is included to highlight conditions that are particularly challenging for the models.

NISQA_TEST_NSC

Figure 6.12 presents scatter plots of predicted versus subjective noisiness scores for the three evaluated models on the NISQA_TEST_NSC database. The 60 degradation conditions in this dataset include a wide range of simulated background noise scenarios, providing substantial variability in perceived noisiness. Overall, all three models demonstrate sensitivity to noisiness-related degradations and are able to capture the general relationship between predicted and subjective noisiness scores.

Among the evaluated models, the multi-task model underpredicts the *Bad* Google Meet condition by approximately one MOS unit and attains the lowest correlation on this database, with $r = 0.936$. In contrast, the NISQA baseline model achieves the highest correlation, with $r = 0.960$, indicating strong alignment with subjective judgments. The single-target model performs comparably, exhibiting stable predictions across most conditions without pronounced outliers.

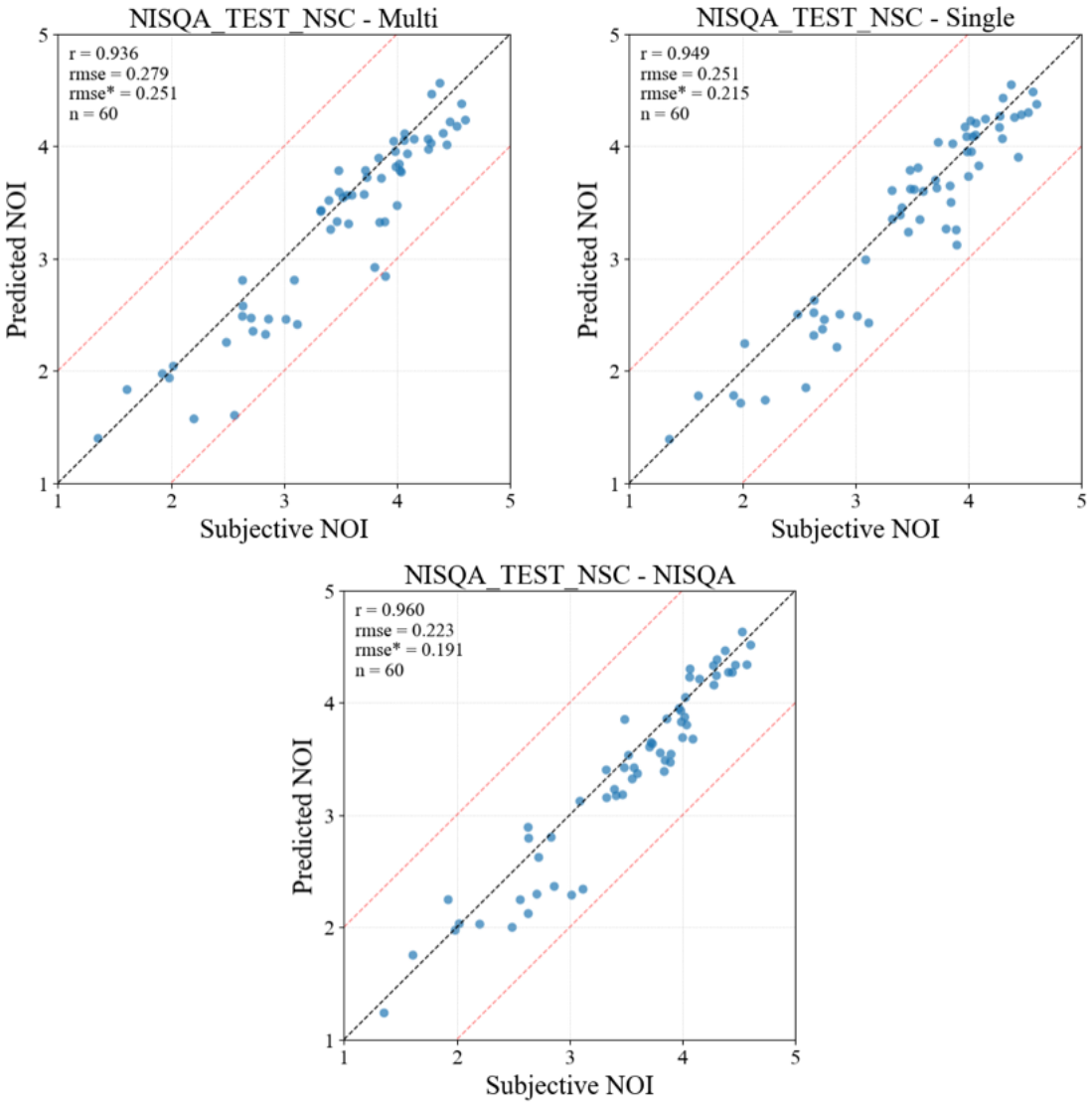


Fig. 6.12: Predicted NOI versus subjective NOI of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_NSC database.

NISQA_TEST_FOR

This database comprises simulated speech conditions that include a range of noisiness-related degradations. Figure 6.13 presents scatter plots of predicted versus subjective noisiness scores for the three evaluated models on the NISQA_TEST_FOR database. All models exhibit positive correlation with the subjective ratings, indicating reliable detection of perceived noisiness across conditions.

Among the evaluated models, the NISQA baseline achieves the highest correlation on this database, with $r = 0.946$. The multi-task AST-based model outperforms the single-target model, suggesting that joint training across perceptual dimensions is beneficial for noisiness prediction in this setting. Across all models, no condition falls outside the ±1

MOS tolerance band, reflecting stable and well-calibrated noisiness predictions. Overall, the results indicate that the evaluated models are robust and sensitive to noisiness-related degradations in this database.

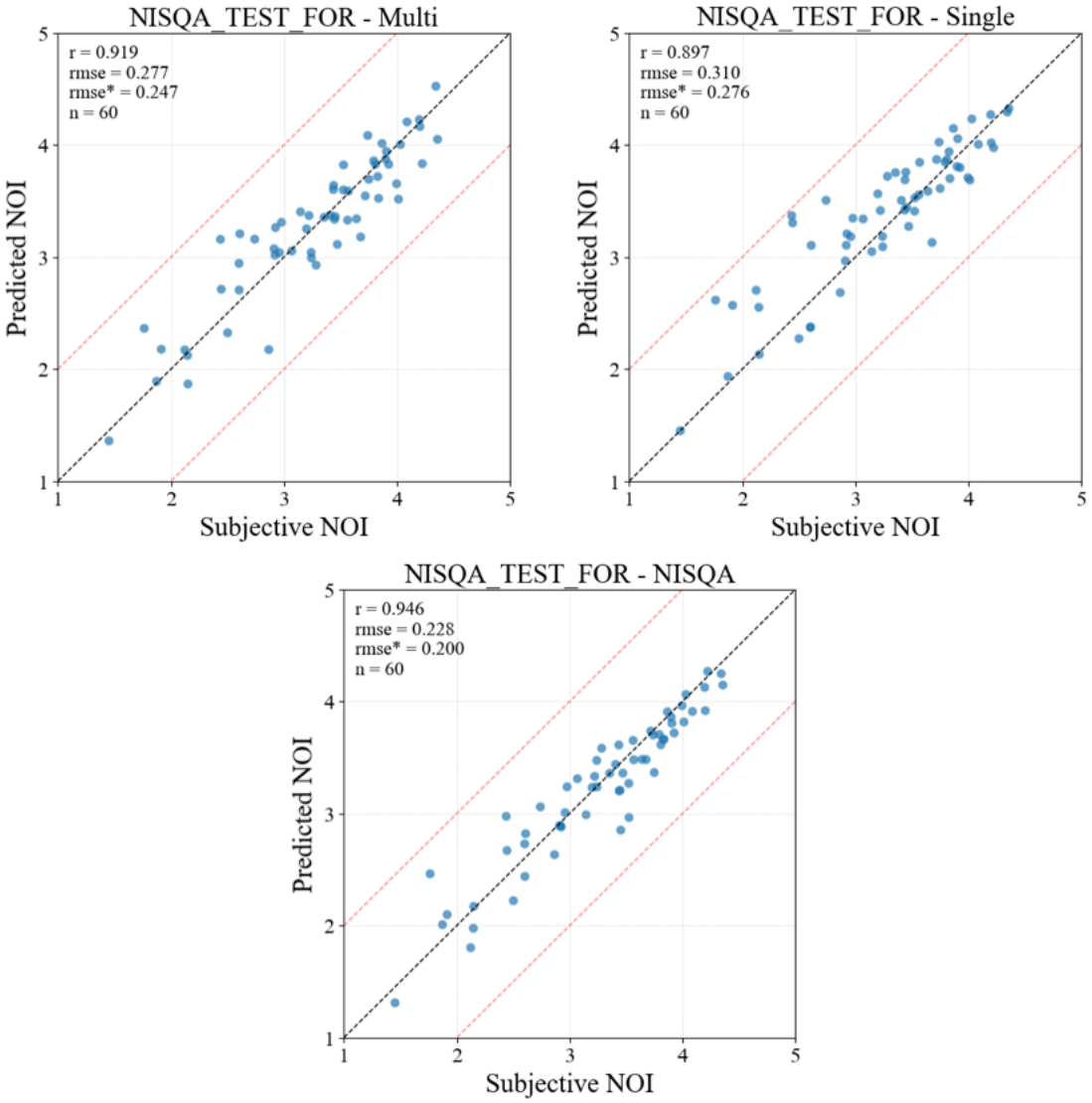


Fig. 6.13: Predicted NOI versus subjective NOI of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_FOR database.

NISQA_TEST_LIVETALK

This database consists of recordings of real telephone and VoIP calls captured across a wide range of acoustic environments and thus reflects realistic, in-the-wild communication conditions. The scatter plots in Figure 6.14 reveal a slight upward shift across all three evaluated models, indicating a general tendency to overpredict perceived noisiness on the ACR scale for this dataset.

All three models find the P50MNRU at 25 dB condition challenging, overpredicting noisiness by more than one MOS unit. Both AST-based models additionally overpredict noisiness for the band-pass filtered condition (500–2500 Hz). The multi-task model further exhibits pronounced overprediction for conditions involving subway station noise under poor reception, recorded on a mobile phone with low speaking effort, as well as for car highway noise under poor reception with a distant talker.

The single-target model overpredicts noisiness for several realistic noise scenarios, including subway station noise and café noise recorded on a mobile phone, particularly when the talker is distant from the microphone, as well as for car highway noise under poor reception. Among the evaluated approaches, the NISQA baseline model shows the greatest difficulty in handling this dataset and attains the lowest correlation, with $r = 0.759$.

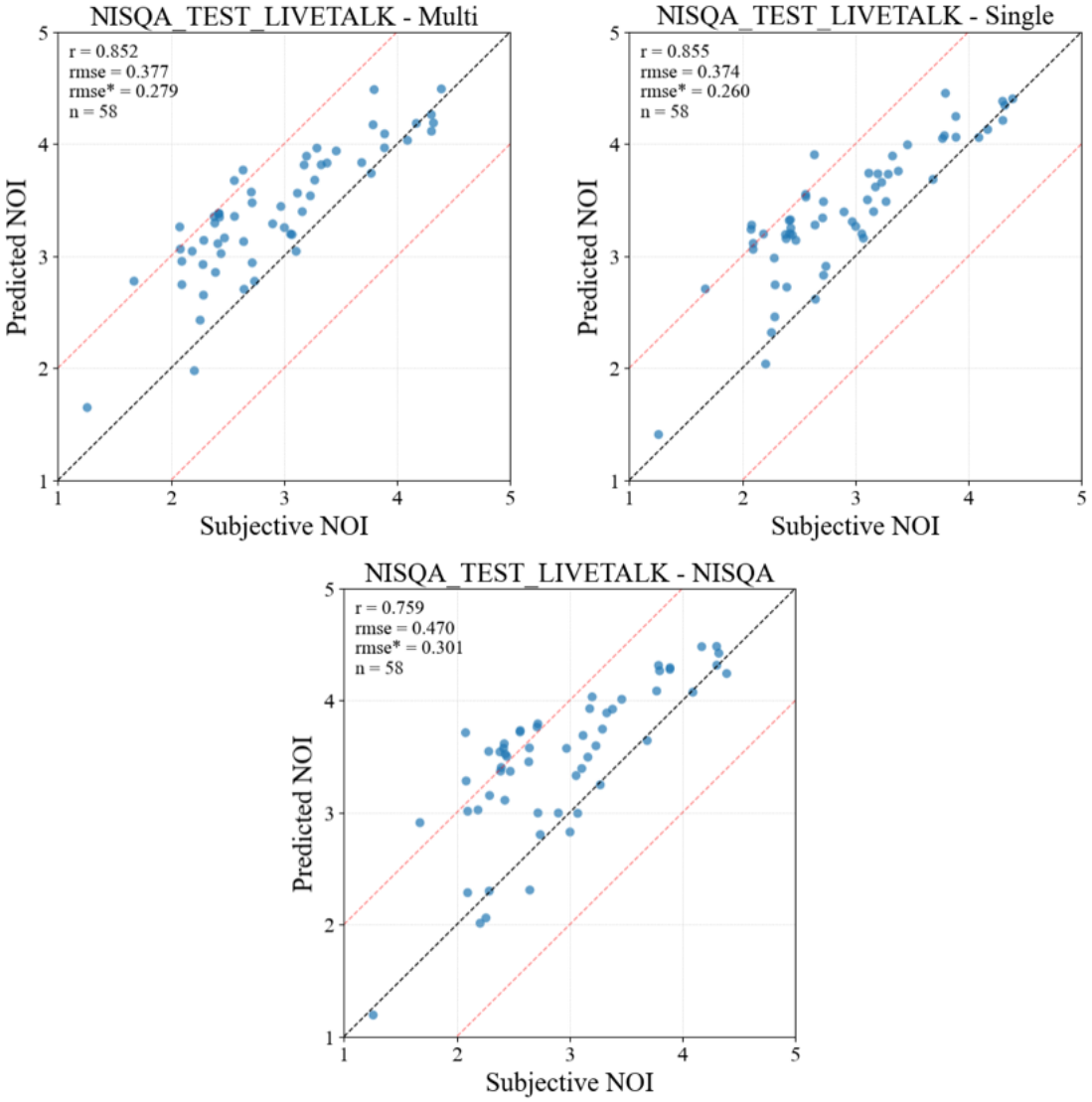


Fig. 6.14: Predicted NOI versus subjective NOI of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_LIVETALK database.

NISQA_TEST_P501

This database comprises simulated speech degradations involving a variety of background noise conditions and additionally includes live communication scenarios such as Skype, Zoom, and WhatsApp, as well as recordings captured over mobile networks. The scatter plots in Figure 6.15 indicate that all evaluated models achieve high correlation with subjective noisiness ratings, with correlation values of at least $r = 0.9$.

For all models, predictions for nearly all 60 conditions fall within the ±1 MOS tolerance band. The only notable exception is observed for the single-target model, which overpredicts noisiness for the condition involving AMR-WB at 23.85 kbps with 6% bursty packet loss and 30 dB background noise by approximately one MOS unit.

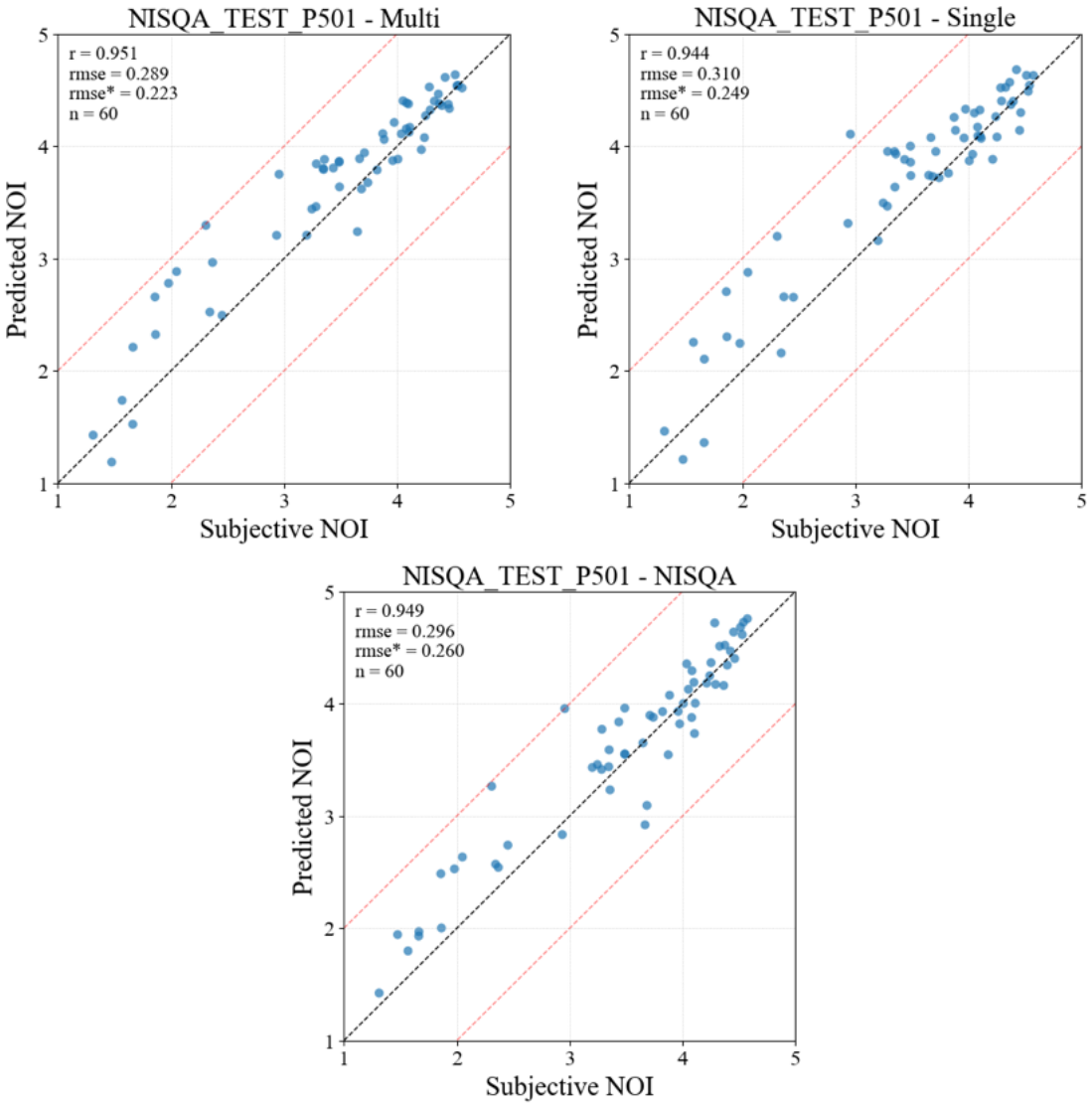


Fig. 6.15: Predicted NOI versus subjective NOI of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_P501 database.

SUB_TUB_2

This database contains handset speech recorded in diverse real-world noise environments and processed using EVS super-wideband and AMR-WB codecs with noise suppression. The multi-task AST model achieves the strongest performance, with a Pearson correlation of $r = 0.924$, and only one of the 25 conditions falling outside the ±1 MOS tolerance band, as shown in Figure 6.16.

The single-target model attains the second-highest correlation, while the NISQA baseline yields the lowest performance with $r = 0.660$. Both the single-target and NISQA models show a general upward shift in their predictions, indicating a tendency to assign higher noisiness scores than observed in the subjective ratings.

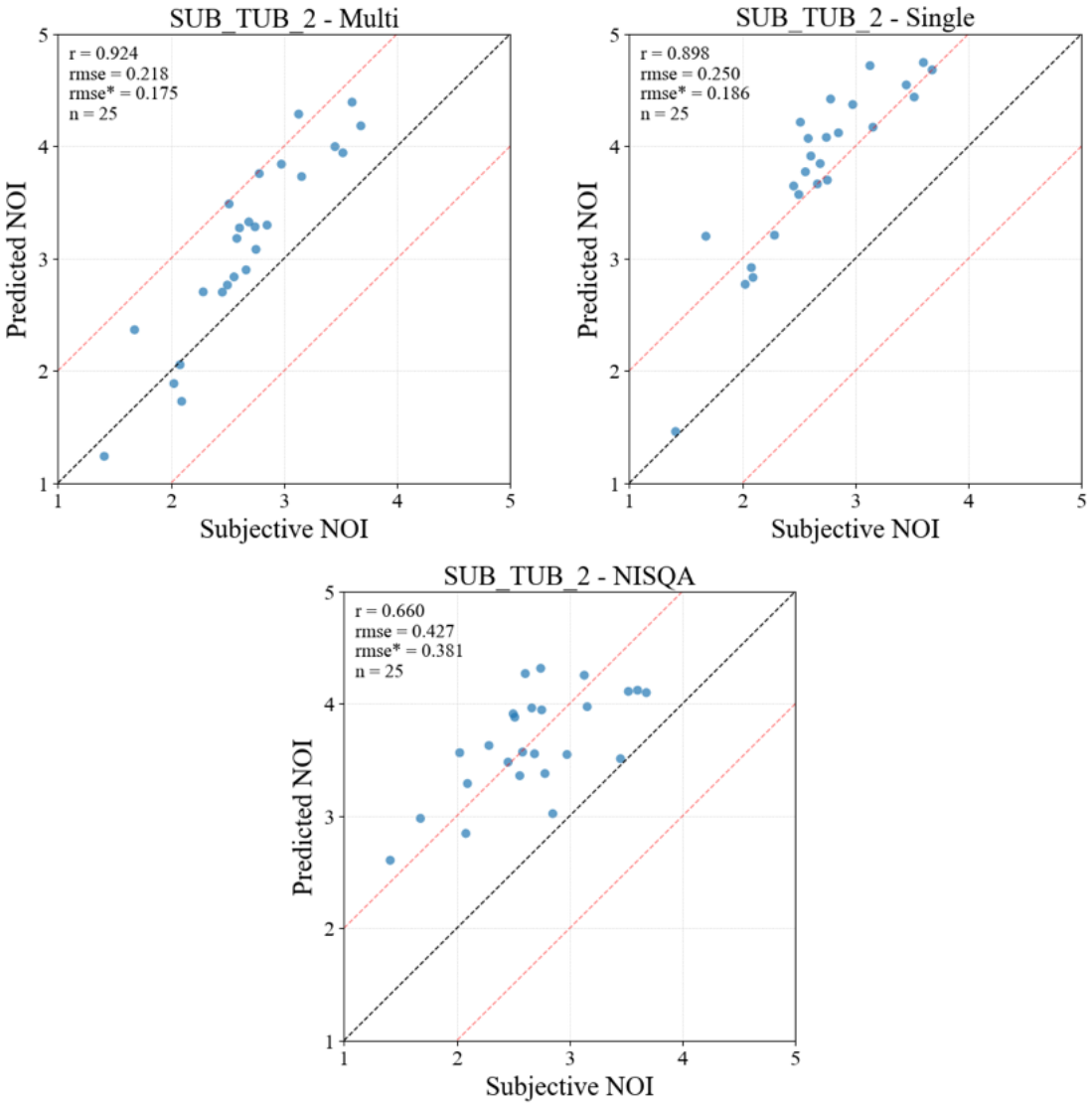


Fig. 6.16: Predicted NOI versus subjective NOI of the Multi-Task AST model, the Single-Target AST model, and the NISQA speech quality model on the SUB_TUB_2 database.

Discussion: Noisiness Prediction

Across the five evaluated test databases, all three models demonstrate the ability to capture perceived noisiness, with consistently positive correlations between predicted and subjective ratings. No single modeling approach dominates across all datasets. The NISQA baseline achieves the strongest performance on databases dominated by simulated degradations, while the multi-task AST model performs best on datasets with diverse real-world noise conditions. The single-target model shows competitive performance and stable behavior across databases, but exhibits increased sensitivity to realistic, in-the-wild conditions. Overall, the results indicate that noisiness prediction remains highly dependent on dataset characteristics and that joint training across perceptual dimensions can be beneficial in settings with heterogeneous noise and processing artifacts.

6.3.4 Discontinuity Prediction

Perceived discontinuity is evaluated on the same five held-out test databases using the three models. As summarized in Table 6.10, the NISQA baseline model achieves the highest correlation between predicted and subjective discontinuity scores on three databases, while the single-target AST-based model attains the strongest performance on the remaining two databases.

Table 6.10: Condition-level performance of the three models for discontinuity (DIS) on held-out test databases. Pearson correlation (r) is computed on raw predictions, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping.

Model	Metric	NSC	FOR	LIVE	P501	TUB_2
Multi	r	0.928	0.908	0.779	0.893	0.718
	RMSE	0.380	0.393	0.448	0.401	0.332
	RMSE*	0.333	0.345	0.367	0.334	0.215
Single	r	0.925	0.892	0.785	**0.912**	**0.799**
	RMSE	0.388	0.425	0.442	0.366	0.286
	RMSE*	0.314	0.372	0.364	0.312	0.175
NISQA	r	**0.956**	**0.965**	**0.831**	0.908	0.348
	RMSE	0.299	0.245	0.397	0.373	0.447
	RMSE*	0.259	0.198	0.312	0.325	0.394

NISQA_TEST_NSC

Figure 6.17 shows scatter plots of predicted versus subjective discontinuity scores for the three evaluated models on the NISQA_TEST_NSC database, which includes simulated packet loss and clipping degradations. All models exhibit positive correlation with subjective ratings, indicating sensitivity to discontinuity-related impairments.

The NISQA baseline model achieves the highest correlation on this database, with $r = 0.956$. The single-target model attains the lowest correlation, with $r = 0.925$, primarily due to underprediction of a high-loss Opus condition. The multi-task model performs comparably and shows stable predictions across most conditions.

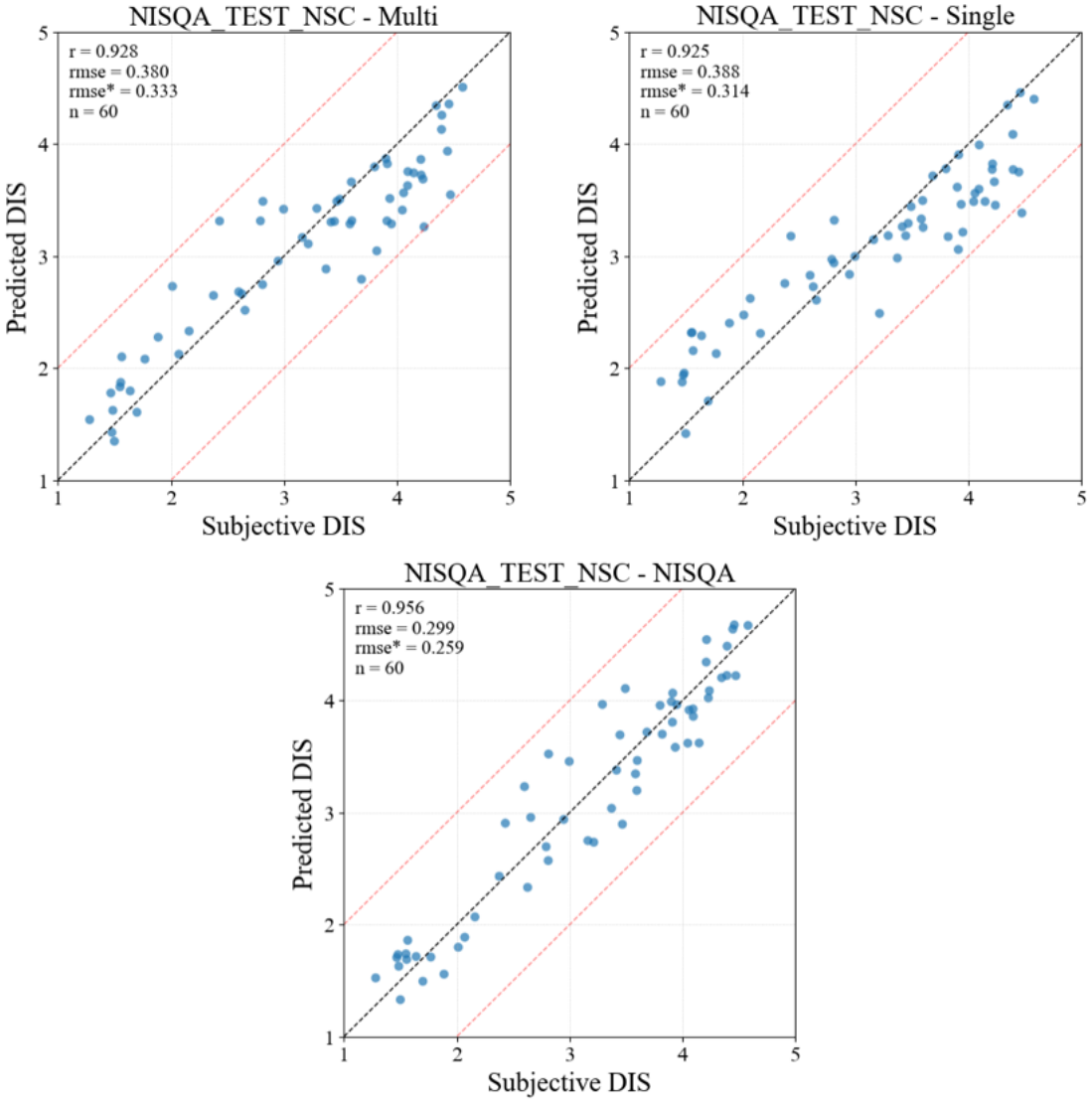


Fig. 6.17: Predicted DIS versus subjective DIS of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_NSC database.

NISQA_TEST_FOR

Figure 6.18 presents scatter plots of predicted versus subjective discontinuity scores for the three evaluated models on the NISQA_TEST_FOR database, which contains simulated packet loss and clipping conditions. All models exhibit positive correlation with the subjective ratings, indicating reliable detection of discontinuity-related impairments.

The NISQA baseline model achieves the highest correlation on this database, with $r = 0.965$. The multi-task model outperforms the single-target model overall, though it overpredicts a severe EVS packet-loss condition beyond the ±1 MOS tolerance band. The single-target model overpredicts several packet-loss conditions, including WhatsApp and Discord scenarios.

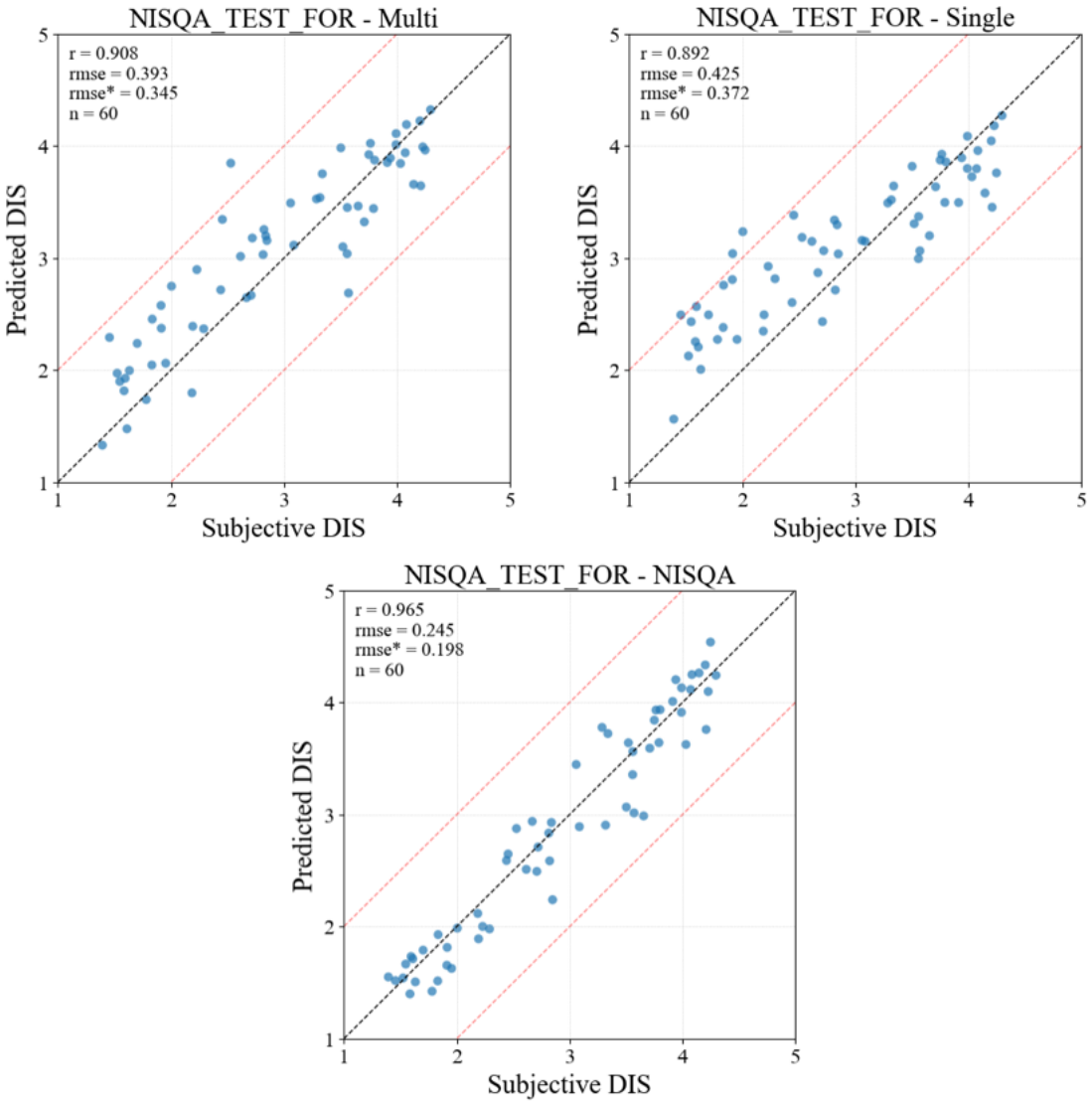


Fig. 6.18: Predicted DIS versus subjective DIS of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_FOR database.

NISQA_TEST_LIVETALK

Figure 6.19 shows scatter plots of predicted versus subjective discontinuity scores for real telephone and VoIP recordings in the NISQA_TEST_LIVETALK database. All models exhibit a slight upward shift, indicating a tendency to overpredict perceived discontinuity under realistic, in-the-wild conditions.

Several challenging scenarios, including subway and café noise recorded on mobile phones under poor reception, are overpredicted beyond the ±1 MOS tolerance band. Among the evaluated models, the multi-task model exhibits the lowest correlation on this database, with $r = 0.779$.

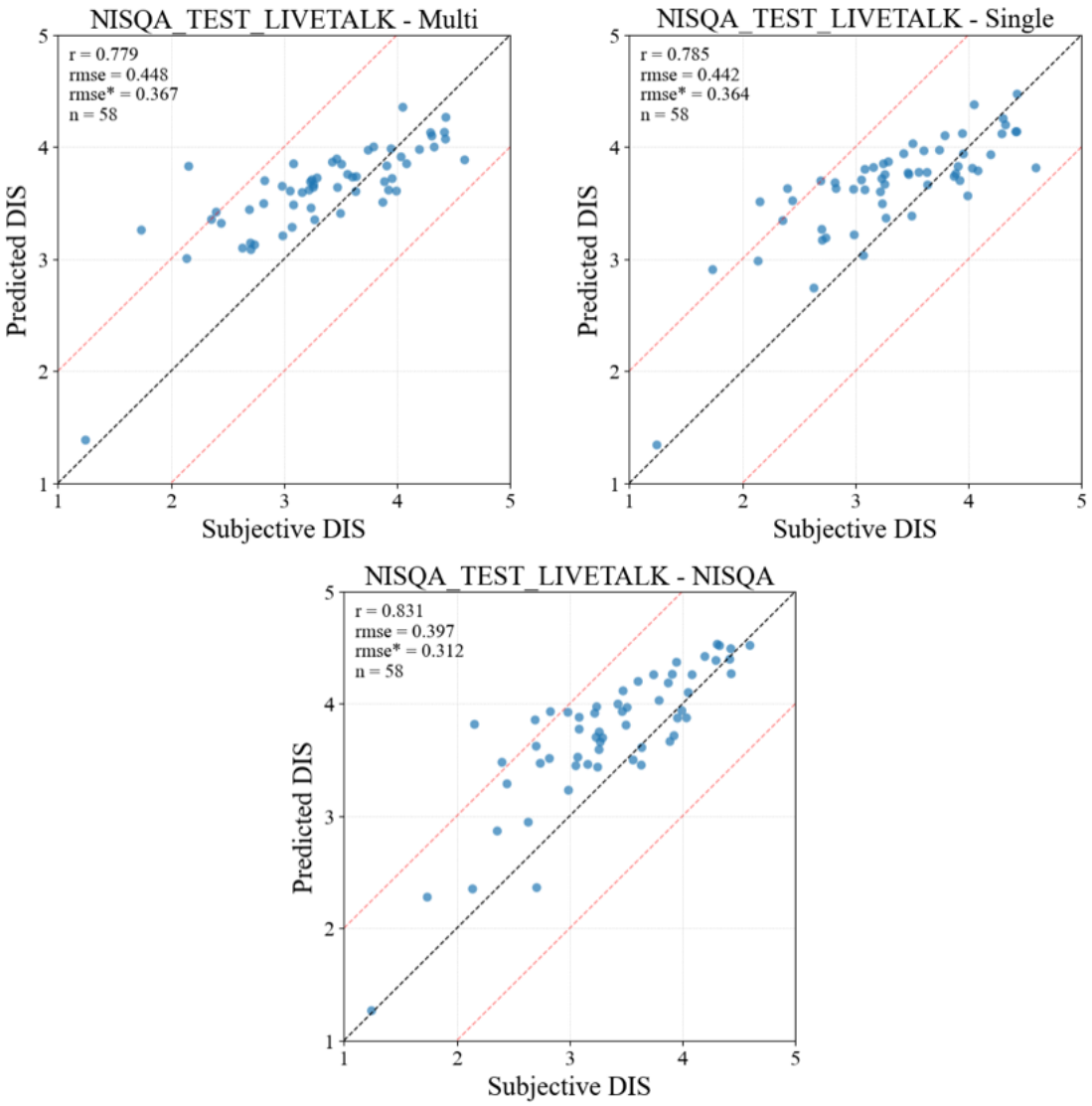


Fig. 6.19: Predicted DIS versus subjective DIS of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_LIVETALK database.

NISQA_TEST_P501

The NISQA_TEST_P501 database includes simulated packet loss and clipping degradations, as well as live communication scenarios such as Skype, Zoom, and WhatsApp. Figure 6.20 shows that all evaluated models achieve high correlation with subjective discontinuity ratings.

The multi-task model overpredicts a severe Opus packet-loss condition beyond the ±1 MOS tolerance band, while the single-target model achieves the highest correlation ($r = 0.912$) but overpredicts one compound AMR-WB/G.722 packet-loss condition. The NISQA baseline similarly overpredicts selected EVS and Zoom conditions.

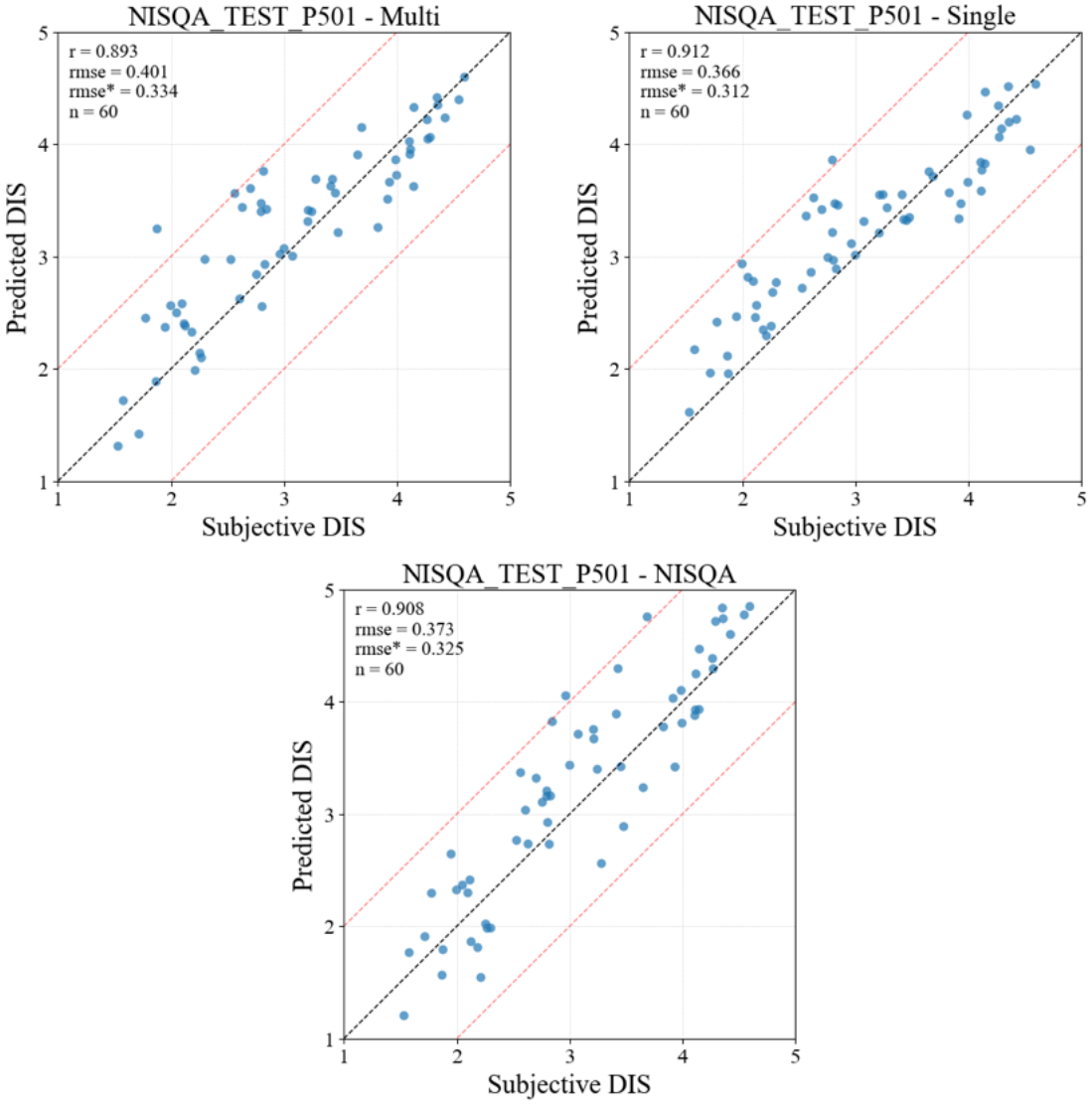


Fig. 6.20: Predicted DIS versus subjective DIS of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_P501 database.

SUB_TUB_2

This database contains handset speech recorded under diverse real-world noise environments and processed using multiple codecs with noise suppression. The single-target AST model achieves the strongest performance on this database, attaining a Pearson correlation of $r = 0.799$, with all 25 conditions falling within the ±1 MOS tolerance band, as shown in Figure 6.21.

The multi-task model attains the second-highest correlation but underpredicts an idle background condition processed with EVS super-wideband coding without noise suppression. In contrast, the NISQA baseline model yields the lowest performance, with a correlation of $r = 0.348$.

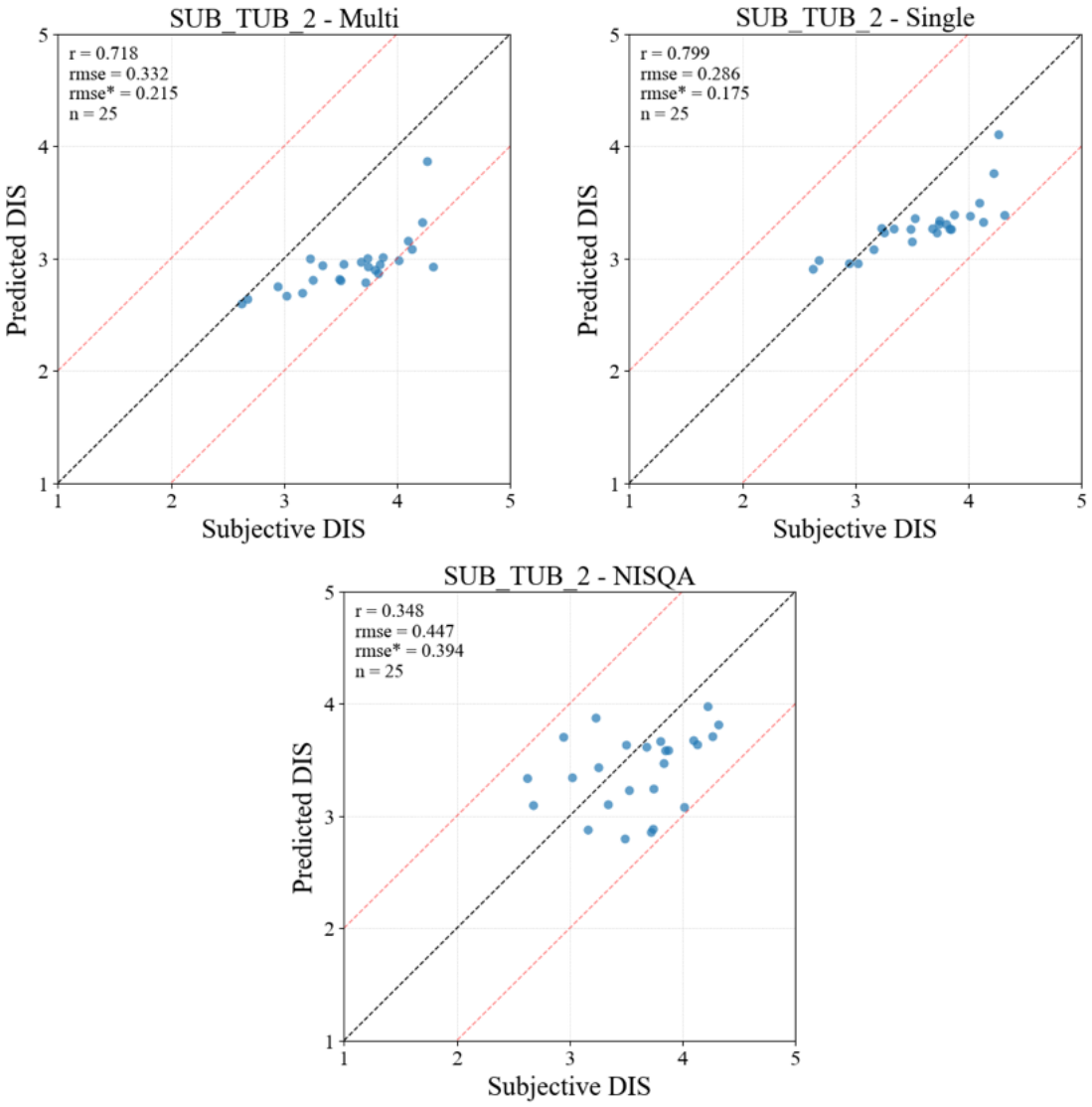


Fig. 6.21: Predicted DIS versus subjective DIS of the Multi-Task AST model, the Single-Target AST model, and the NISQA speech quality model on the SUB_TUB_2 database.

Discussion: Discontinuity Prediction

Across all evaluated databases, the three models consistently capture perceived discontinuity, with positive correlations observed under both simulated and real-world conditions. The NISQA baseline performs strongly on databases dominated by synthetic packet loss and clipping degradations, while the AST-based models demonstrate more robust behavior under heterogeneous and realistic conditions. The single-target model exhibits stable performance and achieves the strongest results on the handset-based SUB_TUB_2 database, whereas the multi-task model shows increased sensitivity to severe packet-loss scenarios. Overall, the results indicate that discontinuity prediction remains highly dependent on degradation type and dataset characteristics, with no single modeling approach uniformly dominating across all conditions.

6.3.5 Coloration Prediction

Perceived coloration is evaluated on the same five held-out test databases using the three models. As summarized in Table 6.11, the NISQA baseline model achieves the highest correlation between predicted and subjective coloration scores on four databases from the NISQA corpus, while the multi-task AST-based model attains the strongest performance on the remaining database.

Table 6.11: Condition-level performance of the three models for coloration (COL) on held-out test databases. Pearson correlation (r) is computed on raw predictions, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping.

Model	Metric	NSC	FOR	LIVE	P501	TUB_2
Multi	r	0.833	0.900	0.789	0.812	**0.618**
	RMSE	0.427	0.310	0.392	0.442	0.303
	RMSE*	0.386	0.275	0.337	0.382	0.254
Single	r	0.875	0.844	0.799	0.802	0.606
	RMSE	0.373	0.381	0.384	0.452	0.307
	RMSE*	0.336	0.335	0.329	0.398	0.263
NISQA	r	**0.932**	**0.941**	**0.875**	**0.913**	0.369
	RMSE	0.280	0.240	0.309	0.310	0.359
	RMSE*	0.249	0.206	0.258	0.246	0.313

NISQA_TEST_NSC

Figure 6.22 shows scatter plots of predicted versus subjective coloration scores for the three evaluated models on the NISQA_TEST_NSC database, which includes a broad range of codec-related degradations. All models exhibit sensitivity to coloration-related impairments.

The NISQA baseline model achieves the highest correlation on this database, with $r = 0.932$. The multi-task model attains the lowest correlation ($r = 0.833$), primarily due to overprediction of amplitude clipping conditions involving Opus and EVS codecs. The single-target model performs comparably, but exhibits similar overprediction for these clipping-related conditions.

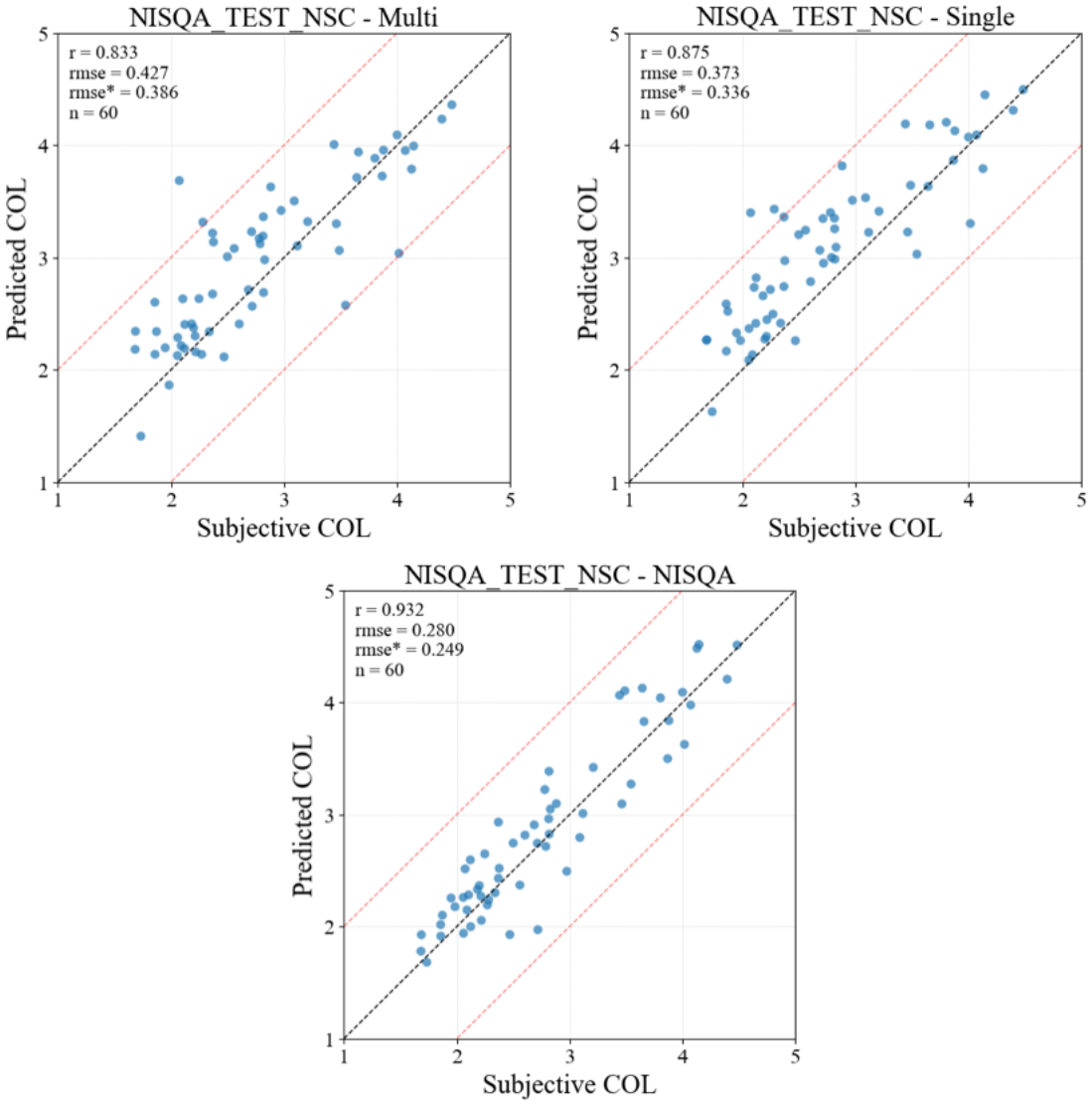


Fig. 6.22: Predicted COL versus subjective COL of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_NSC database.

NISQA_TEST_FOR

Figure 6.23 presents scatter plots of predicted versus subjective coloration scores for the three evaluated models on the NISQA_TEST_FOR database, which contains simulated coloration-related degradations. All models exhibit positive correlation with the subjective ratings.

The NISQA baseline model achieves the highest correlation on this database, with $r = 0.941$. The multi-task AST-based model outperforms the single-target model overall, while the single-target model tends to overpredict coloration for several amplitude clipping and packet-loss conditions involving AMR and Opus codecs.

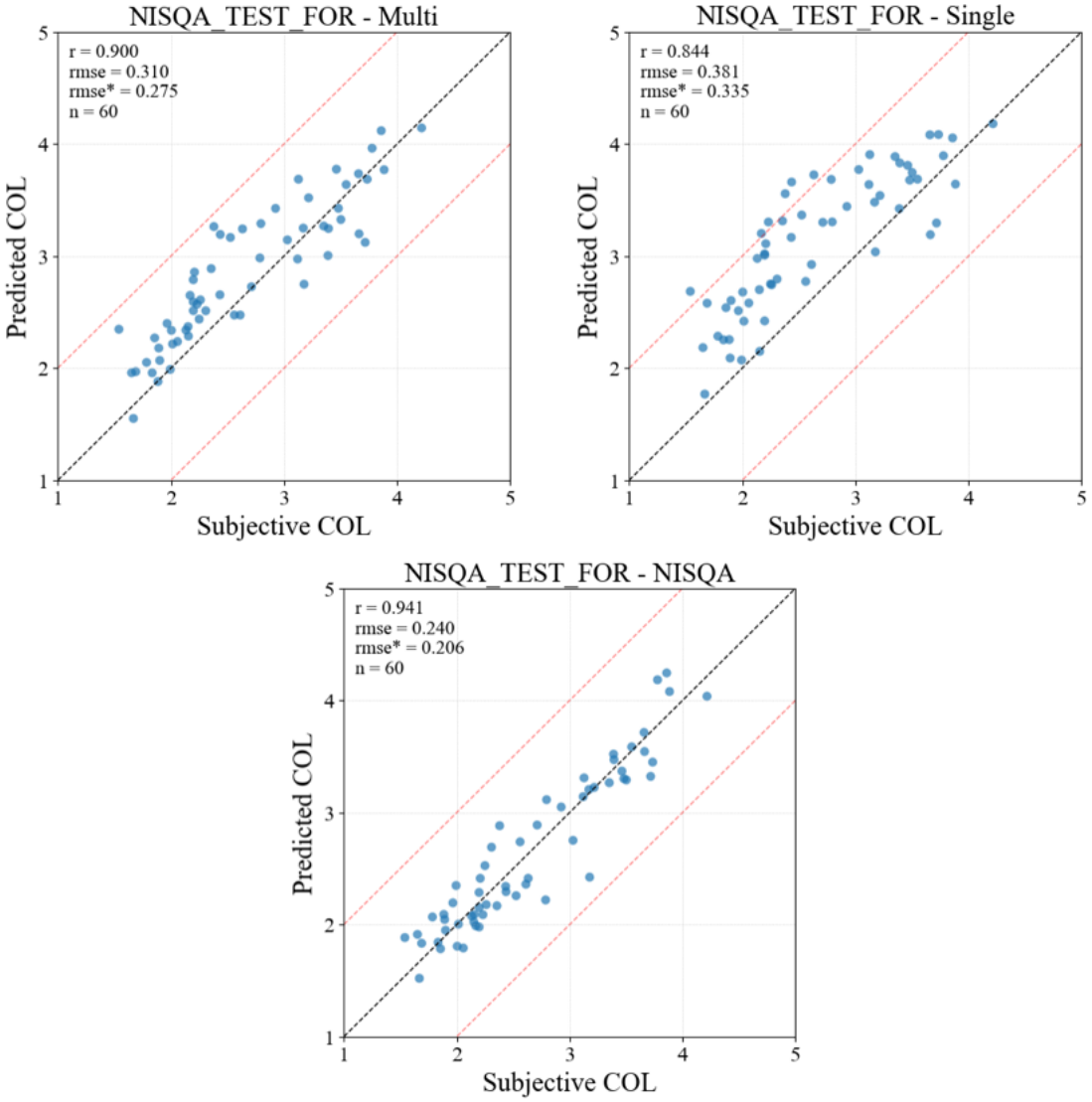


Fig. 6.23: Predicted COL versus subjective COL of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_FOR database.

NISQA_TEST_LIVETALK

This database consists of recordings of real telephone and VoIP calls captured across diverse acoustic environments and thus reflects realistic, in-the-wild communication conditions. The scatter plots in Figure 6.24 show that both AST-based models exhibit positive correlation between predicted and subjective coloration scores. The NISQA baseline model achieves the strongest performance on this database, with a correlation of $r = 0.875$.

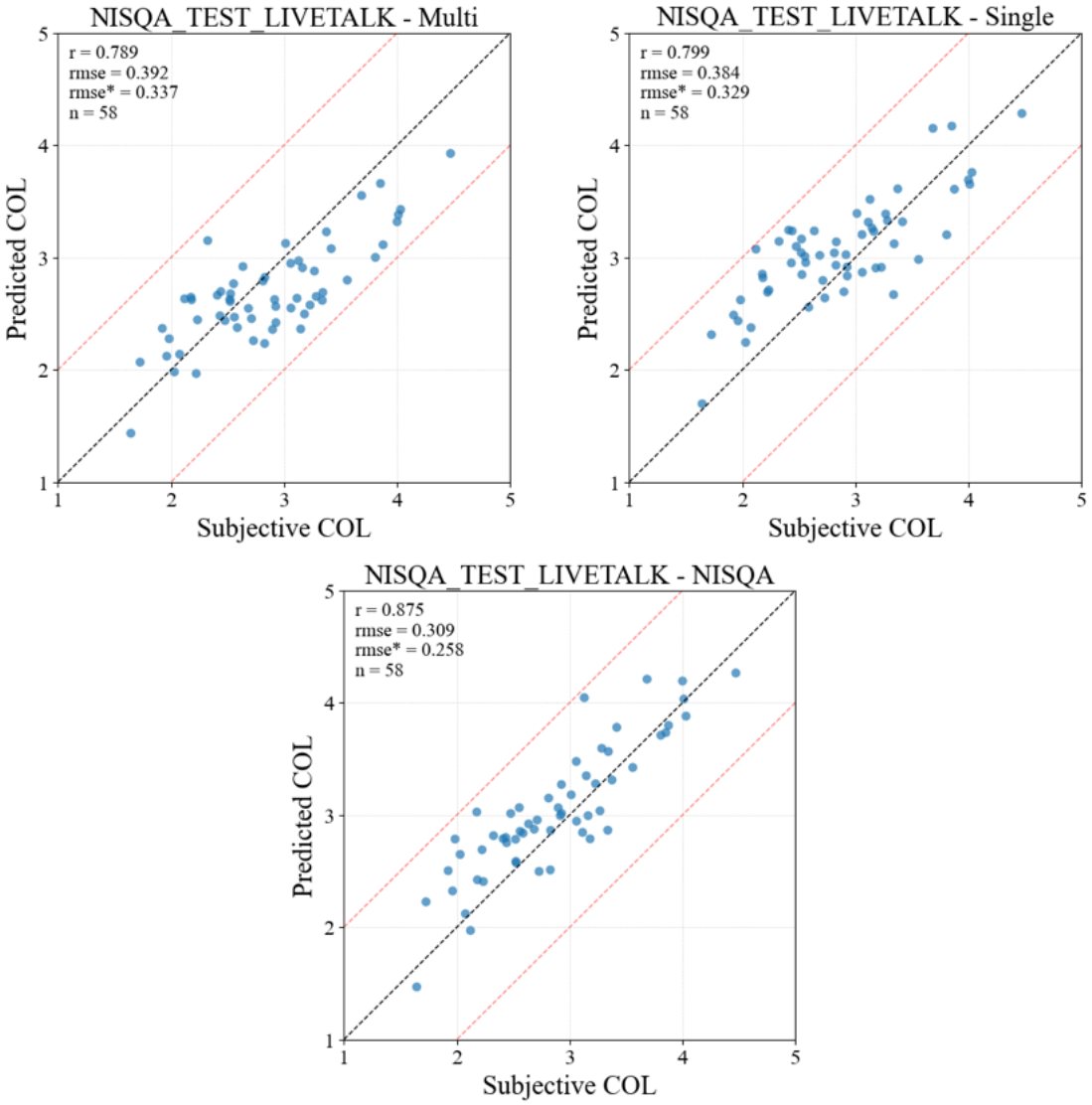


Fig. 6.24: Predicted COL versus subjective COL of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_LIVETALK database.

NISQA_TEST_P501

This database comprises simulated speech degradations involving a variety of codec conditions and additionally includes live communication scenarios such as Skype, Zoom, and WhatsApp, as well as recordings captured over mobile networks. The scatter plots in Figure 6.25 show that all evaluated models achieve high correlation with subjective coloration ratings, with correlation values of at least $r = 0.8$. The NISQA baseline model attains the highest correlation on this database, with $r = 0.913$.

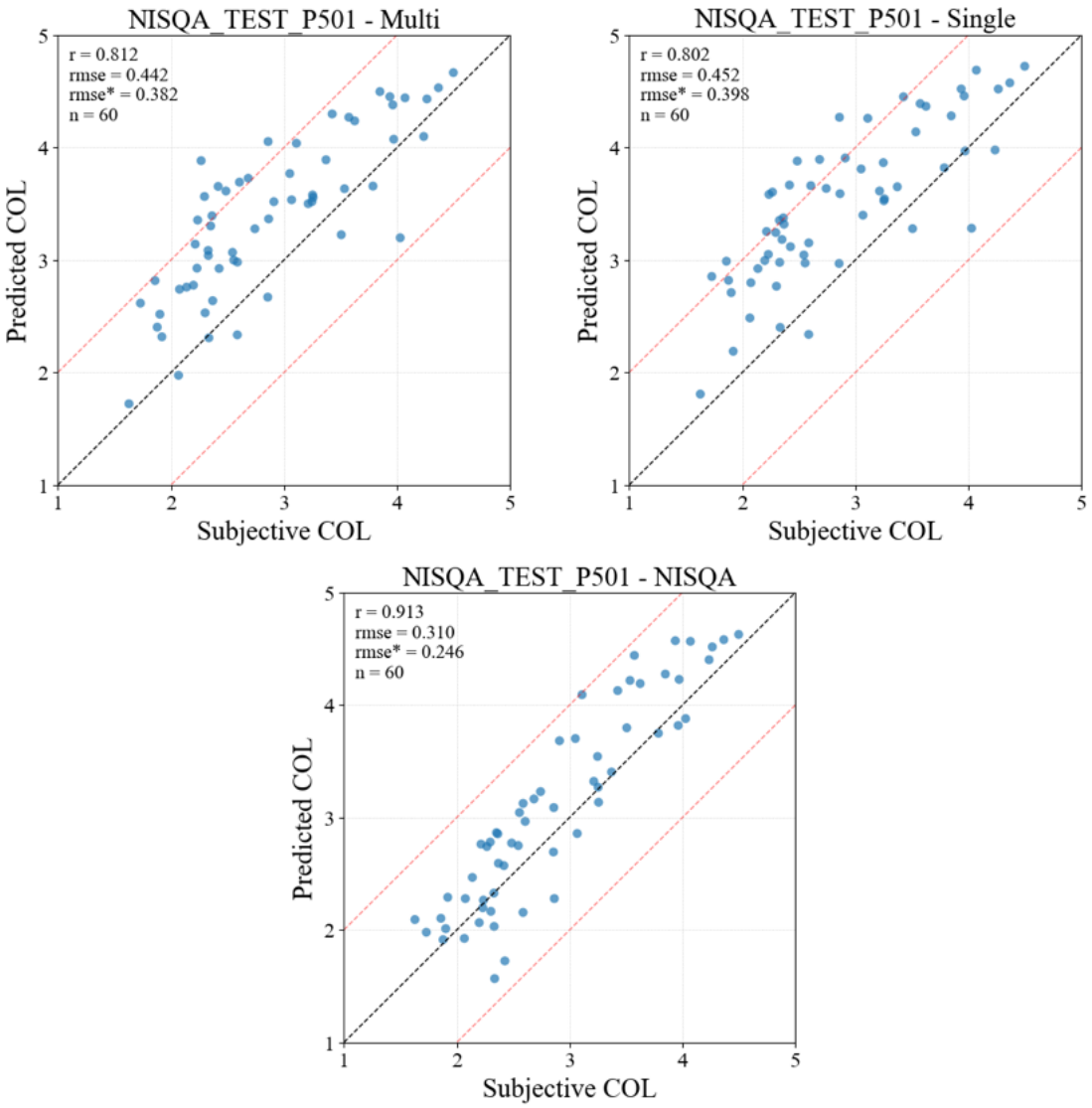


Fig. 6.25: Predicted COL versus subjective COL of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_P501 database.

Both AST-based models exhibit a slight upward shift in their predictions, with larger overpredictions concentrated in conditions involving amplitude clipping, severe packet loss, compound degradations, and degraded live communication scenarios. These results indicate increased difficulty in accurately modeling coloration under artifact-dominated conditions.

SUB_TUB_2

This database contains handset speech recorded under diverse real-world noise environments and processed using EVS super-wideband and AMR-WB codecs with noise suppression, and proves challenging for all evaluated models. The multi-task AST model achieves the strongest performance on this database, with a Pearson correlation of $r = 0.618$. Both AST-based models exhibit an upward shift in their predictions, indicating a tendency to overpredict perceived coloration, as shown in Figure 6.26.

In contrast, the NISQA baseline model yields the lowest performance, with a correlation of $r = 0.037$ as seen in Figure 6.27.

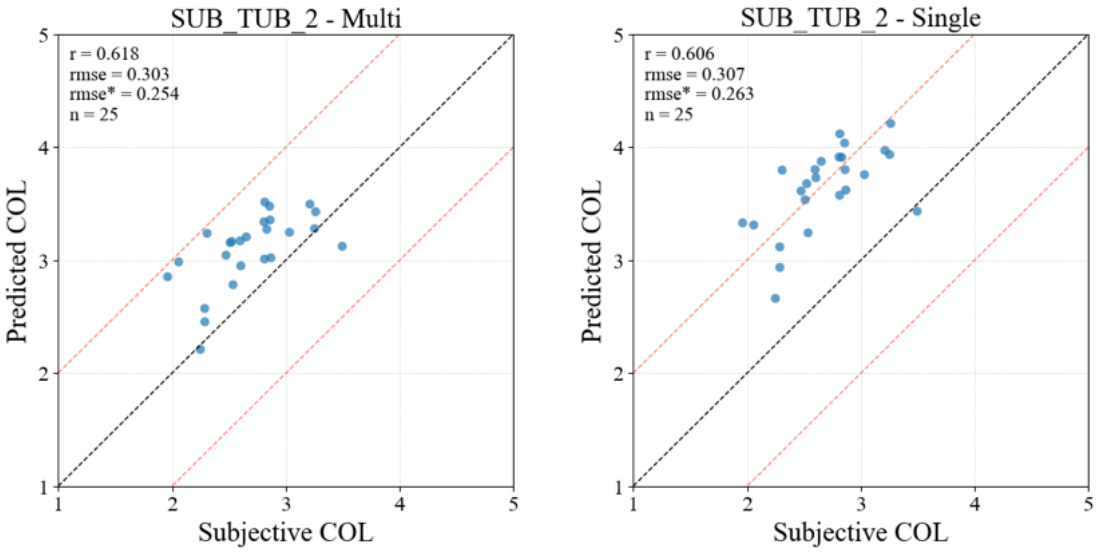


Fig. 6.26: Predicted COL versus subjective COL of the Multi-Task AST model and the Single-Target AST model on the SUB_TUB_2 database.

Discussion: Coloration Prediction

Across the evaluated databases, all models demonstrate the ability to capture perceived coloration, with consistently positive correlations observed under both simulated and real-world conditions. The NISQA baseline model achieves the strongest performance on most databases, particularly those dominated by codec-related and synthetic degradations. The AST-based models show competitive performance but exhibit a systematic tendency to overpredict coloration in artifact-heavy conditions, such as amplitude clipping, severe packet loss, and degraded live communication scenarios. Overall,

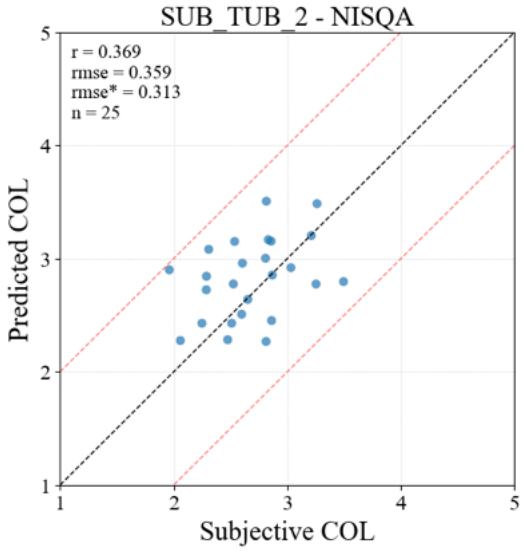


Fig. 6.27: Predicted COL versus subjective COL of the NISQA speech quality model on the SUB_TUB_2 database.

the results indicate that coloration prediction remains sensitive to codec and processing artifacts, and is particularly challenging in realistic handset recordings with noise suppression.

6.3.6 Sub-optimum Loudness Prediction

Perceived loudness is evaluated on the same five held-out test databases using the three models. As summarized in Table 6.12, the NISQA baseline model consistently achieves the highest correlation between predicted and subjective loudness scores across all internal evaluation databases.

Table 6.12: Condition-level performance of the three models for loudness (LOUD) on held-out test databases. Pearson correlation (r) is computed on raw predictions, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping.

Model	Metric	NSC	FOR	LIVE	P501	TUB_2
Multi	r	0.933	0.943	0.608	0.924	-0.455
	RMSE	0.339	0.246	0.403	0.330	0.259
	RMSE*	0.308	0.207	0.311	0.252	0.261
Single	r	0.909	0.893	0.560	0.911	-0.335
	RMSE	0.391	0.332	0.420	0.355	0.274
	RMSE*	0.363	0.300	0.351	0.317	0.261
NISQA	r	**0.964**	**0.961**	**0.709**	**0.955**	**0.685**
	RMSE	0.249	0.204	0.358	0.256	0.212
	RMSE*	0.189	0.170	0.294	0.222	0.142

NISQA_TEST_NSC

Figure 6.28 presents scatter plots of predicted versus subjective loudness scores for the three evaluated models on the NISQA_TEST_NSC database. The 60 degradation conditions include a wide range of level variations and background noise scenarios, resulting in substantial variability in perceived loudness. All models capture the general relationship between predicted and subjective loudness scores.

The NISQA baseline model achieves the highest correlation on this database, with $r = 0.964$. The single-target model overpredicts loudness for the *Mobile-to-Landline* condition with a -10 dB level attenuation.

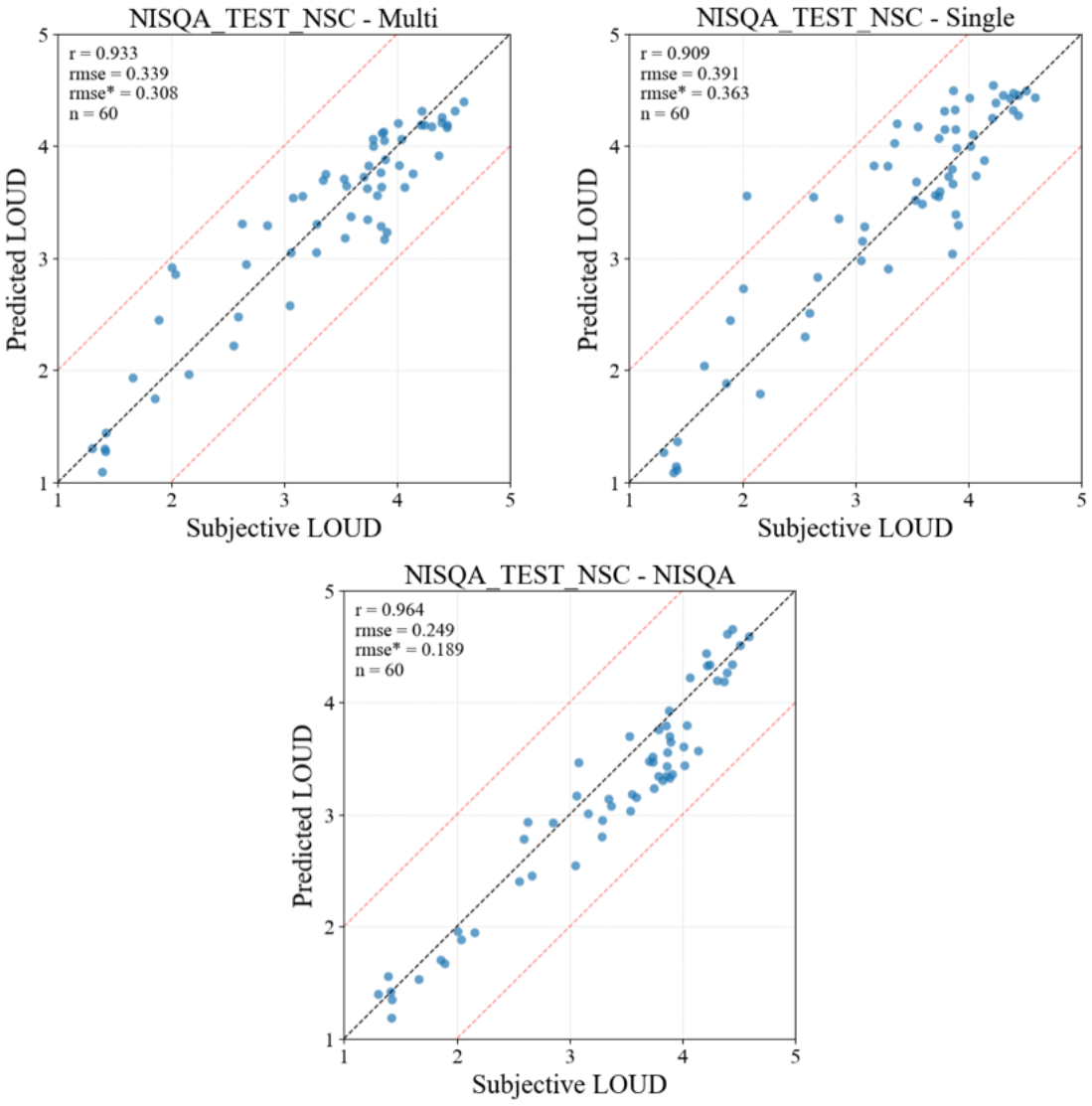


Fig. 6.28: Predicted LOUD versus subjective LOUD of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_NSC database.

NISQA_TEST_FOR

Figure 6.29 presents scatter plots of predicted versus subjective loudness scores for the three evaluated models on the NISQA_TEST_FOR database, which includes a range of simulated degradations. All models exhibit positive correlation with the subjective ratings, indicating reliable loudness prediction across conditions.

The NISQA baseline model achieves the highest correlation on this database, with $r = 0.961$. The multi-task AST-based model outperforms the single-target model overall, while the single-target model tends to overpredict loudness for several band-limited, packet-loss, and low-bitrate communication conditions.

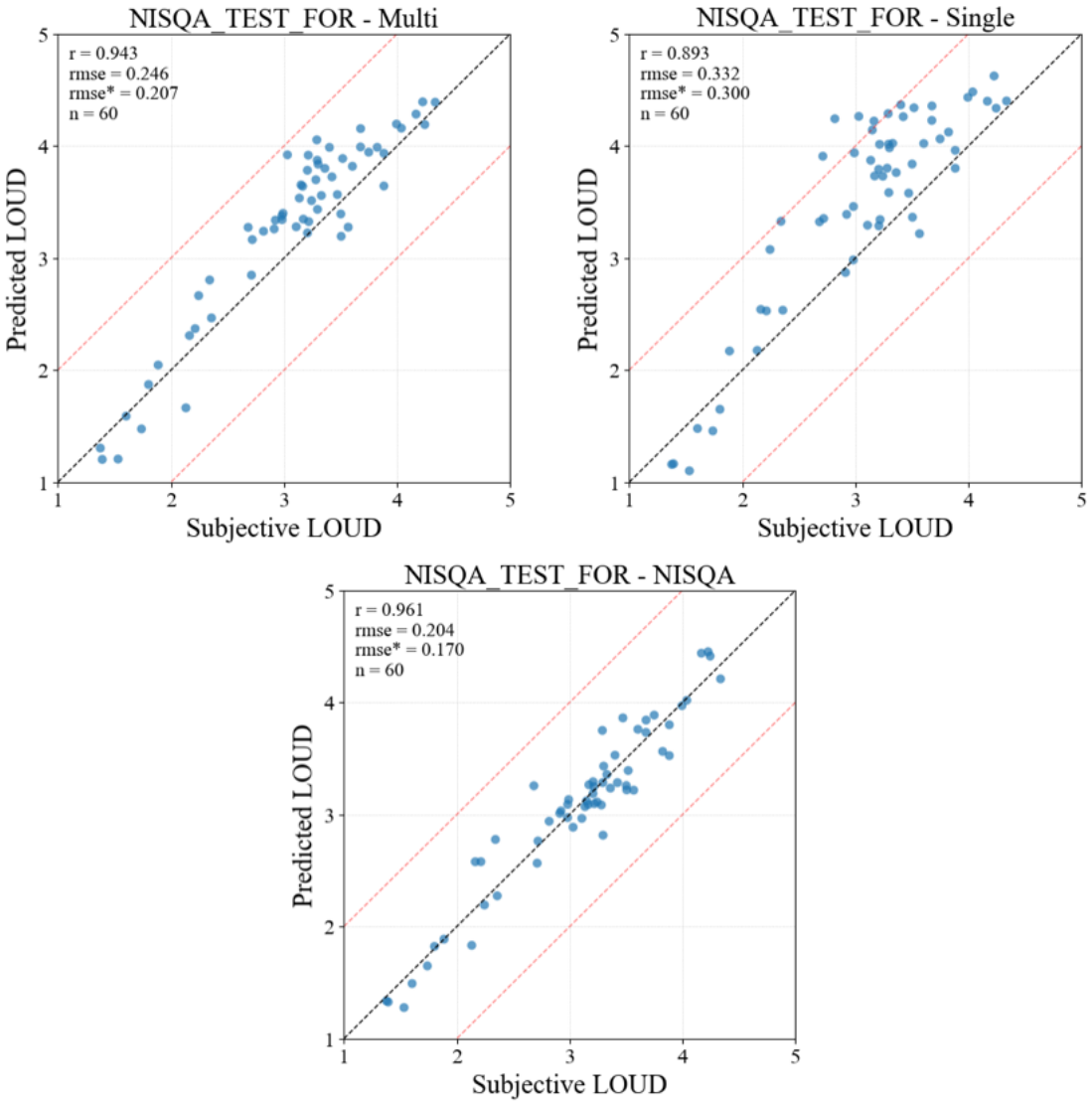


Fig. 6.29: Predicted LOUD versus subjective LOUD of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_FOR database.

NISQA_TEST_LIVETALK

This database consists of recordings of real telephone and VoIP calls captured across diverse acoustic environments and thus reflects realistic, in-the-wild communication conditions. Figure 6.30 shows that the NISQA baseline model achieves the highest correlation on this database, with $r = 0.709$.

The NISQA model nevertheless underpredicts loudness for several challenging conditions involving strong level attenuation, distant talkers, and loudspeaker playback under poor reception. The single-target model additionally underpredicts a band-pass filtered Facebook condition, while the multi-task model shows underprediction for a mobile phone loudspeaker condition.

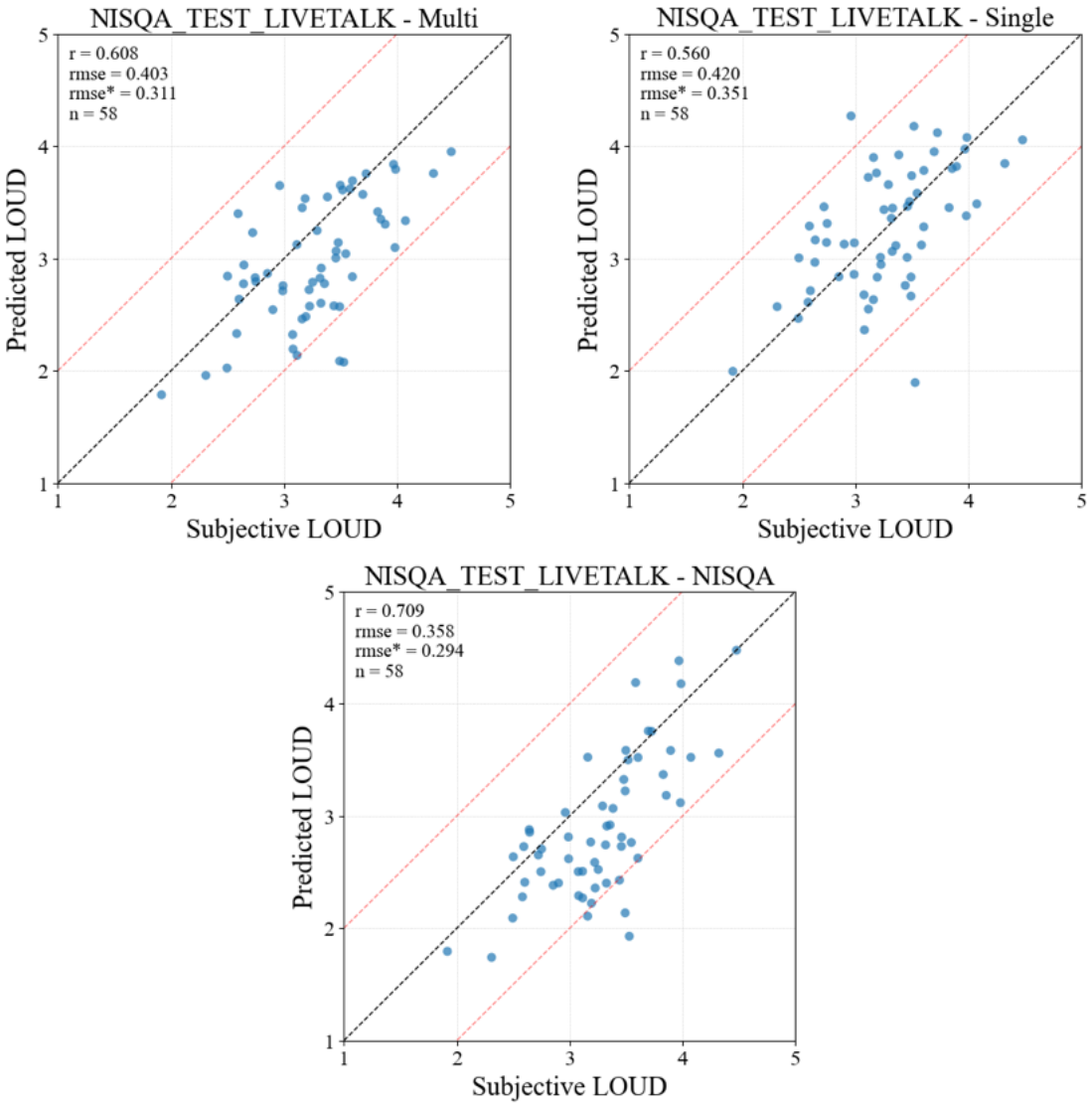


Fig. 6.30: Predicted LOUD versus subjective LOUD of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_LIVETALK database.

NISQA_TEST_P501

This database comprises simulated speech degradations involving different codecs, background noise conditions, packet loss, and clipping artifacts, and additionally includes live communication scenarios such as Skype, Zoom, and WhatsApp, as well as recordings captured over mobile networks. Scatter plots of predicted versus subjective loudness for all evaluated models are shown in Figure 6.31.

All models achieve strong performance on this database, with correlation values of at least $r = 0.911$. The NISQA baseline model attains the highest correlation, with $r = 0.955$. In contrast, both AST-based models exhibit a slight upward shift in their predictions, indicating a tendency to overpredict perceived loudness across conditions.

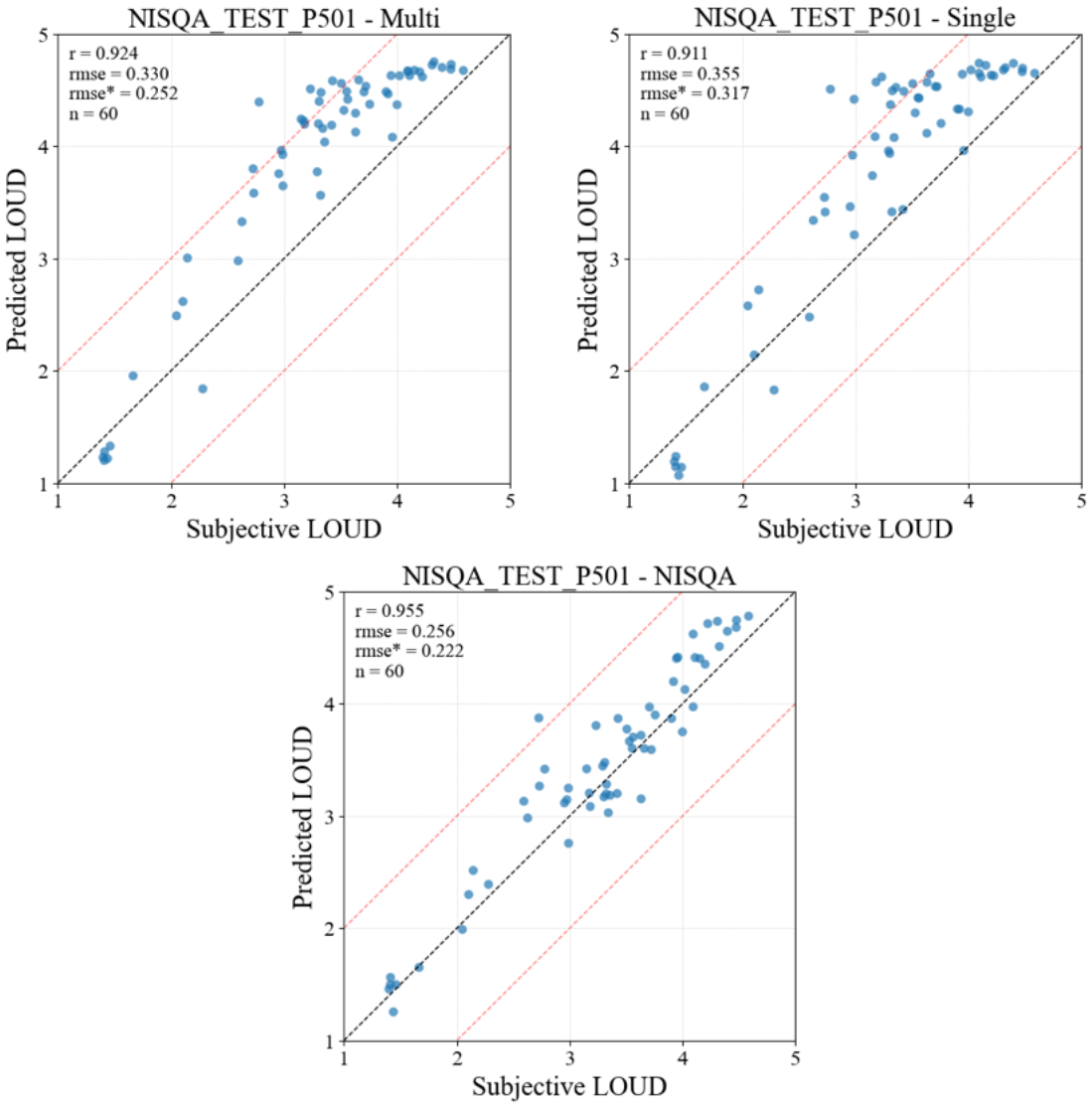


Fig. 6.31: Predicted LOUD versus subjective LOUD of the Multi-Task AST model, the Single-Target AST model set and the NISQA speech quality model on the NISQA_TEST_P501 database.

SUB_TUB_2

This database contains handset speech recorded under diverse real-world noise environments and processed using EVS super-wideband and AMR-WB codecs with noise suppression. The NISQA baseline model achieves the strongest performance on this database, with a Pearson correlation of $r = 0.685$. However, only three of the 25 conditions fall within the ±1 MOS tolerance band, reflecting a systematic tendency to underpredict perceived loudness.

As shown in Figure 6.32, the database exhibits a relatively limited range of loudness variation. Under these conditions, neither of the AST-based models achieves meaningful correlation with the subjective ratings.

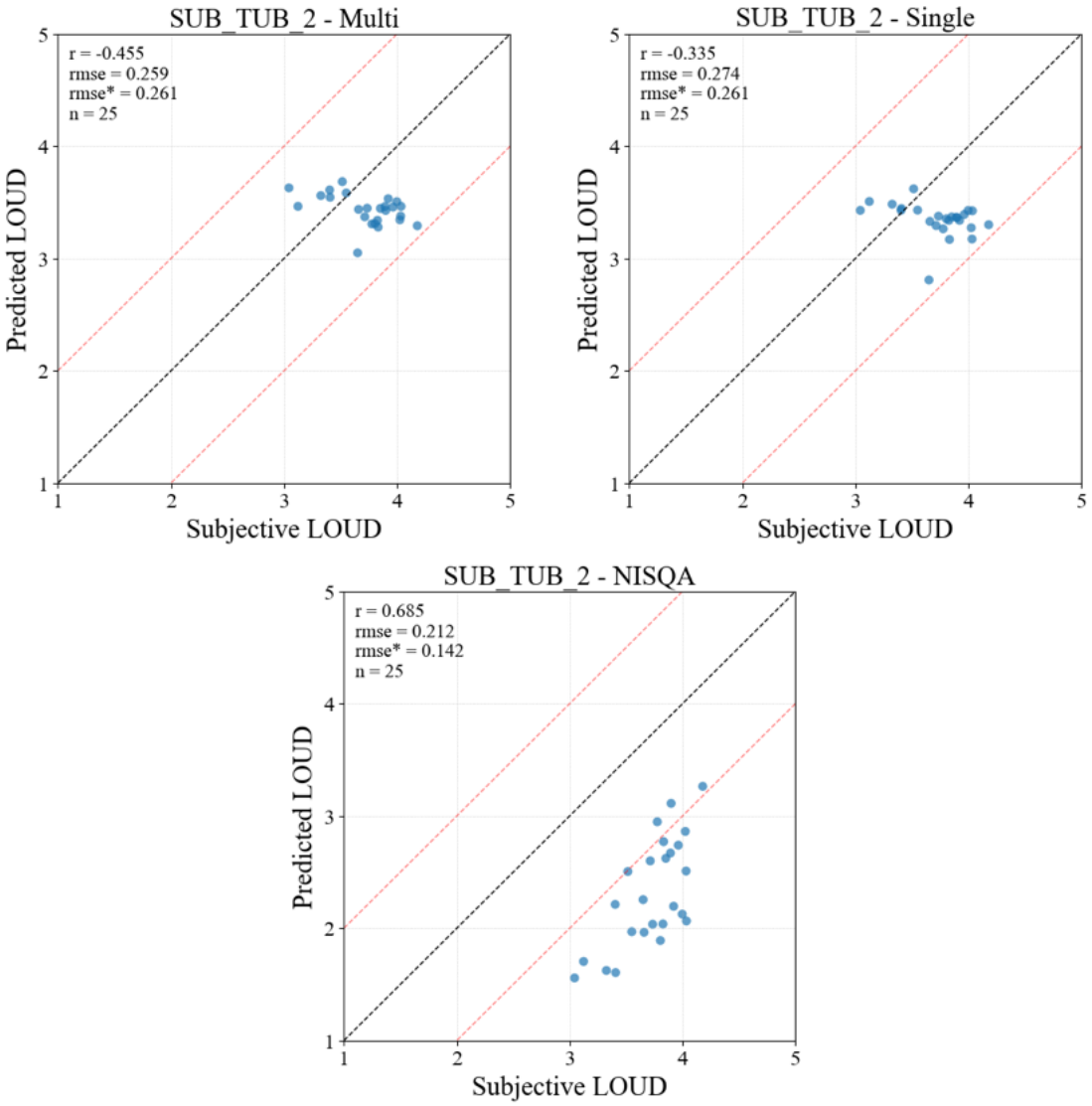


Fig. 6.32: Predicted LOUD versus subjective LOUD of the Multi-Task AST model, the Single-Target AST model, and the NISQA speech quality model on the SUB_TUB_2 database.

Discussion: Loudness Prediction

Across the evaluated databases, perceived loudness is generally well captured by all models when sufficient variability in signal level is present. The NISQA baseline model consistently achieves the highest correlation across datasets, particularly for simulated degradations and mixed communication scenarios. The AST-based models show competitive performance on several databases but exhibit systematic calibration effects, with a tendency to overpredict loudness in simulated conditions and underpredict loudness in realistic, in-the-wild scenarios. Performance degrades notably on databases with a limited dynamic range of loudness, where reliable correlation becomes challenging for all models.

The results reveal a mixed and non-uniform performance across perceptual dimensions and evaluation databases. While the foundation-based models do not consistently outperform the NISQA baseline and therefore do not constitute an immediate replacement or clear new state of the art, they demonstrate notably strong performance on the additional test databases introduced through the ConferencingSpeech Interspeech Challenge. These databases include more heterogeneous and realistic conditions that are representative of contemporary telecommunication scenarios.

This observed robustness under increased condition diversity highlights the potential of foundation-based models as promising candidates for real-world telecommunication applications. At the same time, the results indicate that further investigation, architectural refinement, and targeted optimization are required to fully realize this potential.

6.4 Independent Evaluation and Standardization Context

The single-target dimension-specific model set has undergone external evaluation in the context of the ITU-T P.SAMD work item. For readability, the model is referred to as the *P.SAMD* candidate model for the rest of this chapter. The P.SAMD candidate model is evaluated independently by ITU-T partners on 11 additional heterogeneous datasets using standardized evaluation procedures prescribed in the ITU-T Recommendation P.1401 (2020). These external evaluation results are reported separately and are not used to further modify or tune the selected models.

According to ITU-T requirements, the overall performance for unknown databases is evaluated based on the average RMSE* across all experiments and dimensions, which should ideally remain below 0.35. Additionally, the worst-case performance stipulates that no single RMSE* per experiment and dimension, including overall quality, should exceed 0.5.

The per-condition results reported above demonstrate that the selected *P.SAMD* candidate model meets the performance requirements specified by ITU-T for evaluation on unknown databases. Across the considered datasets and perceptual dimensions, the observed RMSE* values mostly remain within the prescribed thresholds, with no major violations of the worst-case criterion. Although an RMSE* of 0.541 is observed for discontinuity prediction on the RS_PAMD_SetA_SWB_SQ_2 database, this behavior is consistent with the known challenges posed by the database and does not contradict the broader performance trends observed across the evaluation corpus. While perfor-

mance varies across dimensions and database characteristics, the results indicate stable behavior under heterogeneous and previously unseen conditions. Based on these results, the proposed *P.SAMD* candidate model was approved by ITU-T Study Group 12 for standardization, and the corresponding Recommendation text is currently under preparation.

Table 6.13: Independent ITU(-T) partner evaluation results for overall speech quality (MOS) using per-condition metrics. Reported are correlation coefficient, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping (ITU-T Rec. P.1401).

Database	**r**	**RMSE**	**RMSE***	**Information**
RS_P800_2018_FBliveConditions	0.838	0.5469	0.4308	C286 (2018)
RS_P800_2016_FBcodecs	0.915	0.4256	0.3205	C281 (2016)
RS_P800_2015_FBevs	0.944	0.3276	0.2164	C022 (2016)
RS_P800_2025_AI_ULBR_codecs	0.947	0.3662	0.2518	C040 (2025)
RS_P800_2022_FBbandwidth	0.959	0.3178	0.2252	C073 (2023)
ETSI_DB01_MAN_Validation	0.947	0.3650	0.2250	C055 (2025)
ETSI_DB02_GER_Validation	0.942	0.3480	0.2210	C055 (2025)
ETSI_DB03_ENG_Validation	0.944	0.2920	0.2000	C055 (2025)
PSAMD_1	0.928	0.0850	0.0840	Orange (2024)
Neural coders	0.908	0.2510	0.2280	Orange (2023)
EVS-Opus	0.977	0.0800	0.0740	Orange (2022)
PAMD_2	0.742	0.3610	0.3590	Orange (2022)
Average			**0.2363**	

Table 6.14: Independent ITU(-T) partner evaluation results for *Noisiness* using per-condition metrics. Reported are correlation coefficient, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping (ITU-T Rec. P.1401).

Database	**r**	**RMSE**	**RMSE***	**Information**
RS_PAMD_2025_AI_ULBR_codecs	0.936	0.3916	0.2184	C042 (2025)
RS_PAMD_SetA_SWB_SQ_2	0.900	0.5675	0.4208	C573 (2021)
PSAMD_1	0.867	0.1580	0.1540	Orange (2024)
PAMD_2	0.931	0.1910	0.1840	Orange (2022)
Average			**0.2443**	

Table 6.15: Independent ITU(-T) partner evaluation results for *Discontinuity* using per-condition metrics. Reported are correlation coefficient, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping (ITU-T Rec. P.1401).

Database	**r**	**RMSE**	**RMSE***	**Information**
RS_PAMD_2025_AI_ULBR_codecs	0.859	0.5024	0.3829	C042 (2025)
RS_PAMD_SetA_SWB_SQ_2	0.726	0.7016	0.5410	C573 (2021)
PSAMD_1	0.893	0.1750	0.1710	Orange (2024)
PAMD_2	0.901	0.1340	0.1300	Orange (2022)
Average			**0.3062**	

Table 6.16: Independent ITU(-T) partner evaluation results for *Coloration* using per-condition metrics. Reported are correlation coefficient, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping (ITU-T Rec. P.1401).

Database	**r**	**RMSE**	**RMSE***	**Information**
RS_PAMD_2025_AI_ULBR_codecs	0.911	0.2842	0.1898	C042 (2025)
RS_PAMD_SetA_SWB_SQ_2	0.885	0.4248	0.2909	C573 (2021)
PSAMD_1	0.907	0.0830	0.0830	Orange (2024)
PAMD_2	0.761	0.2150	0.2140	Orange (2022)
Average			**0.1944**	

Table 6.17: Independent ITU(-T) partner evaluation results for *Loudness* using per-condition metrics. Reported are correlation coefficient, RMSE after first-order mapping, and RMSE* after third-order monotonic mapping (ITU-T Rec. P.1401).

Database	**r**	**RMSE**	**RMSE***	**Information**
RS_PAMD_2025_AI_ULBR_codecs	0.609	0.3297	0.2331	C042 (2025)
RS_PAMD_SetA_SWB_SQ_2	0.330	0.3445	0.1987	C573 (2021)
PSAMD_1	0.653	0.1970	0.1960	Orange (2024)
PAMD_2	0.709	0.5030	0.5010	Orange (2022)
Average			**0.2822**	

6.5 Summary

This chapter examined single-task learning as an alternative modeling strategy for non-intrusive speech quality assessment based on transformer-based pretrained representations. Using a fixed AST backbone, separate models were trained for overall quality and each perceptual dimension, enabling task-specific optimization without cross-task interference.

A set of controlled training regimes was evaluated to assess the impact of data composition, retraining, and fine-tuning on predictive performance. Internal experiments on unseen test databases showed that single-task AST models achieve strong and stable performance for overall quality, noisiness, and discontinuity across diverse conditions.

In contrast, coloration and loudness exhibited greater sensitivity to dataset characteristics and training configuration, indicating dimension-dependent limits to task-specific optimization. These findings complement the multi-task results presented earlier and clarify the trade-offs between joint and task-specific modeling.

With respect to RQ3, the results show that single-task optimization yields measurable advantages over joint multi-task learning for some perceptual dimensions, on some databases. Most notably overall quality prediction highly benefits from a single-task approach. These empirical gains motivated the adoption of the single-task formulation for standardization, while also indicating that the benefits of task-specific optimization are not uniform across all perceptual attributes or datasets.

Based on the internal evaluation, a single model variant was selected per perceptual dimension for external assessment in the context of the ITU-T P.SAMD work item. Independent evaluation by ITU-T Study Group 12 partners on heterogeneous and previously unseen datasets confirmed that the selected models satisfy the prescribed performance criteria for unknown conditions. As a result, the proposed model was approved for standardization, and the corresponding Recommendation text is under preparation at the time of writing.

With respect to RQ4, the results demonstrate that foundation model-based speech quality predictors generalize effectively across heterogeneous datasets, recording conditions, and domains. In particular, the independent ITU-T evaluations on previously unseen and hidden datasets confirm stable cross-database behavior under realistic telecommunication scenarios, supporting the robustness of the proposed approach beyond the conditions represented in the simulated training data.

Having established the applicability of these models under realistic telecommunication conditions, the next chapter extends the evaluation to out-of-domain scenarios by examining model behavior on synthetic speech.

Chapter 7
Out-of-Domain Transfer to Synthetic Speech

This chapter investigates the transferability of the proposed foundation model-based speech quality prediction framework to out-of-domain assessment tasks. In contrast to the preceding chapters, which focused on real-world telecommunication speech, the present analysis considers synthetic speech generated by text-to-speech and voice conversion systems, whose acoustic and perceptual characteristics differ substantially from those observed during training.

The chapter explores whether models developed for non-intrusive speech quality assessment can be applied to synthetic speech without extensive retraining or task-specific engineering. Two representative out-of-domain scenarios are examined: text-to-speech naturalness assessment and the quality prediction of synthetic speech in voice conversion settings. Model performance is analyzed under controlled experimental conditions to assess robustness, generalization, and limitations of transfer.

Through this analysis, the chapter addresses RQ5 by examining the extent to which foundation model-based speech quality predictors can be effectively reused beyond their original application domain, thereby probing the generality of the learned representations.

7.1 Introduction

Recent advances in text-to-speech (TTS), voice conversion (VC), neural speech coding, AI-based noise suppression, and speaker anonymization have led to a rapid increase in the use of synthetic speech within modern communication systems. As synthetic and human speech increasingly coexist in practical applications, there is a growing demand for objective evaluation methods capable of estimating perceptual speech quality without reliance on extensive and costly subjective listening tests. In the context of TTS research, perceived quality is commonly referred to as *naturalness* (Nusbaum et al., 1997). In this work, however, speech quality is consistently assessed in terms of the mean opinion score (MOS), reflecting the annotation conventions of the datasets considered and enabling direct comparison with natural-speech quality assessment frameworks.

In parallel, the emergence of large-scale foundation models has fundamentally reshaped the field of speech processing. Pretrained on vast and heterogeneous audio corpora, these models learn general-purpose acoustic representations that have demonstrated strong transferability across a wide range of downstream tasks. For speech qual-

ity assessment, such representations offer the potential to capture perceptually relevant characteristics that generalize beyond the specific conditions observed during training, thereby reducing the need for task-specific feature engineering or handcrafted signal descriptors.

Despite these developments, most existing objective speech quality models have been designed and evaluated primarily for natural speech degradations, such as those introduced by coding, transmission impairments, or additive noise. Synthetic speech, by contrast, presents distinct perceptual challenges, including vocoder artifacts, atypical prosodic patterns, spectral over-smoothing, and increased variability in listener judgments. These differences motivate the central research question addressed in this chapter:

Can a foundation model pretrained for non-intrusive quality assessment of natural speech be effectively transferred to the perceptual evaluation of synthetic speech?

While substantial research has been devoted to objective quality prediction for natural speech, comparatively fewer studies have examined synthetic speech scenarios. Existing approaches to TTS naturalness prediction, including NISQA-TTS (Mittag and Möller, 2020), MOSNet-based extensions (Leng et al., 2021), and wav2vec 2.0 fine-tuning strategies (Ragano et al., 2023; Cooper et al., 2022), have demonstrated promising performance within specific datasets. However, these methods often exhibit limited robustness across different synthesis techniques, speakers, and evaluation conditions.

Building on this body of work, the present chapter investigates the transferability of the proposed single-target dimension-specific speech quality model from the preceding chapter to synthetic speech evaluation. Specifically, it examines whether a model trained for non-intrusive quality assessment of natural speech can be applied, with minimal adaptation, to the prediction of TTS naturalness and to quality assessment in voice conversion scenarios. The following sections describe the adaptation strategy, experimental setup, and evaluation methodology used to address this question.

7.2 Synthetic Speech Datasets

The data used for fine-tuning and evaluating the proposed models toward synthetic speech quality prediction comprise several publicly available corpora covering both TTS and VC scenarios. These datasets span a wide range of synthesis systems, languages, evaluation protocols, and listening conditions, and are commonly used benchmarks in the literature on synthetic speech naturalness prediction.

Blizzard Challenge Datasets

A comprehensive collection of Blizzard Challenge datasets from 2008 (Karaiskos et al., 2008) to 2019 (Wu et al., 2019) was included, excluding the 2017 and 2018 editions due to the lack of publicly available subjective ratings. Depending on the challenge year, the datasets comprise between 8 and 24 synthesis systems. For most years, per-file subjective ratings are available, with an average of approximately 13

listener votes per utterance. For the 2014–2016 challenges, only system-level mean opinion scores (MOS) are provided; in these cases, all available stimuli were assigned their corresponding system-level MOS.

Several Blizzard Challenge editions include multiple tasks (e.g., hub and spoke conditions) and different speaker configurations. In the context of the Blizzard and Voice Conversion Challenges, hub tasks evaluate systems trained on a common, fixed dataset, while spoke tasks assess systems trained on participant-specific data, thereby measuring generalization across unseen speakers or conditions. For consistency with prior work and to ensure a fine-grained evaluation, different tasks and speaker conditions within each challenge year were treated as separate datasets.
The Blizzard datasets were split into training and validation sets as follows:

- **Training:** Blizzard 2008 A/B, Blizzard 2009 EH1/EH2/ES1–ES2, Blizzard 2010 EH1/ES1–ES3/MH1/MH2/MS1, Blizzard 2011 EH1, Blizzard 2013 EH1/EH2, Blizzard 2014 IH (system-level), Blizzard 2015 IH (system-level), Blizzard 2019 MH.
- **Validation:** Blizzard 2008 C, Blizzard 2009 MH/MS1–MS2, Blizzard 2010 EH2, Blizzard 2012 EH1, Blizzard 2016 EH (system-level).

Voice Conversion Challenge Datasets

In addition to TTS data, datasets from the Voice Conversion Challenge (VCC) were incorporated to cover voice conversion scenarios, which exhibit perceptual artifacts similar to those found in synthetic speech. The VCC 2018 dataset includes both hub and spoke tasks and provides per-file subjective ratings with an average of approximately four listener votes per sample. To avoid training imbalance due to the large size of this dataset, balanced subsets were constructed for training and validation:

- **Training:** VCC 2018 hub (train), VCC 2018 spoke (train)
- **Validation:** VCC 2018 hub (validation), VCC 2018 spoke (validation)

The earlier VCC 2016 dataset does not provide per-file subjective ratings and was therefore excluded from training and validation. Instead, it was reserved exclusively as an unseen test set to assess the generalization capability of the proposed models across speakers, systems, and evaluation conditions not observed during training.

TU Berlin and Kiel University

Additionally, three listening test datasets collected jointly by TU Berlin and Kiel University were used (Mittag and Möller, 2020). Test 1 consists of 60 speech samples generated by 15 German TTS systems and rated by 30 listeners. Test 2 contains 57 samples from 19 systems, each evaluated by 12 listeners. Test 3 comprises 60 samples generated by six systems, processed to telephone bandwidth (300–3400 Hz), and rated by 17 listeners. These datasets provide complementary coverage of system diversity, language, and transmission conditions.

PhySyQX

Finally, the PhySyQX (Gupta et al., 2015) corpus was included, comprising 36 publicly available speech samples produced by two natural voices and seven commercial TTS systems. Each sample was rated for naturalness by 12 listeners.

To ensure comparability with existing work on synthetic speech quality assessment, the dataset configuration was aligned with that used in NISQA-TTS (Mittag and Möller, 2020). Under this configuration, the resulting fine-tuning dataset comprises 24,144 training samples and 5,132 validation samples.

Evaluation set

Model performance was evaluated exclusively on four test datasets that were fully excluded from training and validation. These comprise Test 2 and Test 3 from the TU Berlin / Kiel University collection (German TTS), the Voice Conversion Challenge 2016 dataset (English voice conversion), and the PhySyQX corpus (English TTS). Together, these evaluation sets cover both text-to-speech and voice conversion scenarios across German and English, and provide a diverse and strictly held-out benchmark for assessing out-of-domain generalization performance.

7.3 Method

This section describes the methodology employed to evaluate and adapt the proposed speech quality prediction framework for out-of-domain synthetic speech assessment. The section first outlines the zero-shot evaluation procedure used to assess direct transferability and to select an appropriate base model. It then describes the fine-tuning strategies investigated for adapting the model to synthetic speech. For the sake of readability, the single-target, dimension-specific speech quality model introduced in the preceding chapter is referred to as *SQ-AST* throughout this chapter.

7.3.1 Zero-Shot Evaluation

A zero-shot evaluation is first conducted to assess the transferability of the dimension-specific SQ-AST models to synthetic speech without any task-specific adaptation. This evaluation serves to identify the most suitable perceptual dimension model for subsequent fine-tuning. Based on the zero-shot performance, a single model is selected and adapted using the fine-tuning strategies described in the following subsection.

7.3.2 Fine-Tuning Strategies

Three fine-tuning strategies are investigated, differing in the extent to which the pretrained SQ-AST parameters are updated during adaptation to synthetic speech.

7.3.2.1 Frozen Backbone

In the frozen-backbone configuration, the AST encoder is kept entirely fixed and only the final prediction head is trained on the synthetic speech data. Let $h \in \mathbb{R}^{768}$ denote the utterance-level embedding produced by the frozen AST backbone for a given input signal. Quality prediction is then obtained via a regression mapping

$$\hat{y} = Wh + b, \tag{7.1}$$

where $W \in \mathbb{R}^{1\times768}$ and $b \in \mathbb{R}$ denote the trainable parameters of the prediction head. This setting represents a minimal adaptation scenario and is referred to as *Min FT SQ-AST* in the results tables.

7.3.2.2 Full Fine-Tuning

In the full fine-tuning configuration, all parameters of the SQ-AST model are updated during training, including both the AST backbone and the prediction head. This approach allows the model to fully adapt its internal representations to the acoustic and perceptual characteristics of synthetic speech, at the cost of increased risk of overfitting given the limited size of the available synthetic speech datasets.

7.3.2.3 Partial Fine-Tuning

The partial fine-tuning strategy employs a two-stage adaptation procedure. In an initial warm-up phase, only the prediction head is optimized while the AST backbone remains frozen. After the first training epoch, the last two transformer encoder blocks (encoder.layer.10 and encoder.layer.11) and the pooling layer are unfrozen, while all earlier layers remain fixed.

Training then continues using a two-group AdamW optimizer, with a higher learning rate assigned to the prediction head and a reduced learning rate applied to the unfrozen backbone layers. This strategy aims to balance representation stability with limited task-specific adaptation.

7.4 Results and Discussion

Table 7.1 reports file-level Pearson correlation coefficients (PCC) obtained in zero-shot transfer to synthetic speech datasets. Among the evaluated SQ-AST dimension modules, the overall MOS model exhibits the most consistent performance across the four test databases, achieving an average PCC of 0.626. In contrast, the perceptual dimension models show substantially greater variability across datasets, with lower and less stable average correlations.

Based on these results, the overall MOS model is selected for subsequent fine-tuning experiments, as it provides the most reliable baseline for transfer to synthetic speech without task-specific adaptation.

Table 7.1: File-level Pearson correlation coefficients (PCC) of SQ-AST dimension-specific models under zero-shot transfer to synthetic speech datasets. The overall MOS model demonstrates the most consistent cross-database performance, motivating its selection for subsequent fine-tuning.

Dimension	Test2	Test3	VCC	PhyS	Avg
MOS	**0.736**	0.445	**0.620**	**0.701**	**0.626**
NOI	0.211	0.035	0.399	0.449	0.274
DIS	0.717	0.547	0.510	0.628	0.601
COL	0.462	0.378	0.504	0.516	0.465
LOUD	0.568	**0.642**	0.128	-0.224	0.279

Table 7.2 compares file-level PCC obtained on synthetic speech test datasets for different fine-tuning strategies applied to the SQ-AST MOS model, alongside the NISQA-TTS baseline. Zero-shot SQ-AST already achieves competitive performance relative to NISQA-TTS, confirming its suitability as a transfer baseline.

Among the investigated adaptation strategies, partial fine-tuning yields the strongest and most consistent performance across test sets, achieving the highest average PCC. This suggests that selectively adapting higher-level encoder representations while preserving lower-level pretrained features provides an effective balance between flexibility and generalization. In contrast, full fine-tuning leads to reduced robustness on some datasets, indicating potential overfitting to the limited synthetic speech training data. The frozen-backbone configuration shows mixed behavior, with improvements on some datasets but degraded performance on others.

Overall, these results demonstrate that limited, targeted adaptation of the pretrained SQ-AST model is sufficient to improve performance on synthetic speech, and that partial fine-tuning offers the most robust transfer among the evaluated strategies.

Table 7.2: File-level Pearson correlation coefficients (PCC) across synthetic speech test databases. Results are shown for zero-shot and fine-tuned SQ-AST variants, alongside the NISQA-TTS baseline.

Model	Test2	Test3	PhyS	Avg
NISQA-TTS	0.610	0.340	0.720	0.557
SQ-AST MOS	**0.736**	0.445	0.701	0.627
Min FT SQ-AST	0.593	0.220	0.782	0.532
Partially FT SQ-AST	0.662	**0.521**	**0.890**	**0.691**
Fully FT SQ-AST	0.607	0.232	0.861	0.567

Tables 7.2 and 7.3 summarize the performance of different SQ-AST fine-tuning strategies in comparison with the NISQA-TTS baseline, evaluated using per-file and per-system Pearson correlation coefficients, respectively.

In the per-file evaluation (Table 7.2), partial fine-tuning achieves the highest average correlation (0.691), improving upon both the zero-shot SQ-AST model (0.627) and the fully fine-tuned configuration (0.567). This indicates that limited adaptation of higher-level representations is sufficient to enhance transfer performance on synthetic speech, while preserving robustness across datasets.

A similar pattern is observed in the per-system evaluation (Table 7.3). Partial fine-tuning yields the strongest overall performance, with an average PCC of 0.863, exceeding both the NISQA-TTS baseline (0.730) and the zero-shot SQ-AST model (0.714). Fully fine-tuned models exhibit reduced average performance, consistent with overfitting to the limited synthetic speech training data, whereas frozen-backbone models do not adapt enough and fail to capture synthesis-specific perceptual characteristics.

Overall, these results show that selectively adapting only the upper transformer layers provides the most effective balance between leveraging pretrained representations and learning synthesis-specific quality cues, leading to improved robustness and generalization in out-of-domain synthetic speech assessment.

Table 7.3: Per-system Pearson correlation coefficients (PCC) across synthetic speech test databases for different SQ-AST fine-tuning strategies and the NISQA-TTS baseline.

Model	**Test2**	**Test3**	**VCC**	**PhyS**	**Avg**
NISQA-TTS	**0.850**	0.380	**0.960**	0.890	0.730
SQ-AST MOS	0.758	0.492	0.892	0.797	0.714
Min FT SQ-AST	0.598	0.246	0.866	0.852	0.641
Part FT SQ-AST	0.708	**0.841**	0.952	**0.950**	**0.863**
Full FT SQ-AST	0.625	0.300	0.958	0.901	0.628

The results reveal distinct adaptation behavior across different synthetic speech scenarios. On the TU Berlin / Kiel TTS test sets (Test 2), the zero-shot SQ-AST model already achieves competitive performance, indicating that representations learned from natural speech quality data capture perceptually relevant cues that generalize to certain text-to-speech conditions. In these cases, fine-tuning yields only limited additional gains, suggesting that direct transfer is sufficient when synthesis artifacts are relatively mild or consistent across systems.

In contrast, on more challenging synthetic speech datasets, including the Voice Conversion Challenge 2016 and PhySyQX, partial fine-tuning of the upper transformer layers consistently yields the largest improvements over both zero-shot SQ-AST and the NISQA-TTS baseline. These results suggest that selectively adapting higher-level representations can be effective for improving performance on synthetic speech quality prediction, while retaining the benefits of pretrained acoustic representations. Fully fine-tuned variants, by comparison, exhibit reduced cross-database robustness, consistent with overfitting to the limited synthetic speech training data.

These observations highlight the importance of controlling adaptation depth when transferring pretrained speech quality models to synthetic speech. While zero-shot transfer can be effective in certain text-to-speech settings, partial fine-tuning provides a more robust strategy for capturing perceptual artifacts that are not well represented in natural speech quality training data.

7.5 Summary

This chapter investigated the transferability of foundation model-based speech quality predictors to out-of-domain synthetic speech assessment, focusing on text-to-speech and voice conversion scenarios. Starting from SQ-AST models trained on natural speech degradations, the chapter evaluated zero-shot transfer as well as several fine-tuning strategies under controlled experimental conditions.

The results show that zero-shot transfer can provide competitive performance for certain text-to-speech conditions, indicating that pretrained representations capture perceptually relevant cues that generalize beyond the original training domain. For more challenging synthetic speech scenarios, particularly voice conversion and diverse TTS systems, partial fine-tuning of the upper transformer layers yields the most consistent performance gains, outperforming both frozen-backbone and fully fine-tuned configurations.

Overall, the findings demonstrate that foundation model-based speech quality predictors can be effectively adapted to synthetic speech assessment with minimal task-specific engineering. Selective fine-tuning of higher-level representations emerges as a principled and robust strategy for capturing synthesis-specific perceptual artifacts, thereby addressing RQ5 and supporting the use of foundation models as a flexible basis for non-intrusive quality evaluation in emerging speech synthesis technologies.

Chapter 8
Conclusion

This thesis investigated transformer-based foundation models for non-intrusive speech quality assessment, with the objective of developing robust, perceptually aligned predictors for overall quality and multiple perceptual dimensions under realistic telecommunication conditions. Through a systematic progression from representation analysis and zero-shot evaluation to single-task modeling, standardization-oriented validation, and out-of-domain transfer, the work established both the capabilities and limitations of foundation model-based approaches for speech quality prediction. This chapter consolidates the main contributions of the thesis, provides concise answers to the research questions posed in Chapter 1, and reflects on the broader implications of the findings.

8.1 Summary of Contributions

This thesis makes the following principal contributions to the field of non-intrusive speech quality assessment:

1. **Foundation model adaptation for speech quality assessment.** The thesis systematically investigates transformer-based foundation models as general-purpose representations for non-intrusive speech quality prediction. It establishes that pretrained transformer encoders can be effectively adapted to predict overall quality and perceptual dimensions without hand-crafted features or intrusive reference signals.

2. **Multi-dimensional modeling of perceptual speech quality.** A unified modeling framework is developed for predicting overall quality alongside established perceptual dimensions, namely noisiness, discontinuity, coloration, and loudness. The work provides a detailed empirical analysis of how different perceptual attributes behave under shared versus task-specific modeling, revealing dimension-dependent sensitivity to data composition and training strategy.

3. **Comparative analysis of multi-task and single-task learning.** The thesis presents a comparison between joint multi-task learning and dimension-specific single-task optimization using a fixed transformer backbone. The results demonstrate that single-task optimization yields measurable advantages for several perceptual dimensions, while multi-task learning offers a stable and coherent representation for joint analysis.

4. **Standardization-oriented evaluation and validation.** A large-scale evaluation is conducted across heterogeneous datasets using protocols aligned with ITU-T Recommendation P.1401. Based on internal and independent external evaluations, the proposed single-task models are selected and approved as candidate models for the ITU-T P.SAMD Recommendation, demonstrating robustness under realistic telecommunication conditions.

5. **Out-of-domain transfer to synthetic speech.** The thesis extends the evaluation beyond natural speech degradations by investigating transfer to synthetic speech generated by text-to-speech and voice conversion systems. It shows that foundation model-based predictors can be transferred with minimal adaptation, and that partial fine-tuning of higher transformer layers provides the most robust performance for modeling synthesis-specific perceptual artifacts.

6. **Reproducible and extensible research artifacts.** The developed models, training procedures, and evaluation protocols are designed with reproducibility in mind. Selected model variants are released as open-source implementations, enabling further research and practical adoption in both academic and industrial contexts.

8.2 Answers to the Research Questions

This section provides concise answers to the research questions posed in Chapter 1, based on the empirical findings presented throughout the thesis.

RQ1: How can transformer-based pretrained foundation models be adapted for accurate and reliable non-intrusive prediction of overall speech quality?

Transformer-based pretrained foundation models can be adapted for non-intrusive speech quality prediction by attaching lightweight regression heads to frozen or fine-tuned encoder representations and optimizing them using subjective quality annotations. The results demonstrate that pretrained representations already encode perceptually relevant information, enabling accurate overall quality prediction under both zero-shot and supervised adaptation settings. Task-specific fine-tuning further improves robustness when sufficient labeled data are available.

RQ2: To what extent can foundation model representations support the prediction of perceptual speech quality dimensions, and how consistently do these predictions reflect subjective human judgments?

Foundation model representations support the prediction of multiple perceptual speech quality dimensions with meaningful alignment to subjective human judgments. The degree of consistency varies across dimensions: noisiness and discontinuity show strong and stable correspondence, while coloration and loudness exhibit greater sen-

sitivity to dataset characteristics and labeling variability. These findings indicate that perceptual dimensions differ in how directly they are captured by pretrained acoustic representations.

RQ3: Does joint multi-task learning of overall quality and perceptual dimensions improve prediction accuracy and robustness compared to single-task modeling?

Joint multi-task learning enables coherent modeling of overall quality and perceptual dimensions using a shared representation, but does not uniformly outperform single-task optimization across all dimensions. Empirical results show that single-task modeling yields measurable advantages for several perceptual targets, particularly overall quality, noisiness, and discontinuity. These gains motivated the selection of single-task models for subsequent evaluation and standardization.

RQ4: How well do foundation model-based speech quality predictors generalize across heterogeneous datasets, domains, and recording conditions, including independent hidden datasets used in ITU-T standardisation?

The proposed models demonstrate strong generalization across heterogeneous datasets, recording conditions, and domains. Independent evaluation by ITU-T Study Group 12 partners on previously unseen and hidden datasets confirms stable cross-database performance under realistic telecommunication scenarios. These results validate the robustness of foundation model-based predictors beyond the conditions represented in the training data.

RQ5: Can a foundation model-based speech quality predictor be effectively transferred to out-of-domain assessment tasks without extensive retraining or task-specific engineering?

Foundation model-based speech quality predictors can be effectively transferred to out-of-domain tasks, including text-to-speech and voice conversion quality assessment, with minimal adaptation. Zero-shot transfer already provides competitive performance in certain synthetic speech scenarios, while partial fine-tuning of higher transformer layers yields the most robust improvements. This demonstrates that extensive retraining or task-specific model redesign is not required for effective out-of-domain transfer.

8.3 Implications and Significance

The findings of this thesis have several important implications for the design, evaluation, and deployment of non-intrusive speech quality assessment systems. At a methodological level, the results demonstrate that transformer-based foundation models provide a viable and robust basis for perceptual speech quality prediction, reducing reliance on hand-crafted features and narrowly specialized model architectures. This supports a shift

toward representation-centric approaches in speech quality modeling, in which general-purpose pretrained encoders serve as the foundation for a wide range of perceptual assessment tasks.

From a modeling perspective, the comparative analysis of multi-task and single-task learning highlights the importance of aligning training strategies with the characteristics of individual perceptual dimensions. While joint learning enables coherent modeling across dimensions, the observed advantages of single-task optimization for several targets indicate that task-specific supervision remains critical for achieving high predictive accuracy. These insights inform principled choices between unified and specialized modeling strategies, depending on application requirements and data availability.

The demonstrated robustness of the proposed models across heterogeneous datasets and recording conditions has direct implications for practical deployment. Independent evaluation on hidden datasets within the ITU-T standardization process confirms that foundation model-based predictors can generalize reliably under realistic telecommunication scenarios. This establishes confidence in their applicability beyond controlled research settings and underscores their suitability for large-scale, real-world quality monitoring.

Finally, the successful transfer of the proposed framework to synthetic speech assessment illustrates the broader generality of foundation model representations. The ability to adapt a model trained on natural speech degradations to text-to-speech and voice conversion scenarios with minimal modification suggests that a unified modeling paradigm can support emerging speech technologies. Together, these implications position foundation model-based speech quality assessment as a scalable, extensible, and standardization-ready approach with relevance across both research and industrial contexts.

8.4 Limitations

While this thesis demonstrates the effectiveness and robustness of transformer-based foundation models for non-intrusive speech quality assessment, several limitations should be acknowledged.

First, the availability and consistency of subjective labels vary substantially across datasets, particularly for perceptual dimensions beyond overall quality. Some dimensions, such as coloration and loudness, are underrepresented or inconsistently annotated, which limits the extent to which their predictive performance can be optimized and compared across datasets. This variability reflects broader challenges in perceptual speech quality data collection rather than shortcomings of the proposed modeling approach.

Second, although extensive evaluation was conducted across a large number of heterogeneous databases, including independent hidden datasets used in ITU-T standardization, the analysis remains constrained to the datasets and conditions available at the time of study. Certain speech scenarios, languages, and emerging synthesis techniques may therefore not be fully represented, and model behavior under such conditions cannot be conclusively assessed within the scope of this work.

Third, the transfer experiments for synthetic speech focus on overall quality and naturalness assessment. While these experiments demonstrate effective out-of-domain adaptation, the applicability of perceptual dimension modeling to synthetic speech remains an open question, as suitable large-scale subjective annotations for such dimensions are currently scarce.

Finally, the architectural choices explored in this thesis are intentionally focused on a single transformer-based backbone to enable controlled comparisons across modeling strategies. While this design choice strengthens the internal validity of the conclusions, it does not address the relative performance of alternative foundation model architectures, which may exhibit different trade-offs between representation capacity, robustness, and computational efficiency.

These limitations define the boundaries of the present study and provide context for interpreting the reported results, while not detracting from the central findings regarding the suitability of foundation models for robust and transferable speech quality assessment.

8.5 Outlook

This thesis has demonstrated that transformer-based foundation models can be effectively adapted for non-intrusive, perceptually aligned speech quality assessment across a wide range of telecommunications scenarios. By systematically investigating zero-shot transfer, multi-task learning, and dimension-specific modeling, the work provides both empirical evidence and methodological insight into the strengths and limitations of foundation-model-based approaches. At the same time, the results highlight several open challenges and promising directions for future research. The following sections outline potential avenues to further extend, refine, and operationalize the proposed approaches.

Representation Analysis and Interpretability

While the experimental results indicate that pretrained transformer representations encode perceptually relevant information, the internal mechanisms by which different perceptual dimensions are captured remain largely unexplored. Future work could focus on analyzing layer-wise representations within the AST backbone to identify which levels contribute most strongly to overall quality, noisiness, discontinuity, coloration, and loudness prediction. Such analyses could employ probing models, attribution methods, or representational similarity analysis to improve interpretability and to inform more principled architectural or training decisions. Improved interpretability would also support greater transparency in standardization and deployment contexts.

Structured Multi-Task Learning and Dimension Interactions

The mixed performance observed across perceptual dimensions suggests that simple joint optimization may not fully capture the relationships between different quality dimensions. Future research could investigate structured multi-task learning strategies that

explicitly model interdependencies between dimensions, for example through hierarchical task formulations, partial parameter sharing, or task-specific routing mechanisms. Such approaches may help mitigate negative transfer effects while preserving the benefits of shared representations, particularly for dimensions that exhibit strong perceptual coupling in human judgments.

Data Coverage and Annotation Strategies

The results of this work underline the critical role of data diversity for robust speech quality prediction. Expanding training data to include a broader range of acoustic conditions, languages, speaker characteristics, and emerging transmission scenarios remains an important direction. In addition, future work could explore targeted data collection strategies, such as active learning or uncertainty-driven sampling, to efficiently improve performance for underrepresented perceptual dimensions. From a standardization perspective, harmonized annotation protocols and shared evaluation benchmarks would further support reproducibility and cross-model comparability.

Condition-Level Modeling and Uncertainty Estimation

Current modeling approaches primarily focus on point estimates of perceptual quality. However, subjective ratings inherently exhibit variability across listeners and conditions. Future work could incorporate explicit uncertainty modeling, such as predictive confidence intervals or probabilistic output representations, to better reflect this variability. In addition, condition-level modeling strategies that operate directly on aggregated degradation conditions, rather than individual utterances alone, may offer improved robustness and closer alignment with evaluation practices defined in ITU-T Recommendation P.1401.

Robustness to Synthetic and Emerging Speech Technologies

The application of dimension-specific models to synthetic speech highlights both the transferability and the limitations of models trained primarily on natural speech degradations. As text-to-speech, voice conversion, and neural speech coding technologies continue to evolve rapidly, future work should investigate adaptive training strategies that explicitly incorporate synthetic speech artifacts. Continual learning or domain adaptation approaches may be required to maintain reliable performance as synthesis methods change over time.

Efficiency and Real-Time Deployment Considerations

For practical deployment in telecommunications systems, computational efficiency and latency constraints play a critical role. Future research could explore model compression techniques, such as pruning, quantization, or knowledge distillation, to enable

real-time or edge-device deployment without substantial loss in perceptual prediction accuracy. Studying trade-offs between model complexity and performance will be essential for large-scale operational use.

Implications for Standardization and Evaluation Protocols

Given the involvement of this work in the ITU-T P.SAMD standardization effort, future research may also consider how foundation-model-based approaches can be systematically integrated into long-term evaluation and benchmarking frameworks. This includes defining stable evaluation protocols, reporting practices, and validation procedures that account for both predictive accuracy and robustness across heterogeneous conditions. Such efforts would support the sustainable adoption of data-driven quality assessment models within international standards.

Beyond MOS: Richer Perceptual Modeling

Finally, while mean opinion score remains a widely accepted and practical target metric, future work could explore richer perceptual representations that capture temporal dynamics, listener-specific factors, or task-dependent quality aspects. Moving beyond static scalar scores may enable more nuanced assessments of speech quality that better reflect real-world user experience.

In summary, the findings of this thesis establish a strong foundation for the use of transformer-based models in perceptual speech quality assessment, while also delineating directions for continued research. Addressing these challenges will further advance the development of robust, interpretable, and standardized speech quality assessment systems capable of meeting the demands of future communication technologies.

Appendix A
Database Collection and Details

This appendix provides an overview of the speech quality databases used in this work. A complete summary of the databases is presented in the tables on the following pages.

Language

This column specifies the language(s) of the speech material contained in each database. Language identifiers follow the ISO 639-1 alpha-2 convention [1]. Databases comprising monolingual speech are listed using the corresponding language code, while multilingual databases are denoted by `mix` and comprise speech in `en`, `de`, `fr`, `ja`, `zh`, `fi`, and `it`. The label `mix+` indicates an extended multilingual set that additionally includes `sv` and `cs`.

Number of Votes

These columns report the number of ratings available per speech file for overall quality (MOS) and each perceptual dimension (noisiness, discontinuity, coloration, and loudness). A vote count greater than one corresponds to ratings obtained from subjective listening tests. Entries with a vote count of one indicate datasets in which the reported quality values are provided as single-score estimates, typically originating from instrumental methods.

For the majority of databases in the latter case, reference signals were available and overall quality labels were therefore derived using POLQA ITU-T Recommendation P.863 (2018). In some cases, perceptual dimension labels were obtained using POLQA-based extensions ITU-T Recommendation P.863.2 (2022). In rare cases where reference signals were not accessible, quality labels were generated using the non-intrusive NISQA model (Mittag et al., 2021).

[1] ISO 639-1, *Codes for the representation of names of languages — Part 1: Alpha-2 code*. International Organization for Standardization.

Database	Lang	BW	Cond Count	File Count	MOS Count	NOI Count	DIS Count	COL Count	LOUD Count	Summary of Degradation Conditions and Other Notes
NB_48kHz104_Ericsson	sv	NB	55	660	8	1	1	1	1	Additive and environmental noise (including MNRU); temporal distortions (clipping, pulse distortions, frame erasures, time scaling); low-bitrate and tandem coding with spectral distortion; AGC and enhancement artifacts; live cellular, POTS, and VoIP scenarios.
NB_48kHz404_PSYTECHNICS	en	NB	48	1151	8	1	1	1	1	Condition descriptions not specified in available metadata.
NB_48kHz504_SWISSQUAL	de	NB	49	196	24	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_ATT_iLBC	en	NB	45	1312	8	1	1	1	1	Coding distortions (G.729E, iLBC 20/30 ms) including tandem coding; additive noise (babble 30 dB SNR, car 15 dB SNR); frame erasures (FE00–FE10); level variations (quiet 16/26/36 dBoV); and MNRU (06–30).
NB_48kHz_BT_P862_BGN_ENG	en	NB	49	196	24	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_BT_P862_PROP	en	NB	50	1200	4	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_DT_P862_1st	de	NB	50	200	24	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_DT_P862_BGN_GER	de	NB	49	196	28	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_DT_P862_Share	de	NB	50	200	24	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_ERIC_AMR_4B	sv	NB	36	432	9	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_ERIC_FIELD_GSM_EU	en	NB	234	234	1	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_ERIC_FIELD_GSM_US	en	NB	372	372	1	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_ERIC_P862_NW_MEAS	sv	NB	46	184	29	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_FT_P563_PROP	fr	NB	50	799	6	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_GIPS_EXP1	en	NB	38	1824	6	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_HUAWEI_1	zh	NB	24	576	8	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_HUAWEI_2	zh	NB	24	576	8	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_ITU_SUPPL23_EXP1a	fr	NB	44	176	24	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_ITU_SUPPL23_EXP1d	ja	NB	44	176	24	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_ITU_SUPPL23_EXP1o	en	NB	44	176	24	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_ITU_SUPPL23_EXP3a	fr	NB	50	200	24	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_ITU_SUPPL23_EXP3c	it	NB	50	200	24	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_ITU_SUPPL23_EXP3d	ja	NB	50	200	24	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_ITU_SUPPL23_EXP3o	en	NB	50	200	24	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_LUC_P563_PROP	en	NB	50	200	24	1	1	1	1	Condition labels not specified in available metadata.

Database	Lang	BW	Cond Count	File Count	MOS Count	NOI Count	DIS Count	COL Count	LOUD Count	Summary of Degradation Conditions and Other Notes
NB_48kHz_NTT_PTEST_1	ja	NB	90	1416	8	1	1	1	1	Coding degradations involving single and tandem speech codecs applied across multiple cascade stages. Exact condition definitions are not fully specified in the available metadata.
NB_48kHz_OPT_P563_PROP	de	NB	50	600	24	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_PSY_P563_PROP	en	NB	54	648	8	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_QUALCOMM_EXP1b	en	NB	32	2048	4	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_QUALCOMM_EXP2b	en	NB	32	2048	4	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_QUALCOMM_EXP3w	en	NB	32	2048	4	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_QUALCOMM_EXP4	en	NB	32	2048	4	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_QUALCOMM_EXP6a	en	NB	48	2304	4	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_QUALCOMM_EXP6b	en	NB	48	2304	4	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_SQ_P563_PROP	de	NB	50	600	8	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_TNO_P862_KPN_KIT97	nl	NB	59	59	60	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_TNO_P862_NW_EMU	nl	NB	50	200	24	1	1	1	1	Condition labels not specified in available metadata.
NB_48kHz_TNO_P862_NW_MEAS	nl	NB	46	184	24	1	1	1	1	Condition labels not specified in available metadata.
SWB_48kHz101_ERICSSON	sv	SWB	57	684	9	1	1	1	1	Additive noise and MNRU distortions; speech coding degradations across NB, WB, and SWB codecs (including tandem coding); packet/frame loss at multiple rates; and presentation-level variations.
SWB_48kHz103_ERICSSON	sv	SWB	54	648	8	1	1	1	1	Additive noise and MNRU distortions; speech coding degradations across NB, WB, and SWB codecs (including tandem coding); temporal impairments (time clipping, time scaling, jitter); packet loss under controlled and live transmission conditions; and level variations.
SWB_48kHz201_FT_DT	fr	SWB	48	192	24	1	1	1	1	Additive ambient and circuit noise; noise-like distortions (MNRU); level variations; temporal distortions (clipping, overload, sampling frequency offsets); packet loss without concealment; and terminal- and filter-dependent transmission conditions.
SWB_48kHz202_FT_DT	fr	SWB	49	196	24	1	1	1	1	Noise and MNRU distortions; speech coding degradations with bitrate variation; signal-processing effects (noise reduction, AGC, time warping); packet loss and wireless transmission errors; and level variations.
SWB_48kHz203_FT_DT	fr	SWB	54	216	24	1	1	1	1	Background noise and MNRU distortions; speech coding degradations with bitrate variation; temporal and transmission impairments; frequency response distortions; level variations; and live transmission conditions.

Database	Lang	BW	Cond Count	File Count	MOS Count	NOI Count	DIS Count	COL Count	LOUD Count	Summary of Degradation Conditions and Other Notes
SWB_48kHz301_OPTICOM	cs	SWB	50	200	24	1	1	1	1	Noise and MNRU distortions; level variations; filtering and bandwidth limitations; temporal impairments; and speech coding degradations (NB/WB/SWB, incl. tandem configurations).
SWB_48kHz302_OPTICOM	en	SWB	44	176	24	1	1	1	1	Noise and MNRU distortions; speech coding degradations with bitrate variation; noise suppression and error concealment; packet loss; temporal modifications; level variations; and filter-dependent transmission conditions.
SWB_48kHz303_OPTICOM	en	SWB	54	216	24	1	1	1	1	Background noise and MNRU distortions; level variations; temporal distortions; speech coding degradations across NB, WB, and SWB; filtering effects; and live acoustic and transmission conditions.
SWB_48kHz401_PSYTECHNICS	en	SWB	48	1152	8	1	1	1	1	Background noise and MNRU distortions; level variations; filtering and bandwidth limitations; packet loss, jitter, and variable delay; speech coding degradations; and VoIP-centric end-to-end transmission scenarios with acoustic interfaces, including routing to PSTN and mobile networks.
SWB_48kHz403_PSYTECHNICS	en	SWB	48	1152	8	1	1	1	1	Noise and MNRU distortions; level variations; filtering effects; temporal distortions; speech coding degradations (NB/WB/SWB); and live VoIP and cellular transmission scenarios.
SWB_48kHz501_SWISSQUAL	de	SWB	50	200	24	24	24	24	24	Noise and MNRU distortions; level variations; filtering effects; temporal distortions; speech coding degradations (NB/WB/SWB); and live cellular, VoIP, and video-call transmission scenarios.
SWB_48kHz502_SWISSQUAL	de	SWB	50	200	24	24	24	24	24	Noise and MNRU distortions; level variations; filtering effects; temporal distortions; speech coding degradations (NB/WB/SWB); and live cellular, VoIP, and video-call scenarios with acoustic conditions.
SWB_48kHz503_SWISSQUAL	de	SWB	54	216	24	24	24	24	24	Noise and MNRU distortions; level variations; filtering effects; temporal distortions; speech coding degradations (NB/WB/SWB); and live VoIP, cellular, and acoustic transmission scenarios.
SWB_48kHz601_TNO	nl	SWB	50	200	24	1	1	1	1	Controlled single-degradation conditions including noise and MNRU distortions, level and filtering effects, temporal and spectral distortions, speech coding degradations, and live VoIP and acoustic transmission scenarios across NB, WB, and SWB.
SWB_48kHz602_TNO	nl	SWB	50	200	24	1	1	1	1	Controlled single-degradation conditions including noise and MNRU distortions, level and filtering effects, temporal and spectral distortions, speech coding and enhancement processing, across NB, WB, and SWB.

Database	Lang	BW	Cond Count	File Count	MOS Count	NOI Count	DIS Count	COL Count	LOUD Count	Summary of Degradation Conditions and Other Notes
SWB_48kHz603_TNO	nl	SWB	48	192	24	1	1	1	1	Noise and MNRU distortions; level variations; filtering effects; temporal distortions; speech coding degradations (NB/WB/SWB); and simulated and live end-to-end transmission scenarios with acoustic conditions.
SWB_48kHz_DT_SWB_Izabela	de	SWB	29	1220	1	1	1	1	1	Speech coding degradations with bitrate variation, packet loss, tandem coding, and bandwidth filtering under primarily clean and lightly noisy conditions.
SWB_48kHz_GIPS_Exp4	en	SWB	36	144	25	1	1	1	1	Speech coding degradations using a single SWB codec under controlled network impairments, including clock drift, jitter, and packet loss, with clean references and MNRU-based anchor conditions.
SWB_48kHz_PAMD_Orange_1	fr	SWB	56	224	1	18	18	18	18	Noise and MNRU distortions; level and gain variations; temporal and spectral distortions; speech coding degradations; and terminal effects across NB, WB, and SWB.
SWB_48kHz_PAMD_Swissqual_1	de	SWB	30	120	1	24	24	24	24	Coding degradations, noise and MNRU distortions, level and temporal impairments, and live cellular, VoIP, and video-call scenarios with acoustic conditions.
SWB_48kHz_TCD	en	SWB	76	304	24	1	1	1	1	Speech database of common Voice-over-IP degradations, focusing on impairments independent of codec, hardware, or network configuration, including clock drift, jitter, and packet loss. (Source: TCD-VoIP; Harte et al., QoMEX 2015)
SWB_TUB_DIS	de	SWB	20	40	41	41	41	41	41	Level variations; spectral band-limiting; additive noise; temporal distortions; and packet-loss effects with and without concealment across NB/WB/SWB speech.
SWB_TUB_LIK	de	SWB	8	96	20	1	1	1	1	Clean and degraded speech with bandwidth limitation, coding-related artifacts, bandpass filtering, and artificial noise on scripted excerpts. (Source: Gallardo et al., QoMEX 2018)
SWB_TUB_VUPL	de	SWB	15	60	36	1	1	1	1	Controlled packet-loss conditions with fixed loss counts applied at different temporal positions (front, middle, end) and speech segments (voiced, unvoiced, and combined). (Source: Liedtke, Bachelor's thesis at TUB, 2018)
WB_48Hz_NTT_PTEST_2	ja	WB	80	1104	8	1	1	1	1	Speech signals affected by controlled transmission impairments, including random and burst packet loss as well as jitter, evaluated under simulated network conditions.
WB_48kHz102_ERICSSON	sv	WB	54	648	14	1	1	1	1	Condition labels not specified in available metadata.
WB_48kHz204_FTDT	fr	WB	53	212	24	1	1	1	1	Wideband speech with noise and MNRU distortions, level and filtering effects, temporal impairments, coding cascades, and live telephony and VoIP transmission conditions.

Database	Lang	BW	Cond Count	File Count	MOS Count	NOI Count	DIS Count	COL Count	LOUD Count	Summary of Degradation Conditions and Other Notes
WB_48kHz402_PSYTECHNICS	en	WB	48	1152	8	1	1	1	1	Narrowband and wideband speech with coding impairments, packet loss, channel interference, and controlled MNRU noise degradations.
WB_48kHz_DTAG_1	de	WB	56	670	3	3	3	3	1	Wideband and narrowband speech with speech coding and transcoding degradations, tandem encoding effects, packet loss impairments, background noise conditions, filtering distortions, and level variations.
WB_48kHz_DTAG_2	de	WB	76	912	4	4	4	4	1	Speech coding degradations combined with background noise at different environments and levels, packet loss impairments, filtering distortions, and modulated noise conditions, evaluated across clean and noisy wideband scenarios.
WB_48kHz_DT_HQ_Voice	de	WB	125	18875	1	1	1	1	1	Condition labels not specified in available metadata.
WB_48kHz_DT_Mobisense	de	WB	26	112	1	1	1	1	1	VoIP transmission conditions covering different network paths, packet loss rates, and codec switching before or after network handovers.
WB_48kHz_GIPS_Exp3	en	WB	36	1728	6	1	1	1	1	Condition labels not specified in available metadata.
WB_48kHz_QUALCOMM_EXP1w	en	WB	40	2560	4	1	1	1	1	Condition labels not specified in available metadata.
WB_48kHz_QUALCOMM_EXP5	en	WB	56	3584	4	1	1	1	1	Condition labels not specified in available metadata.
FB_TUB_AUS	en	FB	50	600	7	7	7	7	7	Additive noise, level and filtering distortions, temporal impairments, packet loss, tandem processing, and live over-the-top communication scenarios across NB, WB, and SWB conditions. (Source: ITU-T Contribution SG12-C440, 2019)
FB_TUB_DEV	de	FB	24	96	22	22	22	22	22	Controlled degradations including noise, level variation, band-limiting, and temporal clipping, as well as real-device recordings covering local recording, telephony, and over-the-top communication scenarios. (Source: Steinmetz, Bachelor's thesis at TUB, 2020)
FB_TUB_SPO	de	FB	27	108	28	28	28	28	28	Spontaneous and planned conversational speech recorded acoustically and via smartphone telephony for speech quality assessment. (Source: Sahin, Bachelor's thesis at TUB, 2020)
NISQA_TEST_FOR	en	FB	60	240	31	31	31	31	31	Contains live and simulated conditions with speech samples from the forensic speech dataset. (Source: NISQA Corpus, Mittag et al., Interspeech 2021)
NISQA_TEST_LIVETALK	de	FB	58	232	24	24	24	24	24	Contains recordings of real phone and VoIP calls. (Source: NISQA Corpus, Mittag et al., Interspeech 2021)
NISQA_TEST_NSC	de	FB	60	240	28	28	28	28	28	Contains live and simulated conditions with speech samples from the NSC dataset. (Source: NISQA Corpus, Mittag et al., Interspeech 2021)
NISQA_TEST_P501	de	FB	60	240	30	30	30	30	30	Contains live and simulated conditions with speech samples from ITU-T Rec. P.501. (Source: NISQA Corpus, Mittag et al., Interspeech 2021)

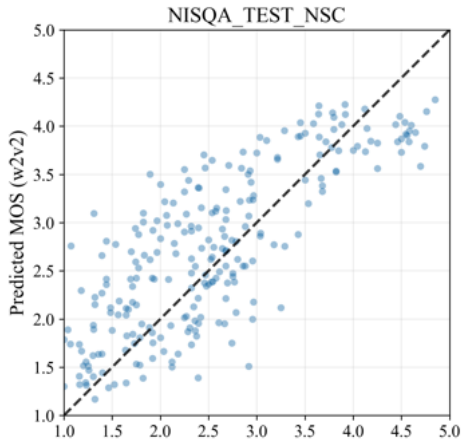
Database	Lang	BW	Cond Count	File Count	MOS Count	NOI Count	DIS Count	COL Count	LOUD Count	Summary of Degradation Conditions and Other Notes
NISQA_TRAIN_LIVE	en	FB	1020	1020	5	5	5	5	5	Live telephone and VoIP call recordings with realistic acoustic and interaction-induced speech degradations. (Source: NISQA Corpus, Mittag et al., Interspeech 2021)
NISQA_TRAIN_SIM	en	FB	10000	10000	5	5	5	5	5	Noise, level, filtering, clipping, coding, packet-loss, and combined speech degradations across NB/WB/SWB conditions. (Source: NISQA Corpus, Mittag et al., Interspeech 2021)
NISQA_VAL_LIVE	en	FB	200	200	5	5	5	5	5	Live telephone and VoIP call recordings with realistic acoustic and interaction-induced speech degradations. (Source: NISQA Corpus, Mittag et al., Interspeech 2021)
NISQA_VAL_SIM	en	FB	2500	2500	5	5	5	5	5	Noise, level, filtering, clipping, coding, packet-loss, and combined speech degradations across NB/WB/SWB conditions. (Source: NISQA Corpus, Mittag et al., Interspeech 2021)
ETSI_ENG_DB01	en	FB	60	960	12	1	1	1	1	Mobile handset and hands-free recordings with acoustic capture, time-variant filtering, reverberation, transcoding artifacts, and background noise. (Source: HEAD Acoustics, ETSI TS 103 558)
ETSI_ENG_DB02_PD	en	FB	48	768	12	1	1	1	1	Mobile handset and hands-free recordings with acoustic capture, time-variant filtering, reverberation, transcoding artifacts, and background noise. (Source: HEAD Acoustics, ETSI TS 103 558)
ETSI_ENG_DB03_PD	en	FB	48	768	12	1	1	1	1	Mobile handset and hands-free recordings with acoustic capture, time-variant filtering, reverberation, transcoding artifacts, and background noise. (Source: HEAD Acoustics, ETSI TS 103 558)
ETSI_GER_DB01_PD	de	FB	48	768	12	1	1	1	1	Mobile handset and hands-free recordings with acoustic capture, time-variant filtering, reverberation, transcoding artifacts, and background noise. (Source: HEAD Acoustics, ETSI TS 103 558)
ETSI_GER_DB02_PD	de	FB	48	768	12	1	1	1	1	Mobile handset and hands-free recordings with acoustic capture, time-variant filtering, reverberation, transcoding artifacts, and background noise. (Source: HEAD Acoustics, ETSI TS 103 558)
ETSI_GER_DB03	de	FB	60	960	12	1	1	1	1	Mobile handset and hands-free recordings with acoustic capture, time-variant filtering, reverberation, transcoding artifacts, and background noise. (Source: HEAD Acoustics, ETSI TS 103 558)
ETSI_MAN_DB01_PD	zh	FB	48	768	12	1	1	1	1	Mobile handset and hands-free recordings with acoustic capture, time-variant filtering, reverberation, transcoding artifacts, and background noise. (Source: HEAD Acoustics, ETSI TS 103 558)
ETSI_MAN_DB02	zh	FB	60	960	12	1	1	1	1	Mobile handset and hands-free recordings with acoustic capture, time-variant filtering, reverberation, transcoding artifacts, and background noise. (Source: HEAD Acoustics, ETSI TS 103 558)

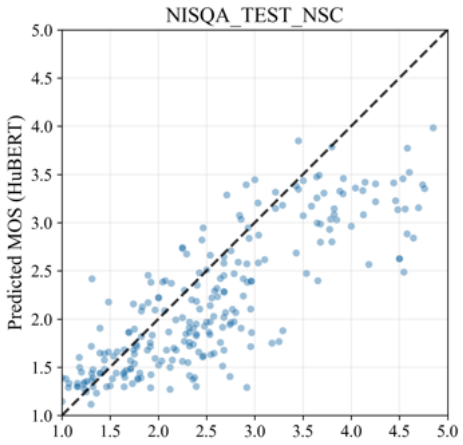
Database	Lang	BW	Cond Count	File Count	MOS Count	NOI Count	DIS Count	COL Count	LOUD Count	Summary of Degradation Conditions and Other Notes
ETSI_MAN_DB03_PD	zh	FB	48	768	12	1	1	1	1	Mobile handset and hands-free recordings with acoustic capture, time-variant filtering, reverberation, transcoding artifacts, and background noise. (Source: HEAD Acoustics, ETSI TS 103 558)
ENG_Automotive	en	FB	314	2497	1	1	1	1	1	(Source: HEAD Acoustics.)
ENG_Handset	en	FB	1020	8151	1	1	1	1	1	(Source: HEAD Acoustics.)
ENG_Headset	en	FB	2295	18266	1	1	1	1	1	(Source: HEAD Acoustics.)
GER_Automotive	de	FB	480	3800	1	1	1	1	1	(Source: HEAD Acoustics.)
GER_Handset	de	FB	995	7937	1	1	1	1	1	(Source: HEAD Acoustics.)
GER_Headset	de	FB	1125	8954	1	1	1	1	1	(Source: HEAD Acoustics.)
MAN_Automotive	zh	FB	478	3805	1	1	1	1	1	(Source: HEAD Acoustics.)
MAN_Handset	zh	FB	1020	8134	1	1	1	1	1	(Source: HEAD Acoustics.)
MAN_Headset	zh	FB	2291	18188	1	1	1	1	1	(Source: HEAD Acoustics.)
Orange_C122_DB01	fr	FB	36	864	1	1	1	1	1	(Source: ITU-T Contribution SG12-C122, 2019)
Orange_C122_DB02	fr	FB	55	220	1	1	1	1	1	(Source: ITU-T Contribution SG12-C122, 2019)
Orange_G711	en	NB	116	116	1	1	1	1	1	Measurements are performed on an enterprise VoIP system using G.711. (Source: Orange Innovation, France.)
Orange_NB_coders	fr	NB	44	44	1	1	1	1	1	(Source: Orange Innovation, France.)
Orange_Neural	mix	FB	53	1484	1	1	1	1	1	(Source: Orange Innovation, France.)
Orange_Teams_Conv	fr	FB	111	111	1	1	1	1	1	(Source: Orange Innovation, France.)
SIM_CODEC	fr	FB	63	1512	1	1	1	1	1	(Source: Orange Innovation, France.)
STORYTIME	de	FB	4	160	7	7	7	7	7	(Source: Storytime Dataset; Spang et al., QoMEX 2022)
SUB_TUB_1	de	FB	54	214	32	32	32	32	32	Additive environmental noise (babble, car, street); level variations; bandwidth limitations and filtering effects; temporal distortions (signal/time clipping); and speech transmission degradations across narrowband, wideband, and super-wideband conditions.
SUB_TUB_2	de	FB	25	100	25	25	25	25	25	Subset of the GER_Handset database rated subjectively at TU-Berlin.
TUB_2_DIS	mix+	FB	5980	5980	1	1	1	1	1	Internal database focused on discontinuity impairements
TUB_2_LOUD	mix+	FB	5220	5220	1	1	1	1	1	Internal database focused on loudness impairements
Vin1008	mix	FB	36	1008	1	1	1	1	1	(Source: Orange Innovation, France.)
Vin700	mix	FB	24	672	1	1	1	1	1	(Source: Orange Innovation, France.)
Vin726	fr	FB	726	726	1	1	1	1	1	(Source: Orange Innovation, France.)

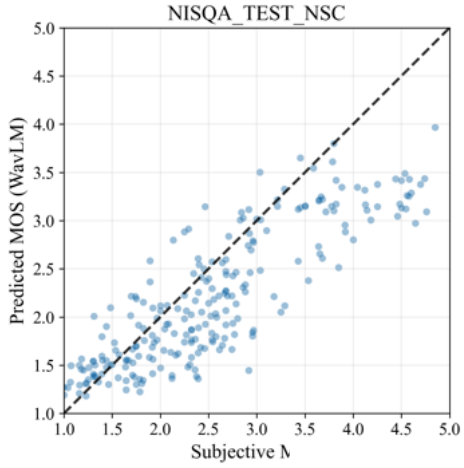
Appendix B
Zero-Shot Evaluation Scatter Plots

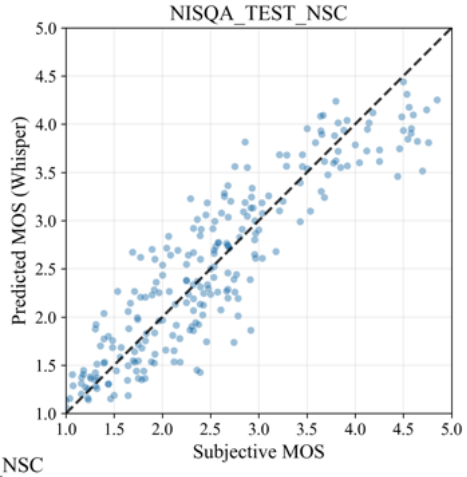
This appendix presents scatter plots of predicted versus subjective speech quality scores for the zero-shot evaluation in Chapter 4. Plots are shown for each test database and pretrained model using frozen encoders and a downstream regression mapping head.

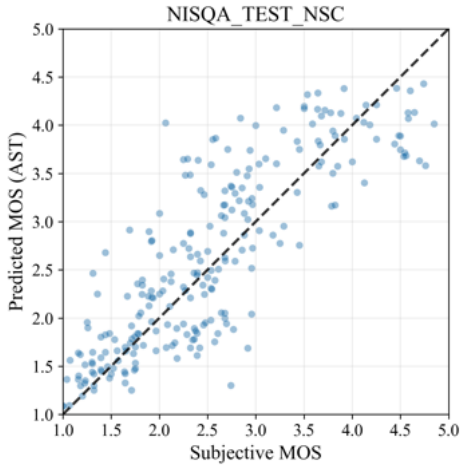
NISQA_TEST_NSC



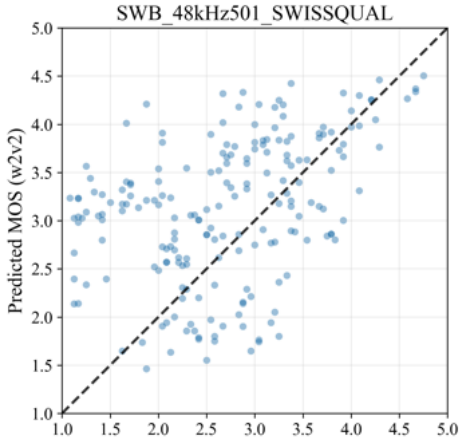


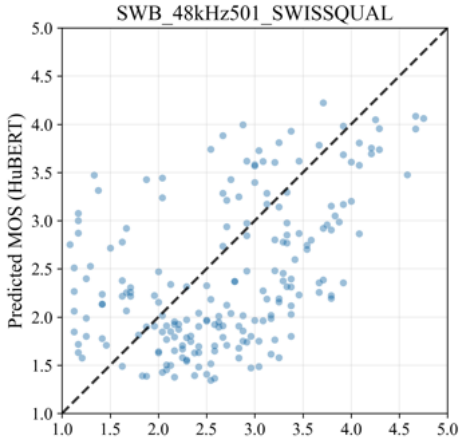


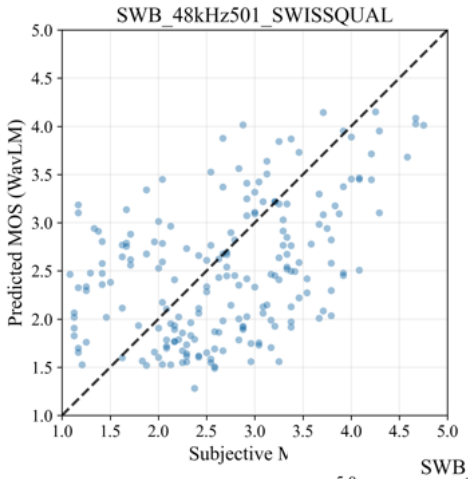


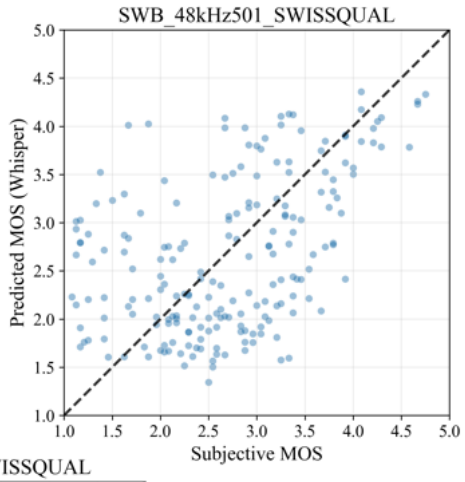


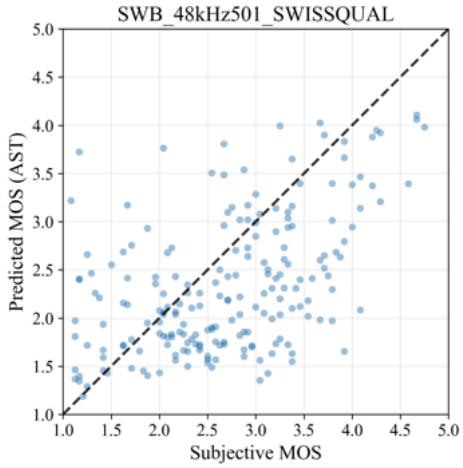
SWB_48kHz501_SWISSQUAL



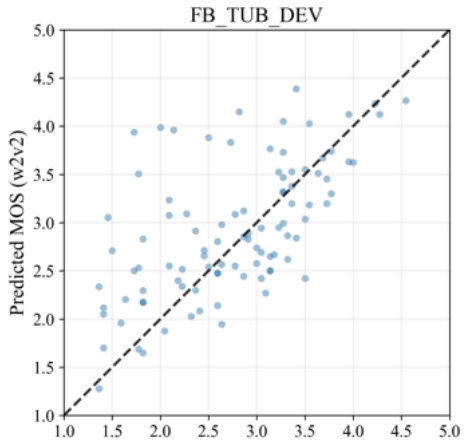


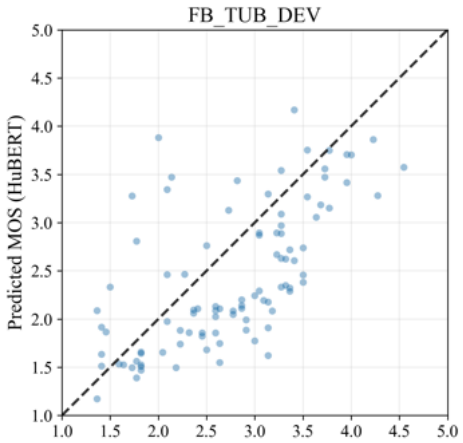


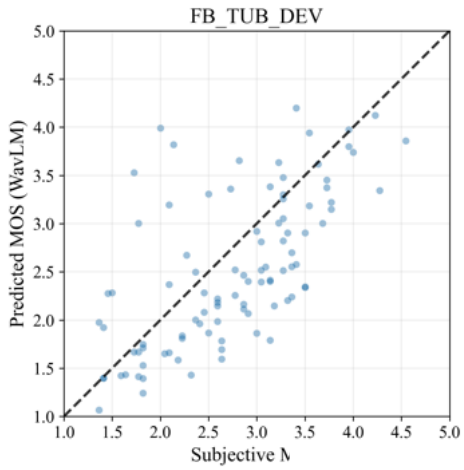


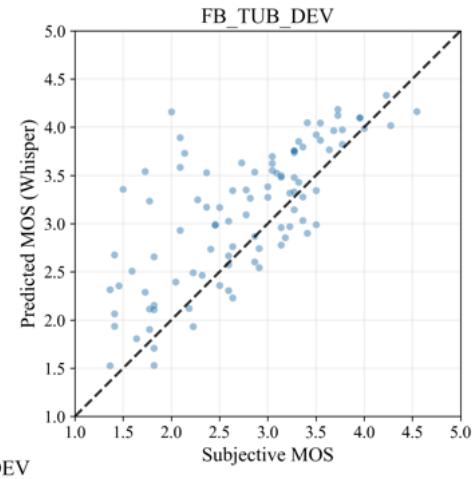


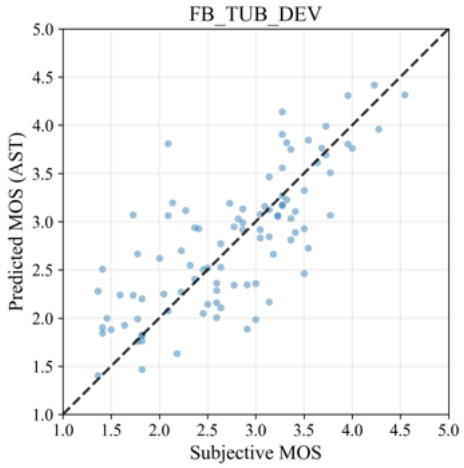
FB_TUB_DEV


FB_TUB_DEV
Predicted MOS (w2v2)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0

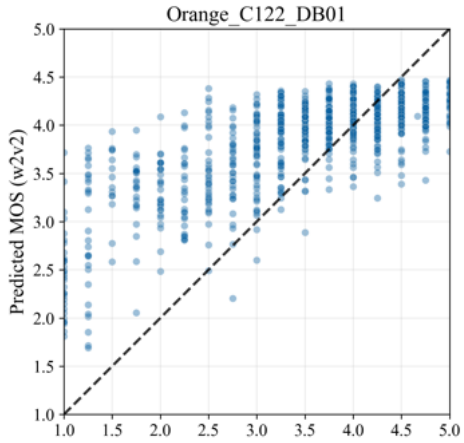

FB_TUB_DEV
Predicted MOS (HuBERT)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0

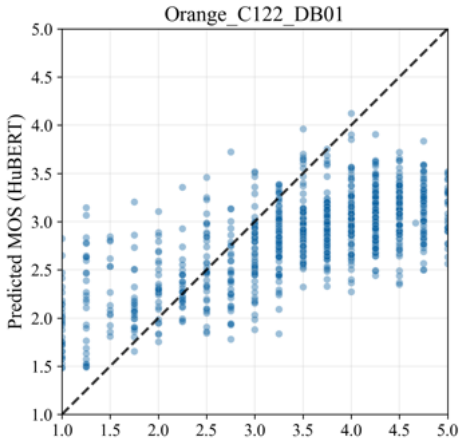

FB_TUB_DEV
Predicted MOS (WavLM)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0
Subjective M

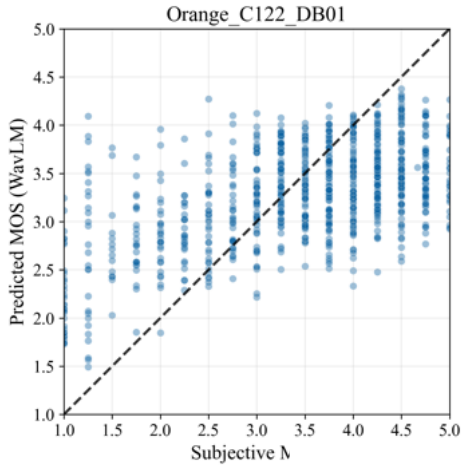

FB_TUB_DEV
Predicted MOS (Whisper)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0
Subjective MOS

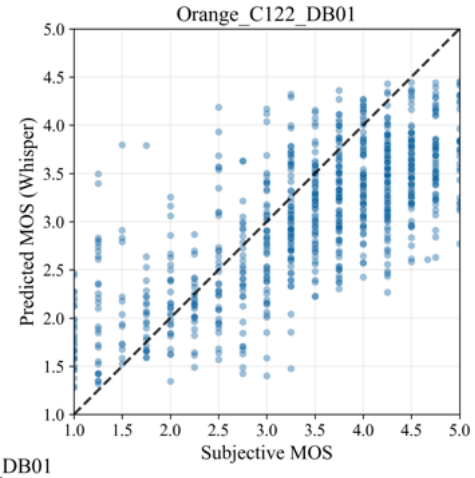

FB_TUB_DEV
Predicted MOS (AST)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0
Subjective MOS

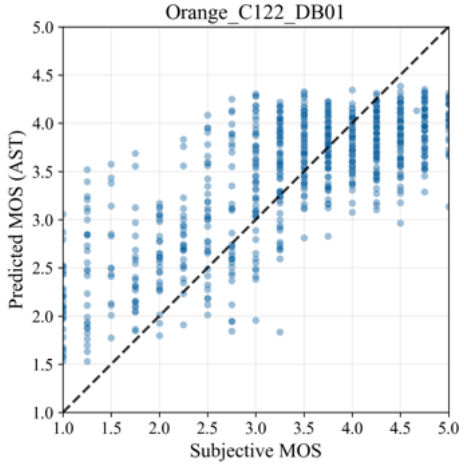
Orange_C122_DB01



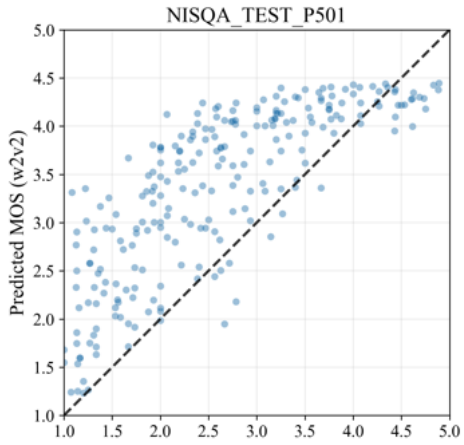


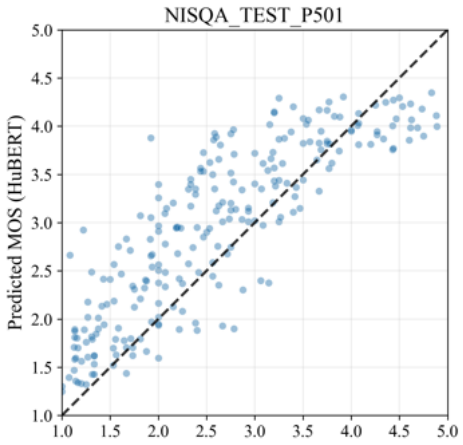


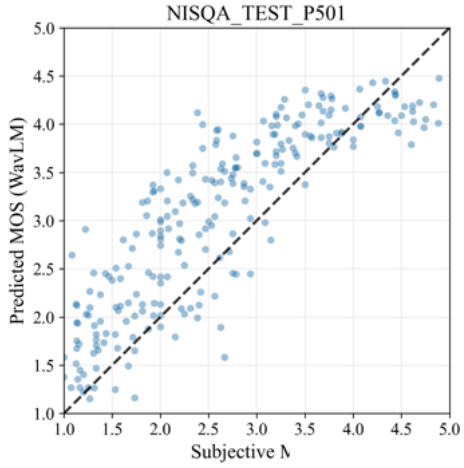


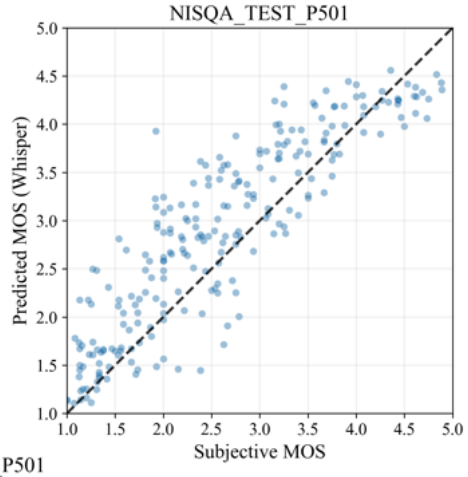


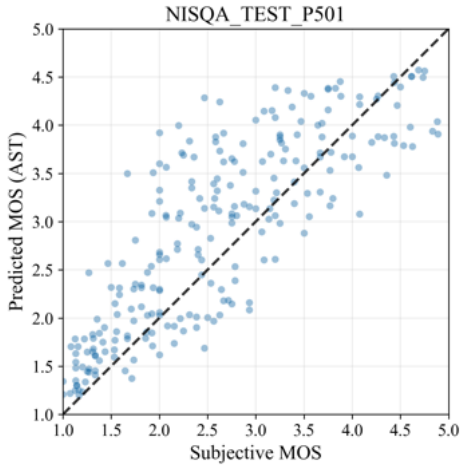
NISQA_TEST_P501



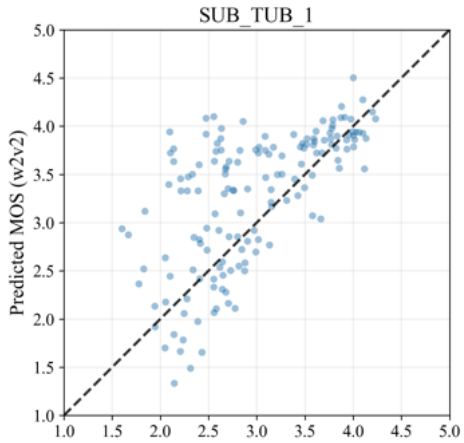


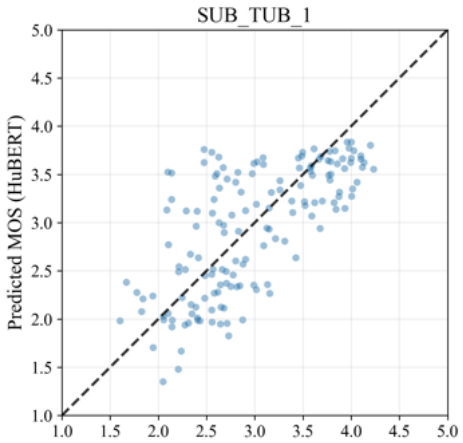


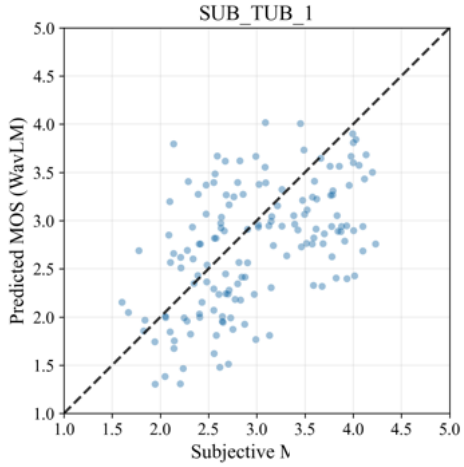


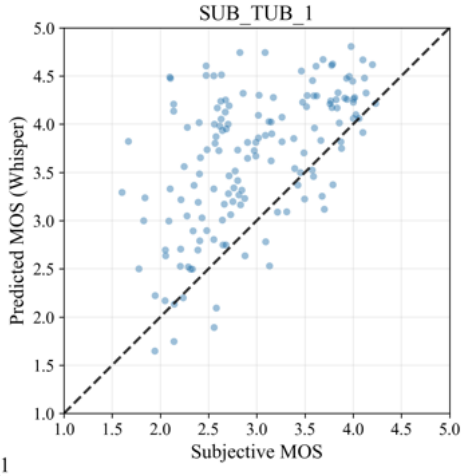


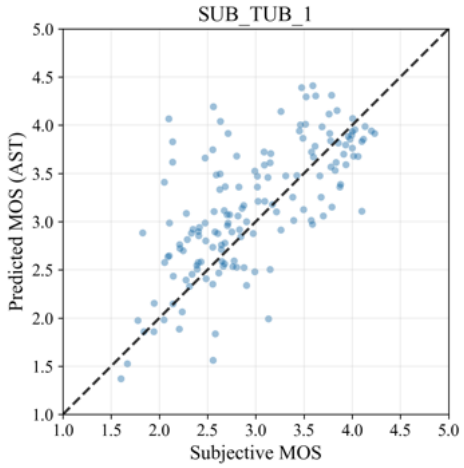
SUB_TUB_1


SUB_TUB_1
Predicted MOS (w2v2)
1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0


SUB_TUB_1
Predicted MOS (HuBERT)
1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0


SUB_TUB_1
Predicted MOS (WavLM)
1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0
Subjective M


SUB_TUB_1
Predicted MOS (Whisper)
1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0
Subjective MOS


SUB_TUB_1
Predicted MOS (AST)
1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0
Subjective MOS

Appendix C
Experimental Data Partitions

Table C.1: Databases used for training the multi-task AST and the single-task AST models ($C1$).

Training Databases for Multi-Task and Some Single-Task Data Split Configurations	
ENG_Handset	NB_48kHz_TNO_P862_NW_EMU
ENG_Headset	NB_48kHz_TNO_P862_NW_MEAS
ETSI_ENG_DB01	NISQA_TRAIN_LIVE
ETSI_ENG_DB02_PD	NISQA_TRAIN_SIM
ETSI_ENG_DB03_PD	Orange_C122_DB01
ETSI_GER_DB02_PD	Orange_C122_DB02
ETSI_GER_DB03	Orange_G711
ETSI_MAN_DB01_PD	Orange_NB_coders
ETSI_MAN_DB03_PD	Orange_Neural
FB_TUB_DEV	Orange_Teams_Conv
FB_TUB_SPO	SIM_CODEC
GER_Automotive	STORYTIME
GER_Headset	SUB_TUB_1
MAN_Automotive	SWB_48kHz101_ERICSSON
MAN_Headset	SWB_48kHz201_FT_DT
NB_48kHz_ATT_iLBC	SWB_48kHz202_FT_DT
NB_48kHz_BT_P862_BGN_ENG	SWB_48kHz301_OPTICOM
NB_48kHz_BT_P862_PROP	SWB_48kHz302_OPTICOM
NB_48kHz_DT_P862_1st	SWB_48kHz401_PSYTECHNICS
NB_48kHz_DT_P862_BGN_GER	SWB_48kHz501_SWISSQUAL
NB_48kHz_DT_P862_Share	SWB_48kHz502_SWISSQUAL
NB_48kHz_ERIC_AMR_4B	SWB_48kHz601_TNO
NB_48kHz_ERIC_FIELD_GSM_EU	SWB_48kHz602_TNO
NB_48kHz_ERIC_P862_NW_MEAS	SWB_48kHz_DT_SWB_Izabela
NB_48kHz_FT_P563_PROP	SWB_48kHz_GIPS_Exp4
NB_48kHz_GIPS_EXP1	SWB_48kHz_PAMD_Orange_1
NB_48kHz_HUAWEI_1	SWB_48kHz_PAMD_Swissqual_1
NB_48kHz_ITU_SUPPL23_EXP1a	SWB_48kHz_TCD
NB_48kHz_ITU_SUPPL23_EXP1d	SWB_TUB_DIS
NB_48kHz_ITU_SUPPL23_EXP3a	SWB_TUB_VUPL
NB_48kHz_ITU_SUPPL23_EXP3c	Vin1008
NB_48kHz_LUC_P563_PROP	Vin700
NB_48kHz_OPT_P563_PROP	WB_48kHz102_ERICSSON
NB_48kHz_PSY_P563_PROP	WB_48kHz204_FTDT
NB_48kHz_QUALCOMM_EXP1b	WB_48kHz402_PSYTECHNICS
NB_48kHz_QUALCOMM_EXP2b	WB_48kHz_DTAG_1

Continued on next page

Training Databases	
NB_48kHz_QUALCOMM_EXP3w	WB_48kHz_DTAG_2
NB_48kHz_QUALCOMM_EXP4	WB_48kHz_DT_HQ_Voice
NB_48kHz_QUALCOMM_EXP6a	WB_48kHz_DT_Mobisense
NB_48kHz_QUALCOMM_EXP6b	WB_48kHz_GIPS_Exp3
NB_48kHz_SQ_P563_PROP	WB_48kHz_QUALCOMM_EXP1w
NB_48kHz_TNO_P862_KPN_KIT97	WB_48kHz_QUALCOMM_EXP5

Table C.2: Databases used for validating the multi-task AST and the single-task AST models ($C2$).

Validation Databases for Multi-Task and Some Single-Task Data Regimes	
ENG_Automotive	NB_48kHz_ITU_SUPPL23_EXP3d
ETSI_GER_DB01_PD	NB_48kHz_ITU_SUPPL23_EXP3o
ETSI_MAN_DB02	NISQA_VAL_LIVE
FB_TUB_AUS	NISQA_VAL_SIM
MAN_Handset	SWB_48kHz103_ERICSSON
NB_48kHz104_Ericsson	SWB_48kHz203_FT_DT
NB_48kHz404_PSYTECHNICS	SWB_48kHz303_OPTICOM
NB_48kHz504_SWISSQUAL	SWB_48kHz403_PSYTECHNICS
NB_48kHz_ERIC_FIELD_GSM_US	SWB_48kHz503_SWISSQUAL
NB_48kHz_HUAWEI_2	SWB_48kHz603_TNO
NB_48kHz_ITU_SUPPL23_EXP1o	SWB_TUB_LIK

Table C.3: New databases used for training the single-task AST models, in addition to all the databases present in C.1.

Additional Training Databases		
TUB_2_DIS	TUB_2_LOUD	Vin_726

Table C.4: Databases used for validating the single-task AST models (model selection) in the $C4$ data split regime.

Validation Databases in the $C4$ Data Split Regime	
FB_TUB_AUS	Orange_C122_DB02
FB_TUB_DEV	SUB_TUB_1
FB_TUB_SPO	SWB_48kHz501_SWISSQUAL
NISQA_VAL_LIVE	SWB_48kHz502_SWISSQUAL
NISQA_VAL_SIM	SWB_48kHz503_SWISSQUAL
Orange_C122_DB01	SWB_TUB_DIS

Table C.5: Held-out databases used for testing the single-task AST models across all the training data split configurations.

Held-Out Databases Used to Test All Data Split Configurations		
NISQA_TEST_NSC	NISQA_TEST_P501	SUB_TUB_2
NISQA_TEST_FOR	NISQA_TEST_LIVETALK	

Appendix D
Multitask Evaluation Scatterplots

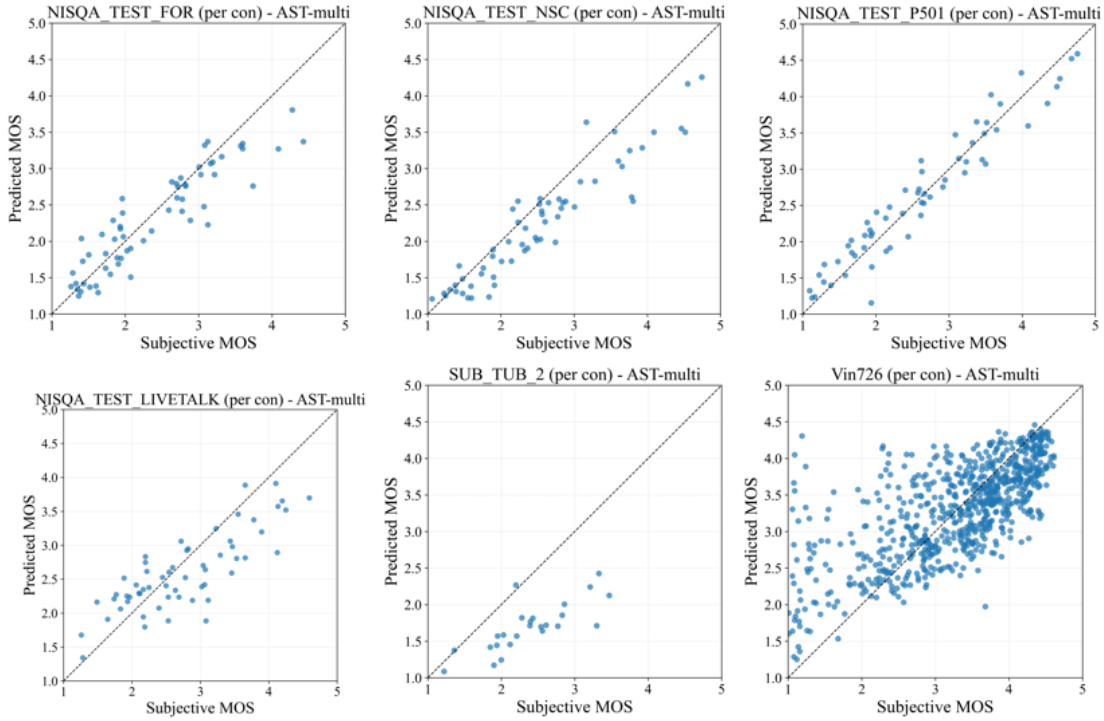


Fig. D.1: Condition-level scatter plots for overall quality (MOS) prediction. Note: Vin726 labels are obtained using POLQA and are not subjective.

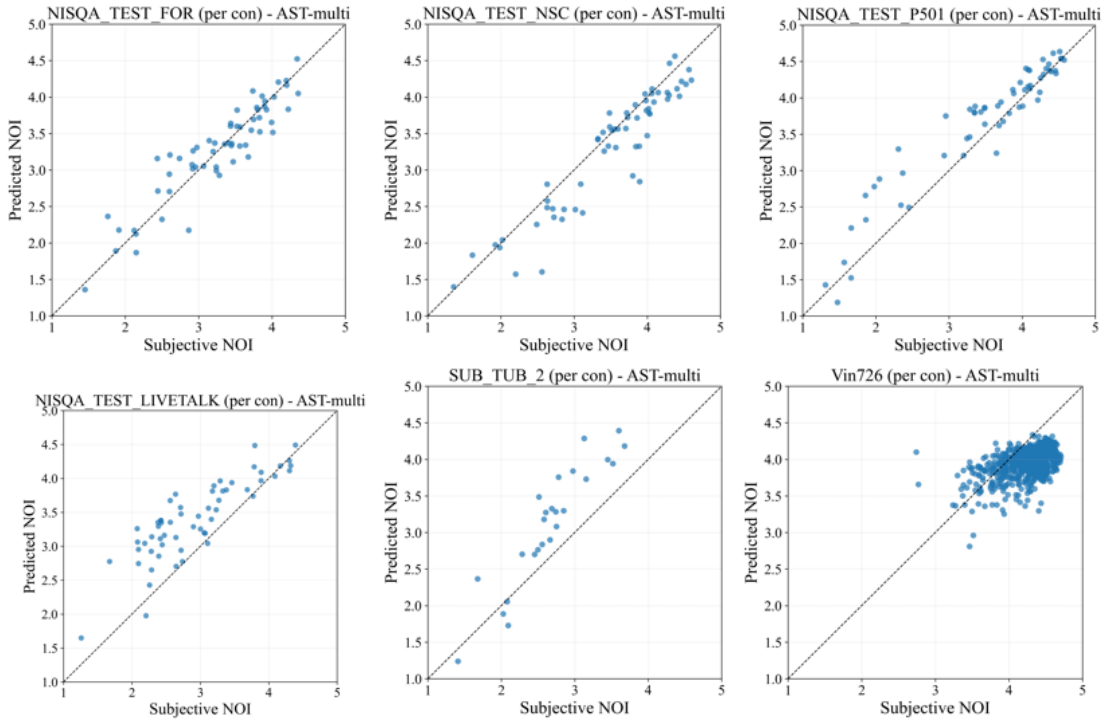


Fig. D.2: Condition-level scatter plots for noisiness (NOI) prediction. Note: Vin726 labels are obtained using NISQA and are not subjective.

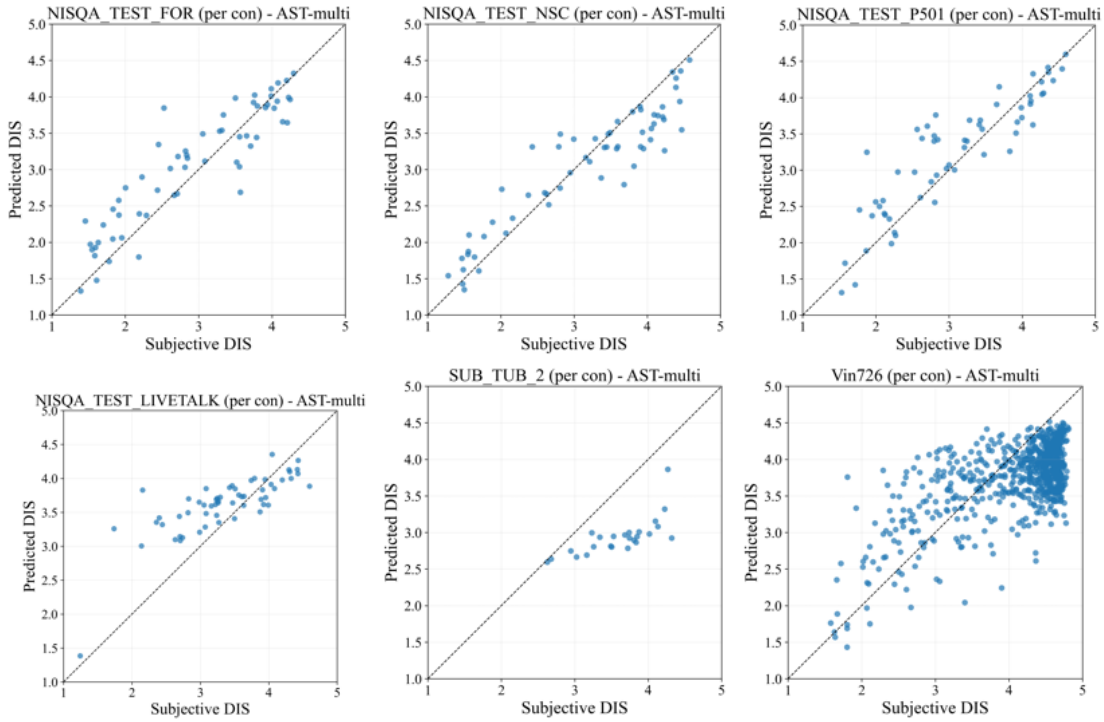


Fig. D.3: Condition-level scatter plots for discontinuity (DIS) prediction. Vin726 labels are obtained using NISQA and are not subjective.

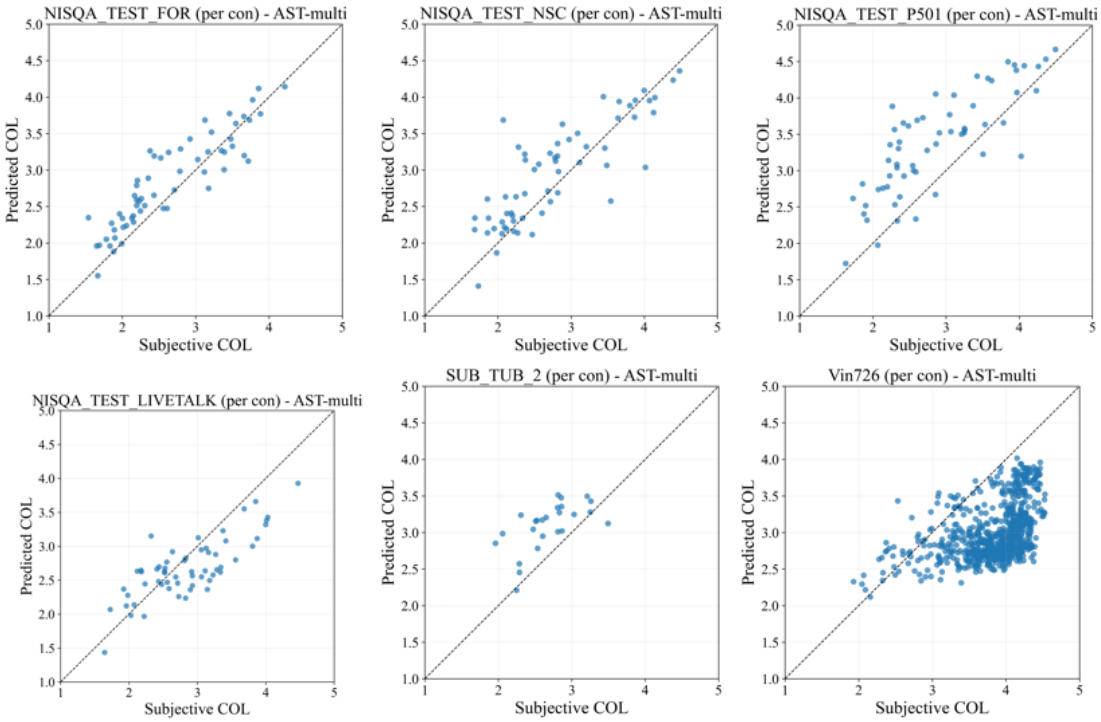


Fig. D.4: Condition-level scatter plots for coloration (COL) prediction. Vin726 labels are obtained using NISQA and are not subjective.

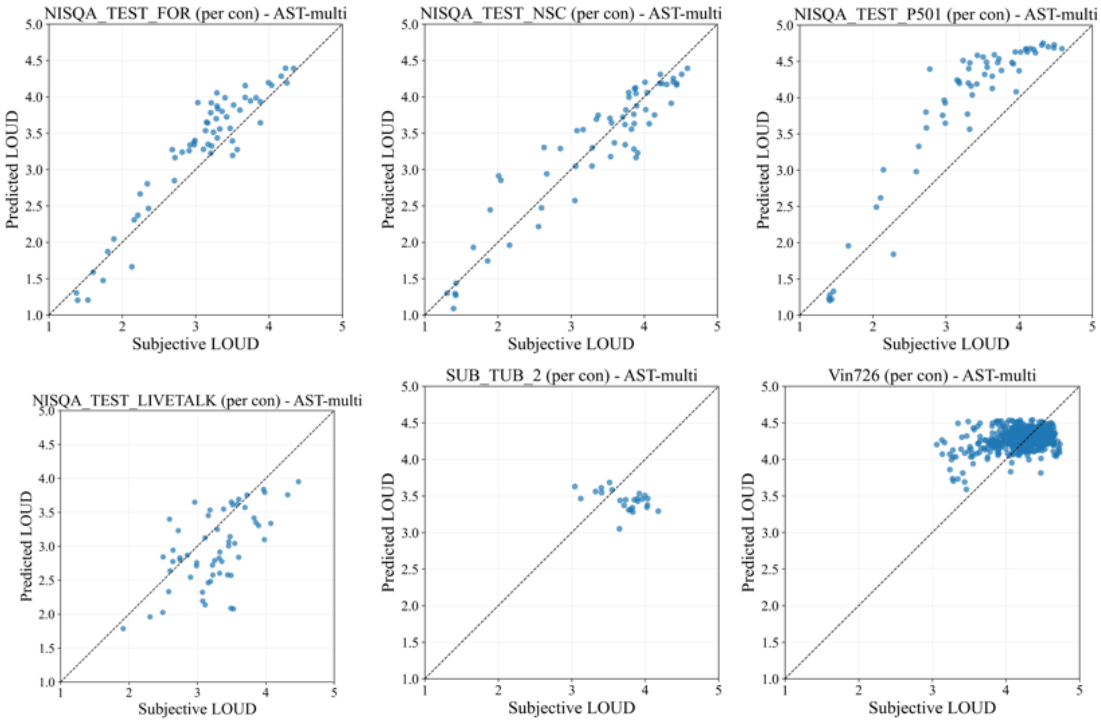


Fig. D.5: Condition-level scatter plots for loudness (LOUD) prediction. Vin726 labels are obtained using NISQA and are not subjective.

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